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1 CHAPTER CLIMATE CHANGE

1. GASES CONTRIBUTING TO GREENHOUSE EFFECT

- **Water vapour:** Water vapour increases as the Earth's atmosphere warms, but so does the possibility of clouds and precipitation, making these some of the most important feedback mechanisms to the greenhouse effect.
- **Carbon dioxide (CO_2):** A minor but very important component of the atmosphere, CO_2 is released through natural processes such as respiration and volcano eruptions and through human activities such as deforestation, land use changes, and burning fossil fuels. This is the most important long-lived "forcing" of climate change. Keeling Curve measures the concentration of CO_2 in the environment.
- **Methane:** A hydrocarbon gas produced both through natural sources and human activities, including the decomposition of wastes in landfills, agriculture, and especially rice cultivation, as well as ruminant digestion and manure management associated with domestic livestock. On a molecule-for-molecule basis, methane is a far more active greenhouse gas than carbon dioxide, but also one which is much less abundant in the atmosphere.
- **Nitrous oxide:** A greenhouse gas produced by soil cultivation practices, especially use of commercial & organic fertilizers, fossil fuel combustion, nitric acid production, and biomass burning.
- **Chlorofluorocarbons (CFCs):** Synthetic compounds entirely of industrial origin used in several applications, but now largely regulated in production and release to the atmosphere by international agreement for their ability to contribute to destruction of the ozone layer. They are also greenhouse gases.
- **Ozone:** Triatomic form of oxygen, and a gaseous atmospheric constituent. In the troposphere, O_3 is created both naturally and by photochemical reactions involving gases resulting from human activities (E.g., smog). Tropospheric O_3 acts as a greenhouse gas (GHG). In the stratosphere, O_3 is created by the interaction between solar ultraviolet radiation and molecular oxygen (O_2). Stratospheric O_3 plays a dominant role in the stratospheric radiative balance. Its concentration is highest in the ozone layer.

GREENHOUSE GAS	GWP	SOURCES AND CAUSES
Carbon dioxide (CO_2)	1	Burning of fossil fuels, deforestation
Methane (CH_4)	12	Growing paddy, excreta of cattle and other livestock, termites, burning of fossil fuel, wood, landfills, wetlands, fertilizer factories.
Nitrous oxides (N_2O)	265	Burning of fossil fuels, fertilizers; burning of wood and crop residue.
Perfluorocarbons (PFCs)	6500	Produced as a by-product in aluminum production and manufacturing of semiconductors.
Hydrofluorocarbons (HFCs)	12400	Used as refrigerants, aerosol propellants, solvents and fire retardants.
Sulphur hexafluoride (SF6)	23500	Used as a tracer gas for leak detection, used in electrical transmission equipment

2. CONCEPTS RELATED TO CLIMATE CHANGE

Tipping Point	A level of change in system properties beyond which a system reorganises, often abruptly, and does not return to the initial state even if the drivers of change are abated. For the climate system, it refers to a critical threshold when global or regional climate changes from one stable state to another stable state.
Net Zero Emissions	Net zero emissions are achieved when anthropogenic emissions of greenhouse gases to the atmosphere are balanced by anthropogenic removals over a specified period. Where multiple greenhouse gases are involved, the quantification of net zero emissions depends on the climate metric chosen to compare emissions of different gases (such as global warming potential, global temperature change potential, and others, as well as the chosen time horizon).
Pathways	It refers to temporal evolution of natural and/or human systems towards a future state. Pathway concepts range from sets of quantitative and qualitative scenarios or narratives of potential futures to solution-oriented decision-making processes to achieve desirable societal goals. 1.5°C pathway: A pathway of greenhouse gas emissions that provides an approximately 1/2 or 2/3rd chance of global warming either remaining below 1.5°C or returning to 1.5°C by around 2100 following an overshoot.
Biophilic urbanism	Designing cities with green roofs, green walls and green balconies to bring nature into the densest parts of cities to provide green infrastructure and human health benefits.
Coping Capacity	The ability of people, institutions, organizations, and systems, using available skills, values, beliefs, resources, and opportunities, to address, manage, and overcome adverse conditions in the short to medium term.
Decarbonisation	The process by which countries, individuals or other entities aim to achieve zero fossil carbon existence. Typically refers to a reduction of the carbon emissions associated with electricity, industry and transport.
Decoupling	Decoupling (for climate change) is where economic growth is no longer strongly associated with consumption of fossil fuels. Relative decoupling is where both grow but at different rates. Absolute decoupling is where economic growth happens but fossil fuels decline.
Co-benefits	The positive effects that a policy or measure aimed at one objective might have on other objectives, thereby increasing the total benefits for society or the environment. Co-benefits are often subject to uncertainty and depend on local circumstances and implementation practices, among other factors. Co-benefits are also referred to as ancillary benefits.
Blue Carbon	Biologically driven carbon fluxes and storage in marine systems that are amenable to management. Coastal blue carbon focuses on rooted vegetation in coastal zones, such as tidal marshes, mangroves and seagrasses. These ecosystems have high carbon burial rates on a per unit area basis and accumulate carbon in their soils and sediments. They provide many non-climatic benefits and can contribute to ecosystem-based adaptation. If degraded or lost, coastal blue carbon ecosystems are likely to release most of their carbon back into the atmosphere.
Total Carbon Budget	Maximum amount of cumulative net global anthropogenic CO_2 emissions that would result in limiting global warming to a given level with a given probability, considering the effect of other anthropogenic climate forces. This is referred to as the Total Carbon

	Budget when expressed starting from pre-industrial period, and as the Remaining Carbon Budget when expressed from a recent specified date.
Carbon Footprint	Measure of the exclusive total amount of emissions of CO_2 that is directly and indirectly caused by an activity or is accumulated over the lifecycle stages of a product. **Household Carbon Footprint:** Carbon footprint of an individual household, inclusive of the direct and indirect CO_2 emissions associated with home energy use, transportation, food provision and consumption of other goods and services associated with household expenditures.
Carbon neutrality	It is a condition in which anthropogenic CO_2 emissions associated with a subject are balanced by anthropogenic CO_2 removals. The subject can be an entity such as a country, an organisation, a district or a commodity, or an activity such as a service and an event. Carbon neutrality is often assessed over a life cycle including indirect emissions but can also be limited to the emissions and removals, over a specified period.
Fugitive emissions	The release of greenhouse gases that occur during the exploration, processing and delivery of fossil fuels to the point of final use. This excludes greenhouse gas emissions from fuel combustion to produce useful heat or power. It encompasses venting, flaring and leaks.
Offsets	The reduction, avoidance or removal of a unit of greenhouse gases emissions by one entity, purchased by another entity to counterbalance a unit of GHG emissions by that other entity. Offsets are commonly subject to rules and environmental integrity criteria intended to ensure that offsets achieve their stated mitigation outcome. Criteria include avoidance of double counting and leakage, use of appropriate baselines, additionality and permanence or measures to address impermanence.
Carbon Leakage & Carbon Border Adjustment Mechanism (CBAM)	Carbon leakage refers to the situation that may occur if, for reasons of costs related to climate policies, businesses were to transfer production to other countries with laxer emission constraints. This could lead to an increase in their total emissions. The risk of carbon leakage may be higher in certain energy-intensive industries. (Trade and Climate Change). To deal with this risk of carbon leakage, the EU as part of the European Green Deal is implementing the Carbon Border Adjustment Mechanism, which would prevent the risk of carbon leakage and support the EU's increased ambition on climate mitigation, while ensuring WTO compatibility. Functioning of CBAM: EU importers will buy carbon certificates corresponding to the carbon price that would have been paid, had the goods been produced under the EU's carbon pricing rules. Conversely, once a non-EU producer can show that they have already paid a price for the carbon used in the production of the imported goods in a third country, the corresponding cost can be fully deducted from the EU importer. The CBAM will help reduce the risk of carbon leakage by encouraging producers in non-EU countries to green their production processes.

3. UNITED NATIONS FRAMEWORK CONVENTION ON CLIMATE CHANGE (UNFCCC)

UNFCCC entered into force in 1994. Today, it has near-universal membership. The 197 countries that have ratified the Convention are called Parties to the Convention. USA has re-joined the UNFCCC after President Biden took over.

Preventing “dangerous” human interference with the climate system is the ultimate aim of the UNFCCC.

The ultimate objective of the Convention is to stabilize greenhouse gas concentrations "at a level that would prevent dangerous anthropogenic (human-induced) interference with the climate system."

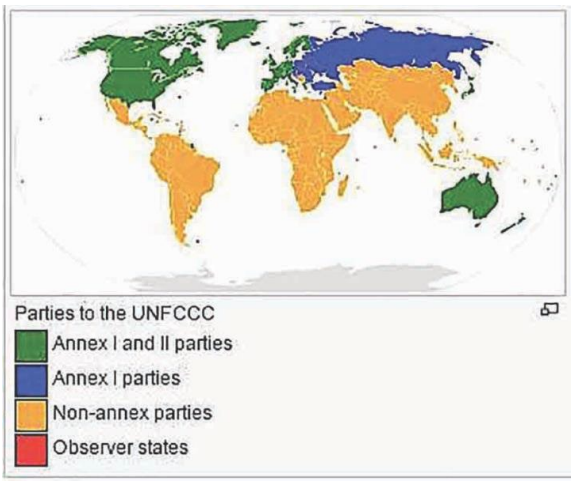
PARTIES TO UNFCCC ARE CLASSIFIED AS:

- **Annex I countries:** Industrialized countries and economies in transition
- **Annex II countries:** Developed countries which pay for the costs of developing countries. Annex II countries are a sub-group of the Annex I countries.
- **Non-Annex I countries:** Developing countries are not required to reduce emission levels unless developed countries supply enough funding and technology.
- **Setting no immediate restrictions under UNFCCC serves these purposes:**
  i. It avoids restrictions on their development because emissions are strongly linked to industrial capacity.
  ii. They can sell emissions credits to nations whose operators have difficulty meeting their emissions targets.
  iii. They get money and technologies for low-carbon investments from Annex II countries.
  iv. Developing countries may volunteer to become Annex I countries when they are sufficiently developed.
  v. India is a non-Annex party to UNFCCC.

GOVERNING BODIES & PROCESS MANAGEMENT BODY

CONFERENCE OF THE PARTIES (COP)

COP is the supreme decision-making body of the Convention. All States that are Parties to the Convention are represented at the COP, at which they review the implementation of the Convention and any other legal instruments that the COP adopts and take decisions necessary to promote the effective implementation of the Convention, including institutional and administrative arrangements.



CONFERENCE OF PARTIES TO KYOTO PROTOCOL (CMP)

Conference of Parties, the supreme body of the Convention, shall serve as a meeting of Parties to the Kyoto Protocol. All States that are Parties to the Kyoto Protocol are represented at the Conference of the Parties serving as the meeting of the Parties to the Kyoto Protocol (CMP), while States that are not Parties participate as observers. The CMP oversees the implementation of the Kyoto Protocol and takes decisions to promote its effective implementation.

CONFERENCE OF PARTIES TO PARIS AGREEMENT (CMA)

Conference of the Parties, supreme body of Convention, shall serve as the meeting of the Parties to the Paris Agreement. All States that are Parties to the Paris Agreement are represented at the Conference of the Parties serving as the meeting of the Parties to the Paris Agreement (CMA), while States that are not Parties participate as observers. The CMA oversees the implementation of the Paris Agreement and takes decisions to promote its effective implementation.

SECRETARIAT

UNFCCC secretariat provides organizational support and technical expertise to the UNFCCC negotiations and institutions and facilitates the flow of authoritative information on the implementation of the Convention, the Kyoto Protocol and the Paris Agreement. This includes the development and effective implementation of innovative approaches to mitigate climate change and drive sustainable development.

CONSTITUTED BODIES

Adaptation Committee (AC): The Adaptation Committee was established by the COP at its sixteenth session as part of the Cancun Agreements to promote the implementation of enhanced action on adaptation coherently under the Convention. The Adaptation Committee also serves the Paris Agreement.

Adaptation Fund Board (AFB): AFB supervises and manages the Adaptation Fund and is fully accountable to the CMP. The Adaptation Fund was established to finance concrete adaptation projects and programs in developing country Parties that are particularly vulnerable to the adverse effects of climate change. The Adaptation Fund is financed by a 2 per cent share of the proceeds from certified emission reductions issued by the Executive Board of the Clean Development Mechanism and from other sources of funding. The Adaptation Fund also serves the Paris Agreement.

Advisory Board of the Climate Technology Centre and Network (CTCN): As the operational arm of the Technology Mechanism, the CTCN stimulates technology cooperation to enhance the development and transfer of technologies and to assist developing country Parties at their request. The Advisory Board gives guidance to the CTCN on how to prioritize requests from developing countries and, in general, it monitors, assesses and evaluates the performance of the CTCN.

CLEAN DEVELOPMENT MECHANISM

Defined in Article 12 of Kyoto Protocol, allows a country with an emission-reduction or emission-limitation commitment under the Kyoto Protocol (Annex B Party) to implement an emission-reduction project in developing countries.

Such projects can earn saleable certified emission reduction (CER) credits, each equivalent to one ton of CO_2, which can be counted towards meeting Kyoto targets.

COMPLIANCE COMMITTEE

The functions of the Compliance Committee of the Kyoto Protocol are to provide advice and assistance to Parties in implementing the Kyoto Protocol, promote compliance by Parties with their commitments and determine cases of non-compliance and apply consequences in cases where Parties are not complying with their commitments under the Kyoto Protocol.

EXECUTIVE COMMITTEE OF WARSAW INTERNATIONAL MECHANISM FOR LOSS & DAMAGE

Established by the COP to guide the implementation of the functions of the Warsaw International Mechanism for Loss and Damage. The Warsaw International Mechanism is anchored in the Paris Agreement by its Article 8.

WARSAW INTERNATIONAL MECHANISM FOR LOSS AND DAMAGE

The COP established the Warsaw International Mechanism for Loss and Damage associated with Climate Change Impacts (Loss and Damage Mechanism), to address loss and damage associated with impacts of climate change, including extreme events and slow onset events, in developing countries that are particularly vulnerable to the adverse effects of climate change at COP19 (November 2013) in Warsaw, Poland.



GLOBAL ENVIRONMENT FACILITY (GEF)

It was established on the eve of the 1992 Rio Earth Summit.

The GEF is a unique partnership of 18 agencies — including United Nations agencies, multilateral development banks, national entities and international NGOs — working with 184 countries to address the world's most challenging environmental issues.

The GEF has a large network of civil society organizations, works closely with the private sector around the world, and receives continuous inputs from an independent evaluation office and a world-class scientific panel.

It is a financial mechanism for five major international environmental conventions: the Minamata Convention on Mercury, the Stockholm Convention on Persistent Organic Pollutants (POPs), the United Nations Convention on Biological Diversity (UNCBD), the United Nations Convention to Combat Desertification (UNCCD) and the United Nations Framework Convention on Climate Change (UNFCCC).

The GEF Trust Fund was established to help tackle our planet's most pressing environmental problems. Funds are available to developing countries and countries with economies in transition to meet the objectives of the international environmental conventions and agreements.

The World Bank serves as the GEF Trustee, administering the GEF Trust Fund (contributions by donors). The Trustee helps mobilize GEF resources; disburses funds to GEF Agencies; prepares financial reports on investments and use of resources; and monitors application of budgetary and project funds. The Trustee creates periodic reports that contain an array of fund-specific financial information.

GREEN CLIMATE FUND (GCF)

GCF is an operating entity of the financial mechanism of the Convention and is accountable to and functions under the guidance of the COP. It is governed by a Board comprising 24 members **(with equal numbers from developed and developing country Parties)** and is intended to be the main fund for global climate change finance in the context of mobilizing USD 100 billion by 2020. The GCF, as an entity entrusted with the operation of the Financial Mechanism of the Convention, also serves the Paris Agreement.

SPECIAL CLIMATE CHANGE FUND

It was established to finance activities, programs and measures relating to climate change, that are complementary to those supported by other funding mechanisms for the implementation of the Convention. The Global Environment Facility (GEF) has been entrusted to operate the SCCF. The SCCF, administered by the GEF, also serves the Paris Agreement.

LEAST DEVELOPED COUNTRIES FUND

The COP established the Least Developed Countries Fund (LDCF) to support the Least Developed Country Parties (LDCs) work program and assist LDCs carry out, inter alia, the preparation and implementation of national adaptation programs of action (NAPAs). The Global Environment Facility (GEF) has been entrusted to operate the LDCF. The LDCF, administered by the GEF, also serves the Paris Agreement.

4. COP 28 OF UNFCCC

- **High level champions:** refer to individuals appointed to mobilize climate action across various stakeholder groups. The High-Level Champions play a crucial role in engaging governments, businesses, cities, and other actors to enhance ambition and implement climate-related initiatives. There are typically two High-Level Champions, one representing a developed country and another representing a developing country. The importance of engaging non-Party stakeholders (businesses, cities, subnational regions, investors, and civil society) in the multilateral process under the United Nations Framework Convention on Climate Change (UNFCCC) has been long recognized by Parties; dating back to the Lima-Paris Action Agenda launched at the 20th session of the Conference of Parties (COP 20), the **appointment of the High-Level Champions since COP 21**, and the Marrakech Partnership for Global Climate Action launched by the High-Level Champions at COP 22. The High-Level Champions and the Marrakech Partnership have led several outcomes on Global Climate Action at COP 27.
- **Fast-tracking a just, orderly, and equitable energy transition.**
- **Global Renewables and Energy Efficiency Pledge launched.**

- To triple the world's installed renewable energy generation capacity to at least 11,000 GW by 2030.
- To collectively double the global average annual rate of energy efficiency improvements from around 2% to over 4% every year until 2030.
- India is not a signatory to this as of now. Not legally binding pledge.

India's Ministry of New and Renewable Energy said that the country was the world's third-largest producer of renewable energy, with 40% of its installed energy capacity coming from non-fossil fuel sources. India has announced its aim to meet 50% of its electricity demands from renewable energy sources by 2030, and reach net zero emissions by 2070.

- **Utilities for net zero alliance** launched by IRENA along with High level champions.
- **'Dubai Consensus':** Transitioning away from fossil fuels to achieve net zero by 2050. First such pledge by countries. In Dubai consensus all fossil fuels have been included but no time limits yet. But it cannot be immediately stopped hence 'transition fuels' are required. (But what are transition fuels is not clear! Generally natural gas is one of them).
- **Phase-down of coal:** Despite being a fossil fuel, just like oil or natural gas, coal has received a separate mention in the agreement. This is because coal was already singled out for phase-down in the Glasgow conference in 2021. There was a move to stipulate that no new coal fired power plants could be opened without an in-built carbon capture and storage facility, but this was strongly resisted by India, China, South Africa and other countries. It was dropped, and finally the Glasgow language was reiterated. There is nothing about how this phase-down is to be measured, or from what baseline.
- **Methane emission cuts:** Methane is the most widespread greenhouse gas apart from CO2, accounting for nearly 25 per cent of all emissions. But several countries, including India, are extremely opposed to any mandate to cut methane emissions, mainly because one of the major sources happens to be agriculture and livestock. The agreement does not mention any targets for methane emission cuts for the year 2030, although a group of about 100 countries had made a voluntary commitment, in Glasgow in 2021, to reduce their methane emissions by 30% by 2030.
- **Global cooling pledge**: to reduce cooling related emissions by 68% from today by 2050, significantly increase access to sustainable cooling by 2030, and increase the global average efficiency of new air conditioners by 50%.
- **Coal transition accelerator:** France, in collaboration with various countries and organizations, introduced the Coal Transition Accelerator. Objectives include knowledge-sharing, policy design, and financial support to facilitate just transitions from coal to clean energy.
- **Declaration to triple nuclear energy**: to work together to advance a global aspirational goal of tripling nuclear energy capacity from 2020 by 2050.

- **Fixing climate finance**
  - **Nature Solutions Hub**: for Asia and the Pacific, launched by the Asia Development Bank, to proactively scale up the flow of public and private finance into conserving nature and biodiversity loss in Asia and the Pacific.
  - **Loss & Damage fund:** Operationalization + capitalization. $750 mn committed to the fund. World Bank = interim host of the fund for four years!
  - **New collective quantified goals:** UNCTAD estimates that wealthy nations owe developing countries USD 500 billion in 2025 under the New Collective Quantified Goal (NCQG) for climate finance. The NCQG was confirmed by developed countries under the Paris Agreement in 2015. The goal is to set a new collective quantified goal before 2025. The goal will start from a floor of USD 100 billion per year. This includes USD 250 billion for mitigation, USD 100 billion for adaptation, and USD 150 billion for loss and damage. The figure is expected to increase to USD 1.55 trillion by 2030.
- **Focusing on people, lives and livelihoods**
  - **UAE Declaration on Climate Relief, Recovery and Peace.** 78 national governments (including the EU) and 40 organizations. (India not as a government but among organizations; All India Disaster Mitigation Institute).

- **Climate & health initiative:** Led by the Asian Development Bank and India with the support of Brazil, the initiative aims to consolidate policies and practices at the nexus of climate change and health through collaboration with governments, international organizations, academia, and the private sector.
- **Coalition for High Ambition Multilevel Partnerships (CHAMP) for Climate Action**: Support from philanthropies. 78 National govt signed it but no India.

INDIA LED INITIATIVES

- **Global River Cities Alliance (GRCA)**, led by the National Mission for Clean Ganga (NMCG) under the Ministry of Jal Shakti, Government of India, was launched at COP28.
  - Unique alliance covering 275+ global river-cities in 11 countries (India, Egypt, Netherlands, Denmark, Ghana, Australia, Bhutan, Cambodia, Japan, Netherlands, and Hungary).
  - International funding agencies the World Bank, Asian Development Bank (ADB), Asian Infrastructure Investment Bank (AIIB)
  - First of its kind in the world. Sustainable river centric development.
  - In 2021, the River Cities Alliance (RCA) was launched as a dedicated platform for river cities in India to ideate, discuss and exchange information for sustainable management of urban rivers.
  - Headquarters in India.
- **Green Credit Initiative**:
  - It envisions the issue of Green Credits for plantations on waste/degraded lands and river catchment areas, to rejuvenate and revive natural eco-systems.
  - Fully digital and tradable instrument for environmental sustainability.

GREEN CREDIT PROGRAMME

- Initiative within the government's broader Lifestyle for Environment or LIFE Movement.
- Market-based instrument to encourage voluntary environmental actions across various sectors, involving individuals, industries, and local bodies.
- It utilizes green credits as distinctive units assigned to specific environmental activities which function as tradable commodities, which can be sold on a domestic market platform.
- While the domestic carbon market focuses solely on CO2 emission reduction, the Green Credit System aims to meet other environmental obligations as well, incentivizing sustainable actions by companies, individuals, and local bodies. Those earning it will be able to put these credits up for sale on a proposed domestic market platform.
- The program identifies eight specific activities: tree plantation, water conservation, sustainable agriculture, waste management, air pollution reduction, mangrove conservation and restoration, Eco Mark (a government scheme for identifying environmentally friendly products), and sustainable building and infrastructure.
- The administration of the Green Credit Program is entrusted to the Indian Council of Forestry Research and Education.

Global Stocktake: The main agenda at COP28 was to carry out a Global Stocktake (GST), a comprehensive assessment of where the world was in its fight against climate change and what more needed to be done to meet the climate objectives. The GST is mandated by the Paris Agreement to be a periodic exercise, the first one in 2023 and every five years thereafter.

5. COP 29 OF UNFCCC

The 29th Conference of the Parties of the UNFCCC (United Nations Climate Change Conference) (COP29) was held in **Baku, Azerbaijan** in November 2024.

KEY OUTCOMES

- **Climate finance:** Governments reached agreement on financing for developing countries, firstly a 'goal' of USD 300 billion by 2035 (New Collective Quantified Goal; NCQG) which represents a tripling of the last goal agreed on at the Copenhagen COP in 2009, and a second wider, 'aspiration' of USD 1.3 trillion to be mobilised by 2035.
- **Carbon markets:** Rules pertaining to international, high-integrity government-backed carbon markets, after a decade of negotiations, were finalized. **COP 29 approved two pathways** for countries and companies to trade carbon offsets.
  - **Article 6.2**: Direct country-to-country trading (bilaterally trade).
  - **Article 6.4**: Separate UN-backed marketplace.
- Countries agreed standards for a centralised carbon market under the UN mechanism. (Article 6.4).
- **The critical role of REDD+** was recognised through a £3 million pledge by the UK International Forest Unit to support UN Climate Change's work over four years. This funding will bolster REDD+ activities in many countries, enabling the secretariat to create dedicated spaces for REDD+ experts to engage in technical dialogue.
- **Global energy storage and grids pledge** was launched: The pledge commits signatories to commit to a collective goal of deploying 1,500 GW of energy storage globally by 2030.
- **Launch of green energy pledge:** The pledge focuses on promoting connecting green energy zones and corridors to the communities most in need by developing larger intraregional and interregional interconnected power grids.
- **Global Renewable Alliance:** GRA called on governments to address critical trade barriers that limit the expansion of renewable energy markets and challenge meeting the goal of tripling renewable energy capacity by 2030.
- **COP29 Hydrogen Declaration**: Endorsers commit to scale up renewable, clean/zero-emission and low-carbon hydrogen production and accelerate the decarbonisation of existing hydrogen production from unabated fossil fuels. It aims to significantly increase green hydrogen production from one million tonnes annually today and reduce the 96 Mt of hydrogen currently produced from unabated fossil fuels.
- **Global Methane Pledge (GMP):** Initially launched at COP26, the pledge aims to cut methane emissions by 30% by 2030 and pledged 500 million dollars for methane reduction. India has not signed the Global Methane Pledge
- **Baku Harmoniya Climate Initiative:** Launched in collaboration with FAO, this initiative aims to bring together other initiatives, coalitions and networks to share experiences, identify synergies and gaps, facilitate finance, and foster collaboration on addressing climate change through adaptation and mitigation in agriculture, including by empowering communities and women in rural areas.
- **Indigenous Peoples and Local Communities:** COP29 adopted the Baku Work plan to elevate voices of indigenous peoples and local communities in climate action. It renewed the mandate of the Facilitative Working Group (FWG) of the Local Communities and Indigenous Peoples Platform (LCIPP). FWG was established at COP24 in Katowice to further operationalize LCIPP. Implementation of the Workplan will begin in 2025 with 3 Focus areas: knowledge exchange; Capacity building for engagement; integrating diverse values and knowledge systems into climate policies.
- **Lima Work programme** which emphasised on gender mainstreaming in climate actions, was extended by COP 29. It was launched in 2014.
- **1.5° Partnership for Action on Climate Transition (PACT) Programme**: launched by United Nations Development Program (UNDP)'s Financial Centres for Sustainability. Foster stronger collaboration between the private sector and governments for achieving the 1.5°C-aligned Nationally Determined Contributions (NDCs) by 2025. o Mobilizing the financial sector and support

INDIAN INITIATIVES AT COP 29

India collaborated with various agencies to organise side events on several aspects of climate action during the CoP29 UN Climate Change Conference. The important side events include the following:

- **Integrating Disaster Resilient Infrastructure into the Adaptation Strategies:** Organised by Govt. of India and Coalition for Disaster Resilient Infrastructure (CDRI).

- **LeadIT (Leadership Group for Industry Transition) Member Meet:** Organised by Govt. of India (MoEFCC), Govt. of Sweden, with support from LeadIT Secretariat. The summit focuses on de-carbonisation of heavy industry. India-Sweden Industry Transition Partnership (ITP) as part of the Lead-IT initiative was launched at COP28.
- **SIDS Adaptation Finance**: India has advocated finance unlocking and disaster-resilient support for Sustainable Infrastructure in Small Island Developing (SIDS).

6. NEW COLLECTIVE QUANTIFIED GOAL (NCQG) ON CLIMATE FINANCE

- By decision, Parties decided that, prior to 2025, the Conference of the Parties serving as the meeting of the Parties to the Paris Agreement (CMA) shall set a new collective quantified goal (NCQG) from a floor of USD 100 billion per year, taking into account the needs and priorities of developing countries.
- A key accompanying decision to the 2015 Paris Agreement stipulated setting a new collective quantified goal (NCQG) for climate finance prior to 2025.
- This new goal is to be built on the foundation of the US$100 billion per year by 2020 commitment and must consider the needs and priorities of developing countries.
- Delegates at COP26 in Glasgow launched an ad hoc work program, which will run until COP29 in 2024, to initiate the deliberations of the NCQG.

7. SANTIAGO NETWORK

Santiago Network is to catalyze the technical assistance of relevant organizations, bodies, networks and experts, for the implementation of relevant approaches for averting, minimizing and addressing L&D at the local, national and regional level, in developing countries that are particularly vulnerable to the adverse effects of climate change. The Santiago Network will connect vulnerable developing countries with providers of technical assistance, knowledge, resources they need to address climate risks comprehensively in the context of averting, minimizing and addressing loss and damage.

TIMELINE

- COP 25 (2019- Madrid): Parties established the Santiago network as part of the WIM (Warsaw international mechanism).
- COP 26 (2021- Glasgow): Parties decided on the functions of the Santiago Network and issued a call for submissions.
- COP 27 (2022- Sharm El-Sheikh, Egypt): Parties agreed on the institutional arrangements to operationalize the Santiago Network for Loss and Damage.

8. CLIMATE TECHNOLOGY CENTRE AND NETWORK (CTCN)

- CTCN is the implementation arm of the Technology Mechanism of the United Nations Framework Convention on Climate Change (UNFCCC) and is hosted by the UNEP in collaboration with UNIDO.
- Arrangements were agreed by the UNFCCC and UNEP governing council in early 2013.
- It is in Copenhagen, Denmark.
- The Centre promotes the accelerated transfer of environmentally sound technologies for low carbon and climate resilient development at the request of developing countries.
- The CTC provides technology solutions, capacity building and advice on policy, legal and regulatory frameworks tailored to the needs of individual countries by harnessing the expertise of a global network of technology companies and institutions.

COP 27 welcomed the first joint work program of the Technology Executive Committee and the Climate Technology Centre and Network, for 2023–2027, and decided that the main challenges identified therein should be considered under the global stock take.

Technology Executive Committee (TEC) and Climate Technology Centre and Network (CTCN) – the two bodies of the Technology Mechanism under the UNFCCC and Paris Agreement – launched their joint work program (5-year work program) to accelerate the deployment of transformative climate technologies that are urgently required to tackle climate change. The new joint work program of the Technology Mechanism covers work from 2023-2027.

9. JUST ENERGY TRANSITION PARTNERSHIP

- JETP is a mechanism evolved by the G7 under which advanced developed countries enter into agreement with developing countries currently overtly dependent on coal power. Countries which enter into JETP will decarbonize their power sector by phasing out of coal power and support will be given for boosting renewable energy. Financial support will be given to the countries in the form of grants, loans from the partner countries and also from multilateral development banks.
- South Africa was the first country to enter into the Just Energy Transition Partnership with the International Partners Group of nations comprising the USA, EU, France, Germany and UK.
- Indonesia was the second country to enter into the Just Energy Transition Partnership with the International Partners Group of nations comprising of USA, Japan, Canada, Denmark, EU, France, Germany, Italy, Norway and UK.
- Vietnam has also entered JETP.
- India has been invited to become a member of Just Energy Transition Partnership. However, India has declined.

10. GLOBAL SHIELD FINANCING FACILITY

- The World Bank has launched a Global Shield Financing Facility (GSFF) to help developing countries access more financing for recovery from natural disasters and climate shocks.
- This facility will support the Global Shield Against Climate Risks, a joint initiative of G7 and V20 to better protect poor and vulnerable people from disasters by pre-arranging more financing before disasters strike.
- GSFF is designed as a disaster risk finance and insurance solution for low-income countries for helping developing countries adapt to natural disasters.
- Channel grants to developing countries through World Bank projects or through projects prepared by other participating partners, including UN agencies and multilateral development banks.
- Finance integrated financial protection packages that offer coordinated and consolidated financial support to those vulnerable to climate shocks and disasters.

11. CLIMATE DAMAGE FUND

The US President Trump has withdrawn from the **United Nations Climate Damage Fund** (Fund for Responding to Loss and Damage- FRLD), which is aimed at providing financial support to countries ravaged by climate change disasters. The US has already pulled out of the **Paris Agreement**, stopped the participation of its scientists in a meeting of the **Intergovernmental Panel on Climate Change (IPCC)**, and cancelled previously committed money for the **Green Climate Fund.**

UN CLIMATE DAMAGE FUND (FUND FOR RESPONDING TO LOSS AND DAMAGE- FRLD)

Background:

- Established under **COP 27 (2022) in Sharm El-Sheikh, Egypt** and further operationalised at **COP 28 (2023) in Dubai**, **UAE.**

- It is created in response to long-standing demands from low income developing nations for **climate justice and financial assistance**.
- It recognises the disproportionate impact of climate change on **developing and vulnerable nations**, particularly Small Island Developing States (SIDS) and Least Developed Countries (LDCs).
- It forms part of the broader **climate finance architecture** under the **United Nations Framework Convention on Climate Change (UNFCCC)** and the **Paris Agreement**.

Objectives and Purpose:

The fund aims to:

1. **Addressing Loss & Damage:** Provide financial support to countries suffering from **climate-induced disasters** (E.g., cyclones, wildfires, droughts).
2. **Support Recovery & Rehabilitation:** Assist in rebuilding communities and infrastructure affected by climate change impacts.
3. **Aid in Non-Economic Losses:** Support responses to **cultural, social, and environmental losses** (E.g., displacement, biodiversity loss, damage to heritage sites).
4. **Enhance Climate Resilience:** Help vulnerable countries develop **risk management strategies** for climate adaptation.

Structure and Governance:

- The **Fund** operates as an entity under the **Financial Mechanism of the UNFCCC and the Paris Agreement**.
- It is **accountable to and functions under the guidance** of both the **Conference of the Parties (COP) and the CMA (Conference of the Parties serving as the Meeting of the Parties to the Paris Agreement).**
- The fund is expected to work **in coordination with other international climate finance mechanisms**, including:
  - Green Climate Fund (GCF)
  - Adaptation Fund
  - Global Environment Facility (GEF)

Funding Sources and Contributions:

- **Developed countries** are expected to **contribute** to the fund as part of their commitments under the UNFCCC and Paris Agreement.
- **Multilateral and private-sector contributions** may also be encouraged.
- **Innovative financing mechanisms** (E.g., levies on fossil fuel companies, climate taxes) can also be explored in future.

Challenges:

- **Sufficient Funding:** Developing countries which are affected the most argue that contributions should be **grants-based** rather than loans.
- **Who Pays?** While developed nations acknowledge the need for funding, discussions continue on how much they should contribute.
- **Coordination with Existing Funds:** Avoiding duplication with **other climate finance mechanisms**.
- **US Withdrawal:** The US, which is historically the largest emitter, has withdrawn from the fund, raising doubts about the future of the fund.

12. CLIMATE PROMISE INITIATIVE

- The UN Development Programme unveiled Climate Promise 2025, next stage of its Climate Promise Initiative. It is an initiative to support developing countries on their climate action.
- It is the world's largest offer of support to developing countries on NDC (Nationally Determined Contribution) enhancement and implementation.

- Each Party to Paris Agreement is required to establish an NDC, which includes targets for mitigating the greenhouse gas emissions and for adapting to climate impacts.
- Climate Promise 2025 aims to align the next generation of developing countries' NDCs to the goals of the 2015 Paris Climate Agreement.

The Government of India has articulated and put across the concerns of developing countries at the COP26 to the UNFCCC, held in Glasgow, United Kingdom 2021.

Further, India presented the following five elements of India's climate action:

i. Reach 500 GW Non-fossil energy capacity by 2030.

ii. 50% of its energy requirement from renewable energy by 2030.

iii. Reduction of total projected carbon emissions by one billion tons from now to 2030.

iv. Reduction of the carbon intensity of the economy by 45% by 2030, over 2005 levels.

v. Achieving the target of net zero emissions by 2070.

The transfer of climate finance and low-cost climate technologies have become more important for implementation of climate actions by the developing countries. The ambitions on climate finance by developed countries cannot remain the same as they were at the time of the Paris Agreement in 2015.

It was emphasised that just as the UNFCCC tracks the progress made in climate mitigation; it should also track climate finance. Further, it was conveyed to the developed countries that India understands the suffering of all other developing countries, shares them, and hence raises the voice of developing countries.

The mantra of **LIFE- Lifestyle for Environment** to combat climate change was also shared in COP 26. It was stated that Lifestyle for Environment must be taken forward as a campaign to make it a mass movement of Environment Conscious Lifestyles. The message conveyed by India was that the world needs mindful and deliberate utilization, instead of mindless and destructive consumption.

13. CARBON MARKET

Carbon markets are one of the tools to tackle the climate change problem. The argument behind carbon trading is that the best way to take climate action is to reduce emissions where it is least costly to do so.

Article 6 under the Paris Agreement contains three separate mechanisms for "voluntary cooperation" towards climate goals: two based on markets and a third based on "non-market approaches".

CARBON MARKETS UNDER THE PARIS AGREEMENT (ARTICLE 6)

- **Market Mechanism 1 (Article 6.2):** It sets up a carbon market which allows countries on voluntary direct bilateral cooperation basis to sell any extra emission reductions {called as Internationally Transferred Mitigation Outcomes (ITMO)} they have achieved compared to their Nationally Determined Contributions (NDCs) target.
- **Market Mechanism 2 (Article 6.4):** A new international carbon market would be created for the trading of emissions reductions created anywhere in the world by the public or private sector. This new market referred to as the "Sustainable Development Mechanism" (SDM) seeks to replace the CDM.
- **The non-Market Approach:** to boost "mitigation, adaptation, finance, technology transfer and capacity building", in situations where no trade is involved. This could involve similar activities without the added element of trading. E.g., a country could support a renewable energy scheme overseas via concessional loan finance, but there would be no trading of any emissions cuts generated.

WHY IS ARTICLE 6 IMPORTANT?

Within the Sustainable Development Mechanism, the notion of Overall Mitigation in Global Emissions (OMGE) has the potential to go beyond the Kyoto markets' "zero-sum game" to accomplish "global" mitigation.

Current international carbon market mechanisms operate under Kyoto Protocol's regulations, which provide that transfers between Parties result in no net decrease in global emissions.

Trading could aid in the reduction of emissions by making it easier and less expensive for governments to accomplish their climate goals, thus encouraging them to set more ambitious targets.

- **According to a World Bank Report,** some 96 country climate pledges - about half of all NDCs - refer to the use of carbon pricing initiatives.
- **According to IETA**, trading could save $250 bn every year by 2030. This could be invested into further emissions cuts to raise ambition.

It also includes a provision that a "share of the proceeds" earned under SDM be used to assist developing nation parties that are particularly vulnerable to the negative consequences of climate change in meeting adaptation expenses. This might become a new way for richer countries to channel climate funds to developing countries, complementing existing initiatives like the Green Climate Fund.

Article 6 might potentially be used to incorporate company climate pledges into the larger UN process. The only component of the legislation that specifically mentions private sector participation in the Paris process is Article 6.

The carbon market system must progress beyond offsets to something better. Rather than offering a cheap way out and substituting someone else's work, it should try to accelerate the change. Offsetting methods should be phased out in favor of climate programs that actually accelerate the zero-carbon transition.

CARBON PRICING: Carbon pricing is a tool that captures the external costs of greenhouse gas (GHG) emissions - costs that the public bears, such as crop damage, health care costs from heat waves and droughts, and property loss from flooding and sea level rise - and ties them to their sources through a price, usually in the form of a price on the CO2 emitted.

TYPES OF CARBON PRICING

There are 2 major types of carbon pricing:

- **Emissions Trading Systems (ETS):** The ETS, also known as a cap-and-trade system, sets a limit on total GHG emissions and permits low-emitting companies to sell their excess allowances to higher-emitting industries.
- **Carbon Tax:** It places a price on carbon by imposing a tax on greenhouse gas emissions or, more typically, the carbon content of fossil fuels. It differs from an ETS in that the carbon tax's emission reduction effect is not pre-determined, but the carbon price is.

OTHER MECHANISMS TO PRICE THE CARBON EMISSION

- **Results-Based Climate Finance (RBCF)** is a funding strategy in which payments are paid in response to pre-determined outputs or outcomes connected to climate change management, such as emission reductions.
  - Many RBCF programs also simultaneously aim to reduce poverty, improve access to clean energy and offer health and community benefits.
- **GHG emission reductions from project** or program-based activities are referred to as offset mechanisms, and they can be traded domestically or internationally.
  - Offset programs have their own register and award carbon credits according to a system.
  - These credits can be utilized to address GHG mitigation compliance requirements under international agreements, domestic policies, or corporate citizenship goals.
- **Internal carbon pricing** is a tool that an organization utilizes to steer its decision-making process in relation to the impacts, risks, and opportunities of climate change.

14. CARBON BORDER ADJUSTMENT MECHANISM

CARBON BORDER ADJUSTMENT TAX?

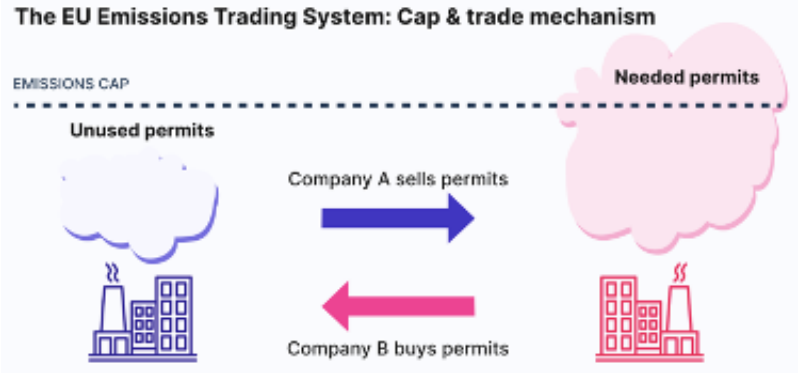
- Duty on imports based on the amount of carbon emissions resulting from the production of the product in question.
- It is aimed at putting a fair price on the carbon emitted during the production of carbon intensive goods that are entering the EU, and to encourage cleaner industrial production in non-EU countries.
- The gradual introduction of the CBAM is aligned with the phase-out of the allocation of free allowances under the EU Emissions Trading System (ETS) to support the decarbonisation of EU industry.

EU Emissions Trading System:

The EU ETS works on the 'cap and trade' principle. A cap is a limit set on the total amount of greenhouse gases that can be emitted by the installations and aircraft operators covered by the system. The cap is reduced annually in line with the EU's climate target, ensuring that emissions decrease over time.

Within the cap, companies primarily buy allowances on the EU carbon market, but they also receive some allowances for free. Companies can also trade allowances with each other as needed. If an installation or operator reduces their emissions, they can either keep the spare allowances to use in the future or sell them.·

Hence, it makes polluters pay for their greenhouse gas emissions, helps bring emissions down and generates revenues to finance the EU's green transition, operates in all EU countries plus Iceland, Liechtenstein and Norway (EEA-EFTA states).



Note: This EU-ETS works for the companies located within the EU borders. CBAM is for those companies which are located outside the E.U.

HOW THE CBAM WILL WORK?

- The CBAM puts an emissions tariff on imports of goods with a high risk of carbon leakage from countries which are not members of the EU Emissions Trading System (ETS).
- Companies importing goods into the EU will have to purchase CBAM certificates and then surrender the required amount to cover the corresponding emissions each year.
- There are no limits on the number of CBAM certificates an importer may purchase so as to avoid imposing restrictions on trade.
- Unlike ETS allowances, CBAM certificates are not tradeable nor bankable.

15. CARBON TRADING & EMISSION TRADING UNDER KYOTO PROTOCOL

Parties with commitments under the Kyoto Protocol (Annex B Parties) have accepted targets for limiting or reducing emissions. These targets are expressed as levels of allowed emissions, or assigned amounts, over the 2008-2012 commitment period (1st round) and 2013-2020 (2nd round). The allowed emissions are divided into assigned amount units (AAUs).

Countries bound to Kyoto targets have to meet them largely through domestic action — that is, to reduce their emissions onshore. But they can meet part of their targets through three "market-based mechanisms".

The Kyoto Flexible Market Protocol mechanisms include:

- Emission Trading
- Joint Implementation (JI)
- Clean Development Mechanism (CDM)

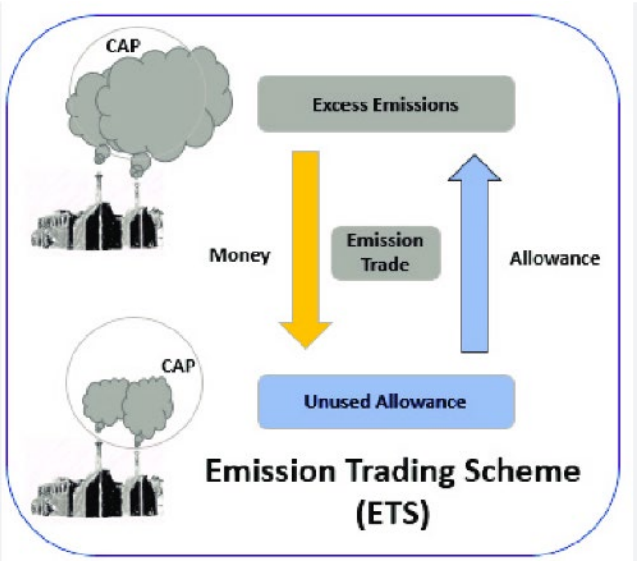
Emissions trading, as set out in Article 17 of the Kyoto Protocol, allows countries with surplus emission units - permitted but not "used" - to sell their excess capacity to countries exceeding their targets.

Thus, a new commodity was created in the form of emission reductions or removals. Since carbon dioxide is the principal greenhouse gas, people speak simply of trading in carbon. Carbon is now tracked and traded like any other commodity. This is known as the "carbon market."

The other units which may be transferred under the scheme, each equal to one tonne of CO2, may be in the form of:

- A removal unit (RMU) based on land use, land-use change and forestry (LULUCF) activities such as reforestation.
- An emission reduction unit (ERU) generated by a joint implementation project.

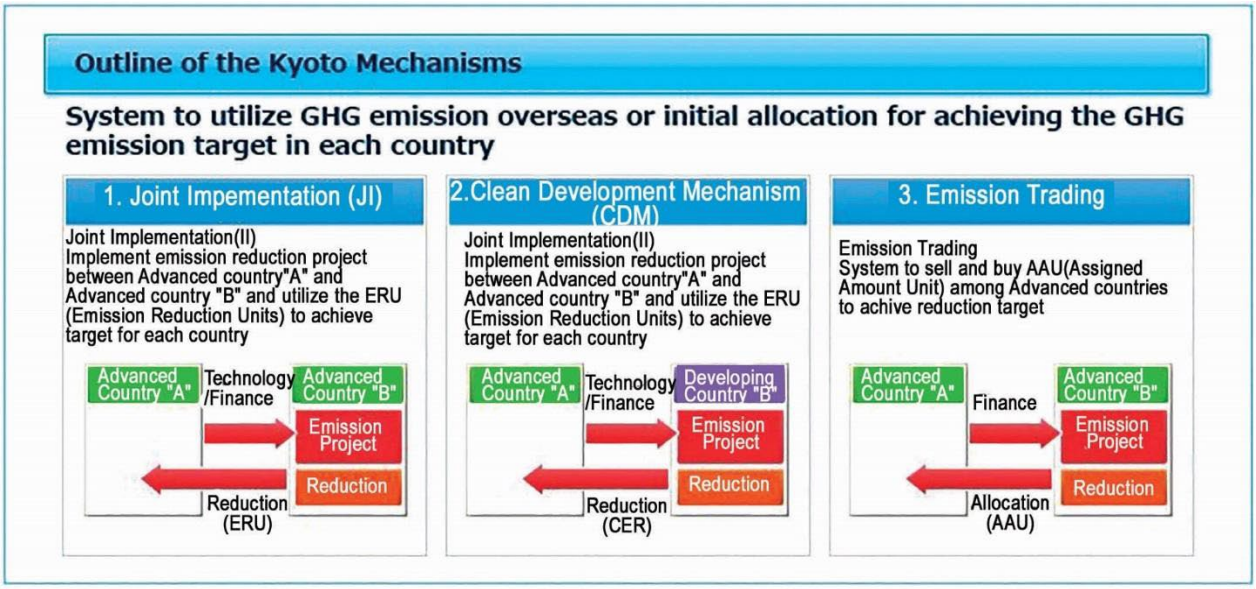
A certified emission reduction (CER) generated from a clean development mechanism project activity.



- The mechanism known as "joint implementation", defined in Article 6 of the Kyoto Protocol, allows a country with an emission reduction or limitation commitment under the Kyoto Protocol (Annex B Party) to earn emission reduction units (ERUs) from an emission-reduction or emission removal project in another Annex B Party, each equivalent to one ton of CO2, which can be counted towards meeting its Kyoto target.
- The **Clean Development Mechanism** (CDM), defined in Article 12 of the Protocol, allows a country with an emission-reduction or emission-limitation commitment under the Kyoto Protocol (Annex B Party) to implement an emission-reduction project in developing countries. Such projects can earn saleable certified emission reduction (CER) credits (Carbon credits), each equivalent to one ton of CO2, which can be counted towards meeting Kyoto targets.
  - Carbon credit is a tradable certificate or permit. One Carbon credit = 1 ton of Carbon Dioxide
  - Carbon trading is the name given to the exchange of emission permits.
- Carbon trading is of two types:
  - Emission trading
  - Offset trading.

Emissions trading allows countries to sell unused emission units to countries that have exceeded their targets. Carbon is tracked and traded like any other commodity in a "carbon market." Under offset trading, carbon credit is to be earned by a country by investing some amount of money in such projects, known as carbon projects, which will emit a lesser amount of greenhouse gas in the atmosphere. For example, suppose a thermal plant of 800 megawatt capacity emits 400 carbon-equivalent in the atmosphere. Now a country builds up an 800 megawatt wind energy plant which does not generate any amount of emission as an alternative to the thermal plant. Then by investing in this project the country will earn 400 carbon-equivalent.

- **Kyoto Protocol emission target gases includes:**
  - Carbon dioxide (CO2),
  - Methane (CH4),
  - Nitrous oxide (N2O),
  - Sulphur hexafluoride (SF6),
  - groups of hydro fluorocarbons (HCFs) and
  - groups of Per fluorocarbons (PFCs).



Transfers and acquisitions of these units are tracked and recorded through the registry systems under the Kyoto Protocol. An international transaction log ensures secure transfer of emission reduction units between countries.

THE COMMITMENT PERIOD RESERVE: To address the concern that Parties could "oversell" units, and subsequently be unable to meet their own emissions targets, each Party is required to maintain a reserve of ERUs, CERs, AAUs and/or RMUs in its national registry. This reserve, known as the "commitment period reserve".

CARBON CREDIT TRADING SCHEME

- Energy conservation amendment act 2022 added new clauses authorizing parliament to establish a domestic carbon credit trading scheme.
- The Bureau of Energy Efficiency (BEE), an agency established by the Ministry of Power launched the Carbon Credit Trading Scheme to address the issue of greenhouse gas (GHG) emissions and mitigation of climate change in the country.
- The scheme is aimed at facilitating trading of carbon credits and incentivizing and rewarding entities that reduce, remove or avoid greenhouse gas emissions.
- It assigns a value, known as a carbon credit, to each ton of carbon dioxide equivalent (tCO2e) reduced or avoided. These credits could be bought, sold and traded within the country's carbon market framework.
- Carbon credit certificates are earned by entities that are covered under the law by reducing emissions. These certificates are tradeable.
- This permit can also be purchased by entities that emit more than a pre-determined threshold or sold by entities emitting less than the threshold under the rules of the trading scheme.
- In other words, companies with low(er) emissions can sell their extra allowance to larger emitters.

Note: As of now it does not include CCUS (Carbon capture utilization and storage) formally, but the government has indicated its willingness to include it in future.

WHAT IS THE RECENT AMENDMENT IN CCTS?

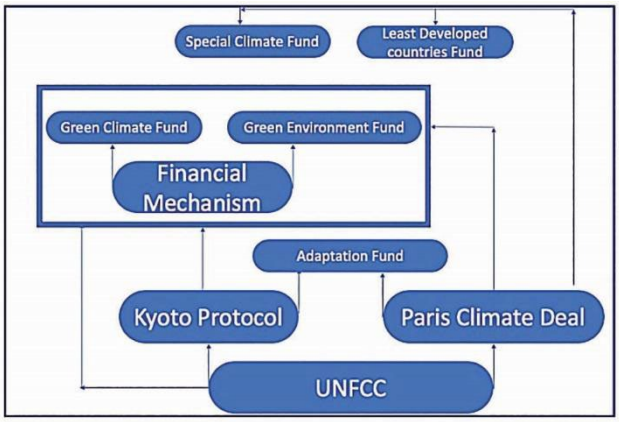
- On account of this amendment, the CCTS now provides for an offset mechanism — in addition to its pre-existing compliance regime.
- The compliance mechanism involves mandatory implementation among 'obligated' entities (which have a legal obligation to reduce emissions).
- Voluntary markets are driven by businesses seeking to decrease their carbon footprint (without a legal mandate to do so), including by 'offsetting' greenhouse gas (GHG) emissions on account of corporate net-zero commitments.
- Under the amended CCTS, nonobligated entities can voluntarily register projects in sectors identified by a national steering committee for securing tradable carbon credit certificates.

16. CLIMATE FINANCE

According to the UNFCCC Standing Committee on Finance, climate finance is "finance that aims at reducing emissions and improving greenhouse gas sinks, as well as reducing vulnerability of, and maintaining and increasing the resilience of, human and ecological systems to negative climate change impacts."

The term has been used in a narrow sense to refer to transfers of public resources from developed to developing countries, in light of their UN Climate Convention obligations to provide "new and additional financial resources", and in a wider sense to refer to all financial flows relating to climate change mitigation and adaptation.

UNFCCC, Kyoto Protocol and Paris Agreement call for financial assistance. In accordance with the principle of "common but differentiated responsibility and respective capabilities" set out in the Convention, developed country Parties are to provide financial resources to assist developing country Parties in implementing the objectives of the UNFCCC.



The Convention has created Financial Mechanisms to offer cash to developing nation Parties to help with this.

- Since the Convention's entrance into force in 1994, the Global Environment Facility (GEF) has acted as the financial mechanism's operating institution.
- **Copenhagen Accord:** Parties agreed for a "goal" for the world to raise $100 billion per year by 2020, from "a wide variety of sources", to help developing countries cut carbon emissions (mitigation). New multilateral funding for adaptation will be delivered, with a governance structure.
- **COP 16 (2010):** Parties established the Green Climate Fund (GCF) and in 2011 (COP 17) also designated it as an operating entity of the financial mechanism.
- **Establishment of special funds:** Special Climate Change Fund (SCCF), the Least Developed Countries Fund (LDCF), both managed by the GEF; and the Adaptation Fund (AF) under the Kyoto Protocol in 2001. (See Diagram)

17. TRUE COST ACCOUNTING

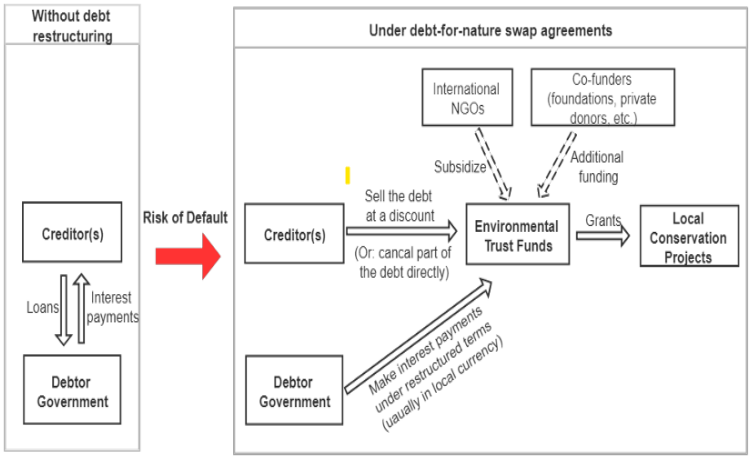
- **United Nations Environmental Programme** defines it as "an evolving holistic and systemic approach to measure and value the positive and negative environmental, social, health and economic costs and benefits to facilitate policy, business, farmer, investor, and consumer decisions.
- True Cost Accounting (TCA) is an accounting approach that measures and values the hidden impacts of economic activities on the environment, society, and health.
- TCA is also referred to as "full cost accounting" (FCA) or "multiple capital accounting (MCA)". The approach moves beyond purely economic thinking with the aim of improving decision-making in commercial organizations and in public policy.
- It includes accounting for natural capital, human capital, social capital and produced capital.

18. DEBT TO NATURE SWAP

Debt-for-nature swaps have been around for decades – as this 1990 paper from the World Bank shows. They were first envisioned by the WWF's Thomas Lovejoy in a New York Times article back in 1982 that advocated conservation groups use debt-equity swaps to raise money locally.

WHAT IS DEBT TO NATURE SWAP

- In essence, they are a financial instrument that allows countries to free up fiscal resources to build resilience against the climate crisis and take action to protect nature while still being able to focus on other development priorities without triggering a fiscal crisis.
- As the IMF's Managing Director, Kristalina Georgieva, explains: "Creditors provide debt relief in return for a government commitment to, say, decarbonize the economy, invest in climate-resilient infrastructure, or protect biodiverse forests or reefs.



Types:

- In a **commercial debt-for-nature swap** or **three-party debt-for-nature swap**, a non-governmental organization (NGO) acts as the funder/donor and purchases debt titles from commercial banks on the secondary market.
  - Since the late 1980s, organizations such **as Conservation International, The Nature Conservancy, and the World Wildlife Fund** have participated in international debt-for-nature swaps.
  - The NGO transfers the debt title to the debtor country, and in exchange the country agrees to either enact certain environmental policies or endow a government bond in the name of a conservation organization, with the aim of funding conservation programs.
- **Bilateral debt-for-nature swaps** take place between two governments. In a bilateral swap, a creditor country forgives a portion of the public bilateral debt of a debtor nation in exchange for environmental commitments from that country.
- **Multilateral debt-for-nature swaps** are similar to bilateral swaps but involve international transactions of more than two national governments.

19. ENVIRONMENTAL, SOCIAL & GOVERNANCE (ESG)

- Investors are increasingly applying above non-financial factors as part of their analysis process to identify material risks and growth opportunities like climate and social unrest.
- **Tragedy of the Horizon:** Climate change as the Tragedy of Horizon was first laid by Mark Carney, Governor of Bank of England in 2015. According to this idea, today's generation is not doing enough to address climate change as they do not feel the gravity of the problem. However, the next generation which will suffer the worst effects of climate change are not in decision making positions.

INITIATIVES FOR SUSTAINABILITY IN FINANCE

- **International Sustainability Standards Board (ISSB):** IFRS Foundation announced creation of a new standard setting board (ISSB) to create international standards with global investment portfolios for high quality, transparent, reliable & comparable reporting by companies on climate & other ESG matters.

THEMATIC AREAS

WHAT IS ONE HEALTH?

One Health is an approach that recognizes that the health of people is closely connected to the health of animals and our shared environment.

Human populations are growing and expanding into new geographic areas. As a result, more people live in close contact with wild and domestic animals, both livestock and pets. Animals play a key role in our lives, whether for food, fibre, livelihoods, travel, sport, education, or companionship. Close contact with animals and their environments provides more opportunities for diseases to pass between animals and people.

The earth has experienced changes in climate and land use, such as deforestation and intensive farming practices. Disruptions in environmental conditions and habitats can provide new opportunities for diseases to pass to animals.

The movement of people, animals, and animal products has increased from international travel and trade. As a result, diseases can spread quickly across borders and around the globe.

These changes have led to the spread of existing or known (endemic) and new or emerging zoonotic diseases, which are diseases that can spread between animals and people. Every year, millions of people and animals around the world are affected by zoonotic diseases. Examples of zoonotic diseases include:

- Rabies
- Salmonella infection
- West Nile virus infection
- Q Fever (Coxiella burnetii)
- Anthrax
- Brucellosis
- Lyme disease
- Ringworm
- Ebola

Animals also share our susceptibility to some diseases and environmental hazards. Because of this, they can sometimes serve as early warning signs of potential human illness. For example, birds often die of West Nile virus before people in the same area get sick with West Nile virus infection.

WHAT ARE COMMON ONE HEALTH ISSUES?

One Health issues include zoonotic diseases, antimicrobial resistance, food safety and food security, vector-borne diseases, environmental contamination, and other health threats shared by people, animals, and the environment. For example:

- Antibiotic-resistant germs can quickly spread through communities, the food supply, healthcare facilities, and the environment (soil, water), making it harder to treat certain infections in animals and people.
- Vector-borne diseases are on the rise with warmer temperatures and expanded mosquito and tick habitats.
- Diseases in food animals can threaten supplies, livelihoods, and economies.
- The human-animal bond can help improve mental well-being.
- Contamination of water used for drinking, recreation, and more can make people and animals sick.
- Even the fields of chronic disease, mental health, injury, occupational health, and non-communicable diseases can benefit from a One Health approach involving collaboration across disciplines and sectors.

20. REDUCING EMISSIONS FROM DEFORESTATION AND FOREST DEGRADATION (REDD)

- A mechanism under UNFCC, since 2005, to mitigate climate change through reducing net emissions of greenhouse gases through enhanced forest management in developing countries.

- Inclusion of reducing emissions from land use change is considered essential to achieve the objectives of the UNFCCC.
- During the negotiations for Kyoto Protocol the inclusion of tropical forest management was debated but eventually dropped due to anticipated methodological difficulties in establishing – in particular – additionality and leakage (detrimental effects outside of project area attributable to project activities).
- REDD+ (Defined in Bali Action Plan, 2007, CoP13)

WHAT CONSTITUTES "+"?

1. sustainable management of forests,
2. conservation of forest carbon stocks and
3. enhancement of forest carbon stocks.

DIFFERENCE BETWEEN REDD AND REDD+?

- **REDD = "reducing emissions from deforestation in developing countries"**
- **REDD+ (or REDD-plus)** to reduce emissions from deforestation and forest degradation in developing countries, and the role of conservation, sustainable management of forests, and enhancement of forest carbon stocks in developing countries.
- REDD+ is essentially a vehicle to financially reward developing countries for their verified efforts to reduce emissions and enhance removals of greenhouse gases through a variety of forest management options.
- India favours REDD+

UGANDA: FIRST AFRICAN COUNTRY TO SUBMIT REDD++ Uganda recently submitted its REDD++ results and has become the first African country to submit the results.

INDIA'S REDD++ India has prepared a National REDD++ strategy complying with the UNFCCC decisions. It is being updated in line with the National Action Plan on Climate Change, Nationally Determined Contributions and Green India Mission. The strategy was prepared by the Indian Council of Forestry Research and Education, Dehradun.

REDD++ HIMALAYAN PROGRAM: REDD++ Himalayan Program was launched in 2016 to address deforestation and forest degradation in the Himalayan states of India. It ends in July 2020. The project was implemented in four countries namely Bhutan, Nepal, Myanmar and India.

INDIA'S PARIS AGREEMENT TARGETS

India ratified the Paris Agreement exactly one year after the submission of its Intended Nationally Determined Contribution (INDC), on 2 October 2016. Since India did not submit an NDC prior to ratification, the INDC became its first NDC. It includes the following main elements (Government of India, 2015):

- To reduce the emissions intensity of GDP by 33%–35% by 2030 below 2005 levels.
- To increase the share of non-fossil-based energy resources to 40% of installed electric power capacity by 2030, with help of transfer of technology and low-cost international finance including from Green Climate Fund (GCF).
- To create an additional (cumulative) carbon sink of 2.5–3 GtCO2e through additional forest and tree cover by 2030.

India does not specify the coverage and metrics of the emissions intensity target in its NDC.

As per the updated NDC, India now stands committed to reduce Emissions Intensity of its GDP by 45% by 2030, from 2005 level and achieve about 50% cumulative electric power installed capacity from non-fossil fuel-based energy resources by 2030.

21. OZONE LAYER DEPLETION

- Ozone is a form of oxygen. Oxygen occurs in three different forms in the atmosphere: as oxygen atoms (O), as oxygen molecules (O2) and as Ozone (O3).

- Ozone's unique physical properties allow the ozone layer to act as our planet's sunscreen, providing an invisible filter to help protect all life forms from the sun's damaging UV (ultraviolet)rays. Most incoming UV radiation is absorbed by ozone and prevented from reaching the Earth's surface. Without the protective effect of ozone, life on Earth would not have evolved the way it has.

ULTRAVIOLET RADIATION

- Ultraviolet radiation is the one form of radiant energy coming out from the sun. The sun emits a range of energy known as the electromagnetic spectrum. The various forms of energy, or radiation, are classified according to wavelength (measured in nanometres where one nm is a millionth of a millimetre).
- The shorter the wavelength, the more energetic the radiation. In order of decreasing energy, the principal forms of radiation are gamma rays, x-rays, UV (ultraviolet radiation), visible light, infrared radiation, microwaves, and radio waves. Invisible Ultraviolet is so named because it occurs next to violet in the visible light spectrum.
- The three categories of UV radiation are:
  - UV-A between 320 and 400 nm
  - UV-B between 280 and 320 nm
  - UV-C between 200 and 280 nm

ROLE OF OZONE IN ABSORBING THESE RADIATIONS

- UV-B and C are highly energetic and are dangerous to life on earth. UV-A being less energetic is not dangerous.
- Fortunately, UV-C is absorbed strongly by oxygen and by ozone in the upper atmosphere. UV-B is also absorbed by the ozone layer in the Stratosphere and only 2-3% of it reaches the earth's surface. The ozone layer, therefore, is highly beneficial to plant and animal life on earth in filtering out the dangerous part of the sun's radiation and allowing only the beneficial part to reach earth. Any disturbance or depletion of this layer would increase UV-B and UV-C radiation reaching the earth's surface leading to dangerous consequences.

OZONE DEPLETION

- Ozone depletion occurs when the natural balance between the production and destruction of stratospheric ozone is tipped in favour of destruction.
- Although natural phenomena can cause temporary ozone loss, chlorine and bromine released from synthetic compounds is now accepted as the main cause of a net loss of stratospheric ozone in many parts of the world since 1980.
- There is strong evidence that global ozone depletion is occurring.
- The evidence is in the observations of the Antarctic ozone "hole" and atmospheric records indicating seasonal declines in global ozone levels.

ABOUT OZONE HOLE

- Ozone layer is the common term for the high concentration of ozone that is found in the stratosphere (layer of the atmosphere between around 10- 50 km altitude).
  - Atmospheric ozone absorbs ultraviolet (UV) radiation from the sun, particularly harmful UVB-type rays.
- Ozone hole refers to a region in the stratosphere where concentration of ozone becomes extremely low.
  - Such holes are spotted over both the Poles.
  - Ozone depletion at the North pole is much smaller in size, owing to warmer temperatures at the North Pole than at the South Pole.
  - Ozone depletion is directly related to the formation of the Polar vortex (in the stratosphere).
    - During winter, temperatures in the vortex usually drop below 195 K (-78°C), and polar stratospheric clouds (PSCs) form.
    - PSCs provide a surface for ozone-depleting substances such as chlorine-containing CFCs, HCFCs, bromine-containing halons etc. to reach the stratosphere.
    - At the poles, ODSs attach to ice particles in PSCs. When the sun comes out again in the polar spring, the ice particles melt, releasing the ozone-depleting molecules from the ice particle surfaces.

- Once released, these ozone-destroying molecules harm and break apart the molecular bonds in UV radiation-absorbing ozone.

o During the Southern Hemisphere spring season (August - October) the ozone hole over the Antarctic increases in size, reaching a maximum between mid-September and mid-October.

o Ozone depletion slows when temperatures in the stratosphere start to rise, in late Southern Hemisphere spring, the polar vortex weakens and breaks down.

- This is because in warmer temperatures fewer PSCs form and they don't persist as long, limiting the ozone-depletion process.

OZONE-DEPLETING SUBSTANCES

Which are the Ozone Depleting Substances (ODS) presently used in India?

- Chlorofluorocarbons (CFCs)–12 for Refrigeration, Chillers and Metered Dose Inhalers.
- Hydrochlorofluorocarbons (HCFCs) - 22 for Air Conditioners.
- Carbon Tetrachloride (CTCs) for as a solvent process agent mainly in the metal cleaning and textile industries. It is also used as feedstock in the manufacture of CFCs and DV Acid Chloride.

Which ODS is no longer produced in India?

- Halons, which were earlier used in fire extinguishers. Halons continue to be used in Defence sector, which is exempt from Montreal Protocol. The production of CFCs has also been stopped since 2008.

What are the commonly used ODS alternatives?

ODS ALTERNATIVES

SUB-SECTOR	APPLICATION	ALTERNATIVE TECHNOLOGY
Domestic Refrigeration	Household Refrigerators And Freezers	HFC-134a, HFC-152a, Blends and mixtures, Hydrocarbons (for refrigerants) and HCFC-22, HCFC-22 + 142b, HCFC-141b, Hydrocarbons for foaming)
Commercial Refrigeration	Refrigerated Cabinets Water Coolers Ice-candy machines Walk-in coolers	HCFC-134a, HFC-152a, Blends and mixtures, Hydrocarbons (for refrigerants) and HCFC-22, HCFC-22 +142b, Hydrocarbons (for foaming) HCFC-22, HFC-134a HCFC-22, HFC-134a (refrigerants) and HCFC-14b (foaming)
Industrial Refrigeration	Cold Storages Process Chillers	HCFC-22, HFC-134a, Ammonia HCFC-22, HFC-134a, Ammonia
Transport Refrigeration	Perishable Transport	HCFC-22, HFC-134a, Blends and mixtures
Air Conditioning	Chillers Automotive-air conditioning	HCFC-123, HFC-134a, HCFC-22 HFC-134a, Blends and mixtures

22. METHANE & CLIMATE CHANGE

- It is a hydrocarbon that is a primary component of natural gas.
- Methane is also a greenhouse gas (GHG), so its presence in the atmosphere affects the earth's temperature and climate system.
- Methane is emitted from a variety of anthropogenic and natural sources.
- Anthropogenic emission sources include landfills, oil and natural gas systems, agricultural activities, coal mining, stationary and mobile combustion, wastewater treatment, and certain industrial processes.

- Methane is the second most abundant anthropogenic GHG after carbon dioxide (CO_2), accounting for about 20 per cent of global emissions.
- Methane is more than 25 times as potent as carbon dioxide at trapping heat in the atmosphere.

Because methane is both a powerful greenhouse gas and short-lived compared to carbon dioxide, achieving significant reductions would have a rapid and significant effect on atmospheric warming potential.

INTERNATIONAL METHANE EMISSIONS OBSERVATORY

- The International Methane Emissions Observatory (IMEO) was launched at the G20 Summit, on the eve of the COP26 UN climate conference in Glasgow.
- It is a data-driven, action-focused initiative by the UNEP with support from the European Commission to catalyse the dramatic reduction of methane emissions, for achieving Paris agreement goals.

GLOBAL METHANE INITIATIVE

- It is an international public-private partnership focused on reducing barriers to the recovery and use of methane as a valuable energy source.
- GMI provides technical support to deploy methane to energy projects around the world that enable Partner Countries to launch methane recovery and use projects.
- GMI advances methane mitigation in 3 key sectors: Oil & Gas, Biogas (including agriculture, municipal solid waste, and wastewater) and Coal Mines.
- **Membership:** GMI Partner Countries account for approximately 70% of global manmade methane emissions. India is a member of this organisation

23. TECHNIQUES OF GEOENGINEERING

1. Solar Radiation Modification:

- This refers to the intentional modification of Earth's shortwave radiative budget with the aim of reducing warming.
- Stratospheric Aerosol Injection: Injection of a gas in the stratosphere, which then converts to aerosols.
- Marine Cloud Brightening: Spraying sea salt or other particles into marine clouds, making them more reflective.
- Cirrus Cloud Thinning: Seeding to promote nucleation, reducing optical thickness and cloud lifetime, to allow more outgoing longwave radiation to escape into space.
- Ground based Albedo Modification: Whitening roofs, changes in land use management (e.g., no-till farming), change of albedo at a larger scale (covering glaciers or deserts with reflective sheeting and changes in ocean albedo).

2. **Ocean Fertilization:** Deliberate increase of nutrient supply to near-surface ocean to enhance biological production through which additional carbon dioxide from the atmosphere is sequestered. This can be achieved by the addition of micro-nutrients or macro-nutrients. Ocean fertilization is regulated by the London Protocol.
3. **Carbon dioxide removal:**

- Bioenergy
- Afforestation and reforestation
- Soil carbon sequestration and biochar

4. **Biochar:** They are stable, carbon rich materials produced by heating biomass in an oxygen-limited environment. They may be added to soils to improve soil functions and to reduce greenhouse gas emissions from biomass and soils and carbon sequestration.
5. **Enhanced Weathering:** Weathering is the natural process of rock decomposition via chemical and physical processes in which CO_2 is spontaneously consumed and converted into solid or dissolved alkaline bicarbonates and carbonates. The process is controlled by temperature, reactive surface area, interactions with biota and in particular water solution composition.

Challenges associated with Enhanced Weathering are:

- Increase in water pH.
- Release of heavy metals like Nickel and Chromium and plant nutrients like K, Ca, Mg, P and Si
- Changes in hydrological soil properties

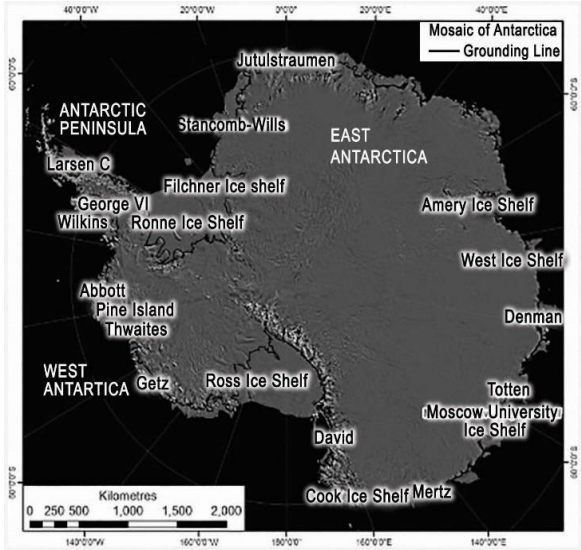
6. **Ocean alkalization:** Ocean alkalization adds alkalinity to marine areas to locally increase CO_2 buffering capacity of the ocean.

24. DOOMSDAY GLACIER (THWAITES)

Scientists studying Antarctica's vast *Thwaites Glacier* —nicknamed Doomsday Glacier — say warm water is seeping into its weak spots, worsening melting caused by rising temperatures.

The size of Britain, Thwaites is one of the world's fastest changing glaciers in Antarctica. Its susceptibility to climate change is a major concern to scientists because if it melted completely, it would raise global sea levels by half a metre.

Ice shelves: Ice shelves are floating ice, connected to the mainland. They receive ice from glaciers flowing into them from the mainland, from accumulation of snow directly onto the ice shelf, and from seawater freezing onto the bottom of the ice shelf. Most mass loss from the Antarctic continent is from ice shelves, and most of this is from just a few small ice shelves around the Antarctic Peninsula and West Antarctica. Ice shelves can collapse dramatically. This can occur over just a few weeks, following progressive thinning by warm ocean waters below, and from excessive melting during a warm summer above. If an ice shelf collapses, it changes boundary conditions for glaciers that flow into the ice shelf.



Landsat Image Mosaic of Antarctica (LIMA) showing location of key ice shelves.



25. VIENNA CONVENTION

- It is a Multilateral Environmental Agreement that was agreed upon at the 1985 Vienna Conference and entered into force in 1988.
- It is one of the most successful treaties of all time.
- It has been ratified by 197 states.
- It acts as a framework for international efforts to **protect the ozone layer**.
- These are laid out in the accompanying Montreal Protocol.
- The Vienna convention is not legally binding.

26. MONTREAL PROTOCOL (MP)

- **Montreal Protocol on Substances that Deplete the Ozone Layer** is a protocol to the Vienna Convention for the Protection of Ozone Layer.
- It is an international treaty that aims to protect the ozone layer by phasing out:
  - Chlorofluorocarbons (CFCs),
  - Hydrochlorofluorocarbons (HCFCs),
  - Hydrobromofluorocarbons (HBFCs),
  - Carbontetrachloride (CCl4),
  - Methylbromide (CH3Br),
  - Bromochloromethane (CH2BrCl),
  - Methyl chloroform (CH3CCl3),
  - Halons.

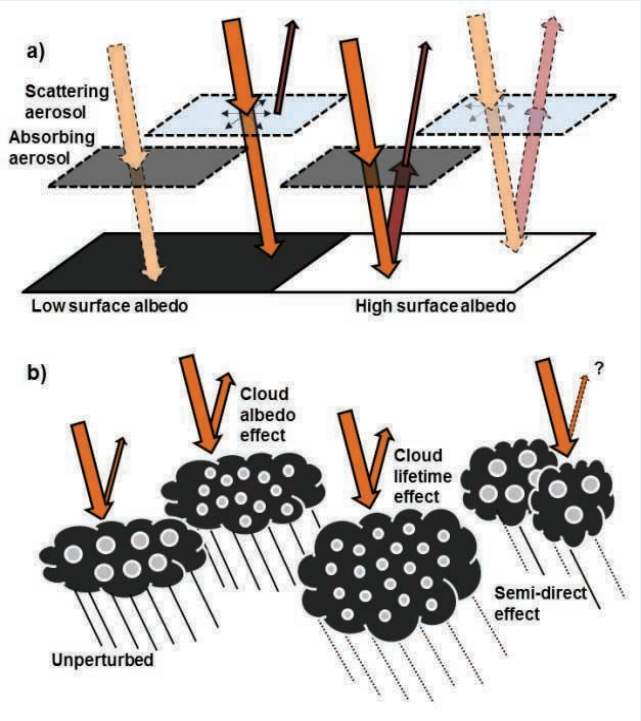
27. KIGALI AGREEMENT (2016)

- The Kigali Agreement amended the 1987 Montreal Protocol that was designed to close growing ozone holes by banning ozone-depleting substances.
- This amended Montreal Protocol which was initially conceived only to plug gases that were destroying the ozone layer now includes HFCs responsible for global warming.
- This move will help to prevent a potential 0.5ºC rise in global temperature by the end of the century.
- All signatory countries have been divided into three groups with different timelines to go about reductions of HFCs.
- The first group includes countries like the US and those in the European Union (EU). They will freeze production and consumption of HFCs by 2018. They will reduce them to about 15% of 2012 levels by 2036.
- The second group includes countries like China, Brazil and all of Africa which will freeze HFC use by 2024 and cut it to 20% of 2021 levels by 2045.
- Third-group countries like India, Pakistan, Pakistan, Iran, Saudi Arabia etc. will be freezing HFC use by 2028 and reducing it to about 15% of 2025 levels by 2047.

28. INTERGOVERNMENTAL PANEL ON CLIMATE CHANGE (IPCC)

- It is a scientific government body under the UN set up at request of member governments, dedicated to providing the world with an objective, scientific view of climate change and its political and economic impacts on the nations.
- First established in 1988 by two UN organizations, the World Meteorological Organization and the United Nations Environment Program, and later endorsed by the United Nations General Assembly.
- Membership of PCC is open to all members of WMO & UNEP.
- IPCC produces reports that support UNFCCC, which is the main international treaty on climate change.
- The main objective of UNFCCC is to stabilize greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system.
- IPCC reports cover the scientific, technical and socio-economic information relevant to understanding the scientific basis of risk of human-induced climate change, its potential impacts and options for adaptation and mitigation.

ANTHROPOGENIC AEROSOL FORCING



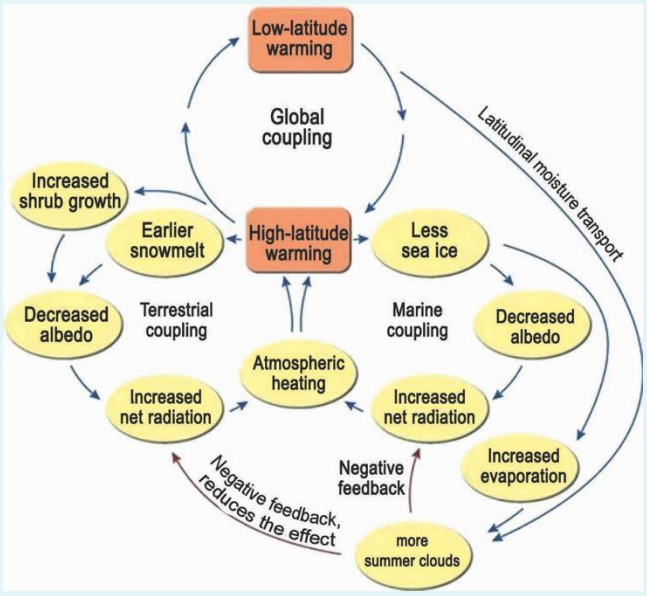
All atmospheric aerosols scatter incoming solar radiation, and a few aerosol types can also absorb solar radiation. BC is the most important of the latter, but mineral dust and some OC components are also sunlight absorbers. Aerosols that mainly scatter solar radiation have a cooling effect, by enhancing the total reflected solar radiation from the Earth.

Strongly absorbing aerosols have a warming effect. In the atmosphere, there is a mixture of scattering and absorbing aerosols, and their net effect on Earth's energy budget is dependent on surface and cloud characteristics. Scattering aerosols above a dark surface and absorbing aerosols above a bright surface are most efficient (see Figure).

Scattering (absorbing) aerosols above a bright (dark) surface are less efficient because the solar radiation is reflected (absorbed) anyway. Absorbing aerosols are particularly efficient when positioned above clouds, which are a main contributor to the total reflection of solar radiation back to space.

ELEVATION-DEPENDENT WARMING

The phenomenon that the warming rate of air temperature is amplified with elevation is termed elevation-dependent warming (EDW).



29. UNITED NATIONS CONVENTION TO COMBAT DESERTIFICATION

Established in 1994, the United Nations Convention to Combat Desertification (UNCCD) is the sole legally binding international agreement linking environment and development to sustainable land management.

The Convention addresses specifically the arid, semi-arid and dry sub-humid areas, known as the drylands, where some of the most vulnerable ecosystems and peoples can be found.

LAND DEGRADATION NEUTRALITY FUND: Land Degradation Neutrality Fund (LDN Fund) was launched at the 13th Conference of the Parties (COP13) to the United Nations Convention to Combat Desertification (UNCCD) in Ordos, China.

ABOUT LDN FUND

- LDN fund is a first-of-its-kind investment vehicle leveraging public money to raise private capital for sustainable land management and landscape restoration activities worldwide.
- It will be independent of the UN and will be managed by a private-sector investment management firm.
- It will invest in bankable projects on land rehabilitation and sustainable land management worldwide, including:
  - Sustainable agriculture, sustainable livestock management,
  - Agro-forestry, sustainable forestry,
  - Renewable energy, infrastructure development, and eco-tourism.

30. PRELIMS SNIPPETS

Vulture Protection	Bangladesh has become the first country that has banned the painkiller ketoprofen.
Omkareshwar Dam	• The Omkareshwar Floating Solar Project was inaugurated. • Established at Omkareshwar Dam on Narmada River, Madhya Pradesh. • Largest floating solar energy project in the world. • Designed to have a total capacity of 600 megawatts (MW).
Bhashan Char Island	• It is a remote uninhabited island of Bangladesh, located in the Bay of Bengal. It is also known as Char Piya. • Around thousands of Rohingyas from Cox's Bazar refugee camp are being shifted to the newly developed facility on the island. Humanitarians oppose this move fearing poor liveable conditions in the new.
Segur Plateau	• A plateau in the Nilgiris Hills of Tamil Nadu. Situated on Moyar River. Important wildlife corridor between the Eastern and Western Ghats. Part of Nilgiris Biosphere Reserve.
Mission Innovation (MI)	It is a global initiative of 24 countries and the European Commission (on behalf of the European Union). Find out more about our members, including annual progress reports.
Great Green Wall Initiative	Aims to transform the lives of 100 million Africans by growing an 8,000 km long and 15 km wide mosaic of trees, vegetation, grasslands, plants. The project aims to restore 100 million hectares of degraded land by 2030. It is an African initiative, started in 2007 by the African Union to combat desertification, land degradation, and drought.
Kaleshwaram lift irrigation project	It is a multipurpose irrigation project on the Godavari River. The project begins at the meeting point of Godavari River and Pranahita river. It was initially called the Pranahita Chevella project. Later it was renamed as Kaleshwaram project in 2014.
Global Initiative to reduce Land Degradation and coral reef program (Global land initiative)	The primary aim of the initiative is to strengthen the implementation of existing frameworks to prevent, halt, and reverse land degradation within G20 member states and globally. Implementation must be done by considering implications on the achievement of other sustainable development goals (SDGs) and adhering to the principle of not harm. The ambition of this Global Initiative is to prevent, halt and reverse land degradation and reduce degraded land by 50 per cent by 2040.

Zig Zag Technology	**In zigzag kilns, bricks are arranged to allow hot air to travel in a zigzag path. The length of the zigzag air path is about three times that of a straight line, and this improves the heat transfer from the fuel gases to the bricks, making the entire operation more efficient.**
Urban Shift Initiative	• It is a program of UNEP to transform cities for people and the planet through sustainable, integrated, zero-carbon urban development. • The program will support 23 cities in Argentina, Brazil, China, Costa Rica, India, Indonesia, Morocco, Rwanda and Sierra Leone (9 countries) to adopt integrated approaches to urban development. • Indian cities participating in the program are Chennai, Pune, Surat, Agra and Puducherry. The Urban Shift project in India is implemented by UNEP and Asian Development Bank.
Coalition for Digital Environmental Sustainability (Codes)	• It is a global multi-stakeholder alliance of governments, companies and civil society established to harness digital transformation to become a positive and exponential force for sustainability and climate action. • Initiated by UNEP, UNDP, International Science Council, German Environment Agency, Kenyan Ministry of Environment, Future Earth and Sustainability in the Digital Age. **Objectives:** • Offer a vision and authoritative framing of environmental sustainability and digitalization nexus. • Establish an acceleration plan for digitalizing environmental sustainability of including immediate priorities and partnerships covering a 2–3-year perspective. • Unite environmental sustainability and digitalization tracks under a common framework and umbrella. • Mobilize scientific community and research agenda.
Alliance for Clean Air	First global corporate initiative to bring together leading businesses to tackle air pollution. They will: • Establish air pollution footprints on nitrogen oxides, Sulphur oxides, particulate matter within 12 months. • Launched at COP 26 of UNFCCC. • Pinpoint where they are being emitted to track human exposure. • Set ambitious targets and objectives to reduce air pollution emissions, with a clear action plan. • Act as champion of clean air by raising awareness about the impact of air pollution and reducing exposure of stakeholders to air pollution. • Use their assets innovatively to accelerate clean air solutions. • **Members:** 10 large companies like Google, Accenture, Bloomberg, IKEA etc. Mahindra & Wipro from India is also a member.
First Movers Coalition	• Global initiative harnessing the purchasing power of companies to decarbonize seven "hard to abate" industrial sectors that currently account for 30% of global emissions: ○ Aluminum, Aviation, Chemicals, Concrete, Shipping, Steel and Trucking • Launched by US President Joe Biden and the WEF at COP26 as a flagship public-private partnership to clean up the most carbon-intensive sectors. • Commitments for the first four sectors (Aviation, Shipping, Steel and Trucking) were launched in November 2021, at COP26 in Glasgow. • Aluminium and Carbon Dioxide Removal sectors were launched at the May 2022 World Economic Forum Annual Meeting in Davos, and the Cement & Concrete sector was

	launched at COP27 in Sharm el-Sheikh. The final sector of Chemicals is planned to launch at COP28 in Dubai. • The 50+ companies that make up the Coalition seek to send a powerful market signal to commercialize zero-carbon technologies. • Government partners: 
Karakoram Anomaly	• Termed as the stability or anomalous growth of glaciers in the central Karakoram, in contrast to the retreat of glaciers in other nearby mountainous ranges of Himalayas and other mountainous ranges of the world.
One Ocean Summit	• Organized by France in February 2022, this brought together 41 heads of state (including India), members of civil society and business. • The goal of the One Ocean Summit is to raise the collective level of ambition of the international community on marine issues and to translate our shared responsibility to the ocean into tangible commitments. **Commitments:** • Protect biodiversity and ocean resources: o Aims at creation of protected areas through the High Ambition coalition for nature and people. o A high ambition coalition on biodiversity beyond national jurisdiction is launched for sustainable use of high seas and protection of their biodiversity. o Commitment to fight against illegal fishing. • Join forces with the oceans to face climate change: France and Colombia launched a global coalition for blue carbon, which will bring together national and multilateral actors in the field to contribute to financing the restoration of coastal ecosystems, using shared and rigorous methodologies • End plastic pollution of the ocean: o Clean ocean initiative was launched, several European banks doubled their commitment to 4 billion euros by 2025 o India and France together launched an initiative on the elimination of single-use plastic pollution, which aims to be multilateral. • Place the ocean at the top of the global political agenda: The European Union has pledged to produce a “Digital Twin of the Ocean”. This “Digital Twin” initiative will inform

	political decisions and track their effects, enable the marine economy to develop with respect for ecosystems and fuel dialogue with stakeholders and the public.
Clean Ocean Manifesto	Clean Ocean Expert Group formed to implement the UN Decade for Ocean Science for Sustainable Development 2021-30. They have released a 'clean ocean manifesto.' **It has essential aims:** 1. Reducing marine debris by 50-90%. 2. A globe circling, high-tech system of monitors. **Objectives:** ● Enlarge understanding of pathways for spread and fates of pollutants. ● Reduce and remove top-priority forms of pollution (Ex. Marine debris) by large amounts, as much as 50% to 90%. ● Improve dramatically the outcomes of control measures. ● Improve monitoring (as part of Global Ocean Observing System (GOOS) for more accurate, precise and real-time tracing of spills and monitoring of ocean sound spaces.
International Monsoons Project Office (IMPO)	● Support activities of Monsoons Panel, jointly established by World Climate Research Program's CLIVAR (Climate & Ocean Variability, Predictability and Change) and GEWEX (Global Energy and Water Exchanges). ● Support to the World Weather Research Program's working group on tropical meteorology research. **Benefits:** ● Understand and predict seasonal variability of monsoons. ● Enhancing prediction skills of monsoons and cyclones. ● Strengthening monsoon research for better support operations and services.
Cyanokhoj	● It is a pan-India map of algal blooms, caused by cyanobacteria in Indian inland water bodies. It has been created using satellite data based on Google Earth. ● It will act as an early warning system and will assist water resource managers to monitor water quality and aid decision-making. ● Algal blooms were most severe in the post-monsoon season when nutrient concentrations are highest. **About Algal Bloom:** ● Decomposition of organic waste increases the nutrient content of water bodies, this enrichment causes profuse growth of algae (algal bloom), especially blue-green algae. ● They cover the entire surface of water bodies and release toxins. Toxins inhibit growth of other algae and aquatic animals may die due to a lack of oxygen and toxicity. ● The process of nutrient enrichment and loss of biodiversity is referred to as Eutrophication.

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2 CHAPTER INTERNATIONAL ORGANISATIONS & INITIATIVES

1. WETLANDS INTERNATIONAL

- Wetlands International is a non-profit organisation established in 1937 as 'International Wildfowl Inquiry'.
- HQ in the Netherlands.
- It is an independent, not-for-profit organization, supported by government and NGO membership from around the world.

2. BIRD LIFE INTERNATIONAL

- World's largest nature conservation partnership. Together there are 120 Bird Life Partners worldwide.
- It strives to conserve birds, their habitats and global biodiversity, working with people towards sustainability in the use of natural resources.
- Bird Life International is the official Red List authority for birds, for the IUCN.
- Identifies sites referred to as 'Important Bird and Biodiversity Areas'.
- For India, the partner organisation is Bombay Natural History Society (BNHS).

IMPORTANT BIRD & BIODIVERSITY AREA

An area identified internationally by a set of criteria for the conservation of bird population. It was developed by Bird Life International. There are 12,000 Important Bird Areas worldwide.

GLOBAL IMPORTANT BIRD AREA CRITERIA

Based on the criteria, the Global Important Bird Areas are classified as follows:

A1 Category: Globally Threatened Species. The sites under this category hold bird populations that are categorised as Critically Endangered, Endangered or Vulnerable by the IUCN Red List of Threatened Species.

A2 Category: Restricted Range species

A3: Biome Restricted Species

A4: Congregations

3. INTERNATIONAL UNION FOR CONSERVATION OF NATURE (IUCN)

IUCN is a membership Union composed of both government & civil society organisations. It is the global authority on the status of the natural world and measures needed to safeguard it. Headquartered in Switzerland.

IUCN RED LIST

- IUCN Red List is a rich compendium of information on threats, ecological requirements, and habitats of species; and on conservation actions that can be taken to reduce or prevent extinctions.
- Based on an objective system for assessing risk of extinction of a species based on past, present, and projected threats using the rigorous IUCN Red List Categories and Criteria.
- There are eight IUCN Red List Categories based on criteria linked to population trend, size and structure, and geographic range.
- Produced by IUCN Global Species Program, Species Survival Commission & IUCN Red List Partnership.

WHAT DOES IUCN MEAN BY "THREATENED"?

Critically Endangered (CR), *Endangered* (EN) and *Vulnerable* (VU) species are considered to be threatened with global extinction.

WHAT DOES EACH CATEGORY MEAN?

In descending order of threat, the IUCN Red List threat categories are as follows:

- ***Extinct or Extinct in Wild***
- ***Critically Endangered, Endangered and Vulnerable*:** Species threatened with global extinction.
- ***Near Threatened:*** Species close to the threatened thresholds or that would be threatened without ongoing conservation measures.
- ***Least Concern:*** Species evaluated with a lower risk of extinction.
- ***Data Deficient:*** No assessment because of insufficient data.

IUCN has UN observer status ensuring nature conservation has a voice at the highest forum of global governance.

COMMISSIONS OF IUCN

Work of IUCN's six Commissions covers a wide range of disciplines: Education & communication, Environmental, Economic & social policy, Ecosystem management, Species survival, Environmental law & Protected areas.

1. **Commission on Education and Communication (CEC):** to raise awareness of the importance of nature and its conservation.
2. **Commission on Environmental, Economic and Social Policy (CEESP):** livelihoods, human rights and responsibilities, human development, and fair and effective governance of natural resources, are some of the main aspects
3. **Commission on Ecosystem Management (CEM)** promotes ecosystem-based approaches for the management of landscapes and seascapes.
4. **World Commission on Environmental Law (WCEL)** serve as the principal source of legal technical advice on all aspects of environmental law to IUCN
5. **The World Commission on Protected Areas (WCPA)** works to develop policy, advice and guidance on issues relating to protected areas.
6. **Species Survival Commission's** more than 9,000 experts work independently and with the IUCN Global Species Program to build knowledge on the status of species and threats to them.

4. IUCN GREEN STATUS OF SPECIES

It aims to complement the IUCN Red List by providing a tool for assessing the recovery of species populations and measuring their conservation success. It also highlights the impact of past conservation and dependence of many species on continued conservation efforts.

Note: IUCN animals with Green Status will continue to be preserved and the IUCN Red List will be published. Since 2012, IUCN has been preparing a Green List of Protected & Conserved Areas.

DEFINITION OF RECOVERY UNDER GREEN LIST

A species is fully recovered if it is present in all parts of its range, even those that are no longer occupied but were occupied before major human impact/disruption.

It is viable in all parts of the range.

It is performing its ecological functions in all parts of the range.

GREEN SCORE

Recovery definition is translated into Green Score, a numeric value. The Green Score calculated at the time of assessment is the Species Recovery Score.

CATEGORIES

- Not evaluated
- Intermediate
- Fully Recovered
- Slightly depleted.
- Moderately depleted
- Largely depleted.
- Critically depleted.
- Extinct in the wild
- Extinct

5. OTHER EFFECTIVE AREA BASED CONSERVATION MEASURES (OECMS)

OECMs are a new conservation approach separate from protected areas, where effective in-situ conservation of biodiversity is achieved mainly as a by-product of other management.

CRITERIA FOR IDENTIFICATION OF OECMS

- The area must not be recognised as a protected area.
- The area should be effectively governed and managed, and it should be geographically defined.
- The area should achieve sustained and effective in-situ conservation of biodiversity.
- The area should deliver associated ecosystem functions and services and cultural, spiritual, socio-economic and other locally relevant values.

Note: Those conservation initiatives which have been notified as 'reserved forests' by the Forest Survey of India will not qualify as potential OECMs in India as they are already being counted as Protected Areas.

POTENTIAL OECM CATEGORIES IN INDIA

Terrestrial: Biodiversity Parks, Industrial estates for conservation purposes, Village commons/lands, Important Bird Area and Important Bird and Biodiversity Areas, Urban Trees and Forests/Urban Greens/City Forest, Urban/City Gardens, Unique Agricultural Systems, Individual Green Trees.

Inland Waterbodies like Lakes and Ponds, Riverine Waterbodies, Artificial Waterbodies.

Coastal Waterbodies like Important Coastal Biodiversity Areas (ICBAs).

Marine areas like Important Marine Biodiversity Areas (IMBAs), Ecologically/Culturally/International Significant Coastal and Marines Areas.

OECMs RECOGNISED IN INDIA

- Aravalli Biodiversity Park, Gurgaon (First OECM to be identified in India).
- Godrej's Pirojshanagar Mangroves, Maharashtra
- Coromandel Birds Paradise, Andhra Pradesh
- TVS Motor Company Nature Conservation Reserve, Tamil Nadu
- Sacred Groves of Nitii (Hong), Arunachal Pradesh.
- Kadwa Kosi Floodplains, Bihar
- Jagatpur Lake, Bihar
- Apatani, Arunachal Pradesh
- Zabo Farming System, Nagaland

6. ECOSYSTEM RESTORATION FLAGSHIPS

- The UN has recognized 10 ground-breaking efforts from around the globe for their role in restoring the natural world.
- The winning initiatives were unveiled at the UN Biodiversity Conference (COP15) in Montreal.
- The initiatives were declared World Restoration Flagships and are eligible to receive UN-backed promotion, advice or funding.
- They were selected under the banner of the United Nations Decade on Ecosystem Restoration, a global movement coordinated by the United Nations Environment Program (UNEP) and the United Nations Food and Agriculture Organisation (FAO).
- Together, the 10 flagships aim to restore more than 68 million hectares − an area bigger than Myanmar, France or Somalia − and create nearly 15 million jobs.
- In revealing the World Restoration Flagships, the UN Decade seeks to honour the best examples of large-scale and long-term ecosystem restoration, embodying the 10 Restoration Principles of the UN Decade on Ecosystem Restoration.

LIST OF THESE EFFORTS

Trinational Atlantic Forest Pact (Brazil, Paraguay and Argentina)	Small Island Developing States Restoration Drive (protecting coral reefs and mangroves - Vanuatu, St Lucia and Comoros)
Abu Dhabi Marine Restoration (Safeguarding the world's second-largest dugong population)	Altyn Dala Conservation Initiative (in Kazakhstan to protect central Asian steppes)
Great Green Wall for Restoration and Peace (restore savannas, grasslands and farmlands across Africa by African Union)	Central American Dry Corridor (six countries: Costa Rica, El Salvador, Guatemala, Honduras, Nicaragua and Panama)
Ganges River Rejuvenation – Namami Ganges Initiative (National mission for clean Ganga)	Building with Nature in Indonesia (for mangrove restoration and protecting Demak, a low-lying coastal community)
Multi-Country Mountain Initiative (Serbia, Kyrgyzstan, Uganda and Rwanda)	Shan-Shui Initiative in China (combines 75 large-scale projects to restore ecosystems)

The UN Decade on Ecosystem Restoration runs until 2030, which is also the deadline for achieving the Sustainable Development Goals. Without halting and reversing the degradation of terrestrial and aquatic ecosystems, 1 million species are at risk of extinction. Scientists say restoring only 15 per cent of ecosystems in priority areas and thereby improving habitats can cut extinctions by 60 per cent.

7. UNITED NATIONS ENVIRONMENT PROGRAMME (UNEP)

UNEP is the leading global environmental authority that sets the global environmental agenda, promotes the coherent implementation of the environmental dimension of sustainable development within the United Nations system, and serves as an authoritative advocate for the global environment.

Headquartered in Nairobi, Kenya, it works through its divisions as well as its regional, liaison and out-posted offices and a growing network of collaborating centres of excellence. It also hosts several environmental conventions, secretariats and inter-agency coordinating bodies.

SECRETARIATS HOSTED WITHIN UNEP

It hosts the secretariats of many critical multilateral environmental agreements and research bodies, bringing together nations and the environmental community to tackle the greatest challenges of our time. These include the following:

- Convention on Biological Diversity
- Convention on International Trade in Endangered Species of Wild Fauna and Flora

- Minamata Convention on Mercury
- Basel, Rotterdam and Stockholm Conventions
- Vienna Convention for the Protection of Ozone Layer and the Montreal Protocol
- Convention on Migratory Species
- Carpathian Convention
- Bamako Convention
- Tehran Convention

Carpathian Convention is a subregional treaty to foster sustainable development and the protection of the Carpathian region. The Carpathians are one of Europe's largest mountain ranges, a unique natural treasure of great beauty and ecological value, and home to the headwaters of major rivers.

Bamako Convention is a treaty of African nations prohibiting the import into Africa of any hazardous (including radioactive) waste.

Tehran Convention or also known as the Framework Convention for the Protection of the Marine Environment of the Caspian Sea is a regional convention signed by the official representatives of the five littoral Caspian states: Azerbaijan, Iran, Kazakhstan, Russian Federation and Turkmenistan in Tehran (Iran) on 4 November 2003.

FUNDING

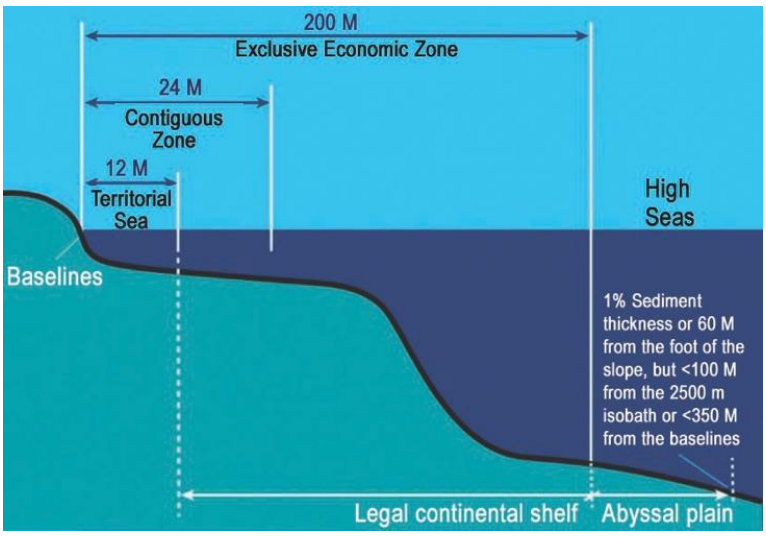
- **Environment Fund** is the core source of flexible funds.
- **Earmarked Fund** (funds given or "earmarked" to a specific project, theme, country etc.) enable us to expand and replicate our program in more countries and with more partners. Main providers of earmarked funds include the Global Environment Facility, the Green Climate Fund and the European Commission.

8. UNITED NATIONS CONVENTION ON THE LAW OF THE SEA (UNCLOS)

- UNCLOS lays down a comprehensive regime of law and order in the world's oceans and seas.
- The 1982 convention was signed by 117 states & it establishes rules governing all uses of the ocean and its resources.
- The convention also provides the framework for the development of a specific area of law of the sea.
- The convention is a lengthy document having 446 articles grouped in 7 parts in 9 annexes.

BASELINE

As otherwise provided in the UNCLOS convention, the normal baseline for measuring the breadth of the territorial sea is the low-water line along the coast as marked on large-scale charts officially recognized by the coastal State.



TERRITORIAL SEA

- Every state has the right to establish the breadth of its territorial sea up to a limit not exceeding 12 nautical miles measured from the baseline determined in accordance with this convention.
- The outer limit of the territorial sea is the line every point of which is at a distance from the baseline equal to the breadth of the territorial sea.

- Where the coasts of two States are opposite or adjacent to each other, neither of the two States is entitled, failing agreement between them to the contrary, to extend its territorial sea beyond the median line every point of which is equidistant from the nearest points on the baselines from which the breadth of the territorial seas of each of the two States is measured.

CONTIGUOUS ZONE

- The contiguous zone generally extends 12 nautical miles beyond the territorial sea limit. It consists of a combination of Revenue and Public health or Quarantine jurisdiction.
- The coastal state can prejudice a foreign flag vessel beyond the territorial see if there are reasonable grounds for assuming they are about to violate Customs or Public Health Regulations

EXCLUSIVE ECONOMIC ZONE

The exclusive economic zone shall not extend beyond 200 nautical miles from the baselines from which the breadth of the territorial sea is measured.

9. WORLD METEOROLOGICAL ORGANISATION (WMO)

As a specialised agency of the United Nations, WMO is dedicated to international cooperation and coordination on the state and behaviour of the Earth's atmosphere, its interaction with the land and oceans, the weather and climate it produces, and the resulting distribution of water resources.

WMO FACILITATES AND PROMOTES

- Establishment of an integrated Earth System observation network to provide weather, climate and water-related data.
- Establishment and maintenance of data management centres and telecommunication systems.
- Activities in operational hydrology as well as closer cooperation between National Meteorological and Hydrological Services in states and territories where they are separate.

WMO STRATEGIC PLAN

- Disaster risk reduction
- The Global Framework for Climate Services (GFCS)
- The WMO Integrated Global Observing System (WIGOS)
- Aviation meteorological services
- Polar and high mountain regions
- Capacity development
- Governance

10. THE ECONOMICS OF ECOSYSTEM & BIODIVERSITY (TEEB)

The Economics of Ecosystems and Biodiversity (TEEB) is a global initiative focused on "making nature's values visible".

Its principal objective is to mainstream the values of biodiversity and ecosystem services into decision-making at all levels. It aims to achieve this goal by following a structured approach to valuation that helps decision-makers recognize the wide range of benefits provided by ecosystems and biodiversity, demonstrate their values in economic terms and, where appropriate, capture those values in decision-making.

11. WORLDWIDE FUND FOR NATURE (WWF)

It is an international non-governmental organization founded in 1961 that works in the field of wilderness preservation and the reduction of human impact on the environment.

WWF is the world's largest conservation organization, with over five million supporters worldwide, working in more than 100 countries and supporting around 3,000 conservation and environmental projects.

The Worldwide Fund for Nature (WWF) is part of the Steering Group of the Foundations Platform F20, an international network of foundations and philanthropic organizations.

12. SEEA-ECOSYSTEM ACCOUNTING

It constitutes an integrated statistical framework for organising data about habitats and landscapes, measuring ecosystem services, tracking changes in ecosystem assets, and linking this information

The UN Statistical Commission adopted the SEEA-Ecosystem Accounting in 2021.

SEEA-EA IS BUILT ON FIVE ECOSYSTEM ACCOUNTS

1. Ecosystem extent
2. Ecosystem condition
3. Ecosystem services (Physical)
4. Ecosystem services (Financial)
5. Monetary Ecosystem Asset

This exercise aims to improve economic benchmarks such as GDP which currently do not take into account the value of nature and other externalities.

13. EDGAR FOOD

It is a database which can be used to assess how changes in consumer behavior or technology may impact food system-derived greenhouse gas emissions.

It incorporates land use data for over 200 countries and goes back to 1990 and spans multiple sectors, which will enable tracking of ongoing and future trends.

This database is an initiative of FAO.

14. ENVIRONMENTAL DNA METABARCODING (EDNA)

It involves taking samples of soil or water and searching for fragments of DNA specific to certain species. This method eliminates the time-consuming process of sorting individual samples and enables us to identify different species present in a river system.

15. GLOBAL EBA FUND

A fund established by UNEP and IUCN, which will provide grants for innovative approaches to ecosystem-based adaptation through seed capital.

OBJECTIVES

1. Build awareness and understanding of the critical role of natural assets.
2. Encourages catalytic initiatives to help overcome identified barriers for upscaling EbA.

ABOUT ECOSYSTEM-BASED ADAPTATION (EBA)

It is the use of biodiversity and ecosystem services as part of an overall adaptation strategy to help people to adapt to the adverse effects of climate change.

16. NATURE-BASED RECOVERY INITIATIVE

- It aims to ensure that at least 10% of overall investments in stimulus packages are channeled to nature, nature-based solutions and interventions that add value to nature.
- It is an initiative of IUCN.

17. POWERING PAST COAL ALLIANCE

It is a coalition that aims to advance the transition from unabated coal power generation to clean energy by 2030 in OECD and EU and by no later than 2050 in the rest of the world. The initiative was launched by UK and Canadian governments at COP23 in 2017.

MEMBERSHIP

- 36 National Governments (India is not a member), Subnational Governments and Organisations.

NATIONAL COAL INDEX

- The concept and design of the Index have been developed by the Indian Statistical Institute, Kolkata.
- NCI is a price index combining the prices of coal from all the sales channels- Notified Prices, Auction Prices, and Import Prices.
- The majority of coal is sold through the Notified Prices. For Non-Coking Coal, CIL fixes notified prices for each grade.
- The Index is meant to encompass all transactions of raw coal in the Indian market.
- This includes coking and non-coking of various grades transacted in the regulated (power and fertilizer) and non-regulated sectors.

18. BIOTRADE INITIATIVE

- UNCTAD's BioTrade Initiative aims to contribute to the conservation and sustainable use of biodiversity through the promotion of trade and investment in BioTrade products and services.
- BioTrade is understood as activities related to the collection or production, transformation and commercialization of goods and services derived from biodiversity (genetic resources, species and ecosystems), under the environmental, social and economic sustainability criteria. These criteria, known as the BioTrade Principles and Criteria (P&C), have been the core foundation that guides the implementation of the activities of the UNCTAD BioTrade Initiative, the BioTrade programs and other related activities since their inception in 2007.

19. GROSS ECOSYSTEM PRODUCT

- Gross Ecosystem Product (GEP) aims to specific indicators to measure the total economic value of all ecosystem products and services.
- GEP is the total value of final ecosystem goods and services supplied to human well-being in a region annually and can be measured in terms of biophysical value and monetary value.
- Ecosystems that can be measured include natural ecosystems such as forests, grassland, wetland, desert, freshwater and ocean, and artificial systems that are based on natural processes like farmland, pastures, aquaculture farms and urban green land, etc.
- IUCN is currently carrying out pilot research projects on GEP with partners.

20. GREEN VOYAGE 2050 PROJECT

- It is a partnership project between Norway and IMO launched in 2019 aiming to transform the shipping industry towards a lower carbon future.
- India has been selected as the first country under IMO Green Voyage 2050 project to conduct a pilot project related to Green Shipping.

GLOBAL INDUSTRY ALLIANCE TO SUPPORT LOW CARBON SHIPPING

Low Carbon GIA was originally established in 2017 under the GEF-UNDP-IMO GloMEEP Project. Following the completion of the GloMEEP Project in December 2019, the Low Carbon GIA was re-established and continues to

operate under the framework of the IMO-Norway GreenVoyage2050 Project. The Low Carbon GIA aims to bring together maritime industry leaders to support an energy efficient and low-carbon maritime transport system.

21. CLYDEBANK DECLARATION

- The UK announced the launch of the Clydebank Declaration on Green Shipping Corridors at the COP26 conference in Glasgow.
- The signatories of the Declaration are to support the establishment of green shipping corridors – zero-emission maritime routes between 2 (or more) ports.
- As part of the declaration, the signatory countries will support the establishment of at least six green shipping corridors by 2050.
- Signatory countries: Around 22 countries have signed the declaration.
- India has not signed the declaration yet.

22. ICIMOD

- ICIMOD stands for international center for integrated mountain development. A regional intergovernmental learning & knowledge-sharing center for Hindu-Kush.
- It has eight regional member countries of the Hindu Kush Himalaya – Afghanistan, Bangladesh, Bhutan, China, India, Myanmar, Nepal, and Pakistan – and is based in Kathmandu, Nepal. Also called 'Third Pole.'
- ICIMOD aims to assist mountain people to understand the changes, adapt to them, and make the most of new opportunities while addressing upstream-downstream issues.
- The ICIMOD has been facilitating the Mount Kailash Sacred Landscape Initiative, involving India, Nepal and China. These countries will collaborate on eco-restoration and bio-diversity management in their parts of the territory.

23. IPBES

IPBES stands for Intergovernmental Science Policy Platform on Biodiversity & Ecosystem Services.

It is an independent intergovernmental body established by States to strengthen the science-policy interface for biodiversity and ecosystem services for the conservation and sustainable use of biodiversity, long-term human well-being and sustainable development.

It was established in Panama City, on 21 April 2012 by 94 Governments.

It is not a United Nations body. However, at the request of the IPBES Plenary and with the authorization of the UNEP Governing Council in 2013, the UNEP provides secretariat services to IPBES.

WHY DOES IPBES MATTER?

Biodiversity and nature's benefits to people underpin almost every aspect of human development and are key to the success of the new Sustainable Development Goals. They help to produce food, clean water, regulate climate and even control disease. Yet they are being depleted and degraded faster than at any other point in human history. IPBES is unique – harnessing the best expertise from across all scientific disciplines and knowledge communities – to provide policy-relevant knowledge and to catalyze the implementation of knowledge-based policies at all levels in government, the private sector and civil society.

WHAT DOES IPBES DO?

The work of IPBES can be broadly grouped into four complementary areas:

- **Assessments**: On specific themes (Ex. "Pollinators, Pollination and Food Production"); methodological issues (Ex. "Scenarios and Modelling); and at both the regional and global levels (Ex. "Global Assessment of Biodiversity and Ecosystem Services").
- **Policy Support**: Identifying policy-relevant tools and methodologies, facilitating their use, and catalyzing their further development.

- **Building Capacity & Knowledge**: Identifying and meeting the priority capacity, knowledge and data needs of our member States, experts and stakeholders.
- **Communications & Outreach**: Ensuring the widest reach and impact of our work.

24. BASEL CONVENTION

Aims at protecting human health and environment against adverse effects of hazardous wastes. Its scope of application covers a wide range of wastes defined as "hazardous wastes" based on their origin and/or composition and their characteristics, as well as two types of wastes defined as "other wastes" - household waste and incinerator ash.

OBJECTIVES OF THE CONVENTION

The Convention covers hazardous wastes that are explosive, flammable, poisonous, infectious, corrosive, toxic, or eco-toxic. For environmentally sound management (ESM) of hazardous waste, the Convention aims towards restricting transboundary movements of hazardous wastes except where it is perceived to be under the principle of ESM.

INDIA AND THE CONVENTION

India ratified Basel convention in 1992. India has been taking measures for the effective management of hazardous wastes and thereby meeting the obligations of the Basel Convention.

Hazardous and Other Wastes (Management & Transboundary Movement) Rules, 2016 was notified to ensure the safe handling, processing, treatment, storage, collection, transportation, collection, and disposal of hazardous waste.

25. ROTTERDAM CONVENTION

The Rotterdam Convention on the prior informed consent procedure for certain hazardous chemicals and pesticides in international trade is a multilateral agreement. The agreement provides obligations on the import and export of certain hazardous chemicals.

Parties are empowered, through virtue of membership, to make informed decisions about the chemicals they want to receive, and to exclude those they believe they cannot manage safely. The convention's objective is to promote shared responsibility and cooperative efforts among parties in the international trade of these chemicals to protect human health and the environment and contribute to the environmentally sound use of chemicals.

PRIOR INFORMED CONSENT (PIC)

It is a mechanism for formally obtaining and disseminating the decisions of importing Parties as to whether they wish to receive future shipments of those chemicals listed in Annex III of the Convention and for ensuring compliance with these decisions by exporting Parties.

26. STOCKHOLM CONVENTION ON PERSISTENT ORGANIC POLLUTANTS

It is a global treaty to protect human health and the environment from chemicals that remain intact in the environment for long periods (persistent), become widely distributed geographically (long range transport), accumulate in the fatty tissue of humans and wildlife (bioaccumulation), and have harmful impacts on human health or the environment (toxic).

PERSISTENT ORGANIC POLLUTANTS

They are also called forever chemicals. They are resistant to environmental degradation through photolytic, chemical and biological processes. Thus, they accumulate in the environment causing adverse impacts on

human health. To eliminate persistent organic pollutants from the environment, the Stockholm convention on persistent organic pollutants was held in 2001. India is a signatory of the convention.

PROPOSED PERSISTENT ORGANIC POLLUTANTS

1. Methoxychlor: It is a pesticide, used as a replacement for DDT against a wide range of pests including biting flies, houseflies, mosquito larvae, and cockroaches on field crops, vegetables, ornamentals, livestock & pets. It is very toxic to invertebrates and fish, has endocrine disrupting effects, has been detected in enviros of Arctic and Antarctica and human serum, adipose tissues, umblical cord and human breast milk.
2. UV-328: Used as an ultra-violet (UV) stabilizer in plastic products such as some personal care products, rubber and coatings. It is the first non-halogenated chemical to be considered to be added under Stockholm convention.
3. Dechlorane Plus: It is a chemical used as flame retardant in electrical wire, cable coatings, plastic roofing materials, connectors in TV and computer monitors.
4. Perfluorohexane Sulphonic Acid (PFHxS): They have high resistance to friction, heat, chemical agents, low surface energy and are used as a water, grease, oil and soil repellent They are group of industrial chemicals used in consumer products as a surfactant and sealant in carpets, leather, clothing, textiles, fire-fighting foams, papermaking, printing inks and non-stick cookware.

The Convention was adopted in 2001 and entered into force in 2004. India ratified the convention in 2006.

OBLIGATIONS UNDER THE CONVENTION

Under the convention, the chemicals can be listed for Elimination (Annex-A), Restriction (Annex-B) or Unintentional production (Annex-C).

The implementation of the convention requires its parties to take measures to eliminate or reduce the release of these POPs into the environment. Till date, 26 chemicals are listed as POPs under the Stockholm Convention. As of now, India has ratified only the 12 initially listed POPs. As a first step for the implementation of the Stockholm Convention in India, a National Implementation Plan has been prepared. India is in process of ratification of selected newly listed POPs.

UNION CABINET APPROVES RATIFICATION OF 7 ORGANIC POLLUTANTS

The Union Cabinet approved ratification of 7 Chemicals that are listed under Stockholm convention on persistent organic pollutants. Further, the cabinet delegated its powers of ratification to the union ministry of external affairs and the ministry of environment forest and climate change. These Ministries are already regulating the persistent organic pollutants under domestic regulations.

7 PERSISTENT ORGANIC POLLUTANTS: The 7 persistent organic pollutants that are currently being ratified by the government of India.

Chlordecone, Hexabromobiphenyl, Hexabromobiphenyl ether and Heptabromodiphenylether (Commercial octa BDE), Tetrabromodiphenyl ether and Pentab-romodiphenyl ether (Commercial penta-BDE), Pentachlorobenzene, Hexabromocyclododecane, Hexa-chlorobutadiene

These pollutants were listed in the Stockholm convention already. In 2018, the Ministry of Environment Forest and Climate Change notified regulation of Persistent organic pollutants rules. Under these rules, the manufacture, trade, use, import and export of these seven Chemicals were prohibited. This was completely domestic and not in accordance with Stockholm Convention.

WHY IS INDIA RATIFYING NOW?

India ratified the convention in 2006. However, India till date stays in an opt-out position following domestic rules on usage of POPs.

OPT-OUT POSITION IN STOCKHOLM CONVENTION

The convention allows its members to stay in such a position under article 25. According to the article, the amendments made to the convention shall not be enforced by its members unless an instrument of acceptance for ratification or approval is deposited with the UN.

The cabinet approval will demonstrate India's commitment towards international obligation in protecting the environment from POPs. This will in turn allow India to access the Global Environment Facility's financial resources by updating the National Implementation Plan.

27. CONVENTION ON BIOLOGICAL DIVERSITY (UNCBD)

It provides a global legal framework for action on biodiversity. It brings together the Parties in the Conference of the Parties (COP) which is the Convention's governing body that meets every two years, or as needed, to review progress in the implementation of the Convention, to adopt programs of work, to achieve its objectives, and provide policy guidance.

It entered into force on 29 December 1993. It has 3 main objectives:

1. Conservation of biological diversity
2. Sustainable use of components of biological diversity
3. Fair and equitable sharing of the benefits arising out of the utilization of genetic resources

The COP is assisted by the Subsidiary Body on Scientific, Technical, and Technological Advice (SBSTTA), which is made up of government representatives with expertise in relevant fields, as well as observers from non-Party governments, the scientific community, and other relevant organizations. SBSTTA is responsible for providing recommendations to the COP on the technical aspects of the implementation of the Convention.

Other subsidiary bodies have been established by the COP to deal with specific issues as they arise. These are called "ad hoc open-ended Working Groups" because they are established for a limited mandate and period of time, and because they are open to all Parties as well as the participation of observers. Current Working Groups are:

- **Working Group on Access & Benefit-Sharing (ABS)** is currently the forum for negotiating an international regime on access and benefit sharing.
- **The Working Group on Article 8 (j)** addresses issues related to protection of traditional knowledge.
- **The Working Group on Protected Areas** is guiding and monitoring the implementation of the program of work on protected areas.
- **Subsidiary Body on Implementation (SBI)** reviews progress in implementing the Convention and identifies strategic actions to enhance implementation, including how to strengthen the means of implementation. It also addresses issues associated with the operations of the convention and the Protocols.
- The Open-ended Ad Hoc Intergovernmental Committee (ICNP) for the Nagoya Protocol on ABS was established as an interim governing body for the Nagoya Protocol until the first meeting of the Parties to the Protocol at which time it will cease to exist.

All living organisms, plants, animals and microbes, carry genetic material that could be potentially useful to humans. These resources can be taken from the wild, domesticated or cultivated. They are sourced from environments in which they occur naturally (in situ), or from human-made collections such as botanical gardens, gene banks, seed banks and microbial culture collections (ex-situ).

WHAT IS ACCESS AND BENEFIT-SHARING?

It refers to the way in which genetic resources may be accessed, and how the benefits that result from their use are shared between the people or countries using the resources (users) and the people or countries that provide them (providers).

WHY IS IT IMPORTANT?

Providers of genetic resources are governments or civil society bodies, which can include private landowners and communities within a country, who are entitled to provide access to genetic resources and share the benefits resulting from their use.

The access and benefit-sharing provisions of the Convention on Biological Diversity (CBD) are designed to ensure that physical access to genetic resources is facilitated and that the benefits obtained from their use are shared equitably with the providers.

In some cases, this also includes valuable traditional knowledge associated with genetic resources that come from ILCs. The benefits to be shared can be monetary, such as sharing royalties when the resources are used to create a commercial product, or non-monetary, such as the development of research skills and knowledge. It is vital that both users and providers understand and respect institutional frameworks such as those outlined by the CBD and in the Bonn Guidelines. These help governments to establish their own national frameworks which ensure that access and benefit-sharing happen fairly and equitably.

HOW DOES IT WORK?

Access and benefit-sharing are based on prior informed consent (PIC) being granted by a provider to a user and negotiations between both parties to develop mutually agreed terms (MAT) to ensure the fair and equitable sharing of genetic resources and associated benefits.

- **Prior informed consent (PIC):** is the permission given by the competent national authority of a provider country to a user prior to accessing genetic resources, in line with an appropriate national legal and institutional framework.
- **Mutually agreed terms (MAT):** is an agreement reached between the providers of genetic resources and users on the conditions of access and use of the resources, and the benefits to be shared between both parties.

These conditions are required under Article 15 of the CBD, which was adopted in 1992 and provides a global set of principles for access to genetic resources, as well as the fair and equitable distribution of the benefits that result from their use

In 2021, the 15th meeting of the Conference of the Parties (COP) to the UNCBD was held virtually in Kunming, China. The COP 15's major goal was to create and accept a post-2020 "Global Biodiversity Framework" to replace and update the Strategic Plan for Biodiversity (SPB) 2011-2020 and the Aichi Biodiversity Targets.

For the next ten years, the framework will comprise a set of global goals, targets, and indicators that will guide biodiversity and ecosystem conservation, protection, restoration, and sustainable management.

The first draft of the GBF was released in July 2021, containing 21 targets for 2030 and 4 Goals to achieve humanity's "living in harmony with nature," vision by 2050.

Parties will reconvene in 2022 for further negotiations and to come to a final agreement on the post-2020 Global Biodiversity Framework.

HIGHLIGHTING FEATURES OF THE COP

- **Kunming Declaration:**
  - The declaration called for immediate and comprehensive action in all areas of the global economy to reflect biodiversity concerns.
  - More than a hundred countries, including India, have pledged to
    - Work together to design, adopt, and implement a viable post-2020 global biodiversity framework.
    - Reverse the current biodiversity loss.
    - Ensure that biodiversity is on the mend by 2030 at the very latest.
- China established the Kunming Biodiversity Fund with a budget of around USD 230 million to fund programs that safeguard biodiversity in underdeveloped nations.
- It also praised many countries' efforts and commitments to safeguard 30% of their land and sea regions by 2030 (30 by 30 objective), which is crucial for reversing a major cause of environmental degradation.

- The summit emphasised the importance of private sector participation, including an open letter from company CEOs to international leaders encouraging decisive action.
- Global Environment Facility, UNDP, and UNEP have pledged to expedite financial and technical assistance to developing countries to help them implement the Global Environment Facility.

GLOBAL BIODIVERSITY FRAMEWORK FUND (GBFF)

- It will scale up financing for the implementation of the **Kunming-Montreal Global Biodiversity Framework**.
- The GBFF aims to help countries achieve the Global Biodiversity Framework goals and targets with a strategic focus on strengthening national-level biodiversity management, planning, policy, governance, and finance approaches.
- The Fund was ratified by 186 countries and launched at the Seventh GEF Assembly in Vancouver, Canada in August 2023.
- Public, private, and philanthropic funding will be disbursed through the GBFF's streamlined funding procedures.
- The Fund will provide enhanced support to Indigenous Peoples and local communities, Small Island Developing States, and Least Developed Countries, according to their own priorities.
- The **1st Global Biodiversity Framework Fund Council Meeting took place in February 2024 in Washington, DC.**

28. UNCBD COP-16

The 16th meeting of the Conference of the Parties (COP) to the **United Nations Convention on Biological Diversity (UNCBD)** was concluded in Cali, Columbia.

MAJOR OUTCOMES

Cali Fund: It aims to **finance biodiversity conservation** and reward those who are frontline protectors of biodiversity such as indigenous people and forest dwellers. It will address how pharmaceutical, biotechnology, animal and plant breeding and other industries benefiting from **digital sequence information on genetic resources** (DSI) should share those benefits with developing countries and Indigenous Peoples and local communities. Large companies and other major entities benefiting commercially from DSI use have to contribute to the Cali Fund, based on a percentage of their profits or revenues.

Entities that fulfil at least two out of three of these thresholds (total assets: $20 million, sales: $50 million, profit: $5 million), should **contribute to the global fund 1%** of their profits or 0.1% of their revenue.

COP 16 at Kali saw the first major outcome in the form of an agreement to share the benefits from the uses of **digital sequence information on genetic resources** (DSI) more fairly and equitably.

The **Kunming-Montreal Global Biodiversity Framework (KMGBF)** 2022, led to an agreement to establish a multilateral mechanism, including a global fund in this regard.

Indigenous People and Local Communities (IPLC):

- COP 16 witnessed the formal incorporation of indigenous communities in the official decision making of the UN biodiversity process. Till now, for the last 20 years, indigenous people and local communities had informal working groups as part of the UN Biodiversity process. For the first time, a permanent subsidiary body is established on Article 8 (j) which means that now such communities do not have to depend on the goodwill of other countries to represent their interest.
- Parties agreed to **develop a new Strategy for Resource Mobilisation** to help secure $200 billion annually by 2030 to support biodiversity initiatives worldwide.
- **New guidelines** were proposed for the **management of invasive alien species** through new databases, improved cross-border trade regulations, and enhanced coordination with e-commerce platforms.
- COP 16 agreed on a new and evolved process to **identify Ecologically or Biologically Significant Marine Areas EBSAs.**

- **The Global Action Plan on Biodiversity and Health** has been designed and approved to help curb the emergence of zoonotic diseases, prevent non-communicable diseases, and promote sustainable ecosystems.

29. CARTAGENA PROTOCOL ON BIOSAFETY TO UNCBD

An international treaty governing movements of living modified organisms (LMOs) resulting from modern biotechnology from one country to another that may have adverse effects on biological diversity, taking also into account risks to human health.

Adopted in 2000 as a supplementary agreement to the Convention on Biological Diversity (UNCBD).

The Protocol seeks to protect biological diversity from the potential risks posed by living modified organisms resulting from modern biotechnology.

It establishes an advance informed agreement (AIA) procedure for ensuring that countries are provided with the information necessary to make informed decisions before agreeing to the import of such organisms into their territory. The Protocol contains a reference to a precautionary approach.

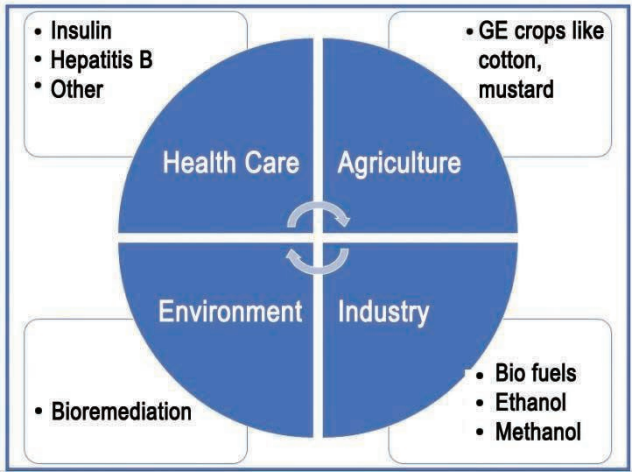
It also establishes a Biosafety Clearinghouse to facilitate the exchange of information on living modified organisms and to assist countries in the implementation of the Protocol.

India has ratified this protocol.

The convention does not apply to Pharmaceuticals for humans that are addressed by other international Agreements or organizations

NAGOYA — KUALA LUMPUR SUPPLEMENTARY PROTOCOL ON LIABILITY AND REDRESS TO THE CARTAGENA PROTOCOL ON BIOSAFETY

The objective of the Supplementary Protocol is to contribute to the conservation and sustainable use of biological diversity by providing international rules and procedures for liability and redress in the event of damage resulting from LMOs.



The Protocol's AIA procedure does not apply to:

1. LMOs in transit.
2. LMOs destined for contained use in the Party of import.
3. LMOs intended for direct use as food or feed or for processing (LMOs-FFP).

Under the Protocol, Advance Informed Agreement (AIA) procedure applies to the first intentional transboundary movement of an LMO for intentional introduction into the environment of the Party of import.

PRECAUTIONARY PRINCIPLE

Proposed as a new guideline in environmental decision-making. It has four central components:

- Taking preventive action in the face of uncertainty.
- Shifting the burden of proof to proponents of an activity.
- Exploring a wide range of alternatives to possibly harmful actions.
- Increasing public participation in decision-making.

30. KUNMING – MONTREAL AGREEMENT

The United Nations Biodiversity Conference (COP15) ended in Montreal, Canada, on 19 December 2022 with a landmark agreement to guide global action on nature. This agreement has been achieved under Convention on Biological Diversity (CBD).

HISTORICAL CONTEXT

- UN Framework Convention on Climate Change (UNFCCC) and UNCBD were both outcomes of 1992 Rio Earth Summit — as was the third member of the family, Convention to Combat Desertification (CCD), which deals specifically with the issue of land degradation. The CBD came into force in 1993; the other two in the following year.
- Signatories to Convention on Biological Diversity (CBD) meet every two years to work on a global plan to halt biodiversity loss and restore natural ecosystems.
- The CBD has given rise to two 'supplementary' agreements — the Cartagena Protocol of 2003 and the Nagoya Protocol of 2014. Both agreements take their names from the places where they were negotiated.
- The Cartagena Protocol on Biosafety seeks to protect biodiversity from genetically modified organisms by ensuring their safe handling, transport and use.
- The Nagoya Protocol on Access and Benefit Sharing deals with the commercial utilisation of biological and genetic resources, for example, by pharma companies. It asks the host countries to provide access to its genetic resources in a legal, fair and non-arbitrary manner and, as mentioned above, offers them a fair and equitable share of benefits arising out of the utilisation of those resources.
- The Montreal meeting was the 15th edition of this conference, hence the name COP15.

- It seeks to respond to the Global Assessment Report of Biodiversity and Ecosystem Services issued by the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES) in 2019 which provides ample evidence that, despite ongoing efforts, biodiversity is deteriorating worldwide at rates unprecedented in human history: An average of around 25% of species in assessed animal and plant groups are threatened, suggesting that around 1 million species already face extinction.
- In total there are 23 targets and 4 goals.
- It sets out targets for 2030:
  - on protection for degraded areas,
  - resource mobilisation for conservation,
  - compensation for countries that preserve biodiversity,
  - halting human activity linked to species extinction,
  - reducing by half the spread of invasive alien species (introduced plants and animals that affect endemic biodiversity),
  - cutting pollution to non-harmful levels and
  - minimising climate change impact and ocean acidification.

Key facts:

- Agreement calls upon members to adopt biodiversity-supporting methods such as agroecology and sustainable intensification.
- It also talks about turning cities into hosts of biodiversity, by expanding urban green and blue spaces. Urban planning should also be biodiversity-inclusive, enhancing native biodiversity, ecological connectivity and integrity, and improving human health and well-being and connection to nature.

30×30 TARGET

This target calls for 30% of the earth's land and sea to be conserved through the establishment of protected areas (PAs) and other area-based conservation measures (OECMs). A related commitment is to ensure that restoration activities would be started on at least 30 per cent of degraded land or marine ecosystems by 2030.

- Four GBF goals for 2050 deal with:
  - Maintaining ecosystem integrity and health to halt extinctions,
  - Measuring and valuing ecosystem services provided by biodiversity,
  - Sharing monetary and non-monetary gains from genetic resources and digital sequencing of genetic resources with indigenous people and local communities,

- Raising resources for all countries to close a biodiversity finance gap of an estimated $700 billion.

- GBF envisages that there will be access to justice and information related to biodiversity for indigenous peoples and local communities, respecting their cultures and rights over lands, territories, resources, and traditional knowledge.

MONITORING AND IMPLEMENTATION

- Member nations need to submit a revised and updated national biodiversity strategy and action plan at the conference to be held in 2024.
- Further, the parties to the CBD should submit national reports in 2026 and 2029 to help prepare global reviews.
- High-level discussions on the progress reviews should be held in 2024 and 2026.

FUNDING ARRANGEMENTS

- By 2030, the GBF hopes to see at least $200 billion raised per year from all sources — domestic, international, public and private — towards implementation of national action plans.
- In terms of international funding, developing countries should get at least $20 billion a year by 2025 and at least $30 billion by 2030 through contributions from developed countries.

BIODIVERSITY CREDIT

- A *'biodiversity credit'* is an economic instrument used to finance activities that deliver net positive biodiversity gains.
- It is a mechanism that allows individuals and companies to invest in environmental projects that contribute to richer biodiversity.
- Different from carbon or biodiversity offset - Unlike carbon or biodiversity offsets, which are payments made by a business to compensate for its damaging impacts on location-specific ecosystems, biodiversity credits allow companies to support nature-positive action, funding long-term conservation and restoration of nature, a higher order contribution than simply offsetting negative impact.

GENETIC ENGINEERING APPRAISAL COMMITTEE

GEAC was established under MoEFCC. It is the Apex body notified under Rules 1989 to accord approval of activities involving large-scale use of hazardous microorganisms and recombinants in research and industrial production from the environmental angle.

COMPOSITION OF GEAC

- **Chairman - Additional / Special Secretary, MoEFCC**
- **Co-Chairman:** Representative of Department of Bio-technology Vice-Chairman —Joint Secretary, MoEFCC

CROPS	MODIFICATION	REMARKS
Bt cotton	crylAc gene	First and only GM crop approved in India. India ranks first in global cotton production
Bt brinjal	Bt brinjal is created by inserting a crystal protein gene (CrylAc) from soil bacteria Bacillus thuringenisus (Bt) and it is resistant to Fruit and Shoot Borer (Leucinodes orbonalis)	In 2009, GEAC recommended the approval of commercial cultivation of Bt brinjal (eggplant) In 2010, MoEFCC announced a moratorium on approval.
GE Mustard, DMH-11	GE Mustard, DMH-11, containing Barnase-Barstar system is under evaluation by the Government of India	Technology developed by Centre for Genetic Manipulation of Crop Plants, University of Delhi.

Six Competent Authorities and their composition have been notified under these Rules which are as follows:

i. Recombinant DNA Advisory Committee (RDAC)

ii. Institutional Biosafety Committees (IBSC)

iii. Review Committee on Genetic Manipulation (RCGM)

iv. Genetic Engineering Appraisal Committee (GEAC)

v. State Biosafety Coordination Committees (SBCC)

vi. District Level Committees (DLC)

While the RDAC is advisory in function, the IBSC, RCGM, and GEAC are of regulatory function; SBCC and DLC are for monitoring purposes.

31. INTERNATIONAL TREATY ON PLANT GENETIC RESOURCES

- The International Treaty on Plant Genetic Resources for Food and Agriculture (ITPGRFA) was adopted by the **Food and Agriculture Organisation of the United Nations** in 2001 during the 31st session of **FAO.**
- Signed in 2001 in Madrid and entered into force in 2004.
- Also known as **Seed Treaty**, **India is a signatory** to it.
- **First legally binding international instrument** to formally acknowledge the enormous contribution of indigenous people and small-holder farmers as traditional custodians of the world's food crops.
- **The treaty was aimed at:**
  - Recognising the significant contribution of farmers to the diversity of crops that feed the world;
  - Establishing a global system to provide farmers, plant breeders and scientists with access to plant genetic materials; and
  - Ensuring that recipients share benefits they derive from the use of these genetic materials with the countries where they have originated.

MULTILATERAL SYSTEM OF ACCESS AND BENEFIT SHARING (MLS)

- The International Treaty created and administered a unique global system that enables countries to exchange much-needed plant genetic material with one another through a Multilateral System of Access and Benefit-sharing.
- Crops grown in different environments develop differently and genetic material from food plants in one country can be essential in another country trying to increase food production, fight plants pests, diseases and the effects of climate change. This is made possible through the MLS of International Treaty.
- To date, the Multilateral System on Access and Benefit-sharing covers 64 of the world's major crops (fundamental to global food security), accounting for about 80% of our food derived from plants.

BENEFIT-SHARING FUND

- The International Treaty established and operated a special initiative operated for the benefit of developing countries – the Benefit-sharing Fund (BSF). This Fund supports agricultural projects for farmers, public institutions and others in developing countries to conserve and use PGRFA (Plant genetic resources for food and agriculture) to improve food crop production, fight plant pests and adapt to the effects of climate change. The high demand for project funding is indicative of the considerable gaps and significant need for support.
- Recipients for material from the Treaty's MLS are to pay a portion of any profits they gain from the use of such material into the BSF. The process of developing and commercialising new varieties takes time, so the amount of user-based payments is still low. Therefore, currently, the BSF rely primarily on voluntary contributions.

SVALBARD GLOBAL SEED VAULT & ITPGRFA

- The International Treaty provided the international legal framework needed for establishment of the **Svalbard Global Seed Vault in Norway**. Both the International Treaty and the Svalbard Global Seed Vault work to conserve and safeguard seeds. Both work to ensure food security sustainably. The Svalbard Global Seed Vault contributes to conservation by providing a safety backup of the world's major crops and plants for future food security.

32. ANTARCTIC TREATY

India signed the **Antarctic Treaty in 1983** and became a Consultative Party with voting rights. The **Indian Antarctic Act, 2022** was enacted to regulate activities at India's research stations in Antarctica.

- **Definition of Antarctica:** All the Land and ice shelves to the south of 60º latitude.
- **Features:** 14 million Sq. km + No Indigenous Population + Global Common + Principles of Peace, Science and Environmental protection.

INDIA AND ANTARCTICA

- India signed the Antarctic Treaty in 1983. Membership: 56 Countries (Including India).
- India ratified the Convention on the Conservation of Antarctic Marine Living Resources in 1985.
- India signed the Protocol on Environmental Protection to the Antarctic Treaty in 1998. (Madrid Protocol)
- **India's research stations in Antarctica:** Dakshin Gangotri (De-commissioned now); Maitri and Bharati. Managed by National Centre for Polar and Ocean Research (NCPOR), Goa under the Ministry of Earth Sciences.
- **Nov 2021**: India launched its 41st Scientific Expedition to Antarctica.

PROVISIONS OF ANTARCTIC TREATY

- Antarctica should be used only for peaceful purposes. No country should build military bases.
- Freedom of scientific investigation in Antarctica.
- No country should claim sovereignty over Antarctica based on setting up of Research stations.
- No Nuclear explosions or disposal of radioactive wastes
- Countries take appropriate efforts to abide by the provisions of the treaty.

COMMISSION FOR THE CONSERVATION OF ANTARCTIC MARINE LIVING RESOURCES

CCAMLR was established by the Convention on the Conservation of Antarctic Marine Living Resources. It was created to address the concern of unregulated increase in krill catches in the Southern Ocean that could be detrimental to Antarctic marine ecosystems, particularly for seabirds, seals, whales and fish that depend on krill for food.

Members: 25 States and EU. India is a member.

KRILL

- Small crustaceans which are found in all the world's oceans. They feed on phytoplankton and zooplankton and are the main source of food for many larger animals.
- In the Southern Ocean, the Antarctic Krill species is one the animals with largest total biomass.
- They are the main prey of baleen whales. Krills display vertical migration during the day, acting as feed for predators at the surface at night and at deeper levels during the day.

33. WINTER ARCTIC EXPEDITION

Himadri, India's Arctic research station at Ny-Alesund in the Norwegian archipelago of Svalbard in the arctic ocean, will now remain operational throughout the year.

ABOUT INDIA'S EXPEDITION

- India sent off its first winter expedition, comprising four scientists from four different institutions, to Himadri.

BASES IN THE ARCTIC

- At least 10 countries have set up permanent facilities at the International Arctic Research base in Ny-Alesund, a small town above the 78th parallel north.
- The Svalbard group of islands is the closest human habitation to the North Pole.
- India will join a small group of countries that operate their Arctic Research bases through the winter.

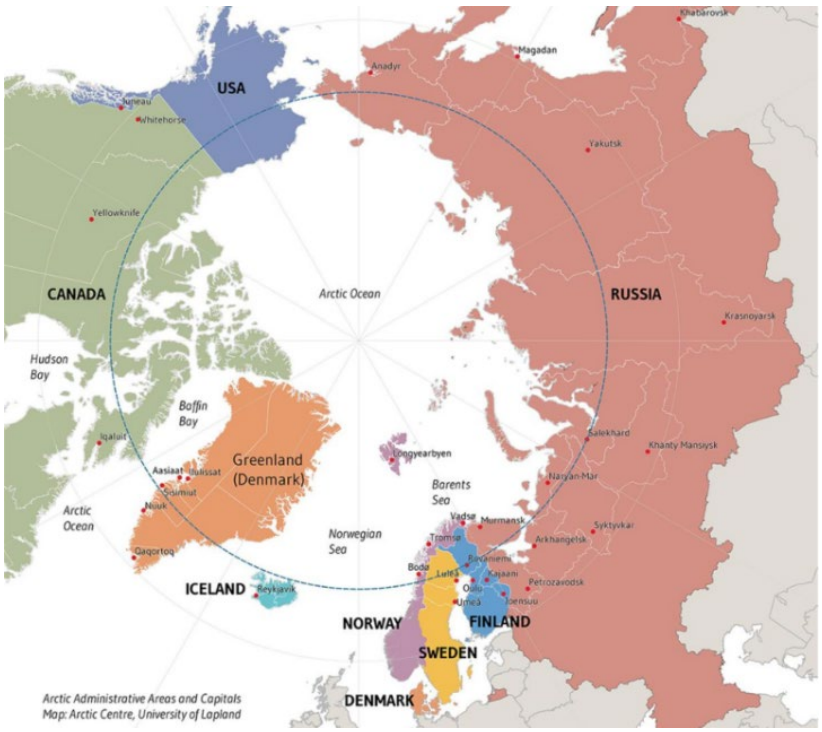
- Due to the extreme cold, taking scientific observation or sampling has remained largely limited to the summer months.

ARCTIC RESEARCH

- The polar regions, Arctic, and Antarctica, offer pristine environments for scientists to study a range of natural phenomena for atmospheric, oceanic, biological, geological, glaciological and earth science research.
- Recently, climate change research has been attracting scientists to the Arctic regions.

ARCTIC REGION

- The area above the Arctic circle, north of 66 degree and 34' minutes is part of eight countries-
  - Canada, Denmark, Finland, Iceland, Norway, Russia, Sweden, and the US form the Arctic Council plus Arctic Ocean.
- Scientific research in the Arctic regions is governed by international legal instruments like the Svalbard Treaty of 1920, and UN convention on the laws of the sea, apart from the Individual jurisdictions of the Arctic countries.



INDIA AND ARCTIC REGIONS

- India signed the Svalbard Treaty in Paris in 1920.
- However, it took until 2007 for the first Indian expedition comprising five scientists to visit the International Arctic research facility at Ny-Alesund with the purpose of initiating studies in Arctic microbiology, atmospheric sciences and geology.
- **Himadri**, India's permanent research station, began operation in July 2008.
- **Dakshin Gangotr**i in Antarctica was set up much earlier in 1983, two years after India's first expedition there. Dakshin Gangotri is now submerged under ice, but India's two other stations, **Maitri and Bharti,** are in use.
- Himadri, which is currently manned for 180 days every year, has so far hosted more than 400 Indian researchers who have made a total of about 200 visits to Svalbard.
- Himadri has been equipped for observations during polar nights and supplied with special winter gear, transport, and logistics support from Norwegian agencies.

IMPACT OF WARMING ON ARCTIC

- Temperature in the Arctic region has risen by about 4 degrees Celsius on average over the last 100 years.
- **2023 was the warmest year on record** with the highest average temperature at 6.4 degrees Celsius.
- According to IPCC, the extent of Arctic sea-ice has been declining at the rate of 13 percent per decade.
- At this rate, the Arctic could become ice free in less than 20 years.

MULTIYEAR ICE

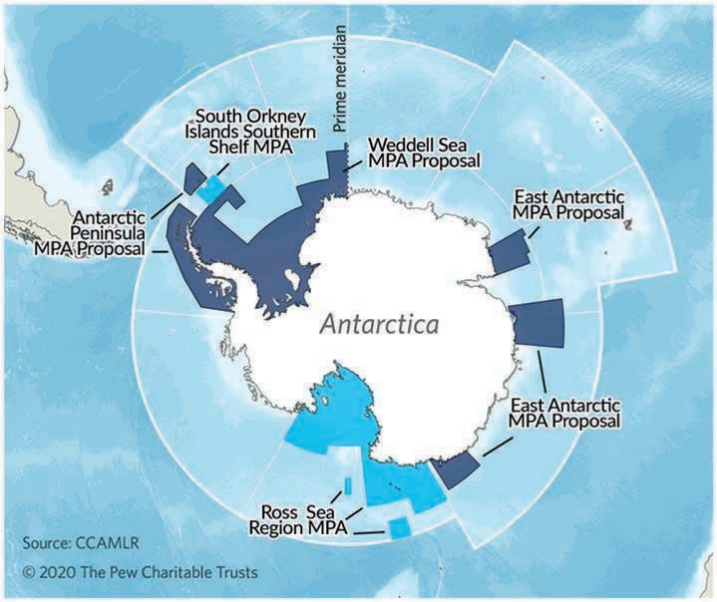
The ice which remains at the end of summer is called multiyear sea ice and is considerably thicker than its seasonal counterpart. It acts as a barrier to the transfer of both moisture and heat between the ocean and atmosphere. Over the past 40 years this multiyear sea ice has shrunk from around 7 million sq. km to 4 million.

BLUE OCEAN EVENT

"Blue ocean event" is defined as when the sea ice area drops below 1 million sq. km. This threshold is used mainly because older, thicker ice along parts of Canada and northern Greenland is expected to remain long after the rest of the Arctic Ocean is ice-free. We can't put an exact date on the last blue ocean event, but one soon would likely mean open water at the North Pole for the first time in thousands of years.

34. MARINE PROTECTED AREAS

India has supported the proposal of the European Union for creation of two marine protected areas (MPAs) in the Antarctic for protecting the Antarctic environment.



The two proposed Marine Protected areas are:

1. East Antarctic
2. Weddell Sea

ABOUT MARINE PROTECTED AREAS

Marine Protected Area is a marine area that protects all or part of the natural resources it contains.

Within an MPA certain activities are limited, or entirely prohibited, to meet specific conservation, habitat protection, and ecosystem monitoring or fisheries management objectives.

MPAs do not necessarily exclude fishing, research or other human activities. Already two MPAs have been agreed by CCAMLR members:

1. South Orkney Islands (World's first international MPA)
2. Ross Sea

ABOUT COMMISSION FOR THE CONSERVATION OF ANTARCTIC MARINE LIVING RESOURCES (CCAMLR)

- The CCAMLR was established by an international convention in 1982 to conserve Antarctic marine life. This was in response to increasing commercial interest in Antarctic krill resources, a keystone component of the Antarctic ecosystem and a history of over-exploitation of several other marine resources in the Southern Ocean.
- Presently, the body has 26 members (25 countries and EU). India is a member of the organisation.
- CCAMLR is a key international instrument part of the Antarctic Treaty System that as a whole for 60 years has ensured peace, freedom of science and protection of the environment in the Antarctic region. The system consists of the Antarctic Treaty and its Environmental Protocol, the Agreement on the Conservation of Albatrosses and Petrels, and the Convention on the Conservation of Antarctic Seals in addition to CCAMLR.

35. CONVENTION ON INTERNATIONAL TRADE IN ENDANGERED SPECIES OF WILD FAUNA AND FLORA (CITES)

CITES is an international agreement between governments to ensure that international trade in specimens of wild animals and plants does not threaten the survival of the species.

PARTIES OF CITES

CITES is an international agreement to which States and regional economic integration organisations adhere voluntarily. States that have agreed to be bound by the Convention are known as Parties.

It provides a framework to be respected by each Party, which must adopt its own domestic legislation to ensure that CITES is implemented at the national level. Members: 185

FUNCTIONING OF CITES

CITES works by subjecting international trade in specimens of selected species to certain controls. All import, export, re-export and introduction from the sea of species covered by the Convention has to be authorized through a licensing system.

Each Party to the Convention must designate one or more Management Authorities in charge of administering that licensing system and one or more Scientific Authorities to advise them on the effects of trade on the status of the species.

Species covered by CITES are listed in three Appendices, according to the degree of protection they need.

APPENDICES UNDER CITES

- **Appendix I** includes species threatened with extinction. Trade in specimens of these species is permitted only in exceptional circumstances.
- **Appendix II** includes species not necessarily threatened with extinction, but in which trade must be controlled to avoid utilization incompatible with their survival.
- **Appendix III** is a list of species included at the request of a Party that already regulates trade in the species and that needs the cooperation of other countries to prevent unsustainable or illegal exploitation.

India Exempted from CITES RST

- The exclusion of India from the 'Review of Significant Trade' for Red Sanders under CITES means that farmers can now increase their income by cultivating and exporting the species.
- Previously, the review subjected Red Sanders trade with India to heightened scrutiny due to concerns about illegal harvesting and smuggling.
- India's compliance with CITES regulations led to its removal from this review process, allowing for easier trade of the endangered species.

36. CONVENTION ON CONSERVATION OF MIGRATORY SPECIES

CMS (Bonn Convention) is an environmental treaty of the UN. It provides a global platform for the conservation and sustainable use of migratory animals and their habitats.

WHAT IS THE MAIN TASK OF THIS CONVENTION?

- CMS brings together the States through which migratory animals pass, the Range States, and lays the legal foundation for internationally coordinated conservation measures throughout a migratory range.
- **As the only global convention specialising in the conservation of migratory species,** their habitats and migration routes, CMS complements and co-operates with several other international organizations, NGOs and partners in the media as well as in the corporate sector.

- CMS acts as a framework Convention. The agreements may range from legally binding treaties (called Agreements) to fewer formal instruments, such as Memoranda of Understanding, and can be adapted to the requirements of particular regions.
- The development of models tailored according to the conservation needs throughout the migratory range is a unique capacity to CMS.

VARIOUS CATEGORIES INTO WHICH THE SPECIES ARE DIVIDED BY THE CONVENTION?

- **Appendix I:**
  - Migratory species threatened with extinction are listed on Appendix I of the Convention.
  - CMS Parties strive towards strictly protecting these animals, conserving or restoring the places where they live, mitigating obstacles to migration and controlling other factors that might endanger them.
  - Besides establishing obligations for each State joining the Convention, CMS promotes concerted action among the Range States of many of these species.
- **Appendix II:**
  - Migratory species that need or would significantly benefit from international cooperation are listed in Appendix II of the Convention.
  - For this reason, the Convention encourages the Range of States to conclude global or regional agreements.

CMS COP13

- The 13th COP to Convention of Conservation of Migratory Species of Wild Animals (CMS) was conducted in Gandhinagar, Gujarat (2020).
- It concluded with the listing of a total of 10 migratory species of the world on Appendix I & II of the convention, along with adoption of several resolutions and decisions to address the needs and threats facing migratory species around the globe. Three of the species listed are from India – Great Indian bustard, mainland Asian elephant and Bengal florican.

GANDHINAGAR DECLARATION

CMS COP13 adopted the Gandhinagar Declaration, which calls for the integration of migratory species conservation and the concept of ecological connectivity into the Post-2020 Global Biodiversity Framework (GBF). This framework was later adopted at the UN Biodiversity Conference (CBD COP15) in December 2022.

CMS COP14

- CMS COP14 was held in Uzbekistan in 2024. It was the first global gathering on biodiversity since the adoption of the Kunming-Montreal Global Biodiversity Framework (GBF) in 2022.
- The first-ever State of the World's Migratory Species report revealed a decline in populations for many migratory species, and indicated the risk of global extinction was increasing.
- **Samarkand Strategic Plan (2024-2032):** Framework for the conservation and sustainable use of migratory species and their habitats, based on the Kunming-Montreal Global Biodiversity Framework.

37. AGENDA 21

Agenda 21 is a voluntary action plan adopted at the Earth Summit (UNCED) in Rio de Janeiro in 1992.

Objective: To promote sustainable development at global, national, and local levels.

Nature: A non-binding action plan for stakeholders to achieve sustainable development. The stakeholders include local and national governments, businesses, international organisations, citizen groups and non-governmental organisations.

India is a signatory to Agenda 21 and has sought to align various parts of its development infrastructure such as energy, transport, industry, water facilities, climate change policy, forests, biodiversity, ecosystems, marine and coastal management, land policy, agriculture, urban governance and human resource development.

38. SPACES COALITION

- SPACES is a coalition that believes in the central role of **spatial intelligence in joint decision-making for nature, climate and sustainable development objectives.**
- SPACES is coordinated by the UN Environment Program World Conservation Monitoring Centre (UNEP-WCMC) and SYSTEMIQ, working with the UN Development Program (UNDP), International Institute for Applied Systems Analysis (IIASA) and International Institute for Sustainability (IIS) amongst other collaborators.

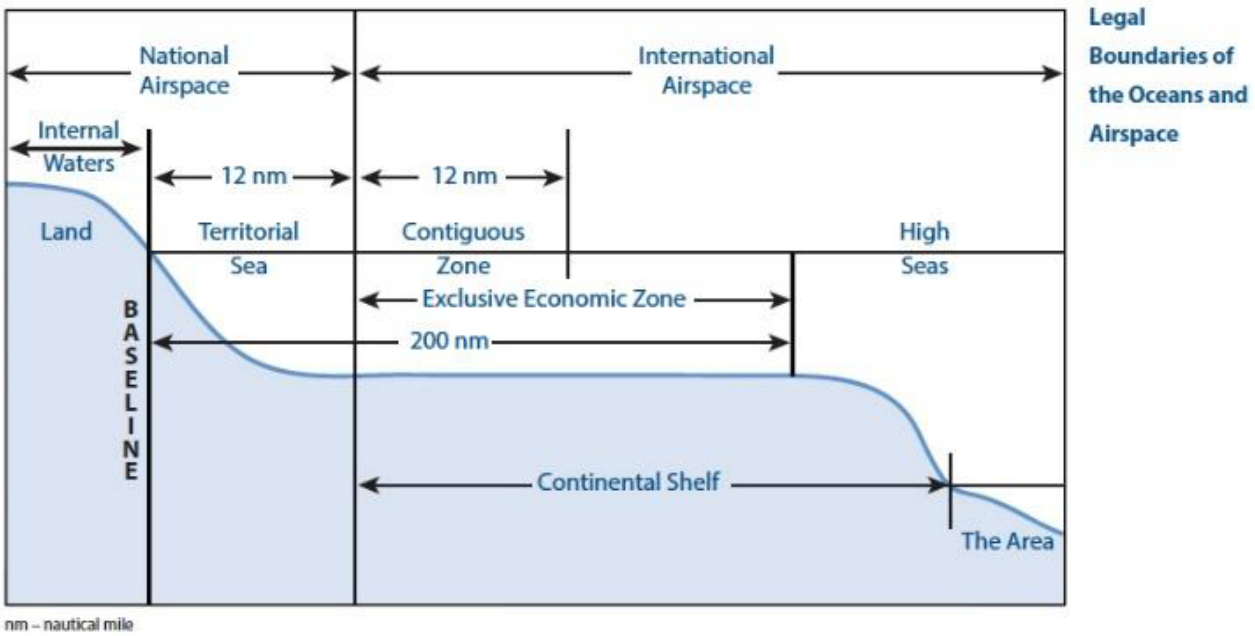
39. HIGH SEAS TREATY

The United Nations has recently adopted the Marine Biodiversity of Areas Beyond National Jurisdiction (BBNJ) or the High Seas Treaty. This historic agreement, the third to be approved under UNCLOS, aims to protect and preserve the marine environment through international cooperation. The process to develop this treaty began in 2002, and after several years of negotiations, it was finally adopted in 2023.

The United Nations Biodiversity of Areas Beyond National Jurisdiction Treaty, also known as the High Seas Treaty or the BBNJ treaty, is a **legally binding instrument for the conservation and sustainable use of marine biological diversity in areas beyond national jurisdiction.**

It is an agreement under the United Nations Convention on the Law of the Sea (UNCLOS).

The high seas refers to the ocean water column that lies beyond the boundaries of any one country, also known as areas beyond national jurisdiction (ABNJ).



KEY POINTS OF THE HIGH SEA TREATY

- **Protecting Marine Biodiversity:** The treaty addresses pressing issues such as rising sea surface temperatures, overexploitation of marine biodiversity, overfishing, coastal pollution, and unsustainable practices beyond national jurisdictions.
- **Establishment of Marine Protected Areas**: The treaty introduces the establishment of marine protected areas through a "three-quarterly majority vote," ensuring that decisions cannot be blocked by a small number of parties.
- **Fair Sharing of Benefits:** The treaty mandates the fair sharing of benefits from marine genetic resources. It includes a "clear house mechanism" to facilitate the sharing of scientific information and monetary benefits among all parties.

- **Open Access to Information:** Parties to the treaty are required to share and exchange information on marine protected areas, scientific and technical tools, and area-based management tools. This promotes open access to knowledge and collaboration.
- **Scientific and Technical Body**: A Scientific and Technical Body will play a crucial role in conducting environmental impact assessments, creating standards and guidelines, and assisting countries with limited capacity.

40. UN WATER CONFERENCE

- **Formally known as:** The 2023 Conference for the Midterm Comprehensive Review of Implementation of the UN Decade for Action on Water and Sanitation (2018-2028).
- **Location:** United Nations Headquarters, New York City

KEY OUTCOMES

- **Adoption of the Water Action Agenda:** A voluntary commitments framework outlining actions to achieve the water-related Sustainable Development Goals (SDGs).
- **Increased awareness of the global water crisis:** The conference highlighted the urgent need to address water scarcity, pollution, and sanitation challenges.
- **Mobilisation of stakeholders:** Governments, businesses, civil society organizations, and individuals were encouraged to take action on water.

KEY THEMES

- **Accelerating progress on integrated water resources management:** This involves managing water resources in a coordinated and sustainable way.
- **Investing in water infrastructure:** This includes building and maintaining water treatment plants, sanitation systems, and irrigation systems.
- **Building capacity and knowledge:** This involves training water professionals and educating the public about water conservation.
- **Ensuring water security for all:** This means guaranteeing access to safe drinking water and sanitation for everyone.

41. NEW GLOBAL FINANCING PACT

It is an initiative aimed at reforming the international financial system to address global challenges like climate change, poverty, and inequality. It was the focus of a summit held in Paris in 2023, co-organised **by France, Barbados, and India**.

The summit brought together heads of state, government, international organizations, civil society, and the private sector to discuss ways to create a fairer and more sustainable financial system.

The discussions focused on four key areas:

- Reforming multilateral development banks (MDBs) to make them more responsive to the needs of developing countries.
- Addressing debt by finding ways to help countries manage their debt burdens and invest in sustainable development.
- Mobilising innovative financing such as green bonds and climate finance mechanisms.
- Reforming international taxation to create a more equitable system.

The summit did not produce any concrete agreements, but it was seen as an important step forward in the discussion about how to reform the global financial system.

Here are some of **the key outcomes of the summit:**

- A commitment to continue working together to reform the MDBs.
- A call for the creation of a new international debt architecture.

- The launch of a new initiative to mobilize private sector investment in climate action.
- A commitment to work towards a more equitable international tax system.

The New Global Financing Pact is a long-term initiative, and it is likely to be the subject of much discussion and debate in the years to come.

42. SPRING INITIATIVE

- UN-backed network Principles for Responsible Investment (PRI) has launched "Spring", a new initiative to halt or reverse nature loss by 2030.
- It is supported by a coalition of 200 investors managing a combined $15trn in assets.
- The objective is to address systemic risk of nature loss to societies and long-term portfolio value creation by enhancing corporate practices on forest loss and land degradation.

43. FISHMIP INITIATIVE

According to the FishMIP, global projections of exploitable fish biomass show a more than 10% decline.

- Fisheries and Marine Ecosystem Model Intercomparison Project was launched in 2013.
- It is an international network of more than 100 marine ecosystem modellers and researchers around the world.
- It is working with FAO to understand the long-term impacts of climate change on marine ecosystems and fisheries
- FishMIP2.0 was established in 2024 to increase the reliability of modelling projections.

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3 CHAPTER POLLUTION & RENEWABLE ENERGY

1. CHEMICALS OF ENVIRONMENTAL CONCERN

CHEMICAL	DESCRIPTION
Arsenic	• Naturally occurring element that is widely distributed in the Earth's crust. • Used in wood preservatives, pesticides, insecticides, herbicides, batteries and semiconductors. • The primary route of arsenic exposure is via ingestion of contaminated food and water. • Highly toxic in its inorganic form and carcinogenic. • Associated with cardiovascular disease, diabetes. • Adverse effects on nervous, respiratory, immune and endocrine systems. • Early childhood exposure affects cognitive development. • Toxic effects in wildlife.
Bisphenol A	• Belongs to a group of synthetic organic compounds primarily used as a building block in production of polycarbonate plastics and epoxy resins, which are used in a wide variety of products including water bottles, sports equipment, medical devices, household electronics, thermal paper receipts, and food and beverage cans. • Polymer degradation is the dominant mechanism responsible for bisphenol releases from products. • The primary source of exposure to BPA for most people is through food and beverages, by migration from containers. • Endocrine disruptor. • May also be linked to cardiovascular diseases, diabetes, metabolic disorders, prostate cancer, and immune system alterations.
Cadmium	• A soft, silver-white metal naturally found in Earth's crust. • Largest use is in batteries, predominantly rechargeable nickel-cadmium batteries. • Also widely used in pigments, coatings, electroplating and solar cells. • Exposure is through ingestion of contaminated food. • Presence in plastic toys and exposure through cadmium wastes have also been noted. • Carcinogenic to humans • Exposure affects kidney function and has been linked to reduced lung function as well as damage to bones, with children particularly at risk. • Adverse effects on animals and plants.
Glyphosate	• Active ingredient in herbicide formulations that are widely used for agricultural, forestry, and residential weed control. • Researchers could not establish any serious threat to human health or environment. • Related to mild skin and eye irritation.

Lead	• Naturally occurring heavy metal found in Earth's crust. • Lead and its compounds are used in gasoline, pipes, paints, toys, ammunition and batteries etc. • Exposure is mainly through inhalation of lead particles in air, drinking water, eating foods or swallowing dust or dirt. • Health effects of exposure to lead include hypertension, renal failure, cardiovascular disease and stroke, especially among workers, while neurodevelopmental toxicity constitutes the most important consequence of lead toxicity in children.
Microbeads	• A type of primary microplastics (particles less than 5 mm in length). • Used in personal care products, other consumer applications and various industrial applications (Ex. scrubs, toothpastes) • The majority of microplastics released to the oceans are secondary microplastics originating from degradation of larger plastic items, in particular textiles and tyres, while microbeads from personal care products and cosmetics represent a relatively small source of microplastics in the environment. • Aquatic organisms may be exposed to microplastics through direct ingestion or consumption of prey that have ingested the plastics. • Humans can be exposed to microplastics through ingestion of contaminated food and water. • Adverse effects on aquatic organisms, continuous release of microbeads may result in long-term effects on biological diversity and ecosystems. • Potential risks of microplastics to human health are largely unknown.
Neonicotinoids	• A class of neuroactive insecticides chemically related to nicotine. • Used as insecticides and biocides. • Adverse impact on pollinators' health and numbers, especially bees. • Impact on human health has not been established yet.
Organotin	• Organic compounds that contain at least one tin-carbon bond. There are four main groups of organotin compounds. • Used in various applications such as biocidal agents in wood preservatives and disinfectants, catalysts, sealants, stabilizers and as antifouling paints on ships. • In addition to occupational exposure, people can be exposed to organotin through ingestion of food and contact with household products containing organotin compounds. • The organotin compound tributyltin (TBT) is considered among the most hazardous substances released into marine environments. • Exposure to organotin can cause skin, eye and respiratory irritation, neurological problems, and effects on the immune system. • Adverse effects on animals like endocrine disruption.
Phthalates	• A group of plasticizers with softening and elastic effects. • Used in products such as vinyl flooring, adhesives, detergents, lubricating oils, automotive plastics, plastic clothing and personal care products. • The main human exposure pathway is oral via food. Other pathways include direct mouthing (toys), house dust ingestion and dermal gaseous absorption. • Linked to developmental toxicity and adverse effects on reproductive function in humans and aquatic invertebrates, fish and birds.

	• While a number of phthalates have so far been found to present a limited risk of harm to human health and environment, others have been shown to be plausible endocrine disruptors.
Polycyclic aromatic hydrocarbons (PAH)	• A group of 100 different chemicals that occur naturally in coal and crude oil but are also formed as a by-product during incomplete burning of coal, oil, gas, wood, garbage and other organic substances. • Carcinogenic, mutagenic and toxic for reproduction. • Persistent, bio accumulative and toxic for humans and other organisms.
Triclosan	• An antibacterial and anti-fungal agent widely used in a variety of products, including cosmetics and toiletries (e.g., toothpaste and soaps). • Can be released to the environment via various pathways and has been detected in surface, ground and drinking water. • Exposure to triclosan occurs primarily through the skin or mouth during the use of triclosan containing products, with only a minor contribution via environmental exposures. • Toxic to aquatic organisms and is an endocrine disruptor in mammals.
Benzene	• A hydrocarbon formed due to natural processes and human activities. • Used to make chemicals for other industries like plastics, resins, and nylon and synthetic fibres. • Also used to make some types of lubricants, rubbers, dyes, detergents, drugs, and pesticides. • Exposure can occur occupationally and domestically due to ubiquitous use of benzene-containing petroleum products, including motor fuels and solvents. • Active and passive exposure to tobacco smoke is also a significant source of exposure. • May cause narcosis: headache, dizziness, drowsiness, confusion, tremors and loss of consciousness. • Moderate eye irritant and a skin irritant. • Associated with long-term adverse health effects and diseases, like cancer and haematological effects
Asbestos	• Groups of naturally occurring fibrous minerals that are resistant to heat and corrosion. • Used as fireproofing and insulation material. • Exposure to asbestos occurs through inhalation of fibres in air. • All types of asbestos cause lung cancer, mesothelioma, cancer of the larynx and ovary, and asbestosis (fibrosis of the lungs).
Dioxins	• Belong to the so-called "dirty dozen" - a group of dangerous chemicals known as persistent organic pollutants (POPs), covered by the Stockholm Convention. • Found throughout the world in the environment and they accumulate in the food chain, mainly in the fatty tissue of animals. • Mainly by-products of combustion and industrial processes but can also result from natural processes, such as volcanic eruptions and forest fires. • These are unwanted by-products of a wide range of manufacturing processes including smelting, chlorine bleaching of paper pulp and the manufacturing of some herbicides and pesticides.

	• More than 90% of human exposure is through food, mainly meat and dairy products, fish and shellfish. • It has been associated with a range of toxic effects, including chloracne; reproductive, developmental and neurodevelopmental effects; immunotoxicity; and effects on thyroid hormones, liver and tooth development. • They are also carcinogenic.
Mercury	• Naturally occurring element which is found in Air, Water and Soil. • Released to the environment from volcanic activity, weathering of rocks and as a result of human activity. • Human activity is the main cause of mercury releases, particularly coal-fired power stations, residential coal burning for heating and cooking, industrial processes, waste incinerators and as a result of mining for mercury, gold and other metals. • Exposure mainly occurs through consumption of fish and shellfish contaminated with methylmercury and through worker inhalation of elemental mercury vapours during industrial processes. Cooking does not eliminate mercury. • Methyl mercury is very different to ethyl mercury. Ethyl mercury is used as a preservative in some vaccines and does not pose a health risk. • Exposure, even small amounts, may cause serious health problems. • Threat to development of children in-utero and early in life. • Toxic effects on the nervous, digestive and immune system on lungs, kidneys, skin and eyes.
Polybrominated Diphenyl Ethers (PBDE) **Polybrominated Biphenyls (PBB)**	Used as fire retardants in furniture foam padding, wire insulation, rugs, draperies and upholstery, plastic cabinets of electronic devices.
Acrylamide	• Formed when carbohydrates rich starchy food is cooked at very high temperatures. • Found in tobacco smoke. • Used to make chemicals to purify water, treat sewage, make paper and cosmetics.
Parabens	• A man-made chemical found in cosmetics such as makeup, moisturisers, haircare products etc; pharmaceuticals, food and beverages. • It can get absorbed through the skin.

2. COMMON POLLUTANTS

POLLUTANTS	SOURCES	HEALTH EFFECTS	OTHER RELATED INFORMATION
Mercury	Coal Combustion, small scale gold mining	Minamata disease (Affects nerves) Mercury reacts inside the body to form Methyl mercury which is fatal for human health	Minamata Convention signed by the UN in 2015 and entered into force in 2017. It has imposed a ban on new mercury mines and declared phasing out of new mines. Kodaikanal Mercury poisoning case that caused poisoning of Kodaikanal lake and led to closure of factory in 2001.

Cadmium	Used in Ni-Cd batteries, coloration of plastics and various discarded electronic products	Itai Itai Disease (Softening of bones and causes kidney stones)	Western U.P is most affected by Cadmium poisoning in India.
Lead	Vehicular emissions, ore and metal processing, lead acid battery manufacturers, paints	Affects Liver and Kidney, mental retardation and abnormality in fertility and pregnancy	MoEFCC has prohibited manufacture, trade, import and export of household and decorative paints containing lead.
Fluorine	Fluoride in air, soil and water	Knock knee syndrome that causes outward bending of knees, stiffness of joints, humped back.	Rajasthan, Gujarat and A.P are amongst the worst affected states in India
Coal Dust	Coal mines, heavy industry	Pneumoconiosis is also known as black lung disease.	Chhattisgarh, Odisha, Jharkhand, West Bengal
Silica	Sand blasting, ship breaking industry	Silico- tuberculosis	Alang in Gujarat is worst affected, Rajasthan, U.P, Bihar, Chhattisgarh, Jharkhand, Odisha, West Bengal
Nitrate	Nitrate contamination of Ground Water due to pesticide in agriculture and vehicular emissions	Blue baby syndrome (Decreased oxygen carrying capacity of hemoglobin)	Rae Bareli district of U.P is worst affected by Nitrate pollution
Arsenic	Ground water contamination	Causes skin cancer, cancer of lungs, affects kidney and in rare cases is linked to diabetes	Ganga Brahmaputra fluvial plains
Hexavalent Chromium	Naturally occurring heavy metal, used in leather and tanneries industry	Respiratory and Gastro-intestinal problems and Carcinogenic	Kanpur is the most affected area along with certain regions of West Bengal. CSIR has recently developed waterless chrome tanning technology.
Dioxins	Production and disposal of Poly Vinyl Chloride, also production of chlorinated paper and biomedical waste is a source for Dioxins	Causes Cardiovascular diseases, diabetes, cancer, early menopause, etc.	

Dichlorine Diphenyl Tricholoroethane	Pesticide usage	It causes vomiting, shakiness and seizures. It is considered as carcinogenic	DDT is to be banned under Stockholm convention from 2020 but India has strongly opposed the move
Brominated Flame Retardants	Used in mattresses and electronic component to reduce fire related injury	Carcinogenic, development retardant	It has a tendency to stay for long in the environment. It is banned under Stockholm Convention on Persistent Organic Pollutant
Aflatoxins	Produced by molds (fungi) where cereals and oilseeds are mishandled.	Carcinogenic & poisonous	ICRISAT has made peanuts free of aflatoxins
Radon	Naturally occurring radioactive gas	Lung cancer	It has spread across few areas of Bangalore
PM 2.5	Vehicular emission	Affects functioning of lung, chronic cough, asthma and heart disease	
PM 10	Vehicular emission	Affects functioning of lung, chronic cough, asthma and heart disease	
Polychlorinated Biphenyl (PCB)	Illegal or improper dumping of PCB waste	It causes rashes in the skin and affects the liver. It is a carcinogenic compound of chlorine.	
Endosulphan	Sprayed as a pesticide on crops such as tea, paddy, cashew, etc.	Causes dysfunctioning of endocrine glands and affects DNA strands in human beings. It causes mental and physical and physical disorders.	Kerala farmers have faced severe toxicity from Endosulphan pollution sprayed on cashew plants. It is banned under Stockholm Convention on Persistent Organic Pollutant
Radioactive Pollution	It is caused by exposure to Radioactive substances Uranium, Thorium, Radium, etc.	Unsafe exposure to radiation leads to various health problems such as cancer, mutation etc.	Rem is the unit of biological damage caused to human beings. It is equivalent to the injury caused by a given amount of X- ray. ATOMIC ENERGY REGULATORY BOARD regulates radioactive substances in India.

3. INDIA'S SOLAR PV WASTE PROBLEM

There has in the last few years been a concerted push from policymakers in India to transition to a circular economy and to, among other things, enable effective waste management. But waste management in the solar photovoltaic (PV) sector still lacks clear directives.

WHAT IS PV WASTE?

- It is the **electronic waste (e-waste) generated by discarded solar panels and Photo-voltaic (PV) devices.**
- Photovoltaic (PV) devices contain semiconducting materials that convert sunlight into electrical energy.
- A single PV device is known as a cell, and these cells are connected in chains to form larger units known as modules or panels.
- Although up to 90% of the components are recyclable, many PV modules contain heavy metals such as **cadmium, copper, lead, antimony or selenium**, and when they are taken out of service or broken, they may be classified as hazardous waste.

4. STANDARDS & LABELING PROGRAM

- The objective of the S&L program is to help consumers make an informed choice about various energy-consuming appliances, in terms of energy savings, that would result based on each appliance's energy efficiency performance.
- This scheme also helps the consumer determine the cost-saving potential of the marketed household and other equipment.
- A star rating, ranging from 1 to 5 in the ascending order of energy efficiency is provided to products registered with the Bureau. An endorsement label is also provided for some products.

WHAT IS A LABEL?

- Describes energy performance (in the form of energy use, efficiency or energy cost)
- "Labels" mainly give consumers the necessary information to make informed purchases.
- There are two types of labels:
  - **a) Comparative label:** allow consumers to compare efficiency of all the models of a product in order to make an informed choice. It shows the relative energy use of a product compared to other models available in the market.
  - **b) Endorsement label:** define a group of products as efficient when they meet minimum energy performance criteria specified in the respective product schedule/ regulation/statutory order.

Comparative Label	
	Allow consumers to compare the energy consumption of similar products, and factor lifetime running cost into their purchasing decision.

Endorsement Label	
	Provide a 'certification' to inform prospective purchasers that the product is highly energy efficient for its category.

MANDATORY APPLIANCES	VOLUNTARY APPLIANCES
Frost free refrigerator	Side by Side/Multi Door Refrigerator
Stationary Storage Type Electric Water Heater	Solar water heater
Colour Television	Computer
Room Air Conditioner (Variable Speed)	Washing machines
Tabular fluorescent lamps	Micro wave oven
LED lamps	Hi energy Li batteries
Deep freezers	Chillers
Ultra-high definition televisions	Tyres
Room air conditioners	Domestic gas stove
Ceiling fans	Pedestal fan, table/walled fan
Direct cool refrigerator	Diesel generator set
Distribution transformer	Submersible Pump Set
	Diesel Engine Driven Monoset Pumps for Agricultural Purposes

5. FROTHING IN RIVER YAMUNA

The primary reason for froth formation are:

1. High phosphate content in the water. It is believed that phosphates from detergent find their way into river water leading to froth formation.
2. Release of surfactants in river water.
3. Release of untreated effluents
4. As the water rich in surfactants, phosphates and other untreated wastes falls down from the barrages, the turbulence of water results in frothing.

CHEMICALS IN DETERGENTS

PHOSPHATES

Phosphates are used in detergents for:

1. Softening hard water by bonding with calcium ions.
2. Stabilise alkalinity of surfactants
3. Keep dissolved dirt in water and prevent it from depositing on clothes.

Issues with phosphates:

1. Results in algal bloom and lead to eutrophication.
2. Inhibits biodegradation of organic substances.

NONYLPHENOL

A hazardous chemical which has a tendency to bioaccumulate and enter the food chain.

It is a hormone disruptor, and can cause many health issues.

Used as a surfactant in detergents, personal hygiene products, automotive & latex paints, lawn care products.

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6. PARTICULATE MATTER

- Particulate matter is a term for a mixture of solid particles and liquid droplets found in the air.
- Some particles, such as dust, dirt, soot, or smoke, are large or dark enough to be seen with the naked eye. Others are so small they can only be detected using an electron microscope.

PARTICLE POLLUTION INCLUDES

- PM10 : inhalable particles, with diameters that are generally 10 micrometers and smaller; and
- PM2.5: fine inhalable particles, with diameters that are generally 2.5 micrometers and smaller. These are more harmful for health as they move freely with air current and block the tiny pores in our lungs.
- Major sources of SPM (suspended particulate matter) are vehicles, power plants, construction activities, oil refinery, railway yard, marketplace, industries, etc.
- According to the Central Pollution Control Board (CPCB), particulates size 2.5 µm or less in diameter (PM 2.5) are responsible for causing the greatest harm to human health. These fine particulates can be inhaled deep into the lungs and can cause breathing and respiratory symptoms, irritation, inflammations and pneumoconiosis (a disease of the lungs due to inhalation of dust, characterized by inflammation, coughing, and fibrosis).

FUGITIVE PARTICULATE MATTER

- Fugitive particulate matter is particulate matter that has not passed through a stack (such as a chimney, pipe, vent, or duct) before being released to the air. It is released into the air by wind or other similar forces.
- The source of which is primarily the Earth's soil.
- High levels of particulate matter in the air can affect human health. It can reach deep into the lungs and cause respiratory problems. Particulate matter is linked to aggravated asthma, chronic bronchitis, and premature death.

SOURCES OF PM POLLUTION

- **Primary sources** cause particle pollution on their own. **E.g.,** wood stoves and forest fires.
- **Secondary sources** let off gases that can form particles. **E.g.,** Power plants and coal fires.
- Common sources of particle pollution can be either primary or secondary. **E.g.,** Factories, cars and trucks, and construction sites. Smoke from fires and emissions (releases) from power plants, industrial facilities, and cars and trucks.

7. FOG

It is one of the major weather hazards, impacting road, aviation transportation, economy, and public life.

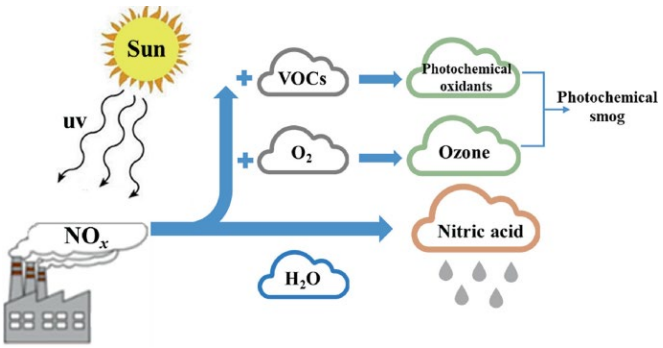
- Fog is a visible mass consisting of cloud water droplets suspended in the air or near the Earth's surface.
- Fog usually appears over a region of high pressure where humidity is greater than 75%. Moisture in the atmosphere could condense around particulate matter of diameter in the range of nanometers, to cause fog.
- Land use changes and increasing pollution are responsible for growing fog occurrence.

8. SMOG

- The word smog is derived from smoke and fog. This is the most common example of air pollution that occurs in many cities throughout the world.
- **Classical smog** occurs in cool humid climates. It is a mixture of smoke, fog and Sulphur dioxide. Chemically, it is a reducing mixture and so it is also called as reducing smog.
- **Photochemical smog** occurs in warm, dry and sunny climates. The main components of the photochemical smog result from the action of sunlight on unsaturated hydrocarbons and nitrogen oxides produced by automobiles and factories. Photochemical smog has high concentration of oxidizing agents and is, therefore, called as oxidizing smog.

SOURCES OF SMOG

- **Nitrogen oxides (NOx):** Emitted from vehicular exhaust, coal power plants, and industrial emissions.
- **Volatile organic compounds (VOCs):** Released from gasoline, paints, and cleaning solvents.
- **Process**: Sunlight triggers chemical reactions between **NOx** and **VOCs** to produce ground-level ozone and particulate matter or smog.



WHY DOES THE INDIAN SUBCONTINENT EXPERIENCE ONE OF THE HIGHEST LEVELS OF AIR POLLUTION?

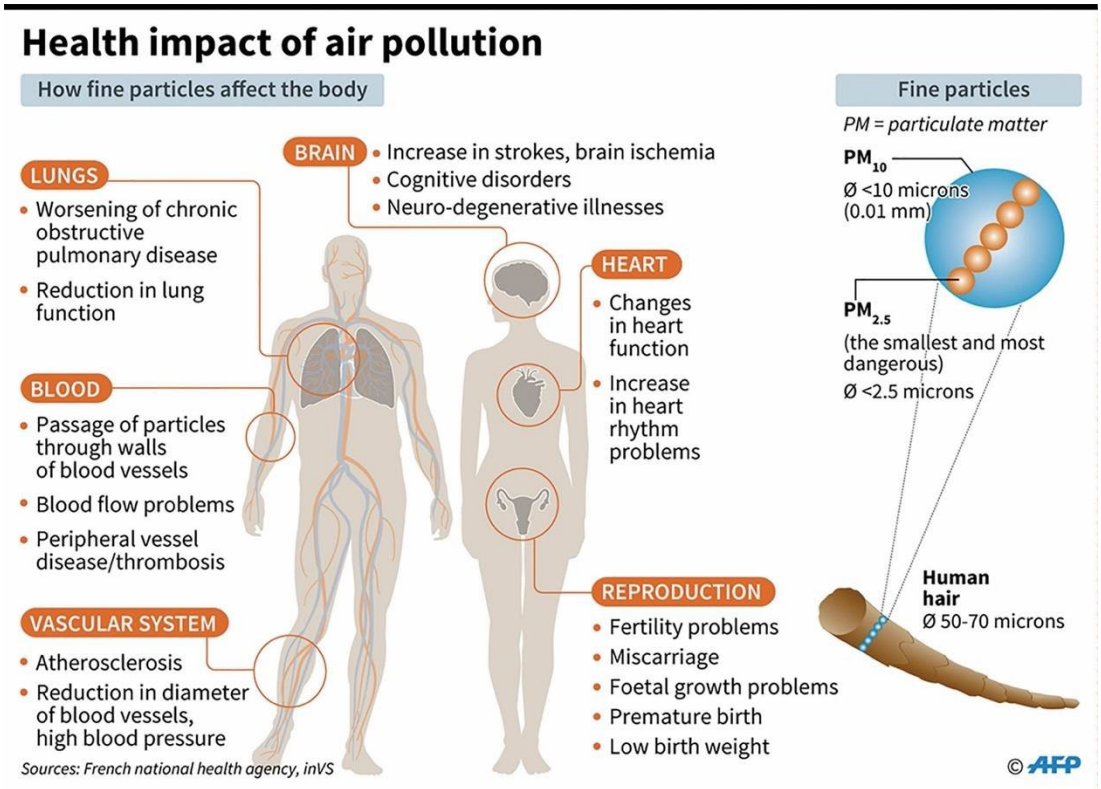
According to the World Air Quality report, India's air had 54.4 micrograms of PM2.5 particles per cubic metre on average. The poor air quality persists in Pakistan as well as Bangladesh.

NATURAL FACTORS RESPONSIBLE

- **Himalayas trap the pollutants** over the subcontinent and do not allow them to disperse.
- **Low lying topography** of Gangetic plains **stabilises the air mass over it**, further accumulating the air pollutants.
- **Continental type of climatic conditions** promote extreme winters in northern Plains spanning across India, Pakistan and Bangladesh. The **winter inversions** promote atmospheric stability, further contributing to air pollution.
- **The presence of the Thar desert** also leads to the **spread of dust particles** in the entire region.

ANTHROPOGENIC FACTORS RESPONSIBLE

- **Emissions from coal-fired power plants,** steel mills, and other industrial facilities release large amounts of pollutants, including sulfur dioxide (SO2), nitrogen oxides (NOx), and particulate matter (PM).
- **Traditional brick kilns** (prevalent in many parts of South Asia) are significant sources of air pollution due to their inefficient combustion processes.
- **High levels of vehicular traffic**, often with poorly maintained vehicles, lead to the release of pollutants such as carbon monoxide (CO), NOx, VOCs, and PM.
- **Farmers often burn crop residues** to clear fields, releasing large amounts of smoke, PM, and greenhouse gases.
- **Decomposing organic waste in landfills** emits methane (CH4) and other VOCs.
- **The use of biomass** (wood, dung, crop residues) and coal for cooking and heating in rural and low-income urban households produces significant air pollution.



9. WINTER HAZE

New research published in Nature geoscience journal has given some important inputs regarding the reasons behind formation of haze in Delhi during winter seasons.

WHAT IS HAZE?

Haze is formed when a cocktail of various gaseous pollutants is oxidized and forms particulate matter diffusing sunlight. Hydroxyl radicals (OH) mainly mediate this process. Haze consists of fine particulate matter containing soot. It is formed when gaseous pollutants, which are from industrial emissions, vehicle exhausts, and other sources, are converted to condensable matter. This condensation is remarkably accelerated under the action of OH radicals.

Although haze is mostly a dry air phenomenon, some of its particles tend to create condensation, subsequently giving form to mist droplets. This phenomenon is known as wet haze, or visibility-reducing aerosols of the wet type. The process of aerosol creation occurs due to the chemical reactions between the atmosphere and the sulphur dioxide gas emissions, which create small droplets of sulfuric acid. Sunlight, high humidity, and stagnant airflow may enhance this process.

WHAT CAUSES HAZE?

1. **Volcanic activity:** In the aftermath of major volcanic eruptions, sulphur dioxide (SO2), hydrochloric acid (HCl), and ashes are distributed into the atmosphere.
2. **Windy weather:** Desert dust blown into the atmosphere by the wind is another form of naturally occurring haze. The minerals contained in the dust and dirt allow sunlight to be absorbed, warming the area. Wind also lifts water from the oceans, causing the salts to be evaporated into the air — accounting for some atmospheric haze. The sea salt acts as cloud condensation nuclei, which accelerates cloud formation.
3. **Human practices:** activities like forest burning, factory emissions, automobile traffic pollution, and the burning of coal and other fossil fuels are said to outweigh natural sources in the production of hazes.

Note: During winters, stubble burning in nearby areas of Punjab, Haryana, Rajasthan, and Uttar Pradesh increase the particulate matter in the atmosphere. Atmospheric stability keeps this pollutant stable over Delhi NCR for a

longer period. Further, burning of wood and waste to keep oneself warm is also common in Northern India during winters. All these factors contribute to sharp rise in haze during winters.

10. BLACK CARBON

India is the second largest contributor to black carbon in the world.

WHAT IS BLACK CARBON?

Black Carbon is produced both naturally and by human activities as a result of the incomplete combustion of fossil fuels, biofuels, and biomass. It is black in colour as its particles strongly absorb sunlight and give soot its black colour.

Primary sources include emissions from diesel engines, cook stoves, wood burning and forest fires.

DIFFERENCE BETWEEN CARBON DIOXIDE AND BLACK CARBON

Reducing CO_2 emissions is essential to avert the worst impacts of future climate change, but CO_2 has such a long atmospheric lifetime that it will take several decades for CO_2 concentrations to begin to stabilize after emissions reductions begin.

In contrast, BC remains in the atmosphere for only a few weeks, so cutting its emissions would immediately reduce the rate of warming, particularly in those areas which have witnessed fast changes in the level of Black Carbon.

11. FLY ASH

Fly ash is a coal combustion product that is composed of the particulates (fine particles of burned fuel) that are driven out of coal-fired boilers together with the flue gases.

IMPORTANT TERMS RELATED TO FLY ASH

- **Bottom Ash:** Ash that falls to the bottom of the boiler's combustion chamber is called bottom ash.
- **Ash Capture:** Fly ash is generally captured by electrostatic precipitators or other particle filtration equipment before the flue gases reach the chimneys.
- **Coal Ash:** Together with bottom ash removed from the bottom of the boiler, it is known as coal ash.
- **Composition:** Depending upon the source and composition of the coal being burned, the components of fly ash vary considerably, but all fly ash includes substantial amounts of –
  - Silicon dioxide (SiO_2)
  - Aluminum oxide (Al_2O_3) and
  - Calcium oxide (CaO), the main mineral compounds in coal-bearing rock strata.

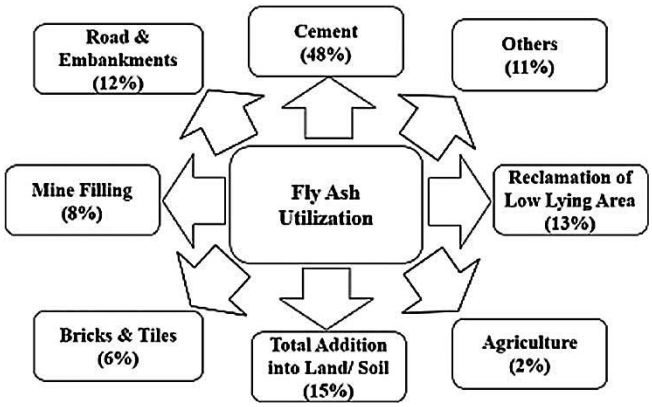
The minor constituents of fly ash depend upon the specific coal bed composition but may include one or more of the following elements or compounds found in trace concentrations - arsenic, beryllium, boron, cadmium, chromium, hexavalent chromium, cobalt, lead, manganese, mercury, molybdenum, selenium, strontium, thallium, and vanadium, along with very small concentrations of dioxins and PAH compounds.

ENVIRONMENTAL HARM CAUSED BY FLY ASH

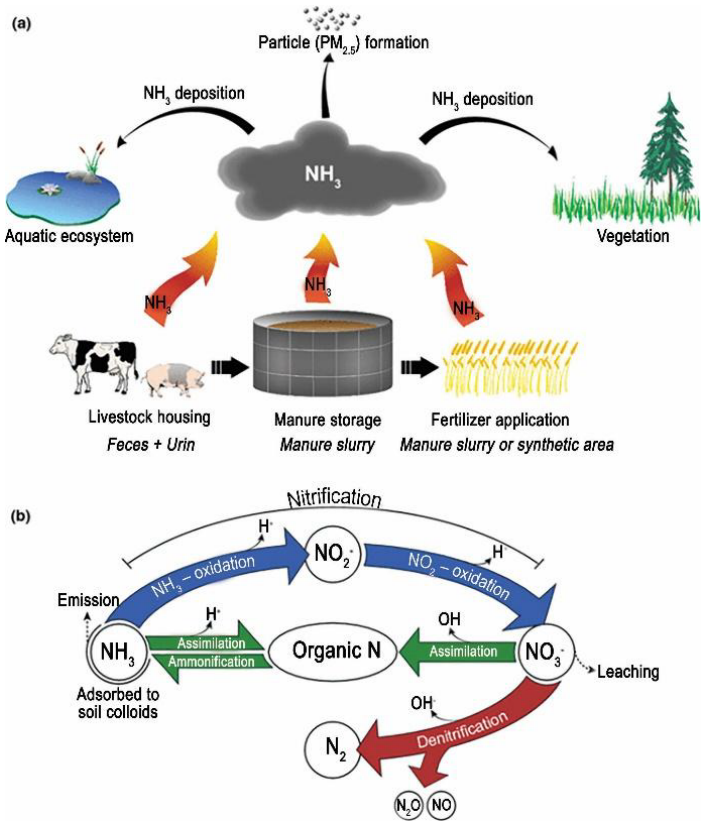
- Fly ash is a major source of PM 2.5 and black carbon. It becomes airborne and gets transported to a radius of 10 to 20 kms.
- It can settle on water and other surfaces. It will contaminate water and soil systems.
- Fly ash contains heavy metals from coal. The wet disposal of Fly ash results in leaching of toxic heavy metals in the ground water system.
- The destruction of mangroves, drastic reduction in crop yields, and the pollution of groundwater in the Rann of Kutch from the ash sludge of adjoining Coal power plants has been well documented.

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USES OF FLY ASH



12. AMMONIA



PRODUCTION OF AMMONIA- A CLIMATE CONCERN

- The process uses natural gas, which leads to 2.6 metric tons of life cycle greenhouse gas (GHG) emissions per metric ton of ammonia produced.
- With ammonia being the second most produced chemical in the world, its production accounts for approximately 2% of worldwide fossil energy use and generates over 420 million tons of CO2 annually.

CONSEQUENCES RELATED TO AMMONIA POLLUTION

Impact on human health:

- After reacting with other particles, ammonia emissions can **lead to the formation of fine particulate matter (PM2.5),** posing serious health risks such as heart and lung diseases.
- Exposure to **high levels of ammonia can cause respiratory problems,** such as asthma, bronchitis, and chronic obstructive pulmonary disease (COPD).

- Direct exposure to Ammonia can also **irritate the eyes, skin, and throat, leading to discomfort and potential health issues.**

Impact on Ecosystem and Biodiversity:

- Ammonia deposition can cause **soil acidification**, which can harm plant growth and reduce soil biodiversity leading to reduced crop yield.
- Excess ammonia in water bodies can lead to **eutrophication**, a process that results in algal blooms and oxygen depletion, disrupting aquatic ecosystems.

Impact on the Climate:

- Ammonia emissions contribute to climate change by releasing **nitrous oxide (N2O) after oxidation,** a potent greenhouse gas **with a global warming potential 298 times that of carbon dioxide (CO2).**
- N2O is involved in the **depletion of stratospheric ozone**, which protects the Earth from harmful ultraviolet radiation.
- Ammonia emissions also contribute to the **formation of aerosols, which can impact cloud properties and potentially influence climate patterns.**

Economic Impacts of Ammonia Pollution: These costs include healthcare expenses, reduced agricultural productivity, and the implementation of pollution control measures.

13. INDIA TO TACKLE LEAD POISONING

Lead poisoning is the one biggest threat to India with 275 million out of 800 million estimated to have high blood lead level (BLL) in 2020.

KEY POINTS ABOUT LEAD POISONING

- A 2023 Lancet report highlights that most children with high blood lead levels (BLLs) are in low- and middle-income countries (LMICs).
- Lead poisoning often goes unnoticed as it can be asymptomatic at lower exposure levels, leading to denial among the public and policymakers.
- India has been studying lead exposure since the 1970s, **with significant findings linking higher BLLs to lower IQ scores in children.**
- India has **standards for lead content in products**, but **enforcement is inconsistent** and requires strengthening.

LEAD POLLUTION IN INDIA

It is ubiquitous and found in the air we breathe, the spices we cook, the toys our children play with, the paint on our walls and many more.

SOURCES OF LEAD POLLUTION

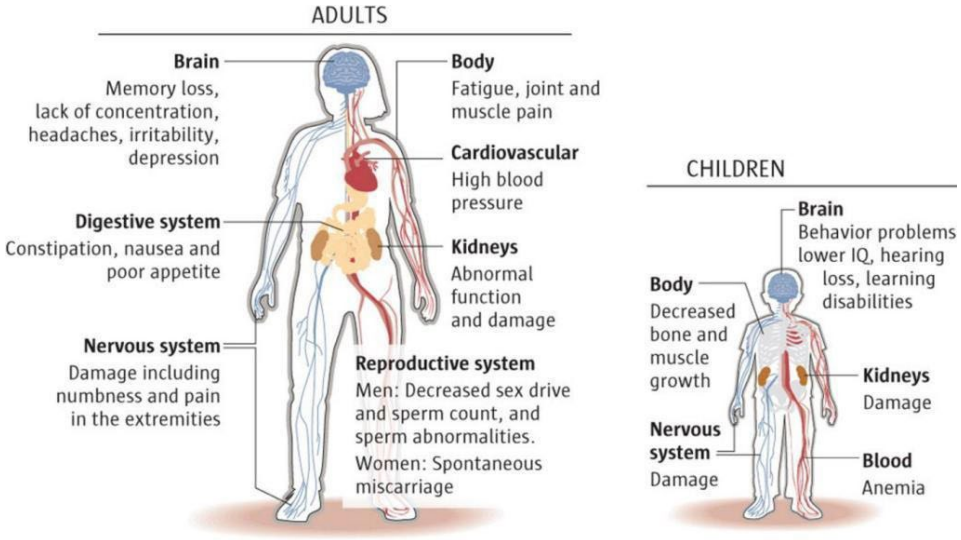
- **Industrial Emissions**: Lead is emitted into the air through various industrial processes such as mining, smelting, refining, and manufacturing of lead-based products like batteries, paints, and ceramics.
- **Automobiles**: Historically, leaded gasoline was a major source of lead emissions. While leaded gasoline has been phased out in many countries, vehicle emissions still contribute to lead pollution, particularly in areas with high traffic density.
- **Lead-Based Paints**: Lead-based paints, commonly used in buildings and infrastructure before regulations restricted their use, can deteriorate over time, releasing lead particles into the environment. This is a significant concern, especially in older homes and buildings.
- **Battery Recycling**: Improper recycling of lead-acid batteries, such as those used in cars and industrial equipment, can lead to the release of lead into the environment, particularly through soil and water contamination.

- **Industrial Waste and Landfills**: Improper disposal of industrial waste containing lead, as well as inadequate management of landfills, can result in leaching of lead into soil and groundwater, posing risks to nearby communities.
- **Lead Pipes and Plumbing Fixtures**: Lead pipes and plumbing fixtures, particularly in older homes and infrastructure, can leach lead into drinking water. Corrosion of these materials can increase lead levels in tap water, especially in areas with acidic water or inadequate water treatment.
- **Mining and Ore Processing**: Mining activities, particularly for lead and other metals, can release lead-containing dust and particles into the air and water, contaminating soil and aquatic ecosystems.
- **Smoking**: Tobacco smoke contains lead, and smoking in indoor environments can lead to the accumulation of lead dust on surfaces, contributing to indoor lead pollution.
- **Food and Water Contamination**: Lead can enter the food chain through contaminated soil, water, and air. Certain food items, particularly those grown in lead-contaminated soil or water, may contain elevated levels of lead.
- **Hobbies and Crafts**: Some hobbies and crafts involve the use of materials containing lead, such as stained glass making, pottery glazes, and ammunition reloading. Improper handling and disposal of these materials can lead to lead exposure.

MAGNITUDE OF THE ISSUE

A 2020 report by UNICEF estimated that nearly 27 crore Indian children have unsafe blood lead levels (BLLs).

- Unsafe concentration of lead in the body can deter their **neuro-cognitive development, leading to decreased IQ, behavioural problems, lowered educational achievement and eventually lowered earning potential.**



14. SULPHUR DIOXIDE POLLUTION

In order to curb emissions from the power sector, the Ministry of Power has proposed incentives worth Rs. 835 billion to step up infrastructure aimed at **cutting Sulphur emissions.**

TROPOSPHERIC AIR POLLUTION

- Atmospheric pollution can be divided as tropospheric and stratospheric pollution.
- Tropospheric pollution occurs due to the presence of undesirable solid or gaseous particles in the air.
- The following are the major gaseous and particulate pollutants present in the troposphere.
- **Gaseous air pollutants:** Oxides of sulphur, nitrogen and carbon, hydrogen sulphide, hydrocarbons, ozone and other oxidants.
- In India, thermal power plants account for 80% of all industrial emissions of particulate matter, sulphur and nitrous oxides in India.

- **Particulate pollutants:** Dust, mist, fumes, smoke, smog etc.

SULPHUR DIOXIDE POLLUTION

- Oxides of sulphur are produced when sulphur containing fossil fuel is burnt.
- The most common oxide of sulphur is sulphur dioxide, SO_2.
- Particulate matter in the air accelerates formation of oxides of sulphur catalyses the process of oxidation.

SOURCES OF SO_2 POLLUTION

- Burning of fossil fuels such as coal, oil and natural gas are the main source of SO_2 emissions.
- Volcanic eruptions are also a major source of SO_2 emissions.
- Hydrogen sulphide, released from biological decay, reacts with O_2 in the atmosphere to produce SO_2.

HARMFUL EFFECTS OF SO_2

- SO_2 is a poisonous gas known to cause respiratory diseases such as asthma, bronchitis, emphysema in human beings, irritation to the eyes.
- High concentration of SO_2 leads to stiffness of flower buds.
- SO_2 is responsible for acid rain.
- SO_2 is the main cause of discoloration of marble in Taj Mahal.

NITRATE RADICALS

Nitrogen trioxide or nitrate radical is an oxide of nitrogen with formula NO3, consisting of three oxygen atoms covalently bound to a nitrogen atom.

- The nitrate radical (NO3) arises from the oxidation of nitrogen dioxide (NO2) by ozone (O3) and occurs principally in the nighttime atmosphere due to its rapid photolysis in sunlight and its reaction with NO.
- This highly unstable blue compound has not been isolated in pure form but can be generated and observed as a short-lived component of gas, liquid, or solid systems.
- Nitrate radicals will oxidize gas pollutants such as volatile organic compounds (VOCs), which will then generate ozone and secondary organic aerosol. So, this deteriorates air quality.
- Ozone is an air pollutant that affects human health and crop yield.
- Secondary organic aerosol is an important component of PM2.5.

15. GLOBAL FRAMEWORK ON CHEMICALS

- Setup during the fifth International Conference on Chemicals Management (ICCM5) in 2023 in Bonn, Germany.
- It complements existing financial mechanisms, like Global Environment Facility etc. and funds that support biodiversity and climate action.
- It aims to support low-and middle-income countries, including small island developing states, in addressing chemicals, including products and waste in line with international standards. o Target medium-scale projects that strengthen national and regional abilities.
- Composition: 2 National representatives of each United Nations region; and Representatives of all donors and contributors.
- It is a Multi-sectoral agreement that outlines a set of 28 targets to address chemicals and waste management including prevention of illegal trade, elimination of highly hazardous pesticides in agriculture by 2035 etc.

16. GRASSHOPPER EFFECT

- Plastic and Toxic chemicals from the world deposit and accumulate in the Arctic, making the Arctic a “hemispheric sink”.
- These are transported on atmospheric and oceanic currents from lower latitudes through a process known as global distillation or the “grasshopper effect.”

- Chemicals threaten the environment and health of Arctic People. E.g. Polyaromatic hydrocarbons (cancer, heart disease), Bisphenols (obesity and cancer), etc.

17. AIR QUALITY INDEX (AQI)

- AQI is an initiative of the Ministry of Environment Forest and Climate Change under Swachh Bharat Abhiyan.
- The index is constituted as a part of the Government's mission to improve the culture of cleanliness and helps the public to judge air quality within their vicinity. It is a colour coded index.
- There are six AQI categories, namely Good, Satisfactory, Moderately polluted, Poor, Very Poor, and Severe.
- The index will measure eight major pollutants, namely, particulate matter (PM 10 and PM 2.5), nitrogen dioxide, sulphur dioxide, ozone, carbon monoxide, ammonia and lead.

18. NATIONAL AMBIENT AIR QUALITY STANDARDS (NAAQS)

- **The Central Pollution Control Board (CPCB)** has notified these standards under powers given to it under the Air (Prevention and Control of Pollution) Act, 1981.
- **It covers 12 pollutants:** Sulphur Dioxide, Nitrogen Dioxide, PM-10, PM-2.5, Ozone, Lead, Carbon Monoxide, Ammonia, Benzene, Benzopyrene, Arsenic, Nickel.
- Whenever monitoring results on two consecutive days of monitoring exceed the limits specified in NAAQS above for the respective category, it is considered adequate reason to institute regular or continuous monitoring and further investigation.

19. NATIONAL CLEAN AIR PROGRAMME

The Centre's ambitious attempt to improve air quality in some of India's most polluted cities suggests that most Indian cities are far from making significant progress.

NATIONAL CLEAN AIR PROGRAMME

- It was launched by **the Ministry of Environment, Forest, and Climate Change** in 2019.
- It is **a long term, time bound, national level strategy** to tackle air pollution problems across the country in a comprehensive manner.
- The **aim** is to **improve air quality in 131 cities** (non-attainment cities and million plus cities) in 24 states/UTs by engaging all stakeholders.

Targets: Initially the target is to **achieve a 20% to 30% reduction of PM10 and PM2.5 concentrations by the year 2024, keeping base year as 2017** for comparison of concentrations.

- The new target is to achieve **reductions of PM10 concentration up to 40%** or **achievement of National Ambient Air Quality standards for PM10 by 2025-2026 (Base year 2017).**

OBJECTIVE OF NCAP

- To **ensure stringent implementation of mitigation measures** for prevention, control, and abatement of air pollution.
- Effective and proficient **ambient air quality monitoring network across the country** for ensuring a comprehensive and reliable database.
- To augment **public awareness and capacity-building measures** for inclusive public participation and for ensuring trained manpower and infrastructure on air pollution.

PORTAL FOR MONITORING

PRANA" – **Portal for Regulation of Air-pollution in non-attainment** cities, is a portal for monitoring implementation of National Clean Air Programme (NCAP).

- It will support tracking of physical as well as financial status of city air action plan implementation.
- It will disseminate information on air quality management efforts under NCAP to the public.

20. AIR QUALITY EARLY WARNING SYSTEM

- The initiative comes under the Ministry of Earth Sciences and Environment.
- The Air Quality Early Warning System for Delhi has been announced by the Central government that can alert, three days in advance, about the likelihood of extreme pollution events & dust storms.
- The air pollution system has been developed jointly by Indian Institute of Tropical Meteorology (IITM), India Meteorological Department, National Centre for Medium Range Weather Forecasting (NCMRWF).
- It intends for real time observations with 72-hour lead time of air quality over Delhi region.
- It provides details about natural aerosols like dust from dust storms and particulate matter using different satellite data sets.
- It will provide warning messages and Alerts to take necessary steps as per Graded Response Action Plan (GRAP).

21. GRADED RESPONSE ACTION PLAN (GRAP)

It specifies actions required for controlling particulate matter (PM) emissions from various pollution sources and prevents PM10 and PM2.5 levels to go beyond 'moderate' national Air Quality Index (AQI) category.

These measures were earlier implemented in Delhi only. However recently GRAP has been extended to the NCR towns also.

- It was planned by the Environment Pollution (Prevention and Control) Authority (EPCA) and approved by the Supreme Court in 2016.
- GRAP works only as an emergency measure. As such, the plan does not include action by various state governments to be taken throughout the year to tackle industrial, vehicular and combustion emissions.
- If air quality reaches the severe stage, GRAP suggests shutting down schools and implementing the odd-even road-space rationing scheme.
- GRAP comprises measures such as prohibition on entry of trucks into Delhi; ban on construction activities, introduction of odd and even scheme for private vehicles, shutting of schools, closure of brick kilns, hot mix plants and stone crushers; shutting down of Badarpur power plant, ban on diesel generator sets, garbage burning in landfills and plying of visibly polluting vehicles etc.

22. HAPPY SEEDER

A machine called the 'Happy Seeder' has been developed in the last few years that can plant the wheat seed without getting jammed by the rice straw.

ABOUT "HAPPY SEEDER"

- Burning of crop residue in the field by farmers in Punjab, Haryana and western Uttar Pradesh is considered as a major source of pollution in Delhi in winters.
- Farmers harvest the rice crop by combining harvesters. This machine leaves rice straw strewn all over the fields.
- Farmers do not use rice straw as animal-feed or for non-feed use.
- The straw clogs the seeder machines that plant the next crop, which is wheat, so farmers need to dispose of the residue before attempting to plant wheat. They do this by burning the residue.
- **The Happy Seeder is a tractor-mounted machine that cuts and lifts rice straw, sows wheat into the bare soil, and deposits the straw over the sown area as mulch.**

23. ARTIFICIAL RAIN OR CLOUD SEEDING

Delhi is planning to induce artificial rains to tackle the rising air pollution.

BRIEF ABOUT CLOUD FORMATION

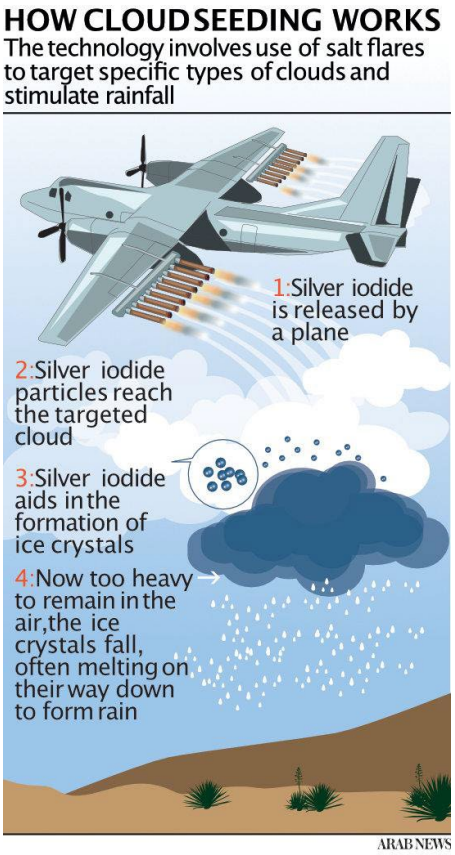
- Clouds are made up of tiny water droplets or ice crystals that form when water vapor in the atmosphere cools and condenses around a tiny particle of dust or salt floating in the atmosphere.

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- Without these particles, known as condensation or ice nuclei, raindrops or snowflakes cannot form, and precipitation will not occur.

CLOUD SEEDING

- Cloud seeding is an artificial method used to increase rainfall by introducing certain substances into clouds.
- During cloud seeding**, chemicals like silver iodide, potassium iodide, and dry ice** are released into the sky using airplanes and helicopters.
- These chemicals attract water vapour, leading to the formation of rain clouds.
- It usually takes about half an hour for this method to make it rain.



TECHNIQUES OF CLOUD SEEDING

According to the ScienceDirect, cloud seeding techniques can be divided into two categories:

1. Hygroscopic cloud seeding aims to accelerate the coalescence of droplets in liquid clouds, resulting in the formation of larger droplets that lead to precipitation.

- The seeding agents used in hygroscopic cloud seeding serve as efficient cloud condensation nuclei (CCN) or GCCN and play a crucial role in strengthening the condensation and collision–coalescence process, thereby widening the droplet size distribution (DSD) and increasing the precipitation efficiency.

2. Glaciogenic cloud seeding, focuses on inducing ice production in supercooled clouds, leading to precipitation.

- Glaciogenic cloud seeding involves dispersing efficient ice nuclei, such as silver iodide particles or dry ice, into the cloud, which enhances ice particle production and increases rain.

24. EFFECTIVENESS OF SUPPRESSANTS TO CONTROL DUST

In response to deteriorating air quality in Delhi, the Government has boosted public transport and used suppressants to control dust.

ABOUT DUST SUPPRESSANTS

- Dust suppressants are substances that are used to control dust emissions from various sources such as construction sites and roads.
- They can be salts of calcium or magnesium that can absorb moisture, or other chemical mixtures.

TYPES OF DUST CONTROL MEASURES

- **Water**: Water is one of the most common and effective dust suppressants. It can be applied as a fine mist, spray, or through irrigation systems to dampen surfaces and keep dust particles from becoming airborne. Water is environmentally friendly and readily available.
- **Chemical Dust Suppressants**: Chemical dust suppressants are substances that are applied to surfaces to control dust. Common chemical suppressants include:
- **Calcium Chloride**: It is often used on roads and construction sites to control dust. It attracts moisture from the air, keeping the road or area damp and dust-free.
- **Magnesium Chloride**: Like **calcium chloride, magnesium chloride** is used for dust control on roads and surfaces. It is less corrosive to metals than calcium chloride.

- **Lignin Sulfonate**: Lignin sulfonate is a **natural biopolymer** that can be used as a dust suppressant, particularly in areas where ecological concerns are a priority.
- **Synthetic Polymers**: Synthetic polymers, like **polyvinyl acetate or polyvinyl alcohol**, can be applied to the surface to create a dust-retaining film.
- **Soil Stabilizers**: They are used to bind the soil particles together and prevent them from becoming airborne.
- **Dust Control Equipment**: Dust control equipment, such as dust collectors, electrostatic precipitators, and baghouses, are used in industrial settings to capture and collect dust particles before they can escape into the air.
- **Vegetation**: Planting vegetation in and around dusty areas can help reduce dust by stabilizing the soil and acting as a natural dust suppressant.
- **Windbreaks**: Installing windbreaks, such as walls or fences, can help reduce the movement of windborne dust by blocking the wind's force.

25. CLIMATE & CLEAN AIR COALITION

- It is a voluntary partnership of governments, intergovernmental organisations, businesses, scientific institutions and civil society organisations committed to protecting the climate and improving air quality through actions to reduce Short lived climate pollutants.
- India became a member of this coalition in 2019.

CURRENTLY IT IS FOCUSED ON 4 SHORT LIVED CLIMATE POLLUTANTS (SLCPS)

- Black Carbon
- Methane
- Hydrofluorocarbons
- Tropospheric Ozone

WHAT ARE SHORT LIVED CLIMATE POLLUTANTS (SLCPS)

- These are powerful climate forces that remain in the atmosphere for a much shorter period of time than carbon dioxide, yet their potential to warn the atmosphere can be many times.
- The SLCPs – Black Carbon, Methane, Hydrofluorocarbons, Tropospheric Ozone are responsible for 45% of current global warming.

26. CLEAN AIR INITIATIVE

- It is an initiative launched by the United Nations, WHO, UNEP and Climate and Clean Air Coalition.
- It calls on national and subnational governments to commit to achieving air quality that is safe for citizens, and to align climate change and air pollution policies by 2030.
- The initiative will be led by the WHO with the support of the Clean Air and Climate Coalition (CCAC).

27. COOL COALITION

- It is a global effort led by the UN Environment, the Climate and Clean Air Coalition, the Kigali Cooling Efficiency Program, and Sustainable Energy for All (SEforALL).
- It was launched at the first Global Conference on Synergies between the 2030 Agenda and Paris Agreement in 2019.
- It is unified front that links action across the Kigali Amendment, Paris Agreement and Sustainable Development Goals
- It aims to inspire ambition and accelerate action on the transition to clean and efficient cooling.
- CO2 is provided to the microalgae by means of a pump that captures polluted air and spurges it through the water to nourish the algae.

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- This system is equivalent to two 10-year-old trees or 200 square meters of grass.
- The advantage of microalgae compared to trees is that they are about 10-50 times more efficient than trees.
- The Bioreactors can be built and installed in a very short time and can start treating polluted air immediately.

28. GLOBAL BIOFUEL ALLIANCE

Leaders of India, Singapore, Bangladesh, Italy, USA, Brazil, Argentina, Mauritius and UAE launched the Global Biofuel Alliance on the sidelines of the G20 Summit in New Delhi.

- **Global Biofuel** Alliance is an initiative by India as the chair of G20 that aims to serve as a catalytic platform, fostering global collaboration for the advancement and widespread adoption of biofuels.
- It is an alliance of governments, international organisations and industry to facilitate adoption of biofuels.
- It brings together the biggest consumers and producers of biofuels aiming to drive biofuels development and deployment and position biofuels as a key to energy transition and economic growth.
- Global Biofuel Alliance intends to expedite the global uptake of biofuels through:
- Facilitating technology advancements
- Intensifying utilisation of sustainable biofuels
- Shaping robust standard setting
- Certification through the participation of a wide spectrum of stakeholders.
- Facilitate mobilising a virtual marketplace to assist industries, countries, ecosystem players in mapping demand and supply and connecting technology providers to end users.
- Facilitate development, adoption and implementation of internationally recognised standards, codes, sustainability principles and regulations to incentivise biofuels adoption and trade.
- Act as a central repository of knowledge and an expert hub.
- GBA Members constitute major producers and consumers of biofuels. USA (52%), Brazil (30%) and India (3%) contribute about 85% of global production of ethanol and 81% of global consumption of ethanol.

29. BIOCHEMICAL OXYGEN DEMAND

- It is the amount of dissolved oxygen needed by aerobic biological organisms to break down organic material present.
- It can be used as a gauge of the effectiveness of wastewater treatment plants.
- The more organic matter there is (e.g., in sewage and polluted bodies of water), the greater the BOD; and the greater the BOD, the lower the amount of dissolved oxygen available for higher animals such as fishes.

30. NATIONAL PLAN FOR CONSERVATION OF AQUATIC ECO-SYSTEMS (NPCA)

- The National Wetlands Conservation Program (NWCP) and 'National Lake Conservation Plan' (NLCP) have been merged into one integrated scheme, National Plan for Conservation of Aquatic Eco-systems (NPCA).
- The scheme aims at holistic conservation and restoration of lakes and wetlands for achieving the desired water quality enhancement, besides improvement in biodiversity and ecosystem through an integrated and multidisciplinary approach and a common regulatory framework.
- The scheme would contribute to reduction of pollution loads in lakes and wise use of wetland resources and their services. NPCA is presently operational on cost sharing between Central and respective state governments.

31. GLOBAL RIVER CITIES ALLIANCE

A nine-country 'Global River Cities Alliance' (GRCA) inspired by India's River Cities Alliance was launched at the Indian Pavilion, Dubai on the sidelines of the ongoing COP28 on Sunday.

GLOBAL RIVER CITIES ALLIANCE (GRCA)

- The GRCA is modelled on the lines of India's River Cities Alliance established by the National Mission for Clean Ganga (NMCG) under the Ministry of Jal Shakti and the National Institute of Urban Affairs under the Ministry of Housing and Urban Affairs in November 2021.
- The National Mission for Clean Ganga, under the Jal Shakti ministry and Housing and Urban Affairs ministry's National Institute for Urban Affairs (NIUA), will take on the role of the Secretariat of the GRCA.
- The nine nations are India, Denmark, Cambodia, Japan, Bhutan, Australia, Netherlands, Egypt, and Ghana.
- Multi-lateral funding agencies such as the World Bank, Asian Development Bank and Asian Infrastructure Investment Bank have also pledged their support for the initiative.

RIVER CITY ALLIANCE

- River Cities Alliance is a platform for river cities in India to ideate, discuss, and exchange information for the sustainable management of urban rivers.
- The Jal Shakti Ministry and Housing & Urban Affairs Ministry has partnered for the initiative.
- The National Institute for Urban Affairs (NIUA) and the National Mission for Clean Ganga (NMCG) have collaborated for launching the River Cities Alliance.
- The Secretariat of the River Cities Alliance will be set up at the National Institute for Urban Affairs (NIUA) with the support of the National Mission for Clean Ganga (NMCG).
- The River Cities Alliance will focus on three themes: Networking, Technical Support, and Capacity Building.
- The participating cities of the River Cities Alliance will work to adopt and localize national policies and instruments with key river-related directions, develop city-specific sectoral strategies for sustainable urban river management, and prepare their Urban River Management Plans.

32. NATIONAL PROJECTS

FUNDING PATTERN FOR NATIONAL PROJECTS

For Projects in 8 North-Eastern States, Himalayan States of Uttarakhand and Himachal Pradesh, UTs of J&K and Ladakh	90 (Centre); 10 (State)
Projects in other States	60 (Centre); 40 (State)

- However, the guidelines say that the funding pattern of a project included in the National Project scheme can change at the discretion of the Central Government to the extent deemed fit for a national project.

LIST OF APPROVED 16 NATIONAL PROJECTS

GOSIKHURD IRRIGATION PROJECT, Maharashtra	On the Wainganga River in Bhandara District
SARYU NAHAR PARIYOJANA, Uttar Pradesh	• It involves interlinking of 5 rivers: Ghaghara, Saryu, Rapti, Banganga and Rohini. • Sub-canals with a length of over 6,600 km for optimum use of the water in Eastern UP.
POLAVARAM IRRIGATION PROJECT, Andhra Pradesh	On the Godavari River in Andhra Pradesh.
SHAHPURKANDI DAM PROJECT, Punjab	• On the Ravi River in Punjab. • Located downstream of the existing Ranjit Sagar Dam.

LAKHWAR PROJECT, Uttarakhand	On the Yamuna River in Dehradun district of Uttarakhand.
UJH PROJECT, J&K	On the Ujh River in Kathua district of J&K. Ujh is a tributary of River Ravi.
RENUKAJI PROJECT, Himachal Pradesh	• On the Giri River in Sirmaur District of J&K. Giri River is a tributary of Yamuna River. • Water from the project is expected to address the drinking water issues of the NCR region.
KEN BETWA LINK PROJECT, MP & UP	• A river linking project where Daudhan Dam (in the Panna Tiger Reserve) will be constructed on the Ken River and via a canal the water will be taken to Betwa River. • Benefit the water scarce Bundelkhand region of UP and MP.
KULSI DAM PROJECT, Assam	• On the Kulsi River, a tributary of Brahmaputra River. • The Dam is located on the boundary of Meghalaya and Assam.
NOA DIHING DAM PROJECT, Arunachal Pradesh	• Proposed on Noa-Dihing river in Arunachal Pradesh. **ABOUT DIHING RIVER** Dihing River originates from Arunachal Pradesh and bifurcates into two rivers in Changlang district. They are: • Noa-Dihing: Flows through Changlang and Lohit district before merging with Brahmaputra. Flows through Namdapha National Park. • Burhi-Dihing: Flows through Dibrugarh and Tinsukia district before merging with Brahmaputra.
Pushpagiri Wildlife Sanctuary	**Pushpagiri Wildlife Sanctuary** • This sanctuary is located in **Kodagu district of Karnataka** and is part of Western ghats. • The Kadamakkal reserve forest is a part of the sanctuary. • **Pushpagiri (Kumara Parvatha)** is the highest peak in it (2nd highest in Karnataka). • The sanctuary adjoins Bisle reserve forest to the north and Kukke Subramanya forest range to the west. • **Mandalpatti peak, Kote betta and Makkalagudi betta** are mountains that fall in the sanctuary. • **Mallalli falls and Kote abbe waterfalls** (also called Mukkodlu falls) are located inside the sanctuary. **Fauna** • A few of the popular animal species are Spotted Deer, Wild Pig, Giant Squirrel, Gaur, Sambar, Asian Elephant. • Orange Flycatcher, Nilgiri Flycatcher, Grey-Breasted Laughing Thrush. • King Cobra, rat, and other variants of snakes can also be found here. • The Nilgiri Pigeon is the most beautiful attraction here.
BURSAR PROJECT, J&K	• On the Marusudar River, a tributary of Chenab. • Located in Kishtwar district of J&K.

KISHAU PROJECT, Himachal Pradesh and Uttarakhand	On the Tons River on the borders between Uttarakhand and Himachal Pradesh.
GYPSA PROJECT, Himachal Pradesh	• On the Bhaga River in Keylong tehsil of Lahaul & Spiti District of Himachal Pradesh. • Bhaga River is a tributary of Chenab river (Also known as Chandrabhaga). Bhaga river originates from Surya Taal near Bara Lacha La Pass in Himachal Pradesh.
Second RAVI VYAS LINK PROJECT, Punjab	Aims to develop a link channel between Ravi and Vyas river to reduce the amount of surplus water in these rivers whose water falls in India's share under the Indus Water Treaty.
UPPER SIANG PROJECT, Arunachal Pradesh	• On the Siang River in Arunachal Pradesh. • Siang is a tributary of Brahmaputra.

33. FORMALIN (METHANAL) CONTAMINATION

- Formalin is the aqueous solution of formaldehyde.
- Pure formaldehyde is a colorless, flammable gas with a strong pungent odor. It is extremely irritating to the mucous membranes and is associated with certain types of cancer in humans and other animals.
- It is mainly used in the production of industrial resins, e.g., for particle board and coatings.
- The Food Safety and Standards Authority of India (FSSAI) has banned formaldehyde in fresh fish, while the International Agency for Research on Cancer labelled the chemical a carcinogen.

34. BUNKER CONVENTION

- International Convention on Civil Liability for Bunker Oil Pollution Damage (BUNKER) is an international treaty listed and administered by the International Maritime Organization, enforced from November 2008.
- The purpose is to adopt uniform international rules and procedures for determining questions of liability and providing adequate compensation.
- The convention covers leakage of that oil and requires signatories to the convention to have their ships appropriately insured against such leakages.
- *India is yet to ratify International Convention on Civil Liability for Bunker Oil Pollution Damage* (BUNKER)

35. HONG KONG CONVENTION

- Adopted by the International Maritime Organization (IMO) in 2009 for Safe and Environmentally Sound Recycling of Ships.
- India is the leader in the global ship recycling industry, with a share of over 30% of the market.
- Indian Parliament passed the Recycling of Ships Bill to enforce this convention.

  NOTE: Alang in Bhavnagar district of Gujarat, has become a major worldwide center for ship breaking.

36. DECARBONIZING SHIPPING: GETTING TO ZERO COALITION

- Aim of the Coalition is to have commercially viable zero emission vessels (ZEVs) operating along deep-sea trade routes by 2030.

- Partnership between the Global Maritime Forum, World Economic Forum and Friends of Ocean Action, whose ambition is to have commercially viable zero emission vessels (ZEVs) operating along deep sea trade routes by 2030
- Alliance of more than 90 companies within the maritime, energy, infrastructure and finance sectors, supported by key governments.

37. CENTRAL POLLUTION CONTROL BOARD (CPCB)

- It is a statutory organisation that was constituted in 1974 under the Water (Prevention and Control of Pollution) Act, 1974.
- It also provides technical services to the Ministry of Environment and Forests of the provisions of the Environment (Protection) Act, 1986.
- Further, CPCB is entrusted with the powers and functions under the Air (Prevention and Control of Pollution) Act, 1981.

AIM

- To promote cleanliness of streams and wells in different areas.
- To improve the quality of air and to prevent, control or abate air pollution in the country.

38. SAFAR

- Introduced by the Ministry of Earth Sciences (MoES), GOI.
- SAFAR for greater metropolitan cities of India is to provide location specific information on air quality in near real time and its forecast 1-3 days in advance for the first time in India.
- It has been combined with the early warning system on weather parameters.
- The implementation of SAFAR is made possible with an active collaboration with local municipal corporations and various local educational institutions and governmental agencies in that Metro city.
- The ultimate objective of the project is to increase awareness among the general public regarding the air quality in their city well in advance so that appropriate mitigation measures and systematic action can be taken up for betterment of air quality and related health issues.

COMPONENTS OF SAFAR

- The development of emission inventory of air pollutants for NCR and defining air quality index for India.
- Network of eleven Air Quality Monitoring Stations (AQMS) equipped with 11 automatic weather stations to provide near real time air quality information.
- 3-D atmospheric chemistry transport forecasting modelling coupled with weather forecasting model to provide 24-hour advance forecast of air pollutant levels.
- Display on LED and LCD screens located at 20 different locations in Delhi in a public friendly format and displaying the online detailed information through the Web portal.
  - Pollutants monitored: PM1, PM2.5, PM10, Ozone, CO, NOx (NO, NO2), SO2, BC, Methane (CH4), non-methane, hydrocarbons (NMHC), VOC's, Benzene, Mercury.
  - Monitored Meteorological Parameters: UV Radiation, Rainfall, Temperature, Humidity, Wind speed, Wind direction, and solar radiation.

39. MINAMATA CONVENTION

- It is an international treaty that aims to protect human health and the environment from anthropogenic emissions and releases of mercury and mercury compounds.
- The Convention is named after the Japanese city Minamata.
- This naming is of symbolic importance as the city went through a devastating incident of mercury poisoning.

40. NITROGEN POLLUTION

FORMS OF NITROGEN

N_2	Un-reactive di-nitrogen; forms 78% of the air we breathe
NR	Reactive nitrogen; fixed in soil by microbes; reacts to form different compounds with various impacts
NH_3	Ammonia; used for making fertilizers; can escape into the air as a pollutant.
NH_4NO_3	Ammonium nitrate, acts as fertilizer; when synthesized in the atmosphere, contributes to particulate matter, water pollution and results in eutrophication.
N_2O	Nitrous oxide, a greenhouse gas; depletes ozone layer
NOx	Mixture of NO and NO2; a major air pollutant.
NO_3	Nitrate: the form in which nitrogen gets fixed in soil; can pollute water sources; forms ozone, which adds to particulate matter load.

41. NOISE POLLUTION

- **World Health Organization (WHO)** guidelines established a health protective recommendation for road traffic noise levels of 53 db.

PERMISSIBLE NOISE LEVEL IN INDIA

- CPCB has laid down permissible noise levels in India for different areas. Noise pollution rules have defined acceptable levels of noise in different zones for both daytime and night-time.
- In industrial areas, the permissible limit is 75 dB for daytime and 70 dB at night.
- In commercial areas, it is 65 dB and 55 dB, while in residential areas it is 55 dB and 45 dB during daytime and night, respectively.

Category of Area / Zone	Limits in L_{eq} dB(A)	
	Day Time*	Nighttime**
Industrial area	75	70
Commercial area	65	55
Residential area	55	45
Silence/Sensitive Zone	50	40
*Daytime shall mean 6:00 A.M. to 10:00 P.M. **Nighttime shall mean 10:00 P.M. to 6:00 A.M.		

REGULATIONS FOR NOISE POLLUTION

- Earlier, noise pollution & its sources were addressed under the Air (Prevention and Control of Pollution) Act, 1981.
- Noise pollution is currently regulated under Noise Pollution (Regulation and Control) Rules, 2000.
- Additionally, noise standards for motor vehicles, air-conditioners, refrigerators, diesel generators and certain types of construction equipment are prescribed under Environment (Protection) Rules, 1986.
- Noise emanating from industry is regulated by State Pollution Control Boards/Pollution Control Committees (SPCBs /PCCs) for state/UTs under the Air (Prevention and Control of Pollution) Act, 1981.

ABOUT NOISE POLLUTION

- Any unwanted sound that causes annoyance, irritation and pain to the human ear is termed noise. It is measured in A-weighted decibels (dB (A)) that indicate loudness of sound.
- Noise level refers to decibel levels of noise produced by any appliance or machine. In general, the human ear can tolerate noise levels up to 85 db. Anything beyond that can affect human productivity & quality of life.
- Decibel levels of common sounds above 80 dB are considered loud, while decibel levels of common sounds between 100-125 dB are termed uncomfortable.
- All machines operating in an area should produce noise within acceptable levels to maintain the well-being of people around.

42. BUREAU OF ENERGY EFFICIENCY

- It is a statutory body established in 2002, under the Energy Conservation Act, 2001.
- It functions under the Ministry of Power. Minister of power shall be the ex-officio chairman of the bureau.
- **Mandate:** It facilitates the implementation of the EC Act by developing policies and strategies which focus on the primary objective of reducing energy intensity of the Indian economy.
- It is responsible for spearheading the improvement of energy efficiency of the economy through various regulatory and promotional instruments.
- It coordinates with State level agencies.
- The members of the governing council of the bureau are appointed by the central government.

43. ENERGY EFFICIENCY SERVICES LIMITED (EESL)

- It is a Joint Venture of NTPC Limited, PFC, REC and POWERGRID to facilitate implementation of energy efficiency projects.
- It will be the first such company exclusively for implementation of energy efficiency in South Asia and amongst a very few such instances in the world.
- EESL will also lead the market-related actions of the NMEEE.

44. ECO MARK

- Eco mark is a certification mark issued by the Bureau of Indian Standards (BEE)to products conforming to a set of standards aimed at the least impact on the ecosystem.
- Under this initiative, all those households and other consumer products which meet certain environmental criteria are labelled with an environment friendly mark.

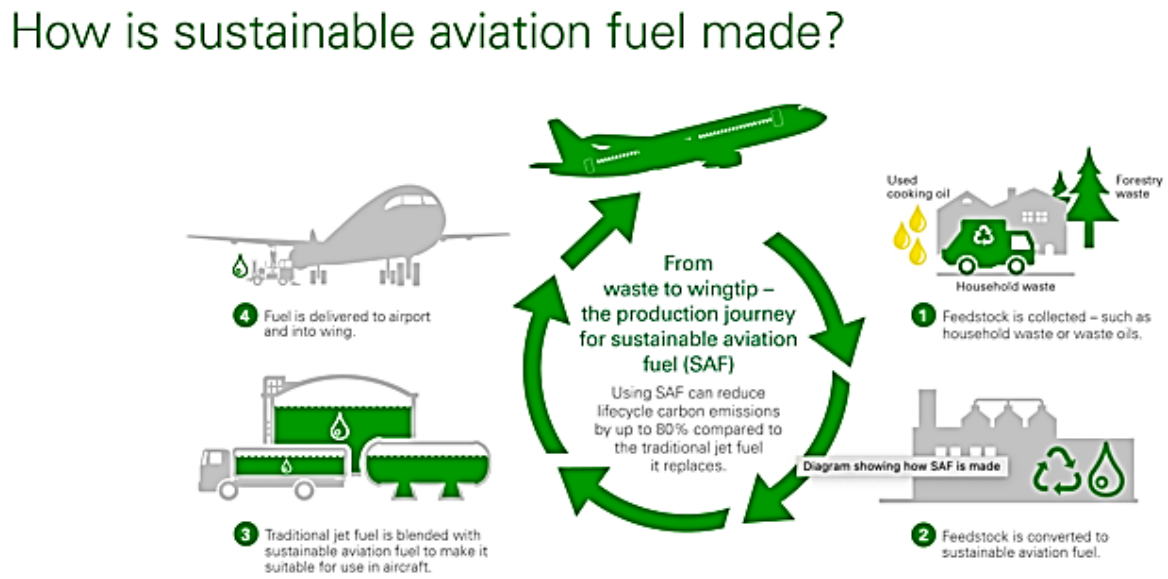
45. BIO-JET FUEL

- It is a type of Biofuel which is produced from biomass resources and used in place of or blended with Aviation Turbine Fuel (ATF).
- Bio jet fuel can be produced from animal fat, used cooking oil, waste dairy fat, sewage sludge, etc.
- The oil needs to have a freezing point below -47 degrees, so it does not freeze at altitudes at which planes fly.
- It should not catch fire on ground when being transferred into a plane.
- It must have the same density as ATF, have a certain calorific value and should not choke the filters.
- It has lower sulphur content which causes less wear and tear.

46. SUSTAINABLE AVIATION FUEL (SAF)

- SAF is a biofuel used to power aircraft that has similar properties to conventional jet fuel but with a smaller carbon footprint.

- SAF has the greatest potential to reduce CO2 emissions from International Aviation. Depending on the feedstock and technologies used to produce it, SAF can reduce life cycle GHG emissions dramatically compared to conventional jet fuel.



- Some emerging SAF pathways even have a net-negative GHG footprint.
- Some typical feedstocks used are cooking oil and other non-palm waste oils from animals or plants; solid waste from homes and businesses, such as packaging, paper, textiles, and food scraps that would otherwise go to landfill or incineration. Other potential sources include forestry waste, such as waste wood, and energy crops, including fast growing plants and algae.
- SAF can be blended at up to 50% with traditional jet fuel and all quality tests are completed as per a traditional jet fuel.
- SAF is currently more costly than traditional fossil jet fuel. It has the potential to provide a lifecycle carbon reduction of up to 80% compared to the traditional jet fuel it replaces.

47. NATIONAL POLICY ON BIOFUELS – 2018

The Union Cabinet approved National Policy on Biofuels – 2018 to encourage the generation and use of biofuels.

FEATURES

- **Categorisation of biofuels** to enable extension of appropriate financial and fiscal incentives under each category. The two main categories are:
- **Basic Biofuels:** First Generation (1G) bioethanol & biodiesel
- **Advanced Biofuels:** Second Generation (2G) ethanol, Municipal Solid Waste (MSW) to drop-in fuels, third Generation (3G) biofuels, bio-CNG etc.
- **Expands the scope of raw material for ethanol production** by allowing use of Sugarcane Juice, Sugar containing materials like Sugar Beet, Sweet Sorghum, Starch containing materials like Corn, Cassava, Damaged food grains like wheat, broken rice, Rotten Potatoes, unfit for human consumption for ethanol production.
- **Allows use of surplus food grains for production of ethanol** for blending with petrol to ensure appropriate price to farmers during surplus.
- **Encourages setting up of supply chain mechanisms** for biodiesel production from non-edible oilseeds, used Cooking Oil, short gestation crops.

ABOUT BIOFUEL

- Biofuel is any hydrocarbon fuel that is produced from organic matter in a short period of time.
- This is in contrast with fossil fuels, which take millions of years to form.

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- Biofuels are considered a renewable form of energy as it emits less than fossil fuels.

DIFFERENT GENERATION BIOFUELS

- **First Generation Biofuels:** It uses the food crops like wheat and sugar for making ethanol and oil seeds for bio diesel by conventional method of fermentation.
- **Second Generation Biofuels:** It uses non-food crops and feedstock such as Wood, grass, seed crops, organic waste are used in fuel preparation.
- **Third Generation Biofuels:** It uses specially engineered Algae whose biomass is used to convert into biofuels. The greenhouse gas emission here will be low in comparison to others.
- **Fourth Generation Biofuels:** It aimed at not only producing sustainable energy but also a way of capturing and storing CO_2.

FLEXIBLE-FUEL VEHICLE

A flexible-fuel vehicle (FFV) or dual-fuel vehicle (colloquially called a flex-fuel vehicle) is an alternative fuel vehicle with an internal combustion engine designed to run on more than one fuel, usually gasoline blended with either ethanol or methanol fuel, and both fuels are stored in the same common tank.

Modern flex-fuel engines can burn any proportion of the resulting blend in the combustion chamber as fuel injection and spark timing are adjusted automatically according to the actual blend detected by a fuel composition sensor.

Flex-fuel vehicles are distinguished from bi-fuel vehicles, where two fuels are stored in separate tanks and the engine runs on one fuel at a time, for example, compressed natural gas (CNG), liquefied petroleum gas (LPG), or hydrogen.

Co-firing is the combustion of two (or more) different types of materials at the same time. One of the advantages of co-firing is that an existing plant can be used to burn a new fuel, which may be cheaper or more environmentally friendly. For example, biomass is sometimes co-fired in existing coal plants instead of new biomass plants.

Substituting 5-7 % of coal with biomass in coal-based power plants can save 38 million tonnes of carbon dioxide emissions, said finance minister Nirmala Sitharaman in her budget speech in 2022. It was also emphasised that the Power Ministry would consider reduction in coal supply for those thermal power plants that do not comply with the policy on biomass cofiring.

48. METHANOL FUEL

- Namrup-based Assam Petrochemicals Limited (APL) has rolled out the country's first methanol-based cooking fuel project- 'Green and Clean Fuel Pilot Project on Methanol Cooking Stove'.
- The project has been promoted by NITI Aayog.

ABOUT METHANOL

- Methanol is a clean-burning fuel that produces fewer smog-causing emissions — such as sulphur oxides (SOx), nitrogen oxides (NOx) and particulate matter — and can improve air quality and related human health issues.
- Methanol is most produced on a commercial scale from natural gas.
- It can also be produced from renewable sources such as biomass and recycled carbon dioxide.
- As a high-octane vehicle fuel, methanol offers excellent acceleration and power. It also improves vehicle efficiency.

49. CLEAN ENERGY

COAL GASIFICATION BASED FERTILISER PLANT

- India's first coal gasification-based fertiliser plant to be set up in Talcher, Odisha.

- It will produce Neem coated urea using coal and pet-coke as feedstock.

COAL GASIFICATION TECHNOLOGY

- It is one of the clean coal technologies and involves the process of converting coal into synthesis gas (also called syngas).
- Syngas is a mixture of hydrogen (H_2), carbon monoxide (CO) and carbon dioxide (CO_2).
- The by-products of coal gasification include coke, coal tar, sulfur, ammonia and fly ash, all having their own potential uses.
- CO_2 and ammonia are further reacted to produce urea.
- Syngas can also be used in a variety of other applications such as in the production of electricity, fuel for IC engines, making plastics, cement etc.

50. LEGACY WASTE

- Legacy wastes are the wastes that have been collected and kept for years at some barren land or a place dedicated for Landfill (an area to dump solid waste).
- They are more dangerous because:
  - They occupy large spaces and make the land unproductive. It is estimated that in India more than 10000 ha of land is a dump site.
  - They become breeding grounds for pathogens and flies.
  - They also lead to water pollution via ground water contamination.
  - They also contribute to generation of greenhouse gases and pose risk of uncontrollable fire.
- **Bio mining (Bio- leaching):** It is a process of using microorganisms (microbes) to extract metals of economic interest from rock ores or mine waste. Metals extracted from bio leaching include gold, silver, uranium, Nickle, copper, cobalt and zinc. Bio-mining techniques may also be used to clean up sites that have been polluted with metals.
- **Bioremediation** is a branch of biotechnology that employs the use of living organisms, like microbes and bacteria, in the removal of contaminants, pollutants, and toxins from soil, water, and other environments. Bioremediation is used to clean up oil spills or contaminated groundwater.

51. e-FUEL

- eFuels are produced with the help of electricity from renewable energy sources, water and CO2 from the air.
- In contrast to conventional fuels, they do not release additional CO2 but are climate neutral.
- With their compatibility with today's internal combustion engines, eFuels can also power vehicles, airplanes and ships, thus allowing them to continue to operate but in a climate-friendly manner.
- The same applies to all heating systems that use liquid fuels. Existing transport, distribution and fuel infrastructures (especially filling stations) can also continue to be used.

PRODUCTION

- eFuel production is based on the extraction of hydrogen. This happens by means of an electrolysis process that breaks down water (Ex. seawater from desalination plants) into its components of hydrogen and oxygen. For this process and further production steps, electricity is required.
- In a second step, with the aid of Fischer-Tropsch synthesis, the hydrogen is combined with CO2 extracted from the air and converted into a liquid energy carrier:
  - Under high pressure using a catalyst, the hydrogen binds with the CO2. Because electricity is used to produce eFuels, the procedure is known as a power-to-liquid (PtL) process: electricity is converted into a synthetic liquid that is easy to store and simple to transport.

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52. ZERO LIQUID DISCHARGE

- An engineering approach to water treatment where all water is recovered, and contaminants are reduced to solid waste.
- While many water treatment processes attempt to maximize recovery of freshwater and minimize waste, ZLD is the most demanding target since the cost and challenges of recovery increase as the wastewater gets more concentrated.
- Salinity, scaling compounds, and organics all increase in concentration, which adds costs associated with managing these increases.
- ZLD is achieved by stringing together water treatment technology that can treat wastewater as the contaminants are concentrated.

53. MICROPLASTICS

Microplastics are small plastic particles less than 0.2 inches (5 mm) long, according to the National Oceanic and Atmospheric Administration (NOAA).

WHERE DO MICROPLASTICS COME FROM?

Microplastics often originate from macroplastics that have broken apart in the ocean. Plastic may break apart due to weathering from the sun, wind, or other causes. The microbeads in many personal care products such as toothpaste and facial scrubs also count as microplastics.

IMPACT OF MICROPLASTICS ON ENVIRONMENT

Though they are small, these bits of plastic bring similar issues that macroplastics do — plus their own set of harms. These small particles serve as carriers for bacteria and persistent organic pollutants (POPs).

- POPs are toxic organic compounds that take years to degrade. They consist of chemicals like pesticides and dioxins, which are hazardous to human and animal health in high concentrations. POPs biomagnifies as they move up the food chain, meaning that larger animals accumulate more of these toxic substances within their fat and tissues than smaller organisms. POPs are not very water-soluble.

WORLD HERITAGE FORESTS

- **Published by:** UNESCO, World Resources Institute & IUCN.
- **Purpose:** Provides first global scientific assessment of greenhouse gas emissions and sequestration by forests in UNESCO World Heritage sites.

54. PLASTIC ROCKS

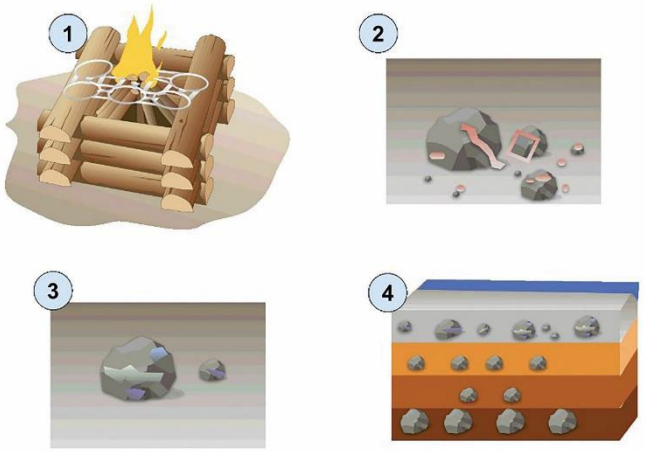
Rocks made of plastic debris have been discovered on the volcanic Trindade Island of Brazil.

WHAT ARE PLASTIC ROCKS?

- Technically they are called plastiglomerate.



- It is formed when plastic trash melts and fuses together with natural materials such as basaltic lava fragments, sand, shells, wood and coral, resulting in a plastic-rock hybrid.
- When the plastic melts, it cements rock fragments, sand, and shell debris together, or the plastic can flow into larger rocks and fill in cracks and bubbles.

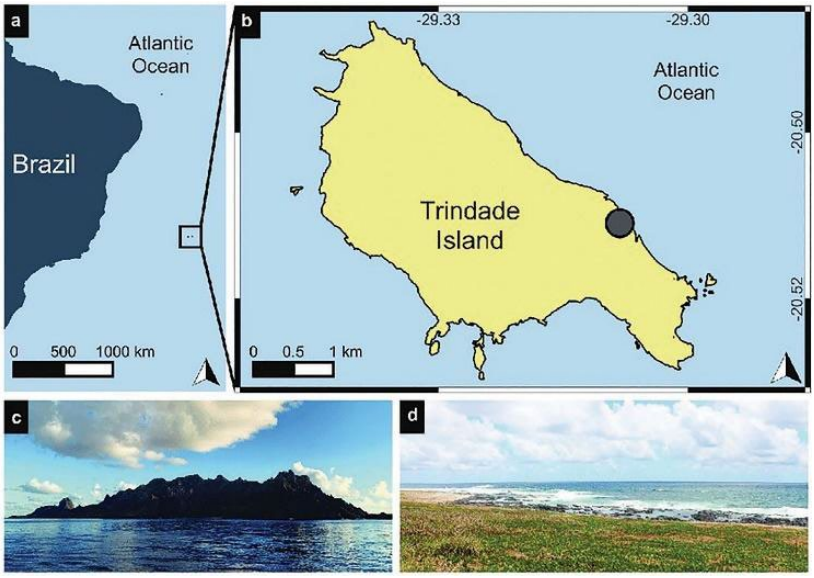


The melting of plastic waste from campfires or high temperatures on beaches(1) is resulting in the formation of a new type of rock known as plastiglomerate (2). Formed plastiglomerate merges with surrounding sediment to create a compositionally different sediment layer(3). The emergence of this new layer is being used as physical evidence of a marker horizon for an Anthropocene Epoch(4).

- Researchers say the new material is likely to last a very long time, possibly becoming a permanent marker in Earth's geologic record.

ABOUT TRINDADE ISLANDS

- **Location** – It is located 1140 Km from the Southeastern state of Espirito Santo of Brazil in South Atlantic Ocean.



- **Ecological significance:** Trindade Island is one of the world's most important conservation spots for green turtles, or Chelonia mydas, with thousands arriving each year to lay their eggs. The only human inhabitants on Trindade are members of the Brazilian navy, which maintains a base on the island and protects the nesting turtles.
- **Plastic threat:** It mainly comes from fishing nets, which is very common debris on Trindade Island's beaches. The (nets) are dragged by the marine currents and accumulate on the beach. When the temperature rises, this plastic melts and becomes embedded with the beach's natural material.

WHAT IS PLASTISPHERE?

- The plastisphere consists of ecosystems that have evolved to live in human-made plastic environments.
- The Plastisphere is a diverse microbial community living on bits of plastic floating in the ocean. These communities are distinct from the surrounding water, suggesting that plastic serves as its own habitat in the ocean.
- It's like a biofilm—a sticky material that traps all sorts of microorganisms, protozoa and fungi. They can multiply and create a unique biome around the plastics.

WHAT IS MARINE SNOW?

- Marine snow is a term popularized by scientific explorer and diver William Beebe in the 1930s.
- It's made up of nutrient-rich, organic material that falls like snow from the ocean's surface to the sea floor.
- The marine snow phenomenon explains why there's life at the bottom of the ocean.
- Also, how carbon can move from top to bottom in the ocean can also be partly explained through marine snow.

55. WASTE TO ENERGY

Government is promoting all the technology options available for setting up projects for recovery of energy in the form of Biogas/BioCNG/Electricity from agricultural, Industrial and urban wastes of renewable nature such as municipal solid wastes, vegetable and other market wastes, slaughterhouse waste, agricultural residues and industrial/STP wastes & effluents.

TYPES OF WASTE

- There are different types of waste which are generated from our daily or industrial activities such as organic waste, e-waste, hazardous waste, inert waste etc.
- Organic waste refers to waste (carbon-based compounds) which degrades or broken down by microorganisms over time.
- Organic waste has a significant portion in overall waste generation in the industrial/urban/ agricultural sector and therefore it can be used for energy generation.

The organic fraction of waste can be further classified as non-biodegradable and biodegradable organic waste.

Biodegradable waste: can be utilised for food by naturally occurring micro- organisms within a reasonable length of time. The biodegradable organic comprises of agro residue, food processing rejections, municipal solid waste (food waste, leaves from garden waste, paper, cloths/ rags etc.), waste from poultry farms, cattle farm slaughter houses, dairy, sugar, distillery, paper, oil extraction plant, starch processing and leather industries.

Non-Biodegradable: resistant to biological degradation or have a very low degradation rate. This primarily includes woody plants, Cardboard, cartons, containers, wrappings, pouches, discarded clothing, wooden furniture, agricultural dry waste, bagasse, rice husk etc.

TECHNOLOGIES AVAILABLE

Waste-to-Energy (WTE) technologies to recover the energy from the waste in the form of Electricity and Biogas/Syngas are given as below:

BIOMETHANATION

Bio methanation is anaerobic digestion of organic materials which is converted into biogas. Anaerobic digestion (AD) is a bacterial fermentation process that operates without free oxygen and results in a biogas containing mostly methane (~60%), carbon dioxide (~40%) and other gases. Bio methanation has dual benefits. It gives biogas as well as manure as product.

This technology can be conveniently employed in a decentralized manner for biodegradation of segregated organic wet wastes such as wastes from kitchens, canteens, institutions, hotels, and slaughterhouses and vegetables markets.

The biogas generated from Bio methanation process can be burned directly in a gas boiler/burner to produce heat for thermal application industries and cooking or burnt in a gas engine to produce electricity. Alternatively, the biogas can be cleaned to remove the carbon dioxide and other substances, to produce BioCNG. This can be injected into the national gas grid to be used in the same way as natural gas or used as a vehicle fuel.

By using Bio methanation process, 20-25kgs of Cattle dung can generate about 1m3 of biogas and further 1m3 of Biogas has potential to generate 2 units of electricity or 0.4kgs of BioCNG.

INCINERATION

Incineration technology is complete combustion of waste (Municipal Solid Waste or Refuse derived fuel) with the recovery of heat to produce steam that in turn produces power through steam turbines.

The flue gases produced in the boilers have to be treated by an elaborate air pollution control system. The resultant ash from incineration of solid waste can be used as construction material after necessary processing while the residue can be safely disposed of in a landfill.

This technology is well established technology and has been deployed in many projects successfully at commercial level in India to treat solid wastes like Municipal Solid Waste and Industrial solid Waste etc. and generate electricity.

GASIFICATION

Gasification is a process that uses high temperatures (500-1800o C) in the presence of limited amounts of oxygen to decompose materials to produce synthetic gas (a mixture of carbon monoxide (CO) and hydrogen (H2). Biomass, agro-residues, Segregated MSW and RDF pellets are used in the gasifier to produce Syngas. This gas further can be used for thermal or power generation purposes.

The purpose of gasification of waste is to generate power more efficiently at lower power level (< 2MW) and also to minimize emissions and hence it is an attractive alternative for the thermal treatment of solid waste.

PYROLYSIS

Pyrolysis uses heat to break down combustible materials in the absence of oxygen, producing a mixture of combustible gases (primarily methane, complex hydrocarbons, hydrogen, and carbon monoxide), liquids and solid residues. The products of the pyrolysis process are: **(i)** a gas mixture; **(ii)** a liquid (bio-oil/tar); **(iii)** a solid residue (carbon black). The gas generated by either of these processes can be used in boilers to provide heat, or it can be cleaned up and used in combustion turbine generators. The purpose of pyrolysis of waste is to minimize emissions and to maximize the gain.

BIO DIGESTORS

The bio-decomposer namely Pusa Decomposer developed by The Indian Council of Agricultural Research (ICAR) has been used by the States of Punjab, Haryana, Uttar Pradesh and NCT of Delhi to total of 978,713 acres (3,91,485 ha) equivalent to about 2.4 million tonnes of straw management in this year.

Human waste disposal in innocuous form is an ever-growing problem leading to aesthetic nuisance, threat of organic pollution & several infectious diseases in epidemic proportions due to contamination of ground water and drinking water resources in highly populated and developing countries, like India.

Less than 30% of Indians have access to the toilets. In rural areas about 10% of houses have toilets and the rest of the people go to open defecation. The population in the cities although have better access to the toilets but only to the tune of ~70%.

Untreated waste is responsible for several diseases like, dysentery, diarrhoea, amoebiasis, viral hepatitis, cholera, typhoid etc. taking the life of lakhs of children annually. Bio decomposers degrade and convert the human waste into usable water and gasses in an eco-friendly manner.

The generated gas can be utilized for energy/ cooking and water for irrigation purposes. The process involves the bacteria which feed upon the faecal matter inside the tank, through anaerobic process which finally degrades the matter and releases methane gas that can be used for cooking, along with the treated water.

56. GLOBAL SOLAR FACILITY

- Global Solar Facility (GSF), a payment guarantee fund formed by International Solar Alliance to stimulate investments into solar power projects**.**
- GSF is needed for both 'quality' (deepening the reach to low-income countries) and 'quantity' (multiplying the global investments) of solar investments.
- **Aim**: To help alleviate the constraints of contractual and financial uncertainty for solar energy.
- GSF, with a fund of $100 million, aims to enable $ 10 billion in investment, providing clean energy access for 35-40 million African households by 2030.

- The Solar Facility would be operationalised to **crowd in investments** from various donors across the globe.
- It will primarily focus on **decentralised solar solutions, rooftop solar, and productive use solar**.
- Through this financing vehicle, which includes payment guarantees, insurance and investment funds, the ISA aims to mitigate project risks, provide technical assistance to address regulatory gaps, reduce currency risks, resolve contractual and **financial** uncertainties in the solar energy sector.
- In 2022, the ISA Assembly approved the creation of a Global Solar Facility (GSF) in Africa, which was **announced** at **COP27 in Sharm-el-Sheikh**.
- After **Africa**, the GSF aims to expand to **regions such as Asia, Latin America and the Middle East,** where the Regional Facilities will be tailored to meet specific requirements.
- The GSF will provide:
  (1) Payment guarantee fund- guarantee mechanisms to ensure payment guarantees.
  (2) Insurance fund- provide mechanisms to mitigate project risks; and
  (3) investment fund- allow Technical Assistance for addressing gaps in the regulatory framework.

Objectives

Provide clean and reliable source of energy to all people across the globe who today do not have access to electricity. Thereby ensuring Green Growth for the world	Enable deeper innovation and capacity building for solar power development	Open up solar markets across the world	Accelerate transition to net zero, by helping catalyse part of the USD 12.5 trillion needed for solar installations globally

57. WORLD COAL ASSOCIATION (FUTURE COAL)

The world coal association has rebranded itself as FutureCoal-The Global Alliance for Sustainable coal and has sought technological intervention for sustainable usage of coal reserves for reliable and affordable supply of energy to a vast majority of the world's population

ABOUT FUTURECOAL

- **It is a nonprofit organisation, headquartered in London.**
- It is a global lobby group for "major international coal producers and stakeholders" with membership open" to companies and not-for-profit organizations with a stake in the future of coal from anywhere in the world**.**
- FutureCoal is the Global Alliance for Sustainable Coal and **the world's only coal multi-lateral and neutral representative organisation.**

58. CLIMATE VULNERABLE FORUM

Former Maldives president Mohamed Nasheed has said that Indian companies must invest in clean energy projects in 'climate vulnerable' countries.

WHAT IS CLIMATE VULNERABLE FORUM?

- The Climate Vulnerable Forum (CVF) is an international partnership of countries highly vulnerable to a warming planet.
- It is composed of 58 members from Africa, Asia, the Caribbean, Latin America and the Pacific.
- It serves as a South-South cooperation platform for participating governments to act together to deal with global climate change.
- The Forum is led by a rotating chair for an ordinary period of two years.

ESTABLISHMENT

- CVF first met near Male', Maldives in November 2009.
- At the CVF High-Level Meeting held during the UNFCCC (COP21), the Manila-Paris Declaration and the 2016-2018 Roadmap of the CVF was adopted.
- At UNFCCC COP22 Forum members committed to ambitious climate action, striving, among others, to achieve maximal resilience and to meet 100% domestic renewable energy production as rapidly as possible.

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4 CHAPTER ENVIRONMENTAL LEGISLATIONS

1. BIOLOGICAL DIVERSITY (AMENDMENT) ACT, 2023

The preamble of the Biological Diversity Act, 2002 has been amended to replace the term "party" with "Party," signifying India's role as a Party to the Nagoya Protocol. This reflects India's commitment to international agreements aimed at biodiversity conservation and sustainable use.

Definitions Amendments: Clarifying Key Concepts

The amendment also introduces several changes to the definitions within the Act, providing clarity and specificity to various terms crucial for the effective implementation of the legislation.

Key amendments include expanding the scope of terms like access, benefit claimers, biological resources, codified traditional knowledge, and derivative, among others.

These changes are aimed at providing a comprehensive framework for regulating access to biological resources and associated traditional knowledge.

Access to Biological Resources Amendment: Regulating Foreign-Controlled Entities

One of the substantial amendments introduced by the Bill is the addition of a new sub-clause governing entities formed or registered in India that are controlled by foreigners. This amendment aims to regulate and monitor access by foreign-controlled businesses to prevent any illicit exploitation of India's biological resources.

Sharing of Research Results: **Regulation and Clearance**

The Bill imposes restrictions on sharing or transferring research findings on biological resources derived from India or traditional knowledge connected to them. It mandates prior written clearance from the National Biodiversity Authority for such sharing, except for certain exemptions outlined in the legislation.

Scope of Approval from NBA: Simplifying Patent Acquisition Process

An important change introduced by the Bill is the expansion of the scope of approval required from the National Biodiversity Authority for obtaining patents. Indian residents and companies not controlled by foreign entities are now required to seek approval only before the grant of a patent, simplifying the process and reducing bureaucratic hurdles.

Benefit Sharing and Exemptions: Ensuring Fair and Equitable Distribution

The Bill mandates fair and equitable benefit-sharing arrangements to ensure that benefits arising from commercialization are shared fairly with appropriate stakeholders. It introduces exemptions for certain categories, such as AYUSH practitioners and users of codified traditional knowledge, aimed at promoting traditional medicine and the cultivation of medicinal plants.

National Authority and Local Community Involvement: Balancing Representation

The Bill introduces provisions for the National Authority to represent and negotiate on behalf of local communities in benefit-sharing arrangements. However, concerns arise regarding the reduced direct involvement of local communities in decision-making processes, raising questions about the adequacy of their representation.

State Biodiversity Board: Empowering Local Governance

Amendments to the composition and functions of State Biodiversity Boards aim to enhance their capacity to make informed decisions related to benefit-sharing and biodiversity conservation at the state level. The inclusion of legal experts, scientists, and conservationists ensures a diverse and knowledgeable composition of the boards.

Biodiversity Management Committee: Strengthening Conservation Efforts

The Bill mandates the establishment of Biodiversity Management Committees at both rural and urban levels to promote habitat preservation and document conservation efforts. This ensures a grassroots approach to biodiversity conservation and management.

Decriminalisation of Offences: Shifting Regulatory Approach

A significant departure from the previous legislation is the complete decriminalization of offenses related to biological resources. Penalties have been revised, with a focus on imposing fines rather than criminal sanctions, aiming to deter violations while avoiding disproportionate punitive measures.

2. WILDLIFE PROTECTION AMENDMENT ACT 2022

POSITIVE KEY FEATURES

- **Holistic approach and widened scope** - In the Wild Life (Protection) Act, 1972, the "protection of wild animals, birds and plants", shall be substituted with the words "conservation, protection and management of wild life" which points towards the change of focus towards the holistic approach.
- **Rationalisation of schedules:** Number of schedules have been reduced from six to four with two schedules for protected animals. Further it has inserted a new schedule (VB) for specimens listed in the Appendices under CITES (scheduled specimens).
- **Invasive alien species** - It empowers the central government to regulate or prohibit the import, trade, possession or proliferation of invasive alien species. The central government may authorise an officer to seize and dispose of the invasive species.
- **Management Authority :** it will be appointed by the central government for the issuance of permits and certificates for trade in scheduled specimens in accordance with the CITES. Until now, CITES provisions were enforced through the Customs Act and not the Wildlife (Protection) Act, which was a major lacunae.
- **Decentralised management:** For sanctuaries falling under special areas, the management plan must be prepared after due consultation with the concerned Gram Sabha.
- **Declaring conservation reserves by Centre:** It empowers both Central and State governments to declare areas adjacent to national parks and sanctuaries as conservation reserve, Earlier only the state government had this power.
- **Surrender of captive animals:** Any person having a certificate of ownership can voluntarily surrender any captive animals or animal products to the Chief Wildlife Warden.

ISSUES REMAINING

- **Commercial trade:** It leaves the door open for commercial trade in elephants. There is concern that the amendments have not defined what purposes elephants can be used for and have perhaps made it easier for the animals to be transported across the country.
- **Dilution of federal framework:** It has also been criticized for diluting the federal framework by seeking to displace the State Boards of Wildlife, which are chaired by the Chief Minister, with a Standing Committee headed by the union Forest minister.
- **Using animals for religious purpose:** In Section 43, elephants, Schedule I animals, are permitted to be used for 'religious or any other purpose'. Experts have also criticized the term 'other purpose'.
- **Lack of ownership:** There are 2,675 captive elephants in India and only 1,251 have ownership certificates. Experts believe that this unclear ownership was an issue and the provisions that are "weak and indeterminate which can cause problems.

3. FOREST CONSERVATION AMENDMENT ACT 2023

The Act amends the Forest (Conservation) Act, 1980 to make it applicable to certain types of land. These include land notified as a forest under the Indian Forest Act, 1927 or in government records after the 1980 Act came into effect. The Act will not be applicable for land converted to non-forest use before December 12, 1996.

KEY FEATURES

- **Exemptions:** It exempts certain types of land, for e.g. land within 100 km of India's border needed for national security projects, small roadside amenities, and public roads leading to a habitation.
- **Approval from Centre:** The state government requires prior approval of the central government to assign any forest land to a private entity. The Bill extends this to all entities, and allows the assignment to be made on terms and conditions specified by the central government.
- **Activities in Forests:** The Act specifies some activities that can be carried out in forests, such as establishing check posts, fencing, and bridges. The Bill also allows running zoos, safaris and eco-tourism facilities.
- **Power to issue directions:** Central government may issue directions for the implementation of the Act to any authority/organisation under or recognised by the centre, state, or union territory (UT).

KEY ISSUES AND ANALYSIS

- The Act excludes two categories of land from the purview of the Act: land recorded as forest before October 25, 1980 but not notified as a forest, and land which changed from forest-use to non-forest-use before December 12, 1996. This provision may go against a 1996 Supreme Court judgement on preventing deforestation.
- Exempting land near border areas for national security projects may adversely impact the forest cover and wildlife in north-eastern states.
- A blanket exemption for projects like zoos may adversely affect forest land and wildlife. The Supreme Court (2023) has remarked that they do not appreciate the necessity of having a zoo inside tiger reserves or national parks.
- Activities like silvicultural operations, safaris, and eco-tourism facilities may help in economic development, and may even contribute to national priorities such as energy security and industrial growth. However, there may be a need to balance economic benefits of such activities with that of conserving forests. It is not clear why the requirement of a case-by-case examination by the central government to determine such balance is being replaced by a blanket exemption.

FOREST CONSERVATION RULES 2022

MOEFCC recently notified the Forest Conservation Rules, 2022, superseding the earlier Forest Conservation Rules, 2003.

These rules govern the process for diversion of forest land for non-forest purposes under the Forest (Conservation) Act, 1980, outlining the approval procedures and roles of various stakeholders.

The rules are based on the principle of polluter pays and requires forest diversion to be compensated by compensatory afforestation. The rules are to operationalise the provision of the Act.

SALIENT FEATURES

- **Project Screening Committee:** The rules propose the constitution of a project screening committee in each State/UT for an initial review of proposals involving diversion of forest land. The 5-member committee will meet at least twice every month and will advise state governments on projects in a time-bound manner. Dilution of Forest Rights: Onus of ensuring forest rights of forest dwellers are rehabilitated is with state governments. Earlier, compliance with FRA was mandatory before the Centre granted stage II forest clearance to any project.
- **Dilution of Gram Sabha powers:** The procedural requirement for consent of Gram Sabha has been done away with. Forest Conservation Rules, 2016 had embedded the need for Gram Sabha consent as a procedural requirement for prior approval by the environment ministry.

- **Undertaking compensatory afforestation in other States/UTs:** Under the rules, compensatory afforestation can be undertaken in another State/UT (having forest cover less than 20% of its geographical area) in case forest land is diverted is in a hilly or mountainous State/UT (having forest cover of more than 2/3rd of its area) or in other State/UT (having forest cover of more than 1/3rd of its geographical area).
- **Accredited compensatory afforestation:** This mechanism allows people to raise vegetation on their land and sell it to persons who need to meet compensatory afforestation targets under the Act. This land must at least be of 10 hectares and comprise trees that are at least 5 years old and have a density of 0.4 or more. Earlier, a project developer had to provide land which is not notified as forest against forest diversion and had to bear the cost of raising compensatory afforestation over the same piece of land.
- **Centre will formulate an accredited compensatory afforestation mechanism** for people interested in using their land for the scheme.
- **The total amount of land** that needs to be compensated can be reduced if the plantations or accredited compensatory afforestation are made available in a wildlife corridor or contiguous to a forest.
- **Land Bank:** States/UTs may for the purpose of compensatory afforestation can create a land bank under control of the forest department. The minimum size of the land bank is 25 hectares. In case the land bank should be in continuity of a land declared or notified as forest, protected area, tiger reserve or within a designated or identified tiger or wildlife corridor, there shall be no restriction on size of the land, there shall be no restriction on size of the land. The lands covered under accredited compensatory afforestation may be included in the land bank.

4. SOLID WASTE MANAGEMENT 2016 RULES

- **Mandatory Segregation:** All waste generators will have to segregate and store the waste generated by them under three separate categories - bio-degradable, non-bio-degradable and domestic hazardous waste - in suitable bins before handing it over to authorised rag pickers or waste collectors.
- **Concept of Extended Producer Responsibility:** Local bodies can charge a fee from the generator of wastes. The new rules have asked all such brand owners who sell products in non-biodegradable packaging material to put in place a system to collect back the packaging waste generated due to their production (ET).
- **Burning of Solid Waste has been prohibited.**
- **Social Dimension has been adequately considered.** Rag pickers are to be integrated in the formal system.
- **Increasing Coverage:** The new rules will now apply much beyond the municipal areas, extending to urban agglomerations, census towns, notified industrial townships, areas under the control of Indian Railways, airports, airbase, port and harbour, defence establishments, special economic zones, State and Central State and Central government organizations, places of pilgrims, religious & historical importance. Event organizers, and new townships and group housing societies have been brought under the system.
- **Waste-processing facilities** to be set up by all local bodies having a population of 1 million or more.

5. BIO-MEDICAL WASTE MANAGEMENT RULES, 2016

- Bio-medical waste has been classified into 4 categories instead of 10 to improve the segregation of waste at source and these 4 categories have colour-codes.
  - **Red Bin** for plastic waste such as bottles, syringes, etc.
  - **Yellow Bin** for infectious wastes such as cotton, bandage, placenta, etc.
  - **Blue Bin** for glass bottles like discarded medicines
  - **Black Bin** for needles without syringes, metal articles, etc.
- Phase-out the use of chlorinated plastic bags, gloves and blood bags within two years.
- The ambit of the rules has been expanded to include vaccination camps, blood donation camps, surgical camps or any other healthcare activity.
- Pre-treatment of the laboratory waste, microbiological waste, blood samples and blood bags through disinfection or sterilization on-site in the manner as prescribed by WHO or NACO.

- State Government to provide land for setting up a common bio-medical waste treatment and disposal facility.
- No occupier shall establish on-site treatment and disposal facility, if a service of `common bio-medical waste treatment facility is available at a distance of seventy-five kilometers.
- The new rules prescribe more stringent standards for incinerators to reduce the emission of pollutants in the environment.
- Inclusion of emissions limits for Dioxin and furans.
- Establish a Bar-Code System for bags or containers containing bio-medical waste for disposal.
- Provide training to all its health care workers and immunize all health workers regularly.

6. E-WASTE (MANAGEMENT) RULES, 2022

These rules will apply to every manufacturer, producer, refurbisher, dismantler and recycler involved in manufacture, sale, transfer, purchase, refurbishing, dismantling, recycling and processing of e-waste or electrical or electronic equipment (EEE) listed in Schedule I, including their components, consumables, parts and spares.

CATEGORIES OF EEE COVERED

- Information technology & telecommunication equipment
- Consumer electrical & electronics & photovoltaic panels
- Large & small electrical & electronic equipment
- Electrical & electronic tools
- Toys, leisure & sports equipment
- Medical devices (with the exception of all implanted & infected products)
- Laboratory instruments

THESE RULES DO NOT APPLY TO

(a) Waste batteries as covered under Battery Waste Management Rules, 2022

(b) Packaging plastics as covered under Plastic Waste Management Rules, 2016

(c) Micro enterprises as defined in MSME Development Act, 2006

(d) Radioactive wastes as covered under Atomic Energy Act, 1962

EXTENDED PRODUCER RESPONSIBILITY FRAMEWORK

- Entities shall register on the portal in any of the following (a) Manufacturer (b) Producer (c) Refurbisher (d) Recycler. If an entity falls in more than one category, then the entity shall register under those categories separately.
- No entity shall carry out any business without registration and shall not deal with any unregistered manufacturer, producer, recycler & refurbisher.
- **Responsibilities of manufacturer:** Register on portal, collect e-waste generated during manufacture of electrical or electronic equipment and ensure its recycling or disposal, file annual and quarterly returns
- **Responsibilities of producers:** Register on portal, obtaining & implementing extended producer responsibility targets, creating awareness, file annual and quarterly returns.
- **Responsibilities of refurbisher:** Register on portal, collect e-waste generated during the process of refurbishing and hand over the waste to registered recycler and upload information on portal, ensure that refurbished equipment shall be as per Compulsory Registration Scheme on MEITY and Standards of BIS, file annual and quarterly returns.
- **Responsibilities of bulk consumer:** Bulk consumers of electrical and electronic equipment shall ensure that e-waste generated by them shall be handed over only to the registered, producer, refurbisher or recycler. However, they do not need to register.

REDUCTION IN USE OF HAZARDOUS SUBSTANCES IN MANUFACTURE OF EEE

- Every producer of EEE and their components etc. shall ensure that new EEE and its components etc. do not contain Lead, Mercury, Cadmium, Hexavalent Chromium, Polybrominated Biphenyls or Polybrominated Diphenyl Ethers beyond a maximum concentration value of:
- 0.1% by weight for lead, mercury, hexavalent chromium, polybrominated biphenyls and polybrominated diphenyl ethers
- 0.01% by weight for cadmium.
- Every producer provides detailed information on constituents of equipment and their components along with a declaration of conformance to reduction of hazardous substances provisions in the product user documentation.
- Imports or placement in the market for new EEE shall be permitted only for those products which comply with maximum concentration of hazardous substances.
- Not required for EEE for defence and other strategic applications.

EPR TARGETS

2023-25	60% of quantity of an EEE placed in market in year Y-X, where X is the average life of product
2025-27	70% of quantity of an EEE placed in market in year Y-X, where X is the average life of product
2027-28 onwards	80% of quantity of an EEE placed in market in year Y-X, where X is the average life of product

- E-Waste recycling targets shall be reviewed and may be increased after FY 2028-29.
- Importers of used EEE shall have 100% EPR obligation for imported material after end of life, if not re-exported.
- E-Waste recycling targets shall not be applicable for waste generated from solar PV modules or panels or cells.

PROCEDURE FOR STORAGE OF E-WASTE

Every manufacturer, producer, refurbishes & recycler may

- Store e-waste for a period not exceeding 180 days.
- Maintain a record of sale, transfer and storage of e-waste.

MANAGEMENT OF SOLAR PHOTOVOLTAIC MODULES OR PANELS OR CELLS

Every manufacturer and producer of solar photovoltaic modules or panels or cells shall:

(i) Ensure registration on portal

(ii) Store Solar PV modules or panels or cells waste generated up to year 2034-35 as per guidelines of CPCB.

(iii) File annual returns

(v) Ensure that inventory of solar photo-voltaic modules or panels or cells shall be put in place distinctly or portal.

(vi) Comply with standard operating procedure.

MODALITIES OF EXTENDED PRODUCER RESPONSIBILITY REGIME

- All producers shall fulfil their extended producer responsibility.
- Extended producer responsibility shall lie entirely on producer only.
- EPR for each product shall be decided based on information provided by producers.

EXTENDED PRODUCER RESPONSIBILITY CERTIFICATE GENERATION FOR RECYCLING

- CPCB shall generate EPR certificate through the portal in favour of a registered recycler in the format laid down in this regard:
- Quantity eligible for generation of EPR certificate shall be calculated by following formula:
- QEPR= QpXCF (QEPR is the quantity eligible for generation of certificate, Qp is the quantity of the end product, Cf is the conversion factor (quantity of inputs required for production of one unit of output)

- Conversion factor (Cf) for each product shall be determined by CPCB with approval from the steering committee.
- Validity of EPR certificate shall be two years from the end of financial year in which it was generated
- Each EPR certificate shall have a unique number containing year of generation, code of end product, recycler code and unique code and EPR certificates shall be in denomination of 100, 200, 500 & 1000 kg or such other denomination as laid down by CPCB.

FOR REFURBISHING

- E-Waste shall also be allowed for refurbishing & refurbishers shall have to get registered on the portal and based on data provided, refurbishing certificate shall be generated in favour of a registered refurbisher.
- On production of refurbishing certificates purchased from registered refurbishers, the EPR of producers shall be deferred by the duration as laid down by CPCB.

TRANSACTION OF EPR CERTIFICATES

- A producer may purchase EPR certificates limited to its EPR liability of current year plus any leftover liability of preceding years plus 5% of current year liability.
- As soon as a producer purchases an EPR certificate, it shall be automatically adjusted against its liability and priority in adjustment shall be given to earlier liability. EPR certificate so adjusted shall be automatically extinguished and cancelled.

7. HAZARDOUS AND OTHER WASTES (MANAGEMENT AND TRANS-BOUNDARY MOVEMENT) AMENDMENT RULES, 2019

- Solid plastic waste has been prohibited from import into the country including in Special Economic Zones (SEZ) and by Export Oriented Units (EOU).
- Exporters of silk waste have now been given exemption from requiring permission from the MOEFCC.
- Electrical and electronic assemblies and components manufactured in and exported from India, if found defective can now be imported back into the country, within a year of export, without obtaining permission from the MOEFCC.
- Industries which do not require consent under Water Act 1974 and Air Act 1981, are now exempted from requiring authorization also under the Hazardous and Other Wastes (Management & Trans-boundary Movement) Rules, 2016 provided that hazardous and other wastes generated by such industries are handed over to the authorized actual users, waste collectors or disposal facilities.

8. PLASTIC WASTE MANAGEMENT RULES AMENDMENT, 2021

- MOEFCC notified the Plastic Waste Management Amendment Rules, 2021, prohibiting identified single-use plastic items by 2022.
- Pollution due to single use plastic items has become an important environmental challenge confronting all countries.
- In the 4th United Nations Environment Assembly held in 2019, India had piloted a resolution on addressing single-use plastic products pollution, recognizing the urgent need for the global community to focus on this very important issue.

SALIENT FEATURES OF AMENDMENT TO PLASTIC WASTE RULES

- **Prohibition:** The manufacture, import, stocking, distribution, sale and use of following single-use plastic, including polystyrene and expanded polystyrene, commodities shall be prohibited with effect from the 1st of July 2022.
- **Compostable plastic:** The provisions will not apply to commodities made of compostable plastic.

- **Timeline:** The government has given industry ten years from the date of notification to comply with any future bans on plastic commodities other than those included in this notification.
- **Thickness of plastic bags:** The allowable thickness of plastic bags will be increased from 50 microns to 75 microns on September 30, 2021, and to 120 microns on December 31, 2022.
- **Monitoring agency:** The Central Pollution Control Board, along with state pollution bodies, will monitor the ban, identify violations, and impose penalties already prescribed under the Environmental Protection Act, 1986.
- **Extended Producer Responsibility (EPR):** According to the Plastic Waste Management Rules, 2016, the plastic packaging waste that is not covered under the phase out of identified single-use plastic items must be collected and managed in an environmentally sustainable manner through the Extended Producer Responsibility (EPR) of the Producer, Importer, and Brand Owner (PIBO).

9. DISASTER MANAGEMENT AMENDMENT BILL 2024

The Disaster Management (Amendment) Bill 2024 to amend the **Disaster Management Act of 2005** has been passed in both the houses of Parliament.

SALIENT FEATURES OF DISASTER MANAGEMENT (AMENDMENT) BILL

- Inclusion of new definitions of disaster risk, evacuation, exposure, recovery, rehabilitation, resilience etc. in line with the Sendai principles on Disaster Risk Reduction.
- Clarification on man-made disasters: The bill proposes to clarify that man-made disasters does not include any law-and-order issues. Thus, loss of lives, displacement, or property damage due to riots, violence etc. would not invoke the DM Act.
- Bring more clarity and convergence in the roles of Authorities and Committees working towards disaster management.
- Provide statutory status to certain pre-DM Act organisations like National Crisis Management Committee (NCMC) and High Level Committee. NCMC is a nodal body to deal with major disasters & High Level Committee is the nodal body for the approval of financial assistance for recovery and rehabilitation from disasters.
- Strengthen and efficient working of National Disaster Management Authority (NDMA) and State Disaster Management Authorities (SDMA).
- Empower the NDMA and the SDMAs to prepare the disaster plans at national level and state level instead of the National Executive Committee (NEC) and State Executive Committee (SEC), presently.
- Provide for creation of a disaster database at national and state level. The database will collect information on assessment of disaster, fund allocation, expenditure, preparation and mitigation plans.
- Provision for constitution of 'Urban Disaster Management Authority' (UDMA) for State Capital and large cities having municipal corporations. Currently, the DM Act only provides for District level District Disaster Management Authority.
- Constitution of 'State Disaster Response Force' by State Governments. Currently, many states have raised disaster response forces on the lines of NDRF. However, the state level disaster response forces have not been mandated under the DM Act, which this amendment aims to provide for.

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5 CHAPTER PROTECTED AREAS

1. WETLANDS

- Transition zones between terrestrial and aquatic ecosystems. Ex. Mangroves, Lake littorals (marginal areas between highest & lowest level of the lakes), floodplains (areas lying adjacent to the river channels beyond the natural levees and periodically flooded during high discharge in the river) & other marshy or swampy areas.
- These habitats experience periodic flooding from adjacent deep-water habitats and support biodiversity specifically adapted to such shallow flooding or water logging.
- Waterlogged soil adapted plant life (hydrophytes) and hydric soils (not enough O_2) are the chief characteristics of wetlands.

IMPORTANCE OF WETLANDS

- Act as "nature's kidneys" by removing polluting nutrients & sediments from surface & groundwater.
- Replenish groundwater, their filtering capacity helps to protect groundwater quality. Helps in nutrient recycling and stabilization of local climate.
- Buffer shorelines against erosion & pollutants.
- Provides habitat for wildlife & fisheries, including threatened habitats. Provides recreational opportunities for birds, amphibians and mammals.
- Acts like a sponge & helps in flood control by soaking extra water from the surroundings.
- Make surroundings fertile for the growth of crops. Provides many opportunities for fishing & other commercial activities.
- Acts as a carbon sink, soils around wetlands can store carbon for many years (climate change mitigation).
- Provides opportunities for tourism and research.

2. RAMSAR CONVENTION

- **Aim:** Conservation & wise use of all wetlands through local, national & international action, as a contribution towards achieving sustainable development throughout the world. Came into force in 1975.
- Only global environment treaties dealing with a particular ecosystem and wetlands.
- **Definition of Wetlands:** Wetlands include lakes, rivers, swamps, marshes, wet grasslands, peatlands, oases, estuaries, deltas, tidal flats, near-shore marine areas, mangroves, coral reefs, human-made sites such as fishponds, rice paddies, reservoirs and salt pans.
- **Wise use is the maintenance of ecological character within the context of sustainable development.**
- The UK has the highest number of wetlands recognized by the Ramsar Convention. Bolivia has the greatest area of listed wetlands.
- Three pillars of the Convention are:
  - Work towards the wise use of all their wetlands.
  - Designate suitable wetlands for the list of Wetlands of International Importance (the "Ramsar List") and ensure their effective management.
  - Cooperate internationally on transboundary wetlands, shared wetland systems & shared species.

CRITERIA FOR STATUS OF WETLAND OF INTERNATIONAL IMPORTANCE

- Sites containing representative, rare or unique wetland types.
- Sites of international importance for conserving biological diversity

- Criteria based on species and ecological communities
- Specific criteria based on waterbirds.
- Specific criteria based on fish.
- If a site supports 1% of the population of a species.

MONTREUX RECORD

- A register of wetland sites on the List of Wetlands of international importance where changes in ecological character have occurred, are occurring, or are likely to occur because of technological developments, pollution or other human interference and therefore in need of priority conservation attention.
- It is maintained as part of the Ramsar List.

TWO MONTREUX RECORD SITES IN INDIA

- **Loktak Lake (Manipur):** Due to deforestation in the catchment area, infestation of water hyacinth & pollution. Construction of a hydroelectric power plant has caused local extinction of several fish species.
- **Keoladeo National Park:** Water shortage and unbalanced grazing regime around it. The invasive growth of the grass and reducing its suitability for certain water-bird species, notably the Siberian Crane.

3. WETLANDS (CONSERVATION & MANAGEMENT) RULES, 2017

MoEF&CC has notified Wetlands (Conservation and Management) Rules, 2017 (Wetlands Rules) under the Environment (Protection) Act, 1986 as framework for conservation & management of wetlands in India.

KEY PROVISIONS

- **Decentralisation:** The management of wetlands has been decentralized. The powers have been given to the State governments so that protection and conservation work can be done at the local level. The central government has mainly retained powers regarding monitoring.
- **State or UT wetland authorities:** States and UTs have been given the responsibility for wetland management by setting up State and UT Wetland Authorities (SWAs).
  - **SWA's** will be headed by the environment minister and include other government officials as well as experts from the fields of wetland ecology, hydrology, fisheries, landscape planning and socioeconomics.

FUNCTIONS OF SWA'S

- SWAs have to identify and notify the wetlands for protection within a stipulated time.
- Develop a comprehensive list of activities to be regulated and permitted within notified wetlands and their zone of influence.
- Recommend additional prohibited activities for specific wetlands, define strategies for conservation and wise use of wetlands, and undertake measures for enhancing awareness within stakeholders and local communities on functions of wetlands.
- State authorities will need to prepare a list of all wetlands of the State or UT within three months, a list of wetlands to be notified within six months, a comprehensive digital inventory of all wetlands within one year which will be updated every ten years.

GUIDELINES FOR IMPLEMENTING WETLANDS (CONSERVATION AND MANAGEMENT) RULES, 2017

The guidelines clarified that all wetlands, irrespective of their location, size, ownership, biodiversity, or ecosystem services values, can be notified under the Wetlands Rules 2017, except:

- river channels
- paddy fields
- human-made water bodies specifically constructed for drinking water.

- aquaculture
- salt production
- recreation
- irrigation purposes
- Wetlands falling within areas covered under the Indian Forest Act, 1927, Forest (Conservation) Act, 1980, Wildlife (Protection) Act, 1972 and the Coastal Regulation Zone Notification, 2011.
- Protected Areas and areas falling within the purview of Coastal Zone Regulation

NATIONAL WETLAND COMMITTEE

- NWC will be headed by the MoEFCC Secretary, to monitor the implementation of these rules.
- NWC has a merely advisory role. These include:
  - Advising central government on proposals received from states/UTs for the omission of prohibited activities.
  - Prescribing guidelines for integrated management of wetlands based on wise-use principles.
  - Recommending transboundary wetlands for notification.
  - Reviewing the progress of integrated management of Ramsar Convention sites.

PROHIBITED ACTIVITIES UNDER THE NEW RULES

- Conversion of wetland for non-wetland uses including encroachment of any kind,
- Setting up of any industry and expansion of existing industries.
- Manufacture or handling or storage or disposal of hazardous substances and construction and demolition waste.
- Solid waste dumping.
- Discharge of untreated wastes and effluents from industries, cities, towns, villages and other human settlements.

The Rules also restrict any kind of encroachment, poaching, or permanent construction, except for boat jetties within 50 metres of the mean high flood level observed in the past 10 years.

4. RAMSAR WETLANDS SITES IN INDIA

- World Wetlands Day is observed on 2nd February every year all over the world to commemorate the signing of the Ramsar Convention on Wetlands of International Importance in 1971.
- India has been a party to the Convention since 1982.
- India has the largest network of Ramsar Sites in Asia.
- India has increased its tally of Ramsar sites to 89, by designating four more wetlands as Ramsar sites. Jharkhand and Sikkim received their first Ramsar designations.

	RAMSAR SITE	DESCRIPTION
1	**Chandratal Lake** Himachal Pradesh Natural Freshwater	• Situated in the Spiti area of Lahaul & Spiti at Samudra Tapu Plateau. Source of Chandra River (One of the headwaters of Chenab River, Chenab is the confluence of Chandra and Bhaga rivers). • High altitude lake on the upper Chandra valley near the Kunzam Pass joining Himalayan and Pir Panjal ranges. • Supports Snow Leopard, Red Fox, Himalayan Ibex, Blue Sheep/Bharal.
2	**Pong Dam Lake** Himachal Pradesh Freshwater Manmade reservoir	• A water storage reservoir created in 1975 on the Beas River in the low foothills of the Himalaya on the northern edge of the Indo-Gangetic plain. • It is located at the trans-Himalayan flyway.

3	**Renuka Wetland** Himachal Pradesh Natural Freshwater	• It is a wetland with springs and inland subterranean karst formations, fed by a small stream flowing from the lower Himalayan to the Giri river. • The lake has high religious significance and is named after the mother of Hindu sage Parshuram, and is thus visited by thousands of pilgrims and tourists.
4	**Chilika Lake** Odisha Natural Lagoon, Brackish Water	• Though added to Montreaux Record in 1993, was removed in 2002. • A brackish lake separated from the Bay of Bengal by a long sandy ridge. • It has seasonal fluctuations in salinity in different sections of the lake and saline areas support aquatic algae. • One of the only two lagoons with a population of Irrawaddy dolphins
5	**Bhitarkanika Wetlands** Odisha Natural mangrove swamps	• One of the finest remaining patches of mangrove forests along the Indian coast. The site's Gahirmatha beach is said to host the largest known Olive Ridley sea turtle nesting beach in the world, with half a million nesting annually, and the site has the highest density of saltwater crocodiles in the country. • Saltwater crocodiles, Gahirmatha beach (largest known Olive Ridley sea turtle nesting in the world)
6	**Deepor Beel** Assam Natural Freshwater	• A permanent freshwater lake in a former channel of the Brahmaputra River. • It is only major stormwater storage basin for the city of Guwahati. It is a staging site on migratory flyways. • Some threatened birds include pot billed Pelican Lesser and Greater Adjutant Stork and Baer's Pochard
7	**East Calcutta Wetlands** West Bengal	• World-renowned as a model of a multiple-use wetland. • The wetland forms an urban facility for treating the city's wastewater & utilizing the treated water for pisciculture and agriculture • The system is described as one of the rare examples of environmental protection and development management where a complex ecological process has been adopted by the local farmers for mastering the resource recovery activities.
8	**Sundarbans Wetland** West Bengal Natural	• Located within the largest mangrove forest in the world. It encompasses hundreds of islands and a maze of rivers, rivulets, and creeks, in the delta of Rivers Ganges and Brahmaputra on the Bay of Bengal in India & Bangladesh. • Sundarbans Tiger Reserve is situated within the Site and part of it has been declared a "critical tiger habitat" under national law and also a "Tiger Conservation Landscape" of global importance. • The Sundarbans are the only mangrove habitat which supports a significant population of tigers, and they have unique aquatic hunting skills. • The Site is also home to a large number of rare and globally threatened species such as the critically endangered northern river terrapin (*Batagurbaska*), the endangered Irrawaddy dolphin (*Orcaellabrevirostris*), and the vulnerable fishing cat (*Prionailurusviverrinus*).

		• It is listed as World Heritage Site and also in UNESCO Biosphere Reserve
9	**Harike Lake** Punjab Manmade Freshwater	• Harike Lake is a shallow water reservoir with thirteen islands, at the confluence of two rivers, i.e., Beas and Sutlej. • Indira Gandhi Canal starts from this place.
10	**Kanjli** Punjab Manmade reservoir, Freshwater	• It is a man-made wetland, with a permanent stream, the Kali Bein, a tributary of Beas River converted by construction of a small barrage in 1870 into a water storage area for irrigation purposes. • The stream is considered to be the most significant in the state from the religious point of view, as it is associated with the first guru of the Sikhs, Shri Guru Nanak Dev Ji
11	**Ropar Lake** Punjab Manmade Freshwater	• A wetland of lake and river formed by the 1952 construction of a barrage for diversion of water from the Sutlej River for drinking and irrigation supplies. • The site is an important breeding place for the nationally protected Smooth Indian Otter, Hog Deer, Sambar, and several reptiles, and the endangered Indian Pangolin is thought to be present.
12	**Keoladeo National Park** Rajasthan Manmade Freshwater Swamps	• It has been in MONTREAUX RECORD since 1990 due to "water shortage and an unbalanced grazing regime". • Included as a World Heritage Site. A complex of ten artificial, seasonal lagoons, varying in size, situated in a densely populated region. • The invasive growth of the grass Paspalum distichum has changed the ecological character of large areas of the site. • The Siberian Crane is found here. • It is only protected by a surrounding wall on all sides.
13	**Sambhar Lake** Rajasthan Natural Saline	• A large lake fed by four streams set in a shallow wetland and subject to seasonal fluctuations. (Inland drainage) • The site is important for a variety of wintering waterbirds, including the second largest breeding ground for flamingos in India.
14	**Kolleru Lake** Andhra Pradesh Natural Freshwater	• A eutrophic lake, situated between basins of Godavari and Krishna rivers. It is fed by two seasonal rivers and several drains and channels, which function as a natural flood-balancing reservoir between the deltas of the two rivers. • Known for its spot-billed pelican sighting.
15	**Loktak Lake** Manipur Natural Freshwater	• This lake was added to MONTREUX RECORD in 1993 as a result of ecological problems such as deforestation in the catchment area, an infestation of water hyacinths, and pollution. Thick, floating mats of weeds covered with soil *(phumdis)* are a characteristic feature • The lake is used extensively by local people as a source of water for irrigation and also plays an important role in flood control.
16	**Nalsarovar** Gujarat Natural Freshwater	• The largest natural wetland in the Thar Desert Biogeographic Province represents a dynamic environment with salinity and depth varying depending on rainfall. • It is an important stopover site within the Central Asia Flyway, with globally threatened species such as the critically endangered Sociable Lapwing.

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		• Lifeline for a satellite population of the endangered Indian Wild Ass.
17	**Point Calimere Wildlife Sanctuary** Tamil Nadu Coastal Swamps & saltpans	• Illegal collection of firewood and forest produce such as fruits (gathered by lopping off tree branches), • The spread of Prosopischilensis (Chilean mesquite), increasingly brackish groundwater caused by expansion of the historical salt works and decreasing inflow of freshwater are all seen as potential causes for concern. • Visitors come to the site both for recreation and for pilgrimage, as it is associated with Lord Rama.
18	**Sasthamkotta Lake** Kerala Natural Freshwater Lake	• The largest freshwater lake in Kerala. • The water contains no common salts or other minerals and supports no water plants; a larva called "cavaborus" abounds and eliminates bacteria in the water, thus contributing to its exceptional purity. • The ancient Sastha temple is an important pilgrimage centre. • Source of drinking water for Kollam.
19	**Vembanad-Kol Wetland** Kerala Natural Brackish water	• The largest brackish, humid tropical wetland ecosystem on the southwest coast of India • Famous for backwater tourism and sub-fossil clam deposits. • Longest lake in India • Thaneermukkom Saltwater barrier is constructed and divides the lake into parts – one with perennial brackish water and other with fresh water from rivers draining into the lake.
20	**Ashtamudi Wetland** Kerala Natural Brackish	An extensive estuarine system, the second largest in Kerala State. The site supports several mangrove species.
21	**Surinsar-Mansar Lakes** Jammu & Kashmir Natural Freshwater	• It is a composite lake in semi-arid Panjab Plains, adjoining the Jhelum Basin with catchment of sandy conglomerate soil, boulders and pebbles. • Important for religious reasons.
22	**Wular Lake** Jammu & Kashmir Natural Freshwater	• Largest freshwater lake in India with extensive marshes of emergent and floating vegetation, particularly water chestnut, that provide an important source of revenue for the State Government and fodder for domestic livestock. • Fed by Jhelum River.
23	**Hokera Wetland** Jammu & Kashmir Natural Freshwater	• Located in the northwest Himalayan biogeographic province of Kashmir, back of the snow-draped Pir Panchal.
24	**Tso Moriri** Jammu & Kashmir Natural Freshwater to brackish	• A freshwater to brackish lake lying at 4,595m above sea level, with wet meadows and borax-laden wetlands along the shores. • The site is said to represent the only breeding ground outside of China for one of the most endangered cranes, the Black-necked crane, and the only breeding ground for Bar-headed geese in India.
25	**Bhoj Wetland** Madhya Pradesh	• Two contiguous human-made reservoirs - "Upper Lake" was created in 11th century by construction of an earthen dam across Kolans River,

	Manmade Freshwater	and lower was constructed nearly 200 years ago, largely from leakage from Upper, and is surrounded by Bhopal city.
26	**Upper Ganga River** Uttar Pradesh River stretch Freshwater	• A shallow river stretch of the great Ganges with intermittent small stretches of deep-water pools and reservoirs upstream from barrages. • Provides habitat for Ganges River Dolphin, Gharial, Crocodile, 6 species of turtles, otters, 82 species of fish and more than a hundred species of birds.
27	**Rudra Sagar Lake** Tripura Natural Freshwater	• A lowland sedimentation reservoir in the northeast hills, fed by three perennial streams discharging to the River Gomti.
28	**Nandur Madhameshwar** Maharashtra Manmade Freshwater	• First Ramsar site in Maharashtra • Developed by making a low dam at the confluence of the Godavari and Kadwa Rivers. • Habitat of critically endangered species including Deolali minnow (a fish), Indian vulture and white-rumped vulture.
29	**Saman Bird Sanctuary** Uttar Pradesh Natural Freshwater	• Seasonal oxbow lake on the Ganges floodplain. • Host over 1% of the South Asian population of graylag goose during winter.
30	**Nawabganj Bird Sanctuary** Uttar Pradesh Natural Freshwater	• Shallow wetland fed by monsoon rain and Sarda canal. • Known to host Siberian cranes during winter.
31	**Samaspur Bird Sanctuary** Uttar Pradesh Natural Freshwater	• Perennial lowland marsh typical of the Indo-Gangetic Plains • Harbours threatened species such as the endangered Egyptian vulture, Pallas's fish eagle and vulnerable common Pochard. • A tall grass called Sarpat is also found in bunches at every spot.
32	**Sandi Bird Sanctuary** Uttar Pradesh Natural Freshwater	• It is a freshwater marsh. • Important Bird Area, declared by Birdlife International. • River Garra passes near the sanctuary. • It hosts common teal, red-crested pochard and ferruginous duck while vulnerable sarus crane. • The Sanctuary dried out leading to a subsequent collapse in Waterbird populations from 2014 to 2015.
33	**Parvati Arga Bird Sanctuary** Uttar Pradesh Natural Freshwater	• Permanent freshwater environment consisting of two oxbow lakes. • It is rainfed has deep natural depression. • Species: critically endangered white-rumped vulture and Indian vulture and the endangered Egyptian vulture.
34	**Sarsai Nawar Jheel** Uttar Pradesh Natural Freshwater	• Fed by precipitation run-off from the Southwest monsoon rains. • It is an example of co-habitation of humans and wildlife: farming practices across most of the Site play important roles in sustaining the Waterbird habitats. • Recognised as an Important Bird Area by Birdlife International.

		• Species: vulnerable sarus crane, critically endangered whiterumped vulture and endangered woolly-necked stork.
35	**Beas Conservation Reserve** Punjab Natural Freshwater	• It is a 185-kilometre stretch of the Beas River majorly in Punjab. • Hosts the only known population in India of the endangered Indus river dolphin. • Other Important species: endangered mahseer and hog deer as well as the vulnerable smooth coated otter. • Here a program was initiated to re-introduce the critically endangered gharial.
36	**Nangal Wildlife Sanctuary** Punjab Manmade Freshwater	• Located in the Shiwalik foothills of Punjab. • It occupies a human-made reservoir constructed as part of the Bhakra-Nangal Project on Sutlej River in 1961. • Historic importance - Indian and Chinese Prime Ministers formalized the "Five Principles of Peaceful Coexistence" there in 1954.
37	**Keshopur-Miani Community Reserve** Punjab Natural Freshwater	• Mosaic of natural marshes, aquaculture ponds and agricultural wetlands maintained by the annual rainfall runoff. • It has a series of managed fishponds and cultivated crops such as lotus and chestnut. • The Site is an example of wise use of a community-managed wetland, which provides food for people and supports local biodiversity. • Species: vulnerable common pochard and the endangered spotted pond turtle
38	**Tso Kar Wetland Complex** UT of Ladakh Natural; One lake saline & one freshwater	• Tso Kar Basin is in a high-altitude wetland complex. It consists of two main waterbodies, Startsapuk Tso and Tso Kar. The Startsapuk Tso is a freshwater lake of 438 hectares to the south. Tso Kar lake is a hypersaline lake of 1800 hectares. • Most important breeding areas of the Black necked Cranes in India.
39	**Lonar Lake** Maharashtra Created by meteorite impact (Natural) Saline	• Scientists in the 1970s confirmed the presence of Maskelynite—a naturally occurring glass that is only formed by extremely high-velocity impacts. • Formed 35,000 to 50,000 years ago, Lonar is the only "fresh" impact structure in basalt on Earth, making it an important analog for impact craters on the surface of the Moon.
40	**Sur Sarovar** Uttar Pradesh Manmade Freshwater	• Also known as Keetham Lake, created by the British to supply water to Agra city. • Migratory birds flock to the site as it is located on Central Asian Flyway. Important for Greater spotted eagle, sarus crane and catfish Wallago attu. Over 1% of the population of graylag geese is present here.
41	**Asan Conservation reserve** Uttarakhand Natural Freshwater	• It is a stretch of Asan river running down to its confluence with Yamuna River in Dehradun district of Uttarakhand. Barrage on the Asan River created it. • Birds spotted: Red-headed vulture (IUCN status: Critically Endangered), White-rumped vulture, Baer's pochard. • Red crested pochard, ruddy shelduck and Putitor mahseer (IUCN status: Endangered) are also found here.

42	**Kabartal Wetland** Bihar Natural Freshwater	• Also known as Kanwar Jheel, located in Indo-Gangetic plains. Plays important role in absorbing surplus waters in monsoons, preventing floods. • Five critically endangered species inhabit the site, including three vultures – the red-headed vulture, white-rumped vulture and Indian vulture – and two waterbirds, the sociable lapwing and Baer's pochard.
43	**Wadhvana Wetland** Gujarat Manmade Freshwater	• This reservoir was created in 1910 by former Baroda State (King Gaikwad). • Located in a semi-arid agricultural landscape and surrounded by wheat and paddy fields and villages. • River Orsang (which joins with the Narmada River at Chandod) flows into the lake. It also provides irrigation to 25 villages.
44	**Thol Lake** Gujarat Manmade Freshwater	• Located in Mehsana district of Gujarat. • It is a shallow freshwater reservoir and a predominantly open water area. It is a man-made wetland, originally constructed for irrigation in 1912.
45	**Bhindawas Wildlife Sanctuary** Haryana Manmade Freshwater	• Located in Jhajjar district, Haryana. • Largest wetland in Haryana. It is a bird sanctuary by Govt. of India. • It was constructed to store the excess waters of the Jawaharlal Nehru canal at the time of power failure of the Lift Canal System. It is an important part of the ecological corridor along the route of Sahibi River which traverses from Aravalli hills in Rajasthan to the Yamuna.
46	**Sultanpur National Park** Haryana Natural Freshwater	• Located in Gurgaon district of Haryana close to the National capital and Aravalli range. • Discovered by IUCN official and bird lover Peter Jackson. • Famous site for the stay of winter migratory birds.
47	**Khijadiya Bird Sanctuary** Gujarat	• A bird sanctuary located in Jamnagar district. About 300 species of migratory birds have been recorded here. • The sanctuary is unique, having freshwater lakes, salt and freshwater marshlands.
48	**Haiderpur Wetland** Uttar Pradesh Manmade Freshwater	• Manmade wetland formed in 1984 due to construction of Madhya Ganga Barrage on a floodplain of the Ganga River.
49	**Bakhira Sanctuary** Uttar Pradesh Natural Freshwater	• Largest natural flood plain (river connected wetland) of India in Sant Kabir Nagar district in Eastern Uttar Pradesh. • A prominent waterbird found in the lake is Indian Purple moorhen or Purple Swamphen & Sarus Crane. • Receives water from Ami River; discharge from its flows into Rapti River. It is rich in perennial reed grasses called Phragmites which attracts many species of the Central Asian Flyway.
50	**Tampara lake** Odisha Natural Freshwater	• Freshwater lakes in Odisha • The wetland is an important habitat for vulnerable species such as Cyprinus carpio, common pochard, and river tern (Sterna aurantia).
51	**Hirakud reservoir**	• The largest earthen dam in Odisha

	Odisha Manmade Freshwater	• The wetland also provides important hydrological services by moderating floods in the Mahanadi delta, the ecological and socio-economic hub of the east coast of India.
52	**Anshupa lake** Odisha Natural Freshwater	• Largest freshwater lake in Odisha • The wetland is an oxbow lake formed by River Mahanadi. The lake is connected with Mahanadi River by a channel known as Kabula Nala through which flood water from Mahanadi enters the lake. • The wetland provides a safe habitat to at least three threatened bird species- Indian Skimmer, Black Bellied Tern and Wagur (fish).
53	**Yashwant Sagar** Madhya Pradesh Manmade Freshwater	• One of the two Important Bird Areas (IBA) in the Indore region as well as one of the most important birding sites in Malwa region of Madhya Pradesh. • It is a stronghold of the vulnerable Sarus Crane in central India. • The lake backwaters have plenty of shallow areas, conducive for waders and other waterfowl. • Due to its vast shallow reed beds, the wetland is considered heaven to a large number of winter migratory birds.
54	**Chitrangda Bird Sanctuary** Tamil Nadu Manmade Freshwater	• Locally known as "Chitrangda Kanmoli" is in Ramavataram district in Tamil Nadu. • Chitrangda Bird Sanctuary is an ideal habitat for winter migratory birds. • Notable waterbirds spotted from the site area include spot-billed pelican, little egret, grey heron, large egret, open billed stork, purple, and pond herons.
55	**Suchindrum Theroor Wetland complex** Tamil Nadu Manmade Freshwater	• Part of Suchindrum-Theroor Manakudi Conservation Reserve in Tamil Nadu. • The Wetland Complex lies at the point of the Kodaiyar river drainage system. • It is declared an Important Bird Area and lies at the southern tip of the Central Asian flyway of migratory birds. • This is a man-made, inland Tank and is perennial. • Copper plate inscriptions from the 9th century mention Pasumkulam, Venchikulam, Nedumarthukulam, Perumkulam, Elemchikulam and Konadunkulam. • Around 250 species of birds have been recorded in the area, of which 53 are migratory, 12 endemic, and 4 threatened.
56	**Vaduvur Bird Sanctuary** Tamil Nadu Manmade Freshwater	• It is composed of small man-made reservoirs interconnected by an ancient network of canals and fed by the Mettur reservoir, in the semi-arid district of Thiruvarur. Tamil Nadu • It is a large human-made irrigation tank and shelter for migratory birds as it provides a suitable environment for food, shelter, and breeding ground. • Indian Pond Heron occurred in most of the surveyed tanks.
57	**Kanjirankulam Bird Sanctuary** Tamil Nadu	• Located near Mudukulathur in Ramanathapuram District, Tamil Nadu. • It is notable as a nesting site for several migratory heron species that roost in the prominent growth of babul trees there.

	Manmade Freshwater	• The site qualifies as an IBA as the threatened Spot-billed Pelican breeds here. • Supports IUCN Red List vulnerable avian species like River Tern. • Recent growth of invasive Prosopis trees has affected the growth of the indigenous babul populations, threatening the habitat of these birds.
58	**Thane Creek** Maharashtra Natural Brackish	• Located in Maharashtra, India. • Thane Creek falls under the Marine Ecoregions of the World (MEOW) but is fed by numerous freshwater sources, making the water brackish. • Fresh water sources include Ulhas River (largest) followed by many drainage channels from various suburban areas of Mumbai, Navi Mumbai & Thane. • Declared as Thane Creek Flamingo Sanctuary. Thane creek is fringed by mangroves on both banks & comprises around 20% of the total Indian mangrove species. • Part of Central Asian Flyway of the birds and an Important Bird Area (IBA).
59	**Hygam Wetland** Jammu & Kashmir Natural Freshwater	• Falls within the River Jhelum basin and is in the Baramulla district. • It lies downstream of another Ramsar Site (Wular Lake) • Consequent to the high rate of siltation, Hygam Wetland has lost its wetland characteristics to a large extent and in many places changed into a landmass. • It lies on the Central Asian Flyway and over 40,000 migratory and resident bird species have been recorded annually.
60	**Shallabug Wetland Conservation Reserve** Jammu & Kashmir Natural Freshwater	• Located in Srinagar district, UT of J&K, located on floodplains of River Jhelum. • It lies west of Anchar Lake. Lies close to another Ramsar Site, Hokera Wetland • This shallow wetland is primarily fed by rainfall, snowmelt from the Kashmir Himalayas, and stream water flowing from the Sindh river and Anchar Lake.
61	**Sirpur wetlands** Madhya Pradesh Natural Freshwater	• Human-made wetland that has stabilized and acquired near-natural characteristics in the last two centuries. • Located in Madhya Pradesh, Commonly named Pakshi Vihar (bird sanctuary). • The Site is a shallow, alkaline, nutrient-rich lake that floods during the monsoon to a maximum depth of two meters.
62	**Sakhya Sagar** Madhya Pradesh Manmade Freshwater	• uman-made reservoir on the outskirts of Shivpuri town within the Madhav National Park, Madhya Pradesh. • Features a mosaic of landforms including open water and surrounding marshes, plantations and a small patch of agricultural land. • Permanent source of water for wildlife, and habitat for animals including thousands of migratory waterfowl and also mugger crocodile whose presence attracts tourists. • The Site is one of the “three lakes of Shivpuri.”
63	**Ranganthittu** Karnataka	• It is a part of Kaveri River on the Southern Deccan Plateau.

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	Natural Freshwater	• This region comprises gently rolling plains traversed by several of the large rivers that rise in the Western Ghats mountain range and flow eastward to the Bay of Bengal. • The Site is an ecologically important riverine wetland. It supports healthy populations of mugger crocodile, smooth-coated otter and hump-backed mahseer. • The Site also supports over 1% of the global population of painted storks, spot-billed pelicans and black-headed ibis.
64	**Nanda Lake** Goa Natural Freshwater	• Nanda Lake comprises intermittent freshwater marshes that lie adjacent to one of the major tributaries of Zuari River, Goa. • They are linked to the adjacent river channel by a sluice gate, which when closed enables the flooding of the marshes. This wetland supports a wide variety of migratory waterbirds.
65	**Pala wetland** Mizoram Natural Freshwater	• Pala Wetland is the largest natural wetland in the state of Mizoram. • Includes a relatively deep lake 16 metres deep on average) and its surrounding forest catchment area which climbs to almost 600m above sea level. • Various species of primate inhabit the wetland, such as the endangered Hoolock gibbon (Hoolock hoolock) and Phayre's leaf monkey. • Pala Wetland is revered by the local Mara people and has a deep connection with their history. The wetland is the major source of freshwater.
66	**Satkosia gorge** Odisha Natural Freshwater	• Satkosia Gorge in Angul district of Odisha State is a mosaic of rivers, marshes and evergreen forests at the meeting point of two biogeographic regions, the Deccan Peninsula and the Eastern Ghats.
67	**Gulf of Mannar marine Biosphere reserve** Tamil Nadu Coast Natural Saline	• Located at the south-eastern tip of India, the Gulf of Mannar Marine Biosphere Reserve is the first Marine Biosphere Reserve in South and South-East Asia. • Globally threatened species include the dugong (Dugong dugon), whale shark, green sea turtle, hawksbill turtle and Indo-Pacific humpback dolphin. • Local communities are mainly *Marakeyars*, with fishing their main livelihood. • It is also recognized as a World Heritage Site.
68	**Pichavaram mangrove** Tamil Nadu Natural Brackish	• One of the largest mangrove ecosystems in India, located between the prominent estuaries of Vellar and Coleroon Rivers • It not only offers waterscape and backwater cruises but also another rare sight – the mangrove trees are permanently rooted in a few feet of water. • Pichavaram Mangrove supports several threatened species such as the critically endangered great, white-bellied heron and spoon-billed sandpiper, the endangered spotted greenshank and the vulnerable olive ridley turtle.
69	**Karikili Bird sanctuary** Tamil Nadu	• Karikili Bird Sanctuary comprises two rain-fed non-perennial irrigation tanks spreading over an area of more than 58 hectares in Tamil Nadu.

	Manmade Freshwater	• The Site is flat or gently undulating and lies near low ridged, rocky hillocks.
70	**Pallikaranai Marsh Reserve Forest** Tamil Nadu Natural Brackish	• This freshwater marsh and partly saline wetland situated about 20 kilometres south of the city of Chennai serves as an aquatic buffer of the flood-prone Chennai and Chengalpattu districts. • The diverse ecosystem of the marshland supports notable species such as Russell's viper and birds such as the glossy ibis.
71	**Udhayamarthandapuram Bird Sanctuary** Tamil Nadu Manmade Freshwater	• The Sanctuary consists of human-made irrigation tanks, interconnected by an ancient network of canals and fed by Mettur dam through the Koraiyar canal. • The site is in Tamil Nadu about 80 kilometres from Thanjavur.
72	**Vedanthangal Bird Sanctuary** Tamil Nadu Manmade Freshwater	• A small irrigation tank in Vedanthangal village surrounded by rocky plains and low-ridged, denuded hillocks in Tamil Nadu. • This Site is an Important Bird and Biodiversity Area (IBA)
73	**Vellode Bird Sanctuary** Tamil Nadu Manmade Freshwater	• Located in Erode district of Tamil Nadu. • This human-made tank is an ideal habitat for birds, due to the abundant food resources it provides from neighbouring agricultural fields, and other aquatic organisms. • It is an important staging and breeding ground for migratory birds on the Central Asian Flyway.
74	**Vembannur Wetland Complex** Kerala Manmade Freshwater	• Human-made irrigation tank situated a few kilometres inland in Kerala near the southernmost point of mainland India. • The Site forms part of an Important Bird and Biodiversity Area (IBA).
75	**Koonthankulam Bird Sanctuary** Tamil Nadu Manmade Freshwater	• Consists of irrigation tanks interconnected by a network of canals built a few centuries ago and fed by the rivers originating from the Western Ghats mountain range. • Largest reserve for breeding resident and migratory waterbirds in South India.
76	**Karaivatti Bird Sanctuary** Tamil Nadu	• Inland wetland. Provides important stopover and foraging ground for birds migrating along the Central Asian Flyway. • Gum Arabic tree provides roosting and nesting grounds for threatened species: spotted eagle, tawny eagle and Indian darter. • Breeding habitat for vulnerable Indian flap-shelled turtles.
77	**Madai Kere Conservation Reserve** Karnataka Manmade	• Constructed to store rainwater for irrigation purposes. Water in the wetland has over time become increasingly alkaline and unsuitable for irrigation. • Designated Important Bird and Biodiversity Area (IBBA).
78	**Longwood Shola Reserve Forest** Tamil Nadu Natural Freshwater	• Intermittent freshwater marshes and streams nested within a 'shola', tropical montane forest located on the eastern slope of Nilgiri mountain range. • Has high endemism, with multiple species of trees, reptiles, frogs and birds.

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		• Species: Globally endangered Black-chinned Nilgiri Laughing thrush, Nilgiri Blue Robin and vulnerable Nilgiri wood-pigeon. • 14 out of 26 endemic bird species of Western Ghats are found in these wetlands.
79	**Ankasamudra Bird Conservation Reserve** Karnataka Manmade Freshwater	• A manmade village irrigation tank built for storing monsoon run-off water coming from the Tungabhadra River. • Several thousands of Gum Arabic trees are found growing from the lakebed. These trees provide nesting and roosting habitat for large colonial waterbirds. • Excessive growth of invasive alligator weed, shrub Prosopis juliflora and African Catfish threaten the native fish and waterbirds.
80	**Aghanashini Estuary** Karnataka Natural Brackish	• Estuary built at the confluence of Aghanashini River & Arabian Sea. • Brackish water from the estuary provides diverse ecosystem services including flood and erosion risk mitigation, biodiversity conservation and livelihood support. • Traditional fish farming in the estuarine rice fields (locally known as Gazni rice fields), bivalve shell collection and salt production.
81	**Nakti Bird Sanctuary** Bihar Man-made freshwater	• Man-made wetland formed after the construction of Nakti Dam on the Nakti River. • Designated as Important Bird and Biodiversity Area. • Important species: Baer's pochard (Critical Endangered) and Steppe Eagle (Endangered). Hosts the largest congregations of bar-headed geese on the Indo-Gangetic Plains.
82	**Nagi Bird Sanctuary** Bihar Man-made freshwater	• Man-made wetland formed after the construction of Nagi Dam on Nagi River. • Provides habitat to over 150 species of birds, mammals, fish, aquatic plants. Acts as wintering habitat for several migratory species. • Species: Indian Elephant (Endangered), Native Catfish (Vulnerable) and largest congregations of red-crested pochard on the Indo-Gangetic plains.
83	**Nanjarayan Bird Sanctuary** Tamil Nadu Man-made freshwater	• Large shallow freshwater lake. The wetland is fed by heavy rain water flow from Nallar drainage. • Also known as Sarkar Periyapalayam Reservoir. The lake has mudflats in the middle which provides habitat for birds.
84	**Kazhuveli Bird Sanctuary** Tamil Nadu Natural Brackish	• Brackish water shallow lake located on Coromandel Coast. Also known as Kaliveli Bird Sanctuary. • Connected to the Bay of Bengal by the brackish Uppukalli creek and Edayanthittu estuary. • One of the largest wetlands in peninsular India. Lies on the Central Asian Flyway.
85	**Tawa Reservoir** Madhya Pradesh Man-made Freshwater	• Constructed on the confluence of Tawa and Denwa Rivers. • Tawa reservoir was mainly built for irrigation purposes, hydropower generation and aquaculture. • Located inside the Satpura Tiger Reserve and forms the western boundary of Satpura National Park and Bori Wildlife Sanctuary.

		• Tawa is the longest tributary of Narmada river (Leftbank tributary). Originates from Mahadeo hills.
86	**Udhwa Lake** Jharkhand	• First Ramsar site in Jharkhand. Falls within the Gangetic Plains bio-geographic zone. • Named after: Saint Uddhava of Mahabharat times. • Identified as an 'Important Bird Area'. Important bird: Oriental white-backed vulture (Critically Endangered)
87	**Khecheopalri Wetland** Sikkim	• Demazong valley, West Sikkim. • Sacred site for both Buddhists and Hindus and is often described as a "wish-fulfilling lake." • Under increased pressure from land-use changes and tourism.
88	**Sakkarakottai Bird Sanctuary** Tamil Nadu	• Peri-urban wetland. Located close to the Gulf of Mannar on the Central Asian flyway. • Supports more than 120 species of birds, abundant babul trees provide ideal nesting habitat for bird species.
89	**Therthangal Bird Sanctuary** Tamil Nadu	• Marshy lake located within Therthangal village. • Lies along Central Asian Flyway. • Important species: Egyptian vulture (Endangered)

5. LIST OF TIGER RESERVES

	TIGER RESERVE	DESCRIPTION
1.	**BANDIPUR TIGER RESERVE** KARNATAKA	• Located in Mysore and Chamarajanagar districts of southern Karnataka. • Ecological confluence of Western and Eastern Ghats. • Part of Nilgiris Biosphere Reserve.
2.	**CORBETT TIGER RESERVE** (UTTARAKHAND) **&** **AMANGARH BUFFER** (UTTAR PRADESH)	• Located on foothills of Himalayas in Uttarakhand in three districts of Pauri, Nainital & Almora. The reserve spreads over Bhabar and lower Shivalik region with a deep-water table. • The rivers Ramganga, Pallaen and Sonanadi flow through it. • **Flora:** Vegetation comprises deciduous sal and mixed forests, interspersed with grasslands and riparian vegetation. Habitat is characterised by open meadows (Chaurs), outcome of abandoned settlements or past clearings. These grasslands constitute about 20% of the area.
3.	**KANHA TIGER RESERVE** MADHYA PRADESH	• Located in Maikala range of Satpura Mountains falling in Mandla and Balaghat district of Madhya Pradesh. • **Fauna:** Critical as last habitat for highly endangered hard ground Barasingha and saving it from extinction.
4.	**MANAS TIGER RESERVE** ASSAM	• Tiger reserve with largest buffer area among tiger reserves in India. • Spreads across districts of Kokrajhar, Chirang, Buxa and Udalguri in North-West Assam. • Separated by River Manas from the Royal Manas National Park in its North. Separated by River Sankosh from the Buxa Tiger Reserve. • Evolutionarily, it is the entry point of tigers into India.

5.	**MELGHAT TIGER RESERVE** MAHARASHTRA	• Fondly known as Kipling Country marked with large tracts of hills and ravines in Satpura range. • Melghat is a raptor's paradise with more than 260 species of birds. • Flora: Characterized by Dry Deciduous Forest with Teak being the dominant species.
6.	**PALAMAU TIGER RESERVE** JHARKHAND	• Located in western part of Chhota Nagpur plateau and spread over two districts of Latemar and Garhwa. • Only tiger reserve in Jharkhand. Formerly used for cattle grazing and camping. • Forms part of Central Indian Tiger Landscape Complex and its habitat contiguity extends to Sanjay Dhubri Tiger Reserve through Guru Ghasidas National Park, all the way to Bandhavgarh Tiger Reserve. • Three rivers i.e., North Koel, Auranga and Burha flow through it. The area is drought prone with Burha being the only perennial river. • Severely impacted by Left Wing Extremism and acute grazing pressure has led to destruction of prime habitats and breeding areas of tigers.
7.	**RANTHAMBORE TIGER RESERVE** RAJASTHAN	• Located at the junction of (Great Boundary Fault) of Aravalli and Vindhyan range. Once considered as the famous and former hunting grounds of Maharajas of Jaipur. Banas River bounds the Ranthambore Tiger Reserve in the North while Chambal River forms the southern boundary of it. • **Flora:** Dry deciduous forests and open grassy meadows. • **Fauna:** Tiger, Leopard, Caracal, Jungle Cat, Elephants etc.
8.	**SIMLIPAL TIGER RESERVE** ODISHA	• Derives its name from red silk cotton trees (Semul) growing in the area. • Tribal groups: Khadia, Mankidia and Lodhas live around STR. • **Fauna:** Bengal Tiger, Asian Elephant, Gaur, Chausingha, Mugger Crocodile. • Only home to unique *melanistic tigers* (Black and Yellow stripes). This is due to a rare genetic phenomenon known as pseudo-melanism, which is characterised by excessive pigmentation of dark colour in the skin of an animal.
9.	**SUNDARBAN TIGER RESERVE** WEST BENGAL	• Sundarbans are the largest deltaic region of the world. The region is named after mangrove tree species 'Sundari'. • **Fauna:** Tiger, Fishing Cat, Estuarine Crocodile, Gangetic and Irrawaddy Dolphins. It harbors a significant population of River Terrapin (Batagur baska). Sundarbans are known as Kingfisher's paradise with 10 species out of 12 kingfishers found in India.
10.	**PERIYAR TIGER RESERVE** KERALA	• PTR along with adjoining protected areas form the largest tiger conservation landscape in southernmost western ghats, extending over 4000 sq km in Tamil Nadu and Kerala. • Spreads across districts of Idukki, Pathanamthitta and Kottayam in Kerala. • **Geography:** Marked by hilly and undulating terrain with a maximum altitude of 2016 metres. Two major rivers namely Periyar and Pamba drain the area. • **Flora:** Tiger, Elephant (Important habitat), Gaur, Sambar Deer, Leopard, Sloth • Tiger reserve with smallest buffer area among tiger reserves in India.

11.	**SARISKA TIGER RESERVE** RAJASTHAN	• It has the unique distinction of successfully reintroducing tigers after its local extinction. • There is no perennial river or water stream. • Flora: Characterised by Dry Deciduous Forests and Tropical Thorn Forest. Anogeissus pendula (also known as Dhok tree or Button tree) is the dominant tree species in the region. • Fauna: Carnivores: Tiger, Leopard, Hyena, Caracal, Jungle Cat, Ratel etc. Herbivores: Nilgai, Sambar deer, Spotted deer, Four horned antelope etc.
12.	**BUXA TIGER RESERVE** WEST BENGAL	• Located in Alipurduar of Jalpaiguri district of West Bengal. • **Flora:** Moist Tropical Forest • **Fauna:** Tiger, Leopard, Leopard Cat, Hog badger, Jungle Cat, Leopard Cat etc.
13.	**INDRAVATI TIGER RESERVE** CHHATTISGARH	• Located in Bijapur district of Chhattisgarh. Indravati River forms the boundary of this reserve on the northern and western side. (Indravati is a tributary of Godavari River). • **Fauna:** Has a small population of Wild Water Buffalos (State Animal of Chhattisgarh, IUCN Status: Endangered), Tiger, Leopard etc. • **Flora:** Dominated by Sal & Teak.
14.	**NAMDAPHA TIGER RESERVE** ARUNACHAL PRADESH	• Large, protected area in Arunachal Pradesh near international border with Myanmar. It is a biodiversity hotspot in the Eastern Himalayas. It is located on the Dapha bum range of Mishmi hills and Patkai range. The Noa Dihing River crosses it from east to west. • Harbors northernmost lowland evergreen rainforests in the world. • **Fauna:** Namdapha flying squirrel (Critically endangered and endemic to this place),
15.	**NAGARJUNASAGAR TIGER RESERVE** ANDHRA PRADESH	• Largest core area and largest overall area among all tiger reserves in India. • Located in Nallamalla range, an extension of Eastern Ghats of Andhra Pradesh spread over three districts of Kurnool, Prakasham & Guntur. • Located on the southern bank of Krishna River. • Features undulating terrain, subterranean valleys and steep cliffs comprising Hardwickia forest and mixed dry deciduous forest.
16.	**DUDHWA TIGER RESERVE** UTTAR PRADESH	• Located on India-Nepal border. Part of Shivalik hills and Gangetic Plains tiger landscape complex in Uttar Pradesh. • It comprises three protected areas: Dudhwa National Park, Katerniaghat Wildlife Sanctuary & Kishanpur Wildlife Sanctuary. • **Fauna:** Provides habitat for unique species including endangered Rhinoceros (successfully reintroduced), Bengal Florican (Critically Endangered) & Hispid Hare (Critically Endangered).
17.	**KALAKAD MUNDANTHURAI** TAMIL NADU	• Located in southern Western Ghats and forms part of Agasthyamalai Biosphere Reserve. • Located adjacent to Kanyakumari district of Tamil Nadu. • Major rivers: Gadananathi river etc.
18.	**VALMIKI TIGER RESERVE** BIHAR	• Located in Bihar in the northern part of West Champaran district along the border of India & Nepal.

		• Located in Terai region of Himalayas, it is connected to Chitwan National Park in Nepal. • Only Tiger Reserve in Bihar. • Sets an excellent example of Shivalik Hills and Gangetic Plains landscape with a mosaic of dense forests, open woodlands, swamps and grasslands.
19.	**PENCH TIGER RESERVE** MADHYA PRADESH	Located on southern slopes of Satpura range in Madhya Pradesh. Pench River flows through the park from north to south and splits the park into two. Meghdoot dam on Pench River is located on the boundary of this Tiger Reserve.
20.	**TADOBA ANDHERI** MAHARASHTRA	• Located in Chandrapur district of Maharashtra in the Vidarbha region. The Andhari River passes through this reserve. Tadao is the name of a lake inside the reserve. • **Flora:** Teak, Bamboo etc. • **Fauna:** Tiger, Leopard, Sloth bear etc.
21.	**BANDHAVGARH TIGER RESERVE** MADHYA PRADESH	• Located between Vindhya and Satpura ranges in Madhya Pradesh. • Characterised by valleys, hills and plains with Bandhavgarh fort being a prominent landmark. Has remnants of archaeological monuments dating to Kalachuri dynasty. • **Flora:** Tropical moist, deciduous forests with sal and grasslands, locally known as 'Vahs'. • **Fauna:** Tigers, Leopard, wild dog etc. Gaur, which became locally extinct here, has been successfully reintroduced from Kanha. *Giant wood spiders* with their enormous webs are a beautiful sight especially in early winter mornings.
22.	**PANNA TIGER RESERVE** MADHYA PRADESH	• Named after the city of diamonds, Panna in the Bundelkhand region in Madhya Pradesh. • Spreads across Panna, Chhatarpur and Damoh districts of Madhya Pradesh. • Located in Vindhyan Range. • It forms the northernmost tip of natural teak forests. • Ken River flows through the tiger reserve.
23.	**DAMPA TIGER RESERVE** MIZORAM	• Located in western edge of Mizoram. It is bound by Chittagong hill tracts of Bangladesh. Drained by River Khawthlangtuipui in the West and the Teirei River to the East. • **Flora:** Tropical Evergreen and Semi-Evergreen forests. Extensive bamboo forests dominate this tiger reserve. • **Fauna:** Hoolock Gibbon, Tiger, Clouded Leopard, Golden Cat, Leopard Cat etc.
24.	**BHADRA TIGER RESERVE** KARNATAKA	• Located in Western Ghats of Karnataka. The habitat is drained by the Bhadra River and its tributaries. • The reserve is shaped in the form of a saucer having mostly undulating terrain with valleys and steep hillocks. • **Flora:** Moist & Dry Deciduous forests and Shola grasslands. • **Fauna:** Tiger, Leopard, Leopard Cat, Dholes, Gaur, Sambhar and Barking Deer.

25.	**PENCH-EXTENSION** MAHARASHTRA	Extension of Pench Tiger Reserve in Maharashtra.
26.	**PAKKE TIGER RESERVE** ARUNACHAL PRADESH	• Located in the foothills of Eastern Himalayas in East Kameng District of Arunachal Pradesh. • Forms a transition zone between Indian and Malayan ecoregions and high endemicity. • Pakke River forms the boundary in the East while Kameng River forms the boundary in the West. • Pakke along with Nameri Tiger Reserve forms one of the largest blocks of semi-evergreen & evergreen forests in the North-East. • **Fauna:** Four species of hornbills; Key habitat for Clouded Leopard
27.	**NAMERI TIGER RESERVE** ASSAM	• Located in the northern part of Sonitpur district of Assam along the foothills of Arunachal Pradesh. This habitat is rich in biodiversity and is famous for white winged wood ducks. • It is bound by Jia Bhoreli River in the West and Bor-Dikorai River in the East. • **Fauna:** Tiger, Leopard, Clouded Leopard, Wild Dog, Asian Elephant etc.
28.	**SATPURA TIGER RESERVE** MADHYA PRADESH	• Prime example of the Central Indian Highlands ecosystem. • Important testimony to human evolution as it houses more than 50 rock shelters about one lakh years old. • Pachmarhi is located inside it. • Located south of Narmada River. • **Flora:** Deciduous forests dominated by Sal. • **Fauna:** Tiger, Gaur, Sambhar, Flying Squirrel, Indian Giant Squirrel
29.	**ANAMALAI TIGER RESERVE** TAMIL NADU	• Largest tiger reserve of Tamil Nadu. Situated at the heart of Anamalai Range (Also known as Elephant hills). Anamalai forms an integral part of Western Ghats after the Palakkad gap.
30.	**UDANTI SITANADI TIGER RESERVE** CHHATTISGARH	• Named after Sitanadi River which originates here and joins Mahanadi River. **Flora:** Teak and Bamboo dominate the landscape. • **Fauna:** Tigers, Leopards, Flying Squirrels, Jackals, Four-horned antelopes, Chinkara, Blackbuck etc.
31.	**SATKOSIA TIGER RESERVE** ODISHA	• Spread over four districts of Angul, Cuttack, Boudh and Nayagarh in the heartland of Odisha. The Mahanadi passes through this tiger reserve and forms a large gorge here, hence also known as Gorge Sanctuary. • **Flora:** Moist deciduous forests. • **Fauna:** Mugger Crocodiles and Indian Skimmers, forms the southernmost limit for Gharials.
32.	**KAZIRANGA TIGER RESERVE** ASSAM	• Located on the southern bank of Brahmaputra River in the Golaghat and Nagaon districts of Assam. • World famous habitat for one-horned rhinoceros. Mary Curzon, wife of Lord Curzon, visited here. The river Diffalu, a tributary of the Brahmaputra, flows through the National Park area (core/critical tiger habitat), while another tributary Moradifalu flows along its southern boundary. • **Flora:** Dominated by Elephant Grass, Marshland and broadleaf forests.

33.	**ACHANAKMAR TIGER RESERVE** CHHATTISGARH	• Located in the Maikala range in Chhattisgarh. It is located close to Amarkantak, the source of Son and Narmada River. • **Flora:** Dense Sal and bamboo forests. • **Fauna:** Tiger, Leopard, Gaur, Chital, Jackal etc.
34.	**KALI TIGER RESERVE** KARNATAKA	• Located in Uttara Kannada district of Karnataka in Western Ghats. It was formerly known as Dandeli Anshi National Park. The Kali River flows through it. • **Fauna:** Deciduous forests, Bamboos. There are extensive monoculture teak plantations here. • **Fauna:** Tiger, Leopard, Dhole, Gaur, Deer, Arboreal Mammals.
35.	**SANJAY DHUBRI TIGER RESERVE** MADHYA PRADESH	• Located in Sidhi and Shahdol districts of Madhya Pradesh on the borders of Chhattisgarh (Close to Guru Ghasidas National Park in Chhattisgarh). • Various perennial rivers like Gopad, • **Flora:** Moist deciduous forests with dense to open sal forests with remains of bamboo stumps. • **Fauna:** Tiger, Elephant, Sloth bear etc. • World famous white tigers "MOHAN" were found and rescued from the forest of this landscape by Maharaja of Rewa in 1951. She delivered the first white tigers in captivity, and most of the white tigers present in the zoo across the world are their progeny only. In remembrance, one of the park Range has been named after this White Tiger, "Mohan."
36.	**MUDUMALAI TIGER RESERVE** TAMIL NADU	• Nilgiris District (TN). It is at the tri-junction of three states, viz, Karnataka, Kerala and Tamil Nadu. Moyar River passes through it. • It is part of Nilgiri Biosphere Reserve (UNESCO World Heritage Site) • It has a common boundary with Wayanad Wildlife Sanctuary (Kerala) on the West, Bandipur Tiger Reserve (Karnataka) on the North, and the Nilgiris North Division on the South and East. • **Flora:** Semi-evergreen, Moist deciduous and Dry deciduous forests. Has tall grasses, referred to as "Elephant Grass", Bamboo of the giant variety, valuable timber species like Teak, Rosewood, etc. • **Fauna:** Tiger, Elephant, Gaur, Panther, Spotted Deer, Barking Deer, Mouse Deer, Common Langur, Malabar Giant Squirrel, Jungle Cat. • Documentary Movie 'Elephant Whisperers' was shot in this Tiger Reserve. Kattunayakan Tribe (PVTG) lives in this tiger reserve.
37.	**NAGARHOLE** KARNATAKA	• Named after a small river 'Nagarhole' (a snake stream in Kannada), which meanders through it, before joining Kabini River. • **Flora:** Eastern stretch of the park has dry deciduous forests while western stretch has semi-evergreen forests. These forests are interspersed with swampy fallows called 'hadlu', which are dominated by grasses and sedges. • **Fauna:** Tiger, Leopard, Wild Dogs etc.
38.	**PARAMBIKULAM** KERALA	• Located in Anamalai landscape of Western Ghats in Palakkad and Thrissur districts of Kerala. • **Flora:** Evergreen, Deciduous and Shola forests. The park has montane and marshy grasslands locally known as 'Vayals'.

39.	**SAHYADRI** MAHARASHTRA	• Located in Sahyadri range of Western Ghats in Maharashtra. These range from a common boundary between Maharashtra, Karnataka & Goa. • First tiger reserve of Western Maharashtra and spreads across two protected areas of Koyna sanctuary and Chandoli National Park. • **Flora:** Comprises Evergreen, Semi-Evergreen and Moist Deciduous Forests.
40.	**BILIGIRI RANGANATHA TEMPLE** KARNATAKA	• Acts as a bio-geographical bridge between Western & Eastern Ghats in South India. • Derives its name from 'Bilgiri', the rocky white rocky cliff which has a temple of Lord Vishnu locally known as Rangaswamy.
41.	**KAWAL TIGER RESERVE** TELANGANA	• Located on the banks of Godavari River and forms part of Deccan peninsula - Central Highlands. • Located on the southernmost tip of Central Indian Tiger Landscape, it has linkages with Tadoba-Andhari (Maharashtra) and Indravati (Chhattisgarh) tiger reserves. • **FLORA:** Characterised by Southern Tropical Dry Deciduous Forests with Teak and Bamboo being dominant species.
42.	**SATHYAMANGALAM TIGER RESERVE** TAMIL NADU	• Located as a transition zone between Western and Eastern Ghats in Erode District of Tamil Nadu. • It is contiguous with other tiger conservation landscapes like BRT, Bandipur, Mudumalai and Nagarhole. • **Flora:** Tropical dry deciduous forests, Evergreen forests.
43.	**MUKUNDARA TIGER RESERVE** RAJASTHAN	• Named after two continuous flat topped, almost parallel hills with narrow central ridges forming part of Vindhyas extending from Chambal to Kali Sindh River. • It has a length of almost 80 km and width ranging from 2-5 km. • **FLORA:** Dhok tree (Anogeissus pendula), Khair (Acacia catechu) etc.
44.	**NAWEGAON NAGZIRA TIGER RESERVE** MAHARASHTRA	• Located in Vidarbha region. • **Flora:** Tropical Dry Deciduous forests. • **Fauna:** Tiger, Panther, Small Indian Civet etc.
45.	**AMRABAD TIGER RESERVE** TELANGANA	• Located on the Nallamalla hills (part of Eastern Ghats) along Krishna River. Sri Sailam dam on Krishna River is located inside the Tiger Reserve. • **Fauna:** Mouse Deer (Indian spotted Chevrotain) which were locally extinct are being re-introduced in this protected area.
46.	**PILIBHIT TIGER RESERVE** UTTAR PRADESH	• Located in Pilibhit district on the international border with Nepal. River Gomti originates from this reserve. The park also forms the catchment of Sharda, Chuka and Mala Khannot rivers. Sharda Sagar dam forms the boundary of the reserve. • Connected to Dudhwa Tiger Reserve through Kishanpur (UP), Shuklaphanta (Nepal) and Nandaur Sanctuary in Uttarakhand. • **Flora:** Dominated by Sal forests, tall grasslands and swamps. • **Fauna:** Critically endangered Bengal florican and hispid hare. Tiger, Leopard, Fishing cat, leopard cat, swamp deer, hog deer, barking deer, blackbuck.
47.	**BOR TIGER RESERVE**	• Located adjacent to Pench Tiger Reserve.

	MAHARASHTRA	• **Fauna:** Tiger, Fox, Hyena, Sloth Bear. Peacocks are in abundance.
48.	**RAJAJI TIGER RESERVE** UTTARAKHAND	• Located in Haridwar, Dehradun and Pauri Garhwal districts of Uttarakhand. The tiger reserve is named after C. Rajagopalachari. • **Fauna:** Forms the northwestern limit of distribution of elephants and tigers in India. Mountain goat Goral etc. Fishes like Mahseer and Trout are found. • **Flora:** Amaltas, Sal, Palash etc.
49.	**ORANG TIGER RESERVE** ASSAM	• Smallest core area and smallest overall area among all tiger reserves in India. • This Tiger Reserve is located on the northern bank of Brahmaputra River in Darrang and Sonitpur districts of Assam. • **Fauna:** Indian rhinoceros, Pygmy Hog (Critically Endangered), Elephant, Wild Water Buffalo, Bengal Tigers. (Only stronghold of rhinoceros on North Bank of Brahmaputra). Bengal Florican (Critically endangered) is one of the flagship birds found in the park.
50.	**KAMLANG** ARUNACHAL PRADESH	• Located in Lohit district. Kamlang River passes through this sanctuary. It is located on the north of Namdapha National Park. • Local inhabitants of the region are Hishmi, Digaru & Mizo. They claim themselves as descendants of King Rukmo of Mahabharat. • **Fauna:** Four big cats (Tiger, Leopard, Clouded Leopard & Snow Leopard), Hoolock Gibbon, Slow Loris, Leopard Cat, Himalayan Palm Civets
51.	**SRIVILLIPUTHUR MEGAMALAI** TAMIL NADU	• Created by combining two wildlife sanctuaries Grizzled Squirrel Wildlife Sanctuary and Megamalai Wildlife Sanctuary. • Located close to Periyar Tiger Reserve. Vaigai River originates from this Tiger Reserve and flows to Palk Strait. • Meghamalai (also known as High Wavy Mountains) is a mountain range located in Western Ghats in Theni district of Tamil Nadu. It is dotted with cardamom and tea plantations. • **Fauna:** Grizzled Giant Squirrel (Near Threatened), Barking deer, Dhole, Elephants, Tigers, Leopard etc.
52.	**RAMGARH VISHDHARI** RAJASTHAN	• Lies in the southeastern part of Rajasthan. • It plays an important role in connecting Ranthambore and Mukundra Tiger Reserves of Rajasthan. • Mez River, a tributary of Chambal River, passes through the tiger reserve. • **Flora:** Characterised by Dry Deciduous Forest and is dominated by Dhok trees (Anogeissus pendula), Khair (Acacia catechu) etc. • **Fauna:** Tiger, Striped Hyena, Leopard, Golden Jackal, Sloth Bear, Indian Wolf, Nilgai, Fox and Chinkara.
53.	**RANIPUR** UTTAR PRADESH	• Located in Bundelkhand Region of Uttar Pradesh. It is located about 150 km from Panna Tiger Reserve in Madhya Pradesh. • This Tiger Reserve has been notified to accommodate the sinking of a part of Panna Tiger Reserve due to the Ken-Betwa Linkage. • **Fauna:** Tigers, Leopard, Sloth bears, Sambhar, Spotted Deer and Chinkara. However, there are no tigers in the Ranipur Tiger Reserve. However, tiger pugmarks are often seen in it.

54.	**Dholpur-Karauli** RAJASTHAN	• Fifth tiger reserve in Rajasthan. • National Chambal Gharial Sanctuary is a part of this tiger reserve. • The Karauli end of this Tiger Reserve is connected to the Ranthambore National Park through Keladevi Wildlife Sanctuary.
55	**Veerangana Durgavati Tiger Reserve** Madhya Pradesh	• This tiger reserve has been notified to accommodate tigers due to the sinking of 1/4th of the Panna Tiger Reserve due to Ken-Betwa River Linking Project. • Located on the higher elevations of Vindhyan Range. • It forms an important corridor connecting Panna, Bandhavgarh and Satpura Tiger Reserves. • Three fourth of the tiger reserve is part of Ganga Basin and one-fourth being part of Narmada Basin.
56	**Guru Ghasidas-Tamor Pingla Tiger Reserve** Chattisgarh	• **Location**: Spread across four districts of Chattisgarh, in Chota Nagpur plateau and partly in Baghelkhand plateau. • It falls between two other important tiger reserves- Bandhavgarh, Madhya Pradesh and Palamau, Jharkhand. • **Third largest tiger reserve in India**, after Nagarjunasagar-Srisailam Tiger Reserve in Andhra Pradesh and Manas Tiger Reserve in Assam.
57	**Ratapani Tiger Reserve** Madhya Pradesh	• Location: Vindhya Range. • It consists of dry deciduous and moist deciduous type forests dominated by Teak trees. • **Bhimbetka rock shelters** are located within the tiger reserve.
58	**Madhav National Park** Madhya Pradesh	• Vegetation: dry deciduous forests, semi-evergreen forests and grasslands. • **Sakhya Sagar,** a man-made reservoir within the park, has been designated as a Ramsar site since 2022. • Fauna: Tiger, leopard, jackals, nilgai, sambar, wild boar • The dense forests of the National Park were the former hunting grounds of Mughals and Scindia rulers.

6. LIST OF BIOSPHERE RESERVES IN INDIA

BIOSPHERE RESERVES	LOCATION	FLORA	FAUNA	TRIBALS
NILGIRI (Included in MAB list of UNESCO)	Part of Wayanad, Nagarhole, Bandipur & Madumalai, Nilambur, Silent Valley & Siruvani hills in Tamil Nadu, Kerala & Karnataka.	Tropical forest; Mixed Mountain and highland systems	Tiger, Elephant, Nilgiri Tahr, Lion-tailed macaque, Nilgiri Langur	Cholanaikans-only surviving hunter-gatherers of the Indian subcontinent
NANDA DEVI (Included in MAB list of UNESCO)	Part of Chamoli, Pithoragarh and Almora districts & Valley of Flowers in Uttarakhand.	Herbaceous species and scrub communities such as Rhododendron. Plant species including lichens,	Snow leopard, Himalayan black bear, Brown Bear, Musk deer, Bharal/blue Sheep, Asiatic black bear,	Bhotia tribe

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		fungi, bryophytes and pteridophytes	Himalayan Tahr, Koklas Pheasant.	
NOKREK (Included in MAB list of UNESCO)	Part of East, West and South Garo Hill districts in Meghalaya.	Evergreen and semi-evergreen deciduous forests dominate the landscape	Slow Loris, Giant flying squirrel, Pig-tailed macaque, Red Panda (Sighted only once), leopards, elephants, Hoolock gibbons.	Garo (Achikmande), Banias or Hajjons
MANAS	Part of Kokrajhar, Bongaigaon, Barpeta, Nalbari, Kamprup and Darang districts in Assam.		Golden Langur, Red Panda, Pygmy Hog, Hispid Hare	
SUNDARBANS (Included in MAB list of UNESCO)	Part of delta of Ganges & Brahmaputra River system in West Bengal.	Tropical humid forest; Mangroves, Sundari Tree	Royal Bengal tiger, Salvator Lizard, Bengal Monitor Lizard.	
GULF OF MANNAR (Included in MAB list of UNESCO)	Part of Gulf of Mannar extends from Rameswaram island in North to Kanyakumari inSouth of Tamil Nadu. There are 21 Islands	Islands including coastal/marine components; coral reefs and mangrove, seagrass beds (Halophila gas), coral reefs	Dugong or Sea Cow, Sea cucumber, Sea Fan	Marakeyars, local people mainly engaged in fishing
GREAT NICOBAR (Included in MAB list of UNESCO)	Southernmost island of A&N Islands. It incorporates two national parks Campbell Bay National Park and Galathea National Park.	Part of Sundaland Biodiversity Hotspot, Tropical Wet Evergreen Forests.	Saltwater Crocodile, Edible-nest swiftlet, Nicobar tailed macaque, Giant Leatherback sea turtle (Only breeding site), Nicobar tree shrew, Nicobar scrub fowl, Serpent Eagle, Crab Eating Macaque	Shompen and Nicobarese
SIMILIPAL (Included in MAB list of UNESCO)	Part of Mayurbhanj district in Orissa.	The park derives its name from the abundance of Semul (red silk cotton trees) that grow here. Orchids, medicinal plants,etc.	Asiatic Elephant,Gaur, Royal Bengal Tiger, Wild elephant. Mugger Crocodile management program was launched here.	Erenga Kharias and Mankirdias, Ho, Gonda and Munda, etc.

DIBRU-SAIKHOVA	Part of Dibrugarh and Tinsukia districts in Assam.	Riverine island amid Brahmaputra	Golden Langur, Only Place in India to find wild horses	
DEHANG-DIBANG	Part of Upper Siang, West Siang and Dibang Valley districts in Arunachal Pradesh.		Mishmi takin, Red goral, musk deer, Red Panda, Asiatic Black bear, Green Pit Viper, Takin.	
PACHMARHI (Included in MAB list of UNESCO)	Satpura Hills runs across it. Covers three protected areas – Satpura National Park, Bori and Pachmarhi Wildlife Sanctuary	Sal Forests	Barasinga, Gaur, Bears, Tigers and leopards, Giant Squirrel and Crested, Flying Squirrel.	Gond, Korkus- tribes introduced the cultivation of potatoes and made use of honeycombs to produce honey in significant quantities for commercial use. Most primitive Bhariya tribes are found here.
KHANGCHENDZONGA (Included in MAB list of UNESCO)	Part of North and West districts in Sikkim.		Snow Leopard, Red Panda	
AGASTHYAMA-LAI (Included in MAB list of UNESCO)	Covers Peppara and Shendurney wildlife sanctuaries and parts of Neyyar sanctuary in Kerala and Kalakad Mundanthurai Tiger Reserve of Tamil Nadu.	Tropical Wet Evergreen Forests.	Lion Tailed Macaque, Slender Loris, Great Pied Hornbill, Nilgiri Tahr, Elephants, Tiger	**Kani tribes** from both Tamil Nadu and Kerala
ACHANAKMAR-AMARKANTAK (Included in MAB list of UNESCO)	Maikala hills of Satpura range pass through it. It separates the rivers that drain into the Arabian Sea and Bay of Bengal. Source of three rivers: Narmada, Son and Johila.		Four horned antelope, Indian wild dog, Saras crane , Asian white-backed vulture, Sacred grovebush frog ,striped Hyaena, , Chital, Wild Bear, Leopard.	Amarkantak, the site of the origin of Son, Johilla and Narmada rivers, is in it.
KACHCHH	Part of Kachchh, Rajkot, Surendranagar	Banni Grasslands, Prosopis juliflora (Native of Central	Indian Wild Ass, Site for Flamingo breeding (Flamingo	Fossil Park at Khadir Bet,

	and Patan districts in Gujarat.	America, Invasive Alien Species).	City), Chinkara, Caracal, Desert Cat, Desert Fox	Maldhari pastoralists
COLD DESERT	Pin Valley National Park, Chandratal, Sarchu Kibber Wildlife sanctuary in Himachal Pradesh.	Deodar tree	Snow Leopard	
SESHACHALA--M	Seshachalam hill ranges in Eastern Ghats.	Tropical dry deciduous forests, Red Sanders	Slender Loris, Indian giant squirrel, Mouse deer Golden Gecko, Yellow throated bulbul.	Tirupati Balaji temple is located here.
PANNA (Included in MAB list of UNESCO)	Part of Panna & Chhatarpur districts in Madhya Pradesh	Dry deciduous forests of Teak, Salai, Kardhai.	Tiger, Chital, Chinkara, Sambhar, Sloth bear	Gond, Famous temple of Prannathji of Pranami Sect.

7. NATURAL WORLD HERITAGE SITES

Created in 1972, the primary mission of the Convention is to identify and protect the world's natural and cultural heritage considered to be of Outstanding Universal Value.

It is governed by the World Heritage Committee supported by UNESCO World Heritage Centre, it meets annually. IUCN is an Advisory Body on natural heritage.

NATURAL WORLD HERITAGE SITES FROM INDIA

- **Great Himalayan National Park Conservation Area, Himachal Pradesh**: This national park is in Western Himalayan region. It is characterised by alpine peaks, alpine meadows & riverine forests. Its glaciers are sources of rivers like Sainj, Jiwa Nal, Tirthan and Parvati Rivers (all tributaries of Beas River). Covers: Great Himalayan National Park, Sainj Wildlife Sanctuary and Tirthan Wildlife Sanctuary. Important Flora & Fauna: Snow Leopard, Musk Deer, Himalayan Serow. Himalayan Yew forms the undergrowth of forests.
- **Kaziranga National Park, Assam:** Last unmodified natural areas in northeast India. This park accounts for 2/3rd of the world population of one-horned rhinoceros. West alluvial grassland formed by elephant grass occupies most of the area. It is also a tiger reserve. Fauna: Tiger, One-Horned Rhinoceros, Wild Water Buffaloes, Ganges River Dolphin, Hoolock Gibbon.
- **Keoladeo National Park: (Covered in Wetlands)**
- **Manas Wildlife Sanctuary:** It is located on the borders of Bhutan and spans districts of Kokrajhar, Chirang, Buxa and Udalgiri in North-West Assam. It is separated from the Royal Manas National Park of Bhutan by the River Manas, and it is separated by Buxa Tiger Reserve of West Bengal by River Sankosh. It forms part of a large conservation landscape which includes the Buxa-Nameri-Pakke-Namdapha tiger and protected areas of Bhutan and Myanmar. Flora: Sal Forests, Bhabar Savannah, Dry deciduous forests. Fauna: Tiger, Pygmy Hog, One-horned Rhinoceros and Elephant, Bengal Florican. The Wild Buffalo population is probably the only pure strain.
- **Nanda Devi and Valley of Flowers National Parks:** Valley of Flowers National Park is renowned for its meadow of endemic alpine flowers and outstanding natural beauty. The landscape is dominated by Nanda

Devi peak which approaches through RishiGanga gorge. Fauna: Asiatic Black Bear, Snow Leopard, Brown Bear and Blue Sheep, Himalayan Musk Deer.

- **Sundarbans National Park:** It is a biosphere reserve, national park and tiger reserve. It houses saltwater crocodiles. Some common species of plants which are found include Sundari tree, Gulati, Champa, Dhundul, Genwa and Hetal. Apart from the Royal Bengal Tiger, other animals found in these areas are fishing cats, macaques, leopard cats, Indian grey mongoose, wild boar, flying fox, pangolin, and Indian grey mongoose. The chital deer and rhesus monkey are common sightings. A species of river turtles called Batagurid Baska (which are classified as endangered by the IUCN) are found on Mecham Beach. They are identified by their small head, a snout which always goes upwards and an olive brown colored carapace. The barking deer also deserves a special mention, as it is found at a place called Halliday Island.
- **Western Ghats:** It has an exceptionally high level of biological diversity and endemism and is recognized as one of the world's eight 'hottest hotspots' of biological diversity. The forests of the site include some of the best representatives of non-equatorial tropical evergreen forests anywhere. The Ghats traverse the States of Kerala, Tamil Nadu, Karnataka, Goa, Maharashtra and Gujarat.
- **MIXED HERITAGE SITE (BOTH NATURAL + CULTURAL):** Khangchendzonga National Park
- **CULTURAL HERITAGE SITE:** Sacred Ensembles of the Hoysalas (2023) – Karnataka.

NATURAL SITES AS PART OF TENTATIVE SITES

These sites do not have World Heritage Status as of now but have been submitted by India for inclusion in the list.

- **Majuli River Island, Assam:** It is a fluvial riverine island. It is the first island to be made a district in India. The island is formed by the Brahmaputra River in the South and Kherkutia Xuti, a branch of Brahmaputra, joined by Subansiri in the north. The island is the cultural capital of Assamese culture whose foundation is laid by Srimanta Sankardeva. Jadhav Payeng, an environmental activist, also known as forest man of India, has planted Molai Forest which one of the largest afforested this forest to protect Majuli from riverine erosion.
- **Namdapha National Park, Arunachal Pradesh:** This national park along with Kamlang Wildlife Sanctuary is in the eastern extremity of Arunachal Pradesh, close to the trijunction of India, China & Myanmar. It is located between Patkai bum and Mishmi hills. The park has northernmost lowland evergreen rainforests in the world along with alpine forests in the higher reaches. It is the fourth largest national park in India by area. Dihing River, a tributary of Brahmaputra passes through it. Dapha Bum (4571 m) is the highest point of this national park. Fauna: Namdapha Flying Squirrel (IUCN status: Critically Endangered) is endemic to it. The park boasts of four large cats: Snow Leopard (IUCN status: Vulnerable), Leopard, Tiger and Clouded Leopard (IUCN status: Vulnerable).
- **Wild Ass Sanctuary, Little Rann of Kutch, Gujarat:** One of the last places where endangered wild ass sub-species Indian Wild Ass (IUCN Status: Near Threatened) can be spotted.
- **Neora Valley National Park, Kalimpong, West Bengal:** It is a compact patch of virgin late succession forests, rich in biodiversity located in the Eastern Himalayan region. Altitude varies between 183 m to 3200 m. It has Temperate & Sub-tropical forests. Fauna: Red Panda (IUCN Status: Endangered)
- **Desert National Park, Rajasthan:** It is one of the Hot deserts of the world with highest human density. It is spread across two districts of Jaisalmer and Barmer. Fauna: Only landscape having breeding population of Great Indian Bustard. Spiny Tailed Lizards. Desert Fox, Chinkara, Desert Cat Flora: Sewan Grass found here which is a major source of nutrition for birds and animals found here. Rohida, the state flower of Rajasthan is found here. Orans are grasslands found here. Khejri tree is commonly found here and protected by a local community called Bishnois. There are settlements in the National Park called Dhanis.
- **Apatani Cultural Landscape: Ziro Valley in Arunachal Pradesh**
- **Chilika Lake (Odisha):** It is a brackish water lake & a shallow lagoon with estuarine character spread across districts of Puri, Khurda & Ganjam. Located on the mouth of Daya River. It is the largest coastal lagoon in India & largest brackish water lagoon in the world and largest saltwater lake in India. Fauna: Irrawaddy Dolphins.

- **Narcondam Island, A&N islands:** It is a separate island part of A&N islands. This volcanic island has an endemic population of Narcondam Hornbill (Only found here) which has the smallest range of all Asian Hornbills.
- **Cold Desert Cultural Landscape of India:** It stretches in the Himalayas from Ladakh to Kinnaur in Himachal Pradesh. This region is known for its harsh climate due to its high altitude and location on the leeward side of Himalayas, which makes it a rain-shadow zone. This area is inhabited by Indo-Mongoloid people who are Buddhist in faith. Buddhist monasteries known as Gompas with a trademark prayer flag fluttering on top. (Important Monasteries in the region are:
- **Satpura Tiger Reserve, MP:** It is a prime example of central Indian highland eco-system. Species from both Himalayan and Western Ghats meet here. Pachmarhi, a famous tourist destination is located near it and Dhoopgarh, the highest peak of Madhya Pradesh, is located in it. Pandav Caves which are a group of 5 caves.
- **Keibul Lamjao:** (Covered in Wetland as Loktak Lake)
- **Garo Hills Conservation Area:** It accompanies three protected areas: Nokrek National Park, Bapakram National Pak & Siju Sanctuary. Garo mountains and its caves give an idea about earth's evolutionary history. Fauna: Insectivorous plants such as Sundew and Pitcher Plant. National Citrus Gene Sanctuary in Nokrek is noted for wild varieties of citrus fruits. Indian Wild Orange found here is considered to be the progenitor of all citrus species in the world. Fauna: Hoolock Gibbon.
- **Bhedaghat-Lametaghat in Narmada Valley, MP:** Bhedaghat (Dhuandhar waterfalls), known as Grand Canyon of India, is in Jabalpur, MP. It appears as if smoke is coming out of the river. The site has outstanding beauty of marble rocks. These magical marble mountains assume different colors and even shapes of animals and other living forms as one moves through them.
- **Living Root Bridges Cultural Landscapes:** Locally known as Jingkieng Jri. They are ficus based rural connectivity and livelihood solutions within dense moist forests in Meghalaya. Roots of Indian Rubber Tree are engineered for the construction of these structures. They are grown by Khasi tribal communities.
- **Geoglyphs of Konkan region:** Geoglyphs are rock art produced on the surface earth (open-air) either by positioning rocks, rock fragments or by carving out or removing part of a rock surface to form a design. The geoglyphs found in the Konkan region are the only evidence of prehistoric human occupation in this region. The geoglyphs have been prominently made in the period starting from Prehistoric to Mesolithic period. Important sites where Geoglyphs are found in Konkan region. Important sites where geoglyphs are found in Konkan region: **(a)** Jambhrun (Maharashtra) **(b)** Ukshi (Maharashtra) **(c)** Kasheli (Maharashtra): Largest rock engraving in India (18X13 in Elephant) **(d)** Rundhetali (Maharashtra) **(e)** Devi Hasol (Maharashtra) **(f)** Barsu (Maharashtra) **(g)** Devache Gothane (Maharashtra) **(h)** Kudopi (Maharashtra).

Note: Despite being older than the Himalayas and the Western Ghats, the Eastern Ghats, an ancient discontinuous low mountain range that spreads along the East coast of the Indian Peninsula, never got its due. The geographical extent of the Eastern Ghats is about 75,000 kms spread over the states of Odisha (25 %), Andhra Pradesh (40%), Telangana (5%), Karnataka (5%) and Tamil Nadu (25%). Though it is bestowed with rich biodiversity and is home to different tribal communities, there has never been a clear policy in place for its conservation.

8. HOLLONGAPAR GIBBON SANCTUARY

Hollongapar Gibbon Sanctuary has been split in two by a railway track running through it. The presence of the railway track splits the gibbons' habitat and impedes their ability to move freely throughout the sanctuary area.

HOLLONGAPAR GIBBON SANCTUARY

- The establishment of the Hollongapar Gibbon Sanctuary in 1997 was largely motivated by the objective of conserving the Hoolock Gibbon species and its natural environment.
- The Hollongapar Gibbon Sanctuary is in the Jorhat district of Assam, near the town of Mariani.
- The geographical extent of the region is roughly 20.98 square kilometres (8.11 square miles).

- The sanctuary exhibits distinct features of semi-evergreen and mixed deciduous forest, hence providing a wide array of habitats that support a broad variety of plant and animal species.

BIO-DIVERSITY

- The sanctuary is well recognized for its significant contribution to the preservation efforts of the Western Hoolock Gibbon (Hoolock hoolock), a species that has been classified as Endangered according to the IUCN Red List.
- In addition to gibbons, the sanctuary accommodates a diverse array of primate species, such as the Stump-tailed Macaque and Capped Langur.
- The avian richness in the area is notable, showcasing a wide array of bird species like the Great Hornbill, Green Imperial Pigeon, and White-cheeked Partridge.

VEGETATION

- The sanctuary's flora comprises semi-evergreen and mixed deciduous woods, characterized by a diverse array of tree species such as Holong (Dipterocarpus macrocarpus), which lends its name to the sanctuary.
- The presence of various forest types plays a crucial role in providing essential habitats for both permanent and migratory species.

9. GREAT NICOBAR ISLAND PROJECT

- The Great Nicobar Island (GNI) Project is a mega project to be implemented at the southern end of the Andaman and Nicobar islands.
- The project will include, among others, an International Container Transhipment Terminal (ICTT), a Greenfield international airport, a township, and a 450 MVA gas- and solar-based power plant over 16,610 hectares in the Great Nicobar Islands.

CONCERNS DUE THE PROJECT

- **Ecological Impact:** Great Nicobar Islands hosts the **Great Nicobar Biosphere Reserve**, recognised by UNESCO in 2013. The infrastructure project entails felling of over 1 million trees and poses risks to endemic species (E.g., Nicobar megapode, leatherback turtles).
- **Tribal Rights:** Affects **Shompen** (PVTG) and **Nicobarese** tribes by disrupting their traditional lifestyle and exposing them to external influence and diseases.

UNESCO'S MAN AND BIOSPHERE PROGRAMME 2013

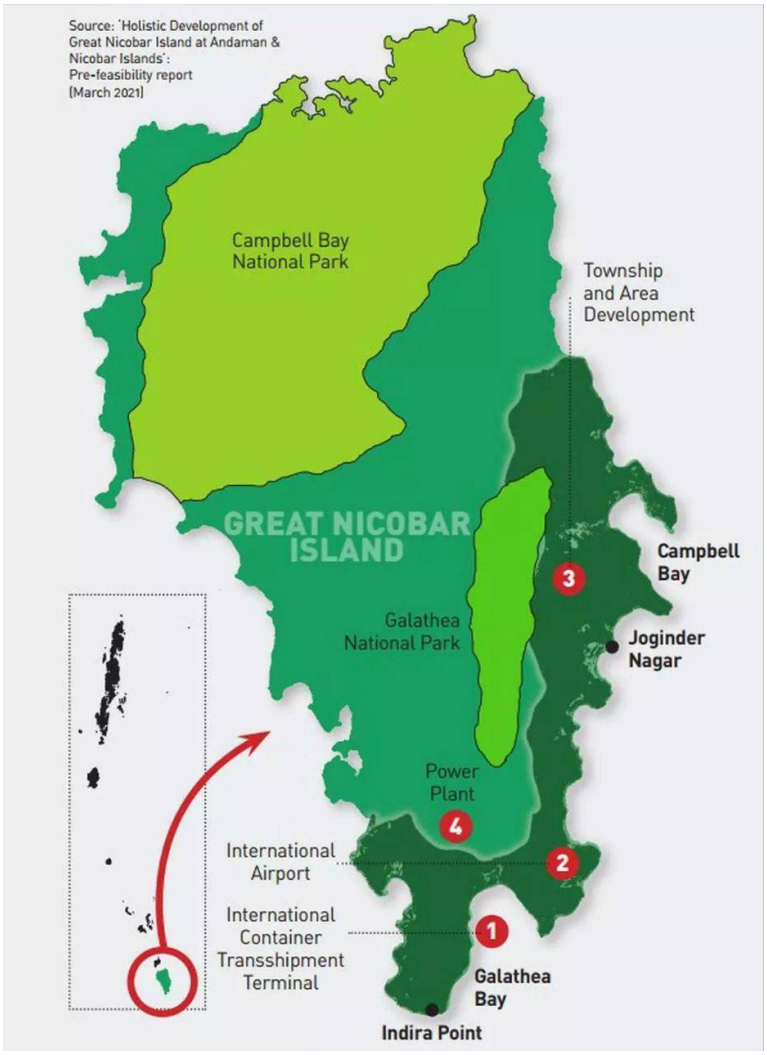
- The GNI was declared a biosphere reserve in 1989, and included in UNESCO's Man and Biosphere Programme in 2013.
- It has an unparalleled array of microhabitats- sandy and rocky beaches, bays and lagoons, littoral patches with mangrove communities, evergreen and tropical forests, and more.
- These habitats host numerous species, including marine animals, reptiles, birds, mammals, trees, ferns, insects, crustaceans, and amphibians.
- Several of these, like the Nicobari Megapode, are endemic to GNI and found nowhere else in the world.

GREAT NICOBAR ISLAND

- Great Nicobar, the southernmost of the **Andaman and Nicobar Islands**, has an area of 910 sq km.
- The Andaman and Nicobar Islands are a cluster of about 836 islands in the eastern Bay of Bengal, the two groups of which are separated by the 150-km wide **Ten Degree Channel.**
- The Andaman Islands lie to the north of the channel, and the Nicobar Islands to the south.
- **Indira Point** on the **southern tip of Great Nicobar Island** is **India's southernmost point,** less than 150 km from the northernmost island of the Indonesian archipelago.

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- The island comprises unique and threatened tropical evergreen forest ecosystems. The region is noted for its rich biodiversity and fosters **several rare and endemic species.**
  - The endemic species comprise 11 species of mammals, 32 species of birds, 7 species of reptiles and 4 species of amphibians.
  - Of these, the well-known Crab-eating Macaque, Nicobar Tree Shrew, Nicobar Megapode, are endemic and/or endangered.
  - It is home to 650 species of angiosperms, ferns, gymnosperms, bryophytes, among others.
- One of the most unique aspects of Great Nicobar is the southernmost point, the **Galathea Bay, a nesting ground for the Leatherback Turtle.**
- The primary human inhabitants of the island are the **Shompen and Nicobarese tribes** who have been living on the island for countless generations.
  - The **Shompen tribe** are an aboriginal people of about 200 to 300 members who inhabit the interiors of Great Nicobar.
  - This **particularly vulnerable tribal group (PVTG)** is a semi-nomadic hunter-gatherer community, practicing basic horticulture and pig rearing, who probably migrated around 10,000 years ago.
  - They are anthropologically completely distinct from the other PVTGs of the Andaman Island as well as from the Nicobarese. Their language, Shompanese, is unrelated to any other existing language.



10. INDIA STATE OF FOREST REPORT 2023

Recently, the Ministry of Environment, Forest and Climate Change has released the **'India State of Forest Report 2023'** (ISFR 2023). The report shows a marginal gain of 156 sq. km in forest cover, and a sizable increase of 1,289 sq km in tree cover since 2021.

What is the India State of Forest Report?

- The India State of Forest Report is a **biennial report** published by the **Forest Survey of India** (FSI), an organisation of the Ministry of Environment, Forest & Climate Change.
  - FSI monitors India's forest and tree resources through periodic assessments and presents the findings in its biennial report.
- **The first State of Forest report was brought out in 1987.** The current report, ISFR 2023 is 18th in the series.

KEY TERMS

- **Tree cover:** It is defined as all tree patches of size less than one hectare occurring outside the recorded forest area. This covers trees in all formations including scattered trees.

- **Forest area:** It denotes the legal status of the land as per the government records, whereas the term 'forest cover' indicates presence of trees over any land.
- **Forest carbon stock:** It is the amount of carbon that has been sequestered from the atmosphere and is now stored within the forest ecosystem, mainly within living biomass and soil, and to a lesser extent also in dead wood and litter.
- **National Forest Inventory:** A systematic approach to generate national level estimates on growing stock, forest area and other forest resources parameters by doing regular inventory in selected sample districts in a cycle of two years.

Table: Forest and Tree Cover of India

in km^2

Class	Area	Percentage of GA
Forest Cover	7,15,342.61	21.76
Tree Cover	1,12,014.34	3.41
Total Forest and Tree Cover	**8,27,356.95**	**25.17**
Scrub	43,622.64	1.33
Non Forest	24,16,489.29	73.50
Geographical Area of the country	**32,87,468.88**	**100.00**

KEY FINDINGS OF ISFR 2023

- The Forest **and Tree cover** of India is **8,27,357 sq km** which is 25.17% of the geographical area of the country, consisting of:
  - 7,15,343 sq km (21.76%) as forest cover.
  - 1,12,014 sq km (3.41%) as tree cover.

For the first time, **India's green cover has exceeded the 25% threshold**. Of this, 49.57% (4,10,175 sq km) is classified as **dense forests.**

- **Increase in tree cover and forest cover:**
  - Sharpest **growth (0.5%) in tree cover** (from 2.91% in 2021 to 3.41% in 2023)
  - Marginal growth in **forest cover (0.05**%) since 2021.
    - Between 2003 and 2013, forest cover increased by 0.61%, from 20.62% to 21.23%.
    - In the next 10 years, it grew by only 0.53% to 21.76%.
- **Total mangrove cover** is 4,992 sq km in the country.
- **Total bamboo bearing area** is 1,54,670 sq km. (an increase from 2021).
- **Total carbon stock in the country's forest** is estimated to be **7,285.5 million tonnes** (an increase of 81.5 million tonnes as compared to 2021).
- **India's carbon stock has reached 30.43 billion tonnes of CO2 equivalent;** which indicates that as compared to the base year of 2005, India has already reached 2.29 billion tonnes of additional carbon sink as against the target of 2.5 to 3.0 billion tonnes by 2030.

Rankings of States & UTs:

- **Top four states showing maximum increase in forest and tree cover:** Chhattisgarh (684 sq km) > Uttar Pradesh (559 sq km) > Odisha > Rajasthan.
- **Top three states showing maximum increase in forest cover:** Mizoram (242 sq km) > Gujarat > Odisha.
- **Area wise top three states having largest forest and tree cover:** Madhya Pradesh (85,724 sq km) > Arunachal Pradesh > Maharashtra.
- **Area wise top three states having largest forest cover**: Madhya Pradesh (77,073 sq km) > Arunachal Pradesh > Chhattisgarh.

- **In terms of percentage forest cover with respect to total geographical area:** Lakshadweep (91.3 %) > Mizoram (85.3 %) > Andaman & Nicobar Island.
- **19 states/UTs** have **above 33%** of the geographical area under forest cover.
  - According to the **National Forest Policy of 1988**, 33% of India's geographical area is required to be forests.
- **8 states/UTs:** Mizoram, Lakshadweep, A & N Island, Arunachal Pradesh, Nagaland, Meghalaya, Tripura, and Manipur **have forest cover above 75%.**

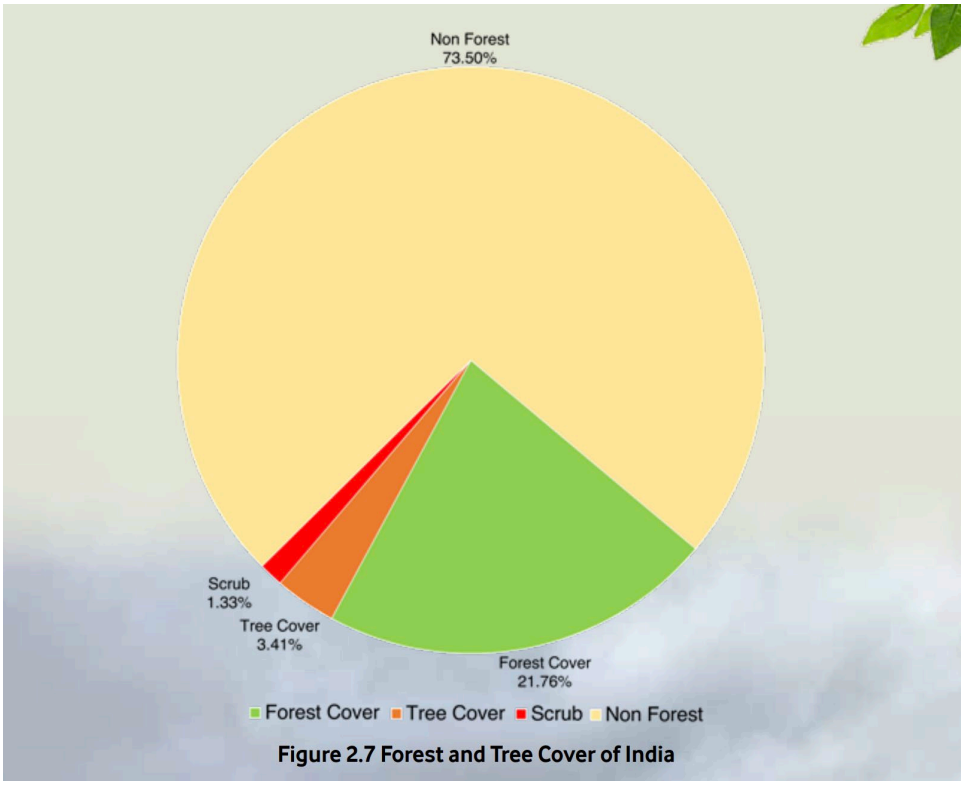


Figure 2.7 Forest and Tree Cover of India

TYPES OF FORESTS

- **Forests:** Irrespective of land use or ownership, **tree patches measuring 1 hectare or more with a minimum canopy cover of 10% are counted as forests in India.**
- **Dense forests**: Areas with a **canopy density of 40% and above are considered dense forests**
- **Open forests**: Areas with **canopy density of 10-40% are open forests (OF).**
- **Very dense forests**: Since 2003, areas with **at least 70% canopy density have been classified as very dense forests (VDF).**

****Canopy density:** It is the percentage of an area that is covered by the crowns of trees. It's also known as forest canopy density (FCD) or crown cover. Canopy density is a key indicator of a forest's health and is used to assess the forest's condition and potential management actions.

CONCERNING FINDINGS OF THE REPORT

- ISFR-2023 shows that **3,913 sq km of dense forests (an area larger than Goa) have disappeared** in India in just two years since 2021.
- India has witnessed the **complete destruction of 24,651 sq km (more than 6.3%) of its dense forests in the two decades since 2003.**
- The **bulk of this loss has been offset by the rapid transformation of 15,530 sq km of non-forested or scantly forested land to dense or even very dense forests** (during 2003-2023).
- However, **this offset has been accomplished through Plantations,** because natural forests do not grow this fast.

CHANGE IN DENSE FOREST COVER 2003-2023

LOST	2021-23	2003-13	2013-23	2003-23
VDF to NF	295	288	1,277	1,565
MDF to NF	3,362	6,714	15,086	21,800
VDF to scrub	24	5	65	70
MDF to scrub	313	144	1,153	1,297
Disappeared	**3,994**	**7,151**	**17,581**	**24,732**
VDF to OF	228	134	1,128	1,262
MDF to OF	5,166	6,414	22,249	28,663
TOTAL LOSS	**9,388**	**13,699**	**40,958**	**54,657**

GAINED	2021-23	2003-13	2013-23	2003-23
NF to VDF	56	43	483	526
NF to MDF	839	3,631	7,554	11,185
Scrub to VDF	1	0	54	54
Scrub to MDF	102	105	1,043	1,148
OF to VDF	496	124	2,567	2,691
Plantations	**1,494**	**3,903**	**11,701**	**15,604**
OF to MDF	8,610	6,122	34,301	40,423
TOTAL GAIN	**10,104**	**10,025**	**46,002**	**56,027**

NET CHANGE | 2021-23: **716** | 2003-13: **-3,674** | 2013-23: **5,044** | 2003-23: **1,370**

Source: ISFR 2003-23

VDF: Very Dense Forest (canopy over 70%); MDF: Moderate Dense Forest (canopy 40-70%); OF: Open Forest (canopy 10-40%); Shrub (canopy under 10%); NF: Non-Forest (no canopy)

CONUNDRUM OF DENSE FORESTS

- **Areas under plantations-as-dense-forests are expanding** as the disappearance of natural dense forests becomes routine.
- Large swaths of **Open forest (OFs) become Moderately dense forest (MDFs) in the last decade** owing to better management. At the same time**, plantations are supplementing these natural gains** to keep the extent of India's dense forest cover stable.
- Though, India's dense forest cover has grown by 6% during 2003-2023. Experts question such an increase as they were brought through a **series of unexplained revisions of data** presented in ISFR adding a total of 20,232 sq km of dense forest to the inventory.

PLANTATIONS AS NATURAL FOREST – ISSUES AND IMPLICATIONS

- Plantations usually have **trees of the same age (and often the same species), are vulnerable to fire, pests and epidemics.**
- Plantations **often act as a barrier to the regeneration of natural forests** which are more biodiverse, perform a wider range of ecological functions, and support numerous species.
- **Old natural forests stock a lot more carbon in their frame and in the soil.** In 2018, the United Nations Framework Convention on Climate Change (UNFCCC) flagged India's assumption that plantations reach the carbon stock level of existing forests in just eight years.
- **Plantations are frequently promoted** for their rapid growth which can achieve carbon targets faster. However, **plantations are often harvested more readily**, defeating climate goals in the long term.

11. SACRED GROVES

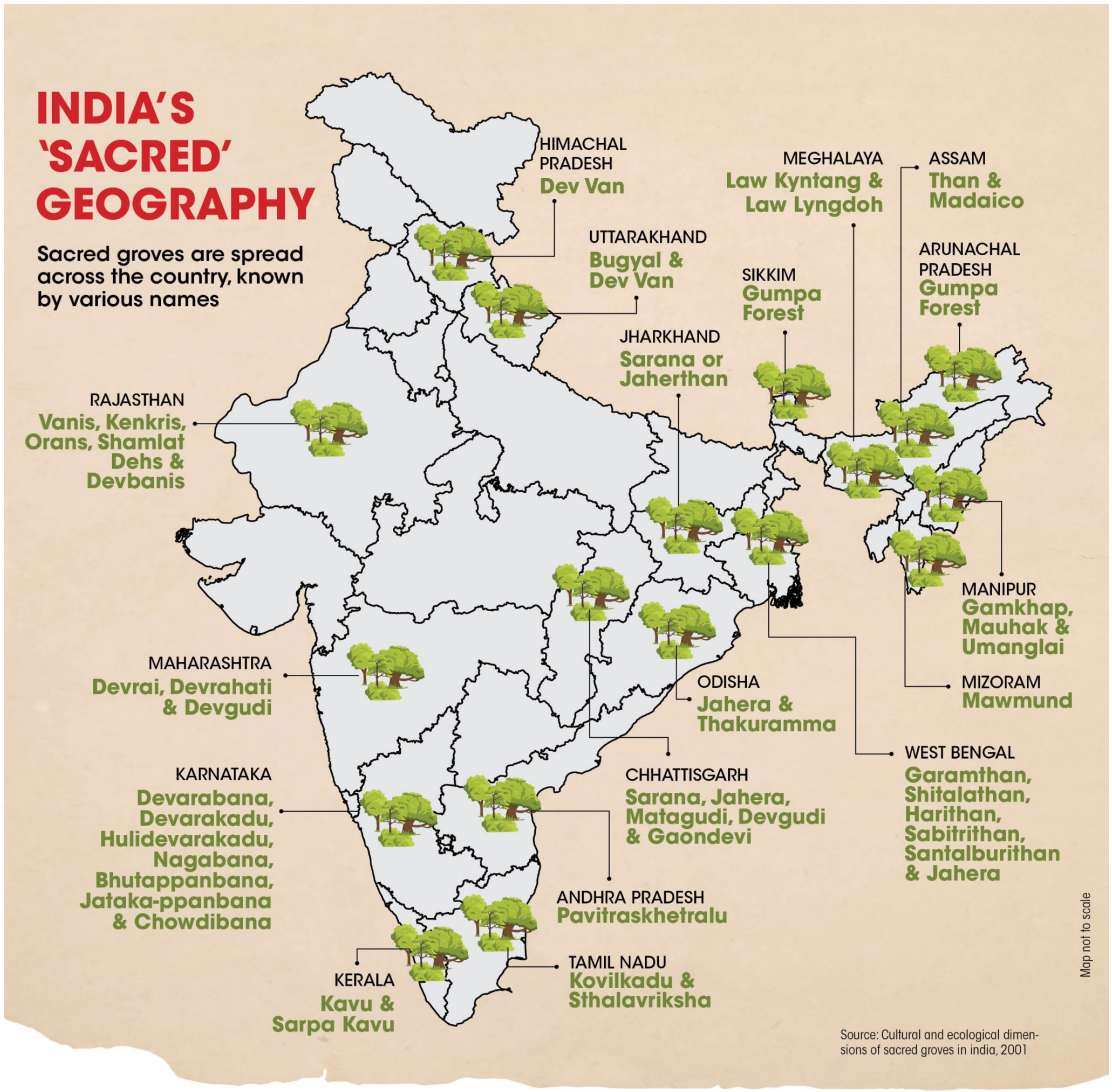
The Supreme Court has directed the Central Government to create a comprehensive policy for the management of sacred groves across the country.

WHAT ARE SACRED GROVES?

- **UNESCO (1996)** defined Sacred Groves as an **area of natural vegetation preserved through local taboos and sanctions** that entail spiritual and ecological values.
  - Protected by local communities for their religious and cultural significance.
  - Important sites for worship and community gatherings.
  - Logging and hunting are strictly prohibited.
- **Found all over India** - especially in Maharashtra, Karnataka, Kerala, and Tamil Nadu. There are around 14,000 sacred groves reported across India.

They are known by different names in different parts of India.

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IMPORTANCE OF SACRED GROVES

- Repository for **various Ayurvedic medicines.** It is a source of replenishable resources like fruits and honey.
- Harbor **great genetic diversity.** They have **become biodiversity hotspots**, as various species seek refuge here, due to habitat destruction and hunting elsewhere.
- Often associated with **ponds and streams**, and meet water requirements of local communities. They help in recharging aquifers as well.
- Prohibition of cutting trees helps **reduce soil erosion** and prevents desertification.
- Act as **lungs in urban landscapes,** providing much needed vegetation cover.

MANAGEMENT OF SACRED GROVES

- Management of wildlife and its habitat is **primarily the responsibility of the States**. The Wild Life (Protection) Act 1972 has empowered States for declaration of any private or community land, as a **community reserve**, for protecting fauna, flora and traditional or cultural conservation values and practices. The preservation of sacred groves is aligned with the **cultural and traditional rights of entire communities.**
- The SC has asked the Ministry of Environment, Forest & Climate Change to develop a **plan for a nationwide survey of sacred groves** to identify the area and extent of the groves. The boundaries of the groves have to be clearly marked but left flexible to accommodate future growth of the forests.

12. AREAS IN NEWS

Papikonda National Park (Andhra Pradesh)	River Godavari passes through it. Part of the National Park is proposed to be submerged due to the Polavaram Hydroelectric Project.
Panna National Park (Madhya Pradesh)	• Some parts of the park will sink due to Ken-Betwa River linkage project and construction of Daudhan dam. • Tiger Reserve
Shoolpaneswar Wildlife Sanctuary (Gujarat)	Located close to Surat city. Located in Western Satpura Range (Rajpipla hills). It is located on the boundary of Madhya Pradesh and Maharashtra. Zarwani Waterfall is in it. It was initially established as a protected area for protection of Sloth Bear. It is also home of Flying Squirrels (IUCN status) The area is predominantly tribal with Vasavas as the main tribal community.
Simlipal National Park (Odisha)	• Famous for sightings of a slightly black coloured Melanistic breed of tigers. • Mankidia tribes live there. • The park derives its name due to red silk cotton trees.
Karlapat Sanctuary (Odisha)	Recently, 7 elephants died in the wildlife sanctuary due to hemorrhagic septicemia which is a bacterial infection.
Marayoor Sandalwood Reserves, Kerala	Situated in Kerala, this reserve is known for its high-quality sandalwood in India. The reserve has many wild animals like elephants and Indian bison. Owing to high oil content and high amount of hardwood, marayoor sandals have huge global demand. This reserve is crucial in connecting several national parks like Eravikulam, Chinnar, Kurinjimala, Anamudi and Pambadum sholai.
Kuno Palpur National Park	Located in Madhya Pradesh. It is widely believed that the Kardhai tree, which is found in abundance here, turns green even with just the presence of humidity in the atmosphere, even before the arrival of first monsoon showers. It has a combination of dry deciduous, dry savannah and tropical riverine forest.
Loktak Lake & Keibul Lamjao National Park (Manipur)	The Loktak Hydroelectric Project has disturbed the cycle of phumdis as it constantly keeps the water level in the lake high. As a result, *phumdis* are unable to feed from the nutrients on the bed, they are thinning out and even breaking away. **Loktak and Phumdis** • Largest freshwater lake in northeast, located in Moirang, Manipur. • It is famous for the floating biomass islands or phumdis: The phumdis float during the rains and sink during the dry months, sucking nutrients from the lakebed to replenish their roots and float again when the next monsoon cycle begins. • Loktak Lake is a 'wetland of international importance' under the Ramsar Convention.

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	• The Ministry of Shipping has given approval for development for the Loktak Inland waterway project. • Keibul Lamjao National Park lies in the heart of the lake. It is the only floating national park in the world. It is home to the endangered brown-antlered deer, sangai, whose habitat is also under threat.
Aravali Safari Park Project (Haryana)	• Ambitious project of the Haryana government. • The project is envisaged to be the world's largest safari park, which will spread across 3,858 hectare (approximately 10000 Acre) across Gurugram and Nuh districts.

13. WETLAND CITY ACCREDITATION (WCA) UNDER RAMSAR CONVENTION

- It is **a voluntary accreditation** system established under **Ramsar Convention during Conference of Parties -12 (COP-12) held in 2015.**
- It r**ecognises** cities which have taken **exceptional steps to safeguard their urban wetlands**.
- This voluntary scheme **provides an opportunity** for cities that value their natural or human-made wetlands **to gain international recognition and positive branding opportunities** for their efforts in demonstrating strong positive relationships with wetlands.
- To be formally accredited, a candidate for the Wetland City Accreditation should satisfy the standards used to implement each of the **six international criteria mentioned in Operational Guidance for WCA** of the Ramsar Convention on Wetlands.
- Criteria for a city or a human settlement to be accredited as Wetland City Accreditation:
  - **Criterion 1:** It has one or more Ramsar Sites, or other wetland conservation sites fully or partly situated within its jurisdiction, which provide(s) a range of ecosystem services to the city.
  - **Criterion 2:** It has adopted measures for conservation of wetlands and their ecosystem services.
  - **Criterion 3:** It has implemented wetland restoration and/or management measures.
  - **Criterion 4:** It considers the challenges and opportunities of integrated spatial/land-use planning for wetlands under its jurisdiction.
  - **Criterion 5:** It has raised public awareness about the values of wetlands by delivering locally adapted information, and enabled participation of local stakeholders in decision making processes.
  - **Criterion 6:** It has established a local committee with appropriate knowledge and experience on wetlands and representation of and engagement with stakeholders to support the preparation work to apply for the Wetland City Accreditation of the Ramsar Convention and the implementation of proper measures to maintain the city's qualifications for the accreditation.

WETLAND ACCREDITED CITIES IN INDIA

- **Indore** and **Udaipur** have become the first two Indian cities to make it to the global list of accredited wetland cities under the Ramsar Convention on Wetlands.
- **Indore:** Founded by Holkar, Indore is the cleanest city in India and the recipient of India's Smart City Award 2023 for its best sanitation, water, and urban environment.
  - **Sirpur Lake**, a Ramsar Site in the city, has been recognised as an important site for water bird congregation and is being developed as a Bird Sanctuary.
  - **Yashwant Sagar** is a Ramsar site located close to Indore.
  - A strong network of more than 200 wetland mitras is engaged in bird conservation and sensitising the local community to protect Sarus Crane.
- **Udaipur:** Located in Rajasthan, the city is surrounded by five major wetlands, namely, **Pichola, Fateh Sagar, Rang Sagar, Swaroop Sagar, and Doodh Talai**.

- These wetlands are an integral part of the city's culture and identity, help maintain the city's microclimate, and provide a buffer from extreme events.

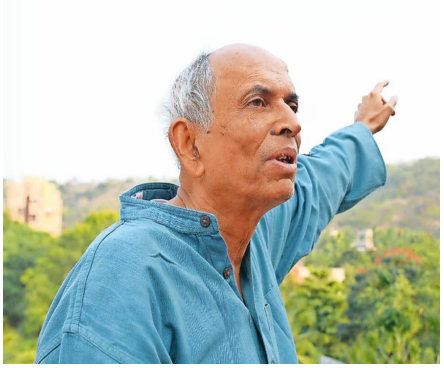
14. CHAMPIONS OF THE EARTH AWARD 2024

Champions of the Earth Award 2024: Awarded to eminent environmentalist Madhav Gadgil.

ABOUT THE AWARD

- The **highest UN honour** in the field of environment.
- It is awarded to outstanding figures from the public sector, private sector and civil society, whose actions have brought a positive impact and transformation on the environment.
- **Awarded by:** United Nations Environment Program.

ABOUT MADHAV GADGIL



- Prominent **Indian Ecologist** with notable contributions to environmental conservation efforts in India. He is a recipient of **Padma Bhushan**.
- His early research helped in the **identification of the Nilgiris** as the **first biosphere reserve in India.**
- He is one of the architects of **India's Biological Diversity Act** and was involved in the implementation of the **Forest Rights Act.**
- He was the chairman of the **Western Ghats Ecology Expert Panel** or the **Gadgil Commission**, which submitted a report (Gadgil Committee Report) in 2011.
  - The report marked around **64% of the Western Ghats region** as an Ecologically Sensitive Area (ESA).
- He has emphasised the **role of local communities** (locals, indigenous populations and gram Sabhas) in protecting the environment.

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6 CHAPTER IMPORTANT SPECIES IN NEWS

1. INVASIVE SPECIES

An alien species is a species introduced by humans – either intentionally or accidentally - outside its natural past or present distribution, however not all alien species have negative impacts. It is these species that are termed '*Invasive alien species*' (IAS).

IMPACT

- Driver of biodiversity loss
- Leads to changes in structure & composition of ecosystems leading to detrimental impact on ecosystem services and humans.

COMMON INVASIVE SPECIES OF INDIA

FLORA	FAUNA
• Prosopis juliflora • Water Hyacinth • Snowflake Corals • Lantana	• African apple snail • Papaya Mealy Bug • Amazon sailfin catfish

FLAGSHIP SPECIES	INDICATOR SPECIES	KEYSTONE SPECIES
Species chosen to represent an environmental cause, such as an ecosystem in need of conservation.	Species whose presence indicates, presence of a set of other species and health of ecosystem.	It is a species whose addition or loss to an ecosystem will lead to major changes in the ecosystem.
Chosen species could be either vulnerable/ attractive/ distinct.	Ex. A species might indicate presence of environmental pollution or arrival of monsoon etc.	As certain species are considered more important in determining the presence of other species.
Ex.: Indian Tiger, African Elephant, Giant Panda of China, etc.	Ex.: Lichens (air quality), most amphibians, fishes, etc.	Ex.: Top predator tiger, lion, crocodile etc. Their removal will distort existing food chains.

2. INDIA'S CRITICALLY ENDANGERED SPECIES

INDIA'S RICH BIODIVERSITY

- Despite having only **about 2.4% of the world's total land surface,** India has **10 biogeographic zones** and is home to **7-8% of all recorded species** (45,000 species of plants and 91,000 of animals). They include-
  - **8.58%** of the documented **mammalian species.**
  - 13.66% avian species
  - 7.91% reptiles
  - 4.66% amphibians
  - 11.72% fish and

- 11.8% plants.

- **Four of the 34 globally identified biodiversity hotspots** are located in India namely,
  - Himalayas
  - Indo-Burma
  - Western Ghats-Sri Lanka
  - Sundaland (Includes Nicobar group of Islands).

Critically endangered species in India:

- The International Union for Conservation of Nature (IUCN) classifies species as critically endangered when they are at the **highest risk of extinction** in the wild.
- As per the latest data (2022) from the Ministry of Environment, Forests and Climate Change, India has **73 species listed as "critically endangered"**. (The number has risen from 47 in 2011)

IMPORTANT SPECIES LISTED AS CRITICALLY ENDANGERED SPECIES

1. Endemic Mammals:

- Eight mammals listed as '**critically endangered' are endemic** to India (their habitat is limited to a small geographic area within India). They include-
  - **Kashmir Stag (Hangul):** Found in Dachigam National Park, Jammu and Kashmir
  - **Namdapha Flying Squirrel:** Namdapha National Park at Arunachal Pradesh.
  - **Large Rock Rat (Elvira Rat):** Endemic to Tamil Nadu
  - **Malabar Large-Spotted Civet:** Western Ghats
  - **Jenkin's Shrew** (small mole like animal): Western Ghats
  - **Nicobar Shrew**: Nicobar islands
  - **Andaman Shrew:** Andaman Islands
  - **Leafletted Leaf-Nosed Bat**: Found in isolated forest patches within India.

2. Birds:

- **Great Indian Bustard:** Rajasthan and Gujarat.
- **White-Bellied Heron:** Arunachal Pradesh and Bhutan foothills.
- **Sociable Lapwing**: Found in parts of Rajasthan during migration.
- **Himalayan Quail**: Last reported in Uttarakhand

3. Reptiles:

- **Gharial:** River systems of Northern India (Chambal River).
- **Red-Crowned Roofed Turtle**: Ganga River.
- **Golden Gecko**: Eastern Ghats.

4. Fish:

- **Ganges Shark:** Ganga and Brahmaputra river systems.
- **Deccan Mahseer:** Endemic to the rivers of the Deccan plateau.

5. Plants:

- **Malabar Lily**: Kerala and Tamil Nadu
- **Rafflesia**: Arunachal Pradesh
- **Kashmir Valley Tulip**: Jammu and Kashmir.

3. ELEPHANT

- Part of Schedule-I, Wildlife (Protection) Act, 1972.
- **IUCN Status:** Endangered.

- Included in Convention on International Trade in Endangered Species of Flora and Fauna (CITES).

ELEPHANT – NATIONAL HERITAGE ANIMAL

The Union Government declared elephants as national heritage animals (2010) on recommendations of a standing committee of National Board for Wildlife.

MIKE PROGRAM (CITES PROGRAM)

- MIKE stands for Monitoring of Illegal Killing of Elephants Program. It was started in South Asia in 2003 after a conference of parties and a resolution of CITES.
- **Aim of MIKE program:** Provide information required by elephant range countries for proper management and long-term protection of elephant populations.

IMPORTANT ELEPHANT RESERVES IN INDIA

ELEPHANT RANGE	ELEPHANT RESERVE WITH DATE OF NOTIFICATION	STATE
Eastern India (South West Bengal-Jharkhand-Odisha)	1. Mayurjharna ER	W. Bengal
	2. Singhbhum ER	Jharkhand
	3. Mayurbhanj ER	Odisha
	4. Mahanadi ER	Odisha
	5. Sambalpur ER	Odisha
	6. Badalkhol-Tamorpingla	Chhattisgarh
North Brahamputra (Arunachal – Assam)	7. Kameng ER	Arunachal
	8. Sonitpur ER	Assam
South Brahamputra (Assam- Arunachal)	9. Dihing-Patkai ER	Assam
	10. South Arunachal ER	Arunachal
Kaziranga (Assam- Nagaland)	11. Kaziranga – Karbi Anglong ER	Assam
	12. Dhansiri-Lungding ER	Assam
	13. Intanki ER	Nagaland
	14. Singphan ER	
Eastern Dooars (Assam- W. Bengal)	15. Chirang-Ripu ER	Assam
	16. Eastern Dooars ER	West Bengal
E. Himalayas (Meghalaya)	17. Garo Hills ER	Meghalaya
Nilgiri –Eastern Ghat (Karnataka- Kerala-Tamilnadu-Andhra)	18. Mysore ER	Karnataka
	19. Dandeli ER	Karnataka
	20. Wayanad ER	Kerala
	21. Nilgiri ER	Tamil Nadu
	22. Rayala ER	Andhra
South Nilgiri (Kerala- Tamilnadu)	23. Nilambur ER	Kerala
	24. Coimbatore ER	Tamil Nadu
Western Ghat	25. Anamalai ER	Tamil Nadu

(Tamilnadu- Kerala)	26. Anamudi ER	Kerala
Periyar (Kerala- Tamilnadu)	27. Periyar	Kerala
	28. Srivilliputtur ER	Tamil Nadu
Northern India (Uttaranchal-U.P.)	29. Shivalik ER	Uttaranchal
	30. Uttar Pradesh ER	U.P.

KEY FACTS ABOUT ASIAN ELEPHANT

- The Asian elephant (Elephas maximus) is the largest mammal living on the land of Asia, and occupies the **Southern** and **Southeastern** parts of the continent.
- This species ranges from Northern India to the Southern tip of the Sumatran islands.
- Asian elephants have three subspecies, based on their distribution and characteristics.
  - Indian Elephant
  - Sumatran Elephant
  - Sri Lankan Elephant

KEY FACTS ABOUT INDIAN ELEPHANT

- Indian elephant (Elephas maximus indicus) is the most widespread of the three subspecies.
- Other countries where they can be found are Bangladesh, Bhutan, Nepal, Cambodia, China, Lao PDR, Malaysia, Myanmar, Thailand, and Vietnam.
- They are declared as I**ndia's National Heritage Animal.**
- **Protection Status:**
  - **IUCN status:** Endangered
  - **CITES:** Appendix I
  - **Wildlife Protection Act, 1972:** Schedule I

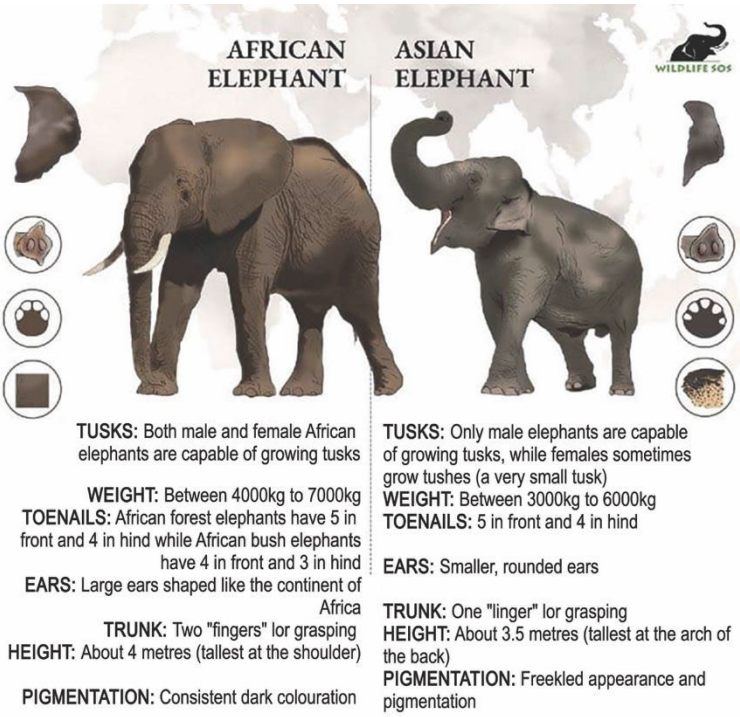
Note:

- IUCN Status of Sumatran Elephant: Critically Endangered
- IUCN Status of Sri Lankan Elephant: Endangered

OTHER FACTS

- Elephants in general are known as 'gardeners of forests'.
  - They are **keystone species** as many trees and plants are known to rely on elephants for pollination and seed dispersal.
  - They are ecosystem engineers, clearing large areas of vegetation, opening up habitats and creating space for new growth.
- Asian Elephants are **highly intelligent** and lead complex social lives. They live in groups of 6-7 related females, and these are led by the eldest female or the **'matriarch'**.
- According to the last national census conducted in 2017, India is home to more than **29,000 elephants.**

DIFFERENCE BETWEEN ASIAN AND AFRICAN ELEPHANTS



4. TIGER

- Tiger, the national animal of India, is a big cat. It is an umbrella species, and its conservation automatically ensures the conversation of a large number of flora and fauna and entire ecosystems.
- India is home to 75% of the **global tiger population.**
  - **National Tiger Conservation Authority** (NTCA) conducts a tiger census across India **every 4 years.**
  - As per the latest Tiger Census Report (2022), India has **3682** tigers (2967 in 2018).
  - **Madhya Pradesh** has the **highest number of tigers** (785) in India, followed by Karnataka and Uttarakhand.
- **Conservation status:**
  - **IUCN Red List:** Endangered
  - **Wildlife Protection Act:** Schedule 1
  - **CITES:** Appendix 1

5. NATIONAL TIGER CONSERVATION AUTHORITY

- A statutory body constituted under the Wildlife (Protection) Act, 1972 for tiger conservation.
- It is headed by the minister of MOEFCC.

OBJECTIVES

- Providing statutory authority to **Project Tiger** so that compliance of its directives becomes legal.
- Fostering accountability of Centre-State in management of Tiger Reserves, by providing a basis for MoU with States within our federal structure.
- Providing for an oversight by Parliament.
- Addressing livelihood interests of local people in areas surrounding Tiger Reserves.
- **Functions**: Its functions are to assist in population assessment of tigers, law enforcement, wildlife forensics, infrastructural development, mitigation, smart patrolling & advisory role in policy formulation.
- **Monitoring System for Tigers:** Intensive Protection and Ecological Status (M-STrIPES) – It is a software-based monitoring system launched across Indian tiger reserves by the NTCA.

INDIAN TIGER

- IUCN Red List: Endangered
- Wildlife protection Act: Schedule 1
- CITES: Appendix 1
- The tiger reserves are constituted on a core/buffer strategy. The core areas have the legal status of a national park or a sanctuary. The buffer or peripheral areas are a mix of forest and non-forest land, managed as a multiple use area.
- India is home to 70% of the global tiger population.

6. CONSERVATION ASSURED TIGER STANDARDS (CA|TS)

- Launched in 2013. Till date 17 tiger reserves from India have been accredited by CA|TS.
- Conservation Assured (CA) is a new conservation tool to set best practice standards for effective management of target species.
- CA fulfills the requirement for protected area management effectiveness in international agreements such as the Convention on Biological Diversity's (CBD) Program of Work on Protected Areas and will help national governments, and their partners in conservation, to meet the CBD's Strategic Plan for Biodiversity.
- CA is also linked to and partnered in the development of IUCN's Green List of Protected and Conserved Areas.

7. ST. PETERSBURG DECLARATION

For doubling tiger population (India was a party to this declaration) by 2022.

NOTE: TX2 program was launched by World Wildlife Foundation (WWF) at the 2010 St Petersburg Tiger Summit held in Russia.

TX2 PROGRAM BY WWF

Nepal is set to become the first country in the world to double its tiger population as part of the World Wildlife Foundation's (WWF) 'Tx2' program.

ABOUT TX2 PROGRAM

- The World Wildlife Foundation had launched its ambitious TX2 program at St Petersburg Tiger Summit in 2010 which aims to double the world tiger population by 2022, which is the year of the tiger in the Chinese calendar.
- **Global Tiger Initiative (GTI):** It was launched in 2008 as a global alliance of governments, international organizations, civil society, the conservation and scientific communities and the private sector and includes organizations like the World Bank, the Global Environment Facility (GEF), etc. It aims to work together to save wild tigers from extinction. In 2013, the scope was broadened to include Snow Leopards. The initiative is led by the 13 tiger range countries (Bangladesh, Bhutan, Cambodia, China, India, Indonesia, Lao PDR, Malaysia, Myanmar, Nepal, Russia, Thailand, and Vietnam).
- **Global Tiger Forum (GTF) is the only inter- governmental international body** established with members from willing countries to embark on a global campaign to protect the Tiger.

8. CANINE DISTEMPER VIRUS

- CDV is a highly **contagious viral disease** that affects a wide range of carnivores (inclusing tigers and lions). It is caused by a **single-stranded RNA virus.**
- Infected dogs living in and around wildlife sanctuaries can spread CDV into Tigers and lions.
- **Impacts:** Several body systems, including the gastrointestinal and respiratory tracts, the spinal cord, and the brain.

- **Symptoms**: Dullness, lacrimation, cough, diarrhea, and seizures; 2 of the 87 captive lions showed lacrimation and respiratory distress. It has a high mortality rate.

9. SUSTAINABLE FINANCE FOR TIGER LANDSCAPES CONFERENCE

- It was hosted by the Bhutan government and supported by the Tiger Conservation Coalition (TCC).
- SFTLC seeks to increase support for tiger protection and its landscape.
- It seeks to mobilise US$1 billion in additional funding by 2034.
- Tiger Conservation Coalition is an Independent group of organizations which brings together leading biologists and experts in wildlife with focus on achieving tiger conservation.
- Members' organizations: International Union for Conservation of Nature, United Nations Development Programme, the Wildlife Conservation Society, the World Wide Fund for Nature, etc.
- Tiger Bonds: By the Asian Development Bank to engage private sector investors.

10. ASIATIC LION

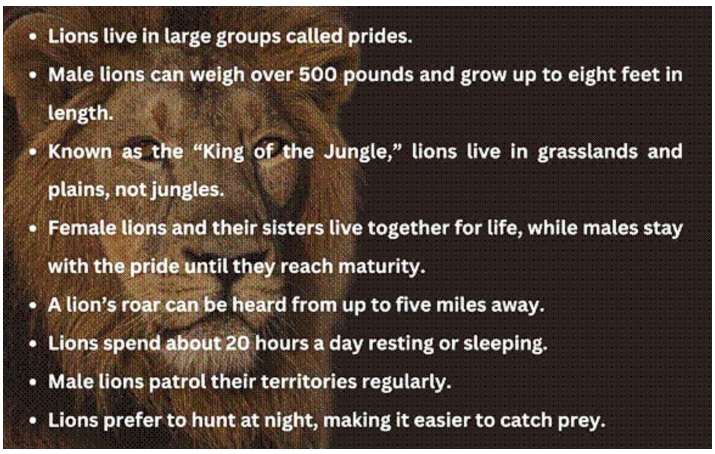
As per the latest government data, **669 Asiatic lions have died** in the last five years (2020-2024). The **Gir forest in Gujarat** is the only remaining natural habitat of **Asiatic lions**.

KEY DATA ON ASIATIC LIONS:

- The **maximum deaths of Asiatic lions** occurred in the year 2024. **Reasons for death include:** Old age, illness, injuries from fights, cub mortality, falling into open wells, electrocution, accidents etc. No incidents of poaching led to lion deaths during these years.
- **Latest Estimate (2020):** Asiatic lion population in Gujarat is 674 in 2020, up from 523 in 2015. A fresh population estimation will be conducted in May 2025.

ASIATIC LIONS

- The Asiatic lion is a **big cat** of the subspecies Panthera leo.
- **Range:** Gir forest, a dry deciduous forest ecosystem in the Saurashtra region of Gujarat, is the abode of the last surviving population.
- **Conservation Status:**
  - **IUCN Status**: Endangered
  - **CITES:** Appendix I
  - **Wildlife Protection Act 1972**: Schedule I



Conservation Initiatives:

- **Project Lion (2020):** Infrastructure development for protection and security of Indian lions, especially health infrastructure. Focus on conflict mitigation, habitat improvement, radio-collar monitoring, and camera traps, and addressing human-wildlife conflict.
- **International Big Cats Alliance (**2023**):** Foster international cooperation among 97 range countries for conservation of big cats.

Concerns: Experts have called for the **translocation of lions in India**, as the big cats remain geographically isolated in Gir. A second habitat would help protect them from extinction in case of an epidemic, a sharp decline in prey or natural disasters. **E.g.,** In 2018, 27 lions in Gir died due to **canine distemper virus (CDV)**.

11. CHEETAH

The Cheetah is the **world's fastest land animal.**



- **Characteristics:**
  - Cheetahs have a slender body with long legs, a small round head, and distinctive black spots on their tan fur.
  - They are **carnivorous animals** and hunt mainly small to medium-sized antelopes such as gazelles and impalas.
- **Distribution of Cheetahs:**
  - Historically, Asiatic Cheetahs had a very wide distribution in India, occurring from as far north as Punjab to Tirunelveli district in southern Tamil Nadu, from Gujarat and Rajasthan in the west to Bengal in the east.

ORIGINAL GEOGRAPHICAL RANGE

- Thus, the Cheetah's **habitat was very diverse** - scrub forests, dry grasslands, savannas and other arid and semi-arid open habitats.
- **In 1952,** the cheetah **was declared officially extinct** in India. Today, Cheetahs are found in only 9% of their historic range, occurring in a variety of habitats such as savannahs in the Serengeti(Protected area in Tanzania), arid mountain ranges in the Sahara and hilly desert terrain in Iran.
- **Namibia** has the largest population of Cheetahs in the world, earning it the title **"The Cheetah Capital of the World".**

CONSERVATION: Cheetahs (African) are listed as **Vulnerable** on the International Union for Conservation of Nature **(IUCN) Red List.**

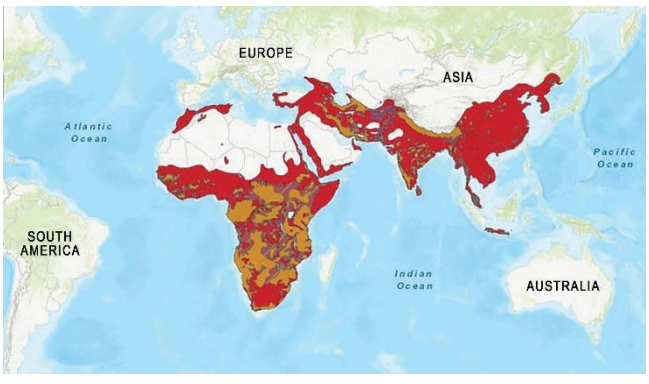
CHEETAH CONSERVATION FUND

- It is an international non-profit organization headquartered in Namibia, founded in 1990.
- Its mission is to be the internationally recognized center of excellence in the conservation of cheetahs and their ecosystems.
- CCF was the result of efforts of Dr. Laurie Marker who moved to Namibia to fight farmer-cheetah conflict in the region.

12. LEOPARD

- It occurs in a wide range in sub-Saharan Africa, in small parts of Western and Central Asia, a small part of European Russia, and on the Indian subcontinent to Southeast and East Asia.
- **Natural habitat:** Forest, Savanna, Shrubland, Grassland, Rocky areas (e.g., inland cliffs, mountain peaks), Desert.



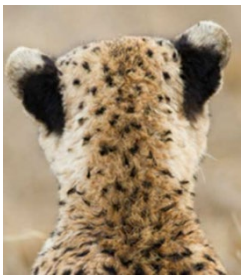
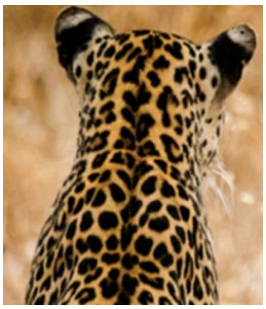


- It is listed as Vulnerable on the IUCN Red List.

- Contemporary records suggest that the leopard occurs in only 25% of its historical global range.

INDIAN LEOPARD

- Indian leopard is a leopard subspecies widely distributed on the Indian subcontinent.
- The species Panthera pardus is listed as Vulnerable on the IUCN Red List
- Leopards have survived outside protected areas in many parts of India since historical times.

Cheetahs	**Leopards**
	
Identifiable by their black "tear" line which runs down their face. Cheetahs have a solid round or oval spot shape. 	The leopard has rosettes which are rose-like markings. 
They prefer to hunt in the day	They usually hunt in nights
Cheetahs are the fastest land animals. They can reach speeds of up to 120 km/h (75 mph) and can accelerate from 0 to 103 km/h (64 mph) in three seconds.	Leopards, on the other hand, only reach a top speed of about 58 km/h (37 mph) but they are super climbers (as they have retractable claws) and good swimmers.
Cheetah is much lighter and taller	Leopards are the shortest of the big cats, although they are strong and bulky - strong enough to pull their prey up trees in order to protect their meal.
IUCN - Vulnerable	IUCN - Vulnerable

NICHE PARTITIONING

- **Niche partitioning** is an important aspect of species survival in the wild with subordinate predators adjusting their behaviour to avoid overlapping space and time with dominant carnivores.
- A study from Manas Tiger Reserve finds that- **Clouded leopards** have developed **nuanced co-existence strategies** to survive in the wild. The forest of Manas National Park supports a diverse carnivore community of **five large carnivores** — tiger, common leopard, clouded leopard, dhole and Asiatic black bear.
- The strategy is having a **peak activity time that does not overlap** with other carnivores and by potentially playing "hide-and-seek" in spaces they share with stronger & larger competitors.

- Although clouded leopards and other carnivores **shared overlapping activity times, each species had distinct peak activity times**. This indicates that they may avoid direct conflicts or competition during their most active periods.
- The study found that there was **temporal overlap** but **no spatial overlap** between clouded leopards and other carnivores, which is a **nuanced coexistence strategy** for survival.

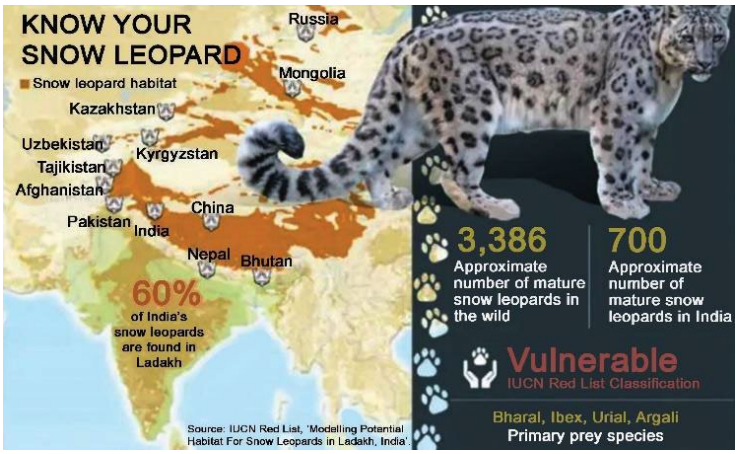
13. SNOW LEOPARD

SNOW LEOPARD



They are also called the **'ghosts of the mountains'**.

- The snow leopard's fur is white to brown and has black spots around its head and neck.
- It preys on Himalayan Blue Sheep, Himalayan Tahr, Tibetan Argali and Himalayan Ibex etc.
- Snow leopards are sparsely distributed **in 12 countries of Central Asia, from southern Russia to the Tibetan Plateau, including Mongolia, China, Afghanistan, Pakistan, India and Nepal.**
- In India, it is found **in Jammu and Kashmir, Ladakh, Uttarakhand, Arunachal Pradesh, Sikkim and Himachal Pradesh.**
- **Accommodation Type:** Forests, shrubs, grasslands, rocky areas



Conservation Status:

- Appendix I: Convention on International Trade in Endangered Species (CITES)
- Schedule I: Wildlife (Protection) Act 1972
- IUCN Status: **- 'Vulnerable'**

National and International Efforts for the Protection of Snow Leopards

- **Indian Efforts: Project Snow Leopard** (PSL): The project was **launched in 2009** to promote an inclusive and participatory approach to the conservation of snow leopards and their habitat.
- **International Efforts:** On the **Bishkek Declaration for the conservation of snow leopards** by 12 'Snow Leopard' Range Countries (Afghanistan, Bhutan, China, India, Kazakhstan, Kyrgyzstan, Mongolia, Nepal, Pakistan, Russia, Tajikistan and Uzbekistan) in 2013. During this, the **'Global Snow Leopard and Ecosystem Protection'** (GSLEP) program was also launched. **India is a signatory to the programme**.

ABOUT DARJEELING ZOO

- **The official name of Darjeeling Zoo is Padmaja Naidu Himalayan Zoological Park (PNHZ).** It is located in the town of Darjeeling in the Indian state of **West Bengal.**

- The zoo has been a pioneer for **conservation breeding of endangered Eastern Himalayan Species in India.**
- The **first ex-situ conservation breeding program started in 1986 as a Snow Leopard** conservation breeding project.
- The **Red Panda project** was started in 1990. In 2003-04 the park released 4 female Red Pandas into the Singalila National Park. **This was the first reintroduction programme for the species in the world and was immensely successful.**
- In the year 2022, **the Central Zoo Authority of India (CZA) ranked PNHZ Park First in all categories among all zoos in the country.**

Central Zoo Authority of India

- It was established as a **statutory body** under the Ministry of Environment & Forest in 1992.
- It was established under the Wildlife (Protection) Act 1972.
- The main **objective** is to **complement and strengthen the national efforts** in conservation of the rich biodiversity of India, **particularly the fauna as per the National Zoo Policy of 1988**.
- Enforcing the **minimum standards and norms for upkeep and healthcare of animals** in Indian Zoos and **to control the mushrooming of unplanned and ill-conceived zoos.**

14. INTERNATIONAL BIG CAT ALLIANCE

The Union Environment Ministry plans to set up and coordinate an International Big Cat Alliance (IBCA), along the lines of the International Solar Alliance, an India-headquartered initiative to promote solar installations globally.
International Big Cat Alliance (IBCA)

OBJECTIVE

- The alliance aims to reach out to 97 range countries covering the natural habitats of Tiger, Lion, Leopard, Snow Leopard, Puma, Jaguar, and Cheetah.
- Out of Seven big cats, five are found in India (Tiger, leopard, snow leopard, lions, and Cheetah)
- It aims to provide a platform for "dissemination of information on benchmarked practices, capacity building, resources repository, research and development, awareness creation", etc., on the protection and conservation of big cats.

ABOUT

- The IBCA, which will be headquartered in India and to which India has already committed ₹150 crore [for five years- 2023-24 to 2027-28) will lead initiatives to disseminate good practices in conserving big cats.
- It has been conceived as a multi-country, multi-agency coalition that aims for mutual cooperation among countries for mutual benefit in furthering the conservation agenda.
- The 16 nations that have expressed interest in joining the alliance are Armenia, Bangladesh, Bhutan, Brazil, Cambodia, Egypt, Ethiopia, Ecuador, Kenya, Malaysia, Mongolia, Nepal, Nigeria, and Peru.
- In addition, nine international organizations, including the International Union for Conservation of Nature (IUCN) and World Wildlife Fund (WWF), have given consent to join the IBCA.

15. ILLEGAL WILDLIFE TRAFFICKING

WILDLIFE CRIME CONTROL BUREAU

- It is a statutory body under MOEFCC established to combat organized wildlife crime under Wildlife Protection Act, 1972.
- Wildlife Protection Act prohibits illegal capturing, killing, buying, selling of animals & animal parts. Any person committing a breach of any of the conditions listed in the Act shall be punished with imprisonment which may extend up to 3 years, or with a fine which may extend to Rs 25,000 or both.

FUNCTIONS

- Collect & collate intelligence related to wildlife crime.
- Disseminate information to States and other enforcement agencies for immediate actions to apprehend criminals.
- Create a centralized wildlife crime data bank.
- Coordinate and assist foreign authorities/ organisations to curb illegal wildlife trade and wildlife products.

TRAFFIC

- It was established by WWF and IUCN in 1976 as a wildlife trade monitoring network to undertake data collection, analysis and recommendations on sustainable wildlife trade.
- It became an independent NGO in 2017, with WWF and IUCN sitting on its Board along with independent Board members.

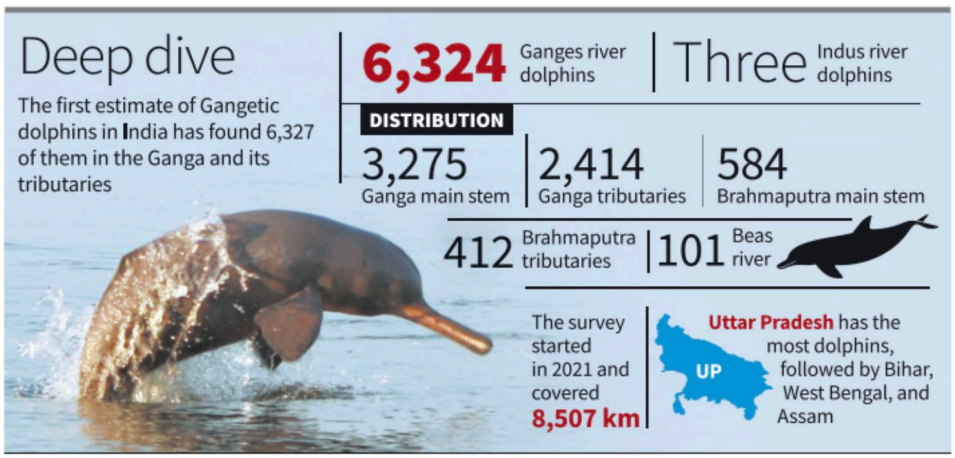
16. DOLPHIN

The first-ever comprehensive survey of river dolphins was led by the Wildlife Institute of India and state forest departments of certain states. It was conducted under Project Dolphin.

MAJOR HIGHLIGHTS OF THE SURVEY

- There were **6,324 Ganges river dolphins**, and **three Indus river dolphins** (a related species) in India, across eight states. The Gangetic dolphins were spread in river Ganga, tributaries of Ganga, Brahmaputra, tributaries of Brahmaputra and Beas river.
- The **highest number of dolphins** were reported in **Uttar Pradesh** (highest concentration in Chambal River), followed by Bihar, West Bengal, and Assam.
- **Acoustic Survey**: Researchers travelled by boat at a constant speed using **acoustic hydrophones** (essentially underwater microphones) that **pick up sounds emitted by the dolphins**.

ABOUT GANGETIC DOLPHIN AND INDUS DOLPHIN



- India is home to **two species** (out of five species of river Dolphin globally) **of freshwater dolphins** — Gangetic Dolphin and Indus river Dolphin.
- **Found mainly** in the Indian subcontinent, particularly in Ganga-Brahmaputra-Meghna and Karnaphuli-Sangu River systems.
- **Key Characteristics:**
  - An adult gangetic dolphin could weigh between 70 kg and 90 kg.
    - Male dolphins as being about 2-2.2 metres long
    - Females are a little longer at 2.4-2.6 m.
  - The **breeding season** extends from January to June.
  - The species is **shy** (avoids human contact) and feeds on several species of fishes, invertebrates etc. Dolphins stay under water and surface only **sporadically to breathe.**

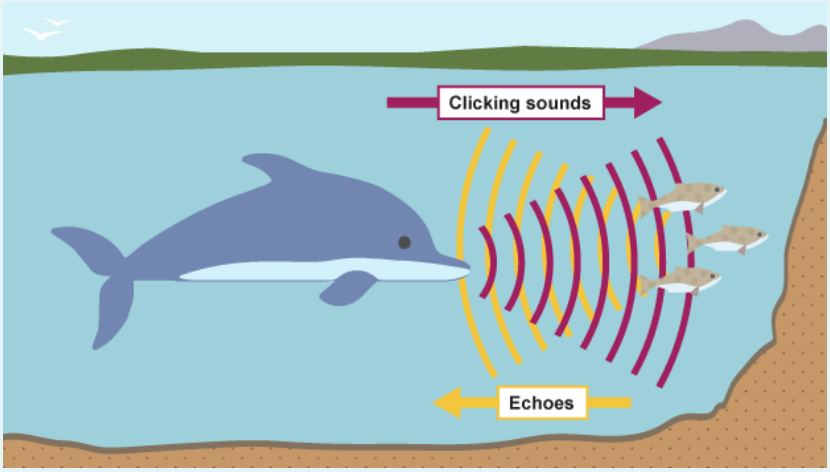
- **Significance:** Dolphins act as an **umbrella species** and their conservation has direct impacts on their habitat and biodiversity. They serve as **bio-indicators** of healthy river ecosystems.
- **Conservation Status** (Both dolphins):
  - **IUCN Status:** Endangered
  - **CITES**: Appendix I
  - **Wildlife Protection Act, 1972**: Schedule I
- **Threats:** Getting entangled in fishermen's nets, River Pollution, Habitat loss due to construction of dams and barrages, Poaching.

DOLPHIN CONSERVATION EFFORTS

- **Project Dolphin (2020)** to promote awareness and involve local communities in conservation efforts.
- **Comprehensive Action Plan** (2022-2047) to ensure well being of the river dolphin and aquatic habitats.
- **Recognition of Ganges river dolphin** as the **National Aquatic Animal of India.**
- MoEFCC has included Ganges river dolphin as **one of the 22 critically endangered species** for providing financial assistance to States under the Centrally sponsored scheme 'Development of Wildlife Habitats'.
- **Important habitats of Gangetic river dolphins** along the Ganges river have been notified as **Protected Areas**, such as Vikramshila Dolphin Sanctuary, Bihar.

ECHOLOCATION

- Echolocation is the **use of sound as a form of navigation.** Echolocation (bio sonar) is a **biological active sonar** used by several animal groups, both in the air and underwater.
- Echolocating animals **emit calls and listen to the echoes of those calls** that return from various objects near them.
- They use these echoes to for **navigation, foraging and hunting prey.**



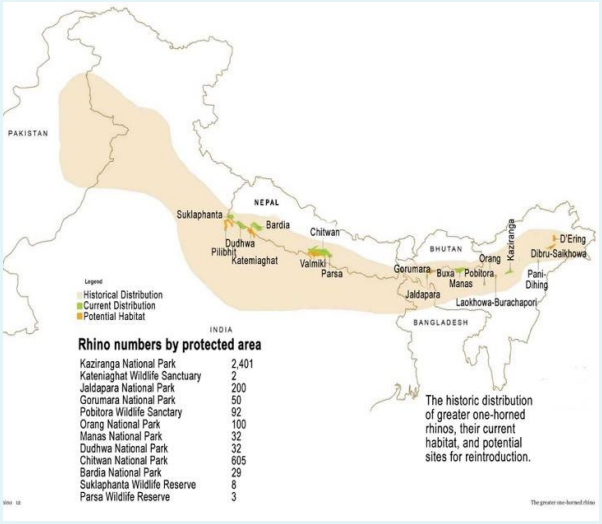
GANGES DOLPHIN AND ECHOLOCATION

- Ganges river dolphin **most typically makes echolocation sounds** such as clicks, bursts, and twitters. As the dolphin is blind, it produces an **ultrasonic sound** that is **echoed off** other fish and water species which allows it to **identify prey.**
- Echolocation is **used for population counts** by using acoustic surveying.

17. RHINOCEROS

GREATER ONE HORNED RHINO





- It is the **largest of the rhino species and identified by a single black horn** and a grey-brown hide with skin folds.
- The **species is solitary**, except when adult males or rhinos nearing adulthood gather at wallows or to graze.
- They primarily graze, with a diet consisting of grasses as well as leaves, branches of shrubs and trees, fruit, and aquatic plants.
- IUCN Status: **Vulnerable**

SUMATRAN RHINO



- They are the **smallest species** of Rhino.
- They **have two horns** that are dark grey to black in colour.
- They are fast runners and are very agile.
- IUCN Status: **Critically Endangered**

JAVAN RHINO



- They are **most threatened** of the five rhino species, with **only around 60 individuals** that live only in Java, Indonesia.

- It has a **dusky grey colour** and a **single horn** of up to about 10 inches.
- It is very **similar in appearance** to the closely-related **greater one-horned rhinoceros**.
- IUCN Status: **Critically Endangered**

BLACK RHINO



- Among black and white rhinos, black rhinos are the smaller of the two African rhino species.
- It is native to eastern and southern African countries.
- Black and white rhinos can be **distinguished by the shape of their lips.**
- **Black rhinos are browsers**, rather than grazers, meaning **they are herbivores** who do not feed on low-growing vegetation.
- They **have two horns**, which grow continually from the skin at their base throughout the rhino's life (like human fingernails).
- IUCN Status: **Critically Endangered**

WHITE RHINO

- They are the **second-largest land mammal** and also known as the **square-lipped rhinoceros**.
- The majority (98.8%) of the southern white rhinos occur in just four countries: **South Africa, Namibia, Zimbabwe, and Kenya**.
- They are the **only of the five rhino** species **that are not endangered**.
- IUCN Status: **Near Threatened**



18. CROCODILES

CROCODILES IN INDIA

- India is home to **three crocodile species** – Gharial, Mugger, and Saltwater Crocodiles.
- Usually spotted on banks of the aquatic systems, crocodiles are **cold-blooded animals**.
- Crocodiles are **apex predators.** These reptiles play a critical role in the ecosystem by maintaining biodiversity and ecological balance: they control the population of other aquatic animals.
- India launched the **Crocodile Conservation Project** in Odisha's Bhitarkanika National Park in 1975, with the aid from the United Nations Development Programme.
- All three crocodile species in India are placed under the **Schedule I of the Wildlife Protection Act, 1972.**

1. Mugger:

- They have a **broad-snouted nose.**



- They are also known as Marsh crocodiles. They are found in **freshwaters** like rivers and also in estuaries and marshy areas.
- Mugger has a diverse and broad diet. They are known to dig burrows or holes for **nesting purposes.**
- Muggers have **stronger legs** which allow them to bask mainly on river banks choosing steeper slopes and elevated platforms.
- Their **tough keratin scales** are known to be sensitive to even the slightest motion in the water and this helps them detect prey easily.
- **IUCN status:** Vulnerable

2. Gharials:

- They are **endemic** to the **Indian subcontinent.**
- They are shy-natured and the most aquatic of all the species.
- They are the **longest living crocodile species,** native to northern India and are distinguished by their long, narrow snouts. The gharials are predominantly a **fish-eater**.
- Gharials prefer to bask on mid-river sand islands on gentle slopes as they have **weaker legs** and can only crawl.
- **Chambal River** (tributary of river Yamuna) holds the **largest population of Gharials** in the wild.
- **IUCN status:** Critically Endangered



Note: Both the species are **freshwater** and have overlapping habitats in the northern rivers (Ganges, Chambal, Son, Ramganga and Girwa) and eastern (Mahanadi) river systems of India. They have shown systematic **resource partitioning** in their aquatic environments.

3. Saltwater crocodiles:



- Largest crocodile species, also recognised as man eater. Largest reptile in the world.
- Also known as **estuarine crocodiles.** Its habitat includes regions such as Odisha's Bhitarkanika National Park, West Bengal's Sundarbans, and the Andaman and Nicobar Islands.
- **IUCN status:** Least concern.

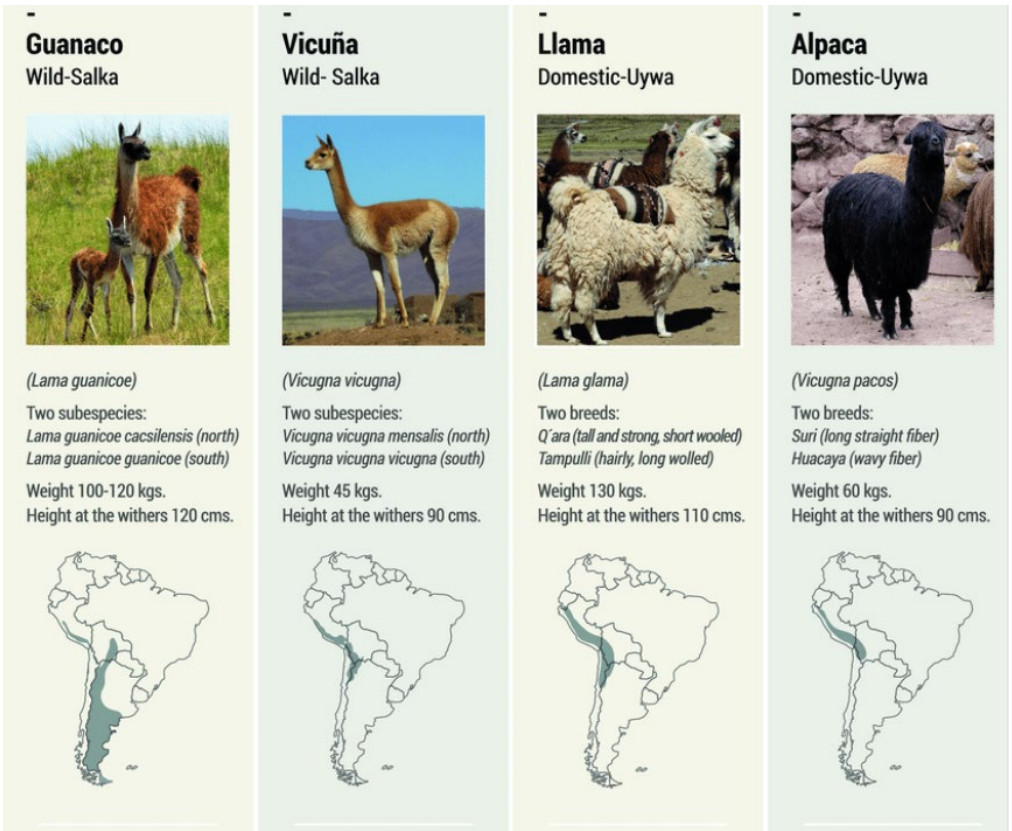
19. BLACKBUCK

- Found only in the Indian subcontinent, mainly in three countries: India (95% of population is present); Nepal (small population survives in the arid part or Terai), Pakistan (extinct as a free-ranging animal but an introduced population is found in Lal Suhanra National Park in Bahawalpur)
- Male Blackbuck are strikingly handsome animals with long, spiralling horns marked with rings and distinctly marked faces with white patches around their eyes. Their bodies have contrasting countershading, with white underparts against their dark back and legs. Females and juveniles are much less conspicuous, fawn in colour with white underparts which resemble the widespread Impala from Africa.



- Due to co-evolution with cheetahs as their main predator, Blackbuck are extremely fast and rely on speed to escape from their main predators. (Considered to be the fastest animal next to Cheetah).
- The species inhabits open grassland, dry thorn scrub, scrubland and lightly wooded country as well as agricultural margins, where it is often seen feeding in fields. Blackbucks require water daily, which restricts distribution to areas where surface water is available for the greater part of the year. Blackbuck are primarily grazers, but browse when lack of grasses forces a greater dependency on leaf litter, flowers and fruits. They are mainly sedentary, but in summer may move longer distances in search of water and forage.
- Found in arid and semi-arid short grass plains once abundant in undivided Punjab, Haryana, parts of UP, Rajasthan, MP, Gujarat and down south up to Tamil Nadu. Lives in open countryside, avoiding forest and hilly areas.
- Included in Appendix III of CITES & Schedule I of the Wildlife Protection Act of 1972. IUCN Status: Least Concern.
- Inhabits several protected areas of India, such as:
  - **Gujarat:** Velavadar Wildlife Sanctuary, Gir Forest National Park.
  - **Bihar:** Kaimur Wildlife Sanctuary.
  - **Maharashtra:** Great Indian Bustard Sanctuary.
  - **Madhya Pradesh:** Kanha National Park.
  - **Rajasthan:** Tal Chhapar Sanctuary, National Chambal Sanctuary, Ranthambhore National Park.
  - **Karnataka:** Ranibennur Blackbuck Sanctuary.
  - **Tamil Nadu:** Point Calimere Wildlife and Bird Sanctuary, Vallanadu Wildlife Sanctuary, Guindy National Park.

20. CAMELIDS



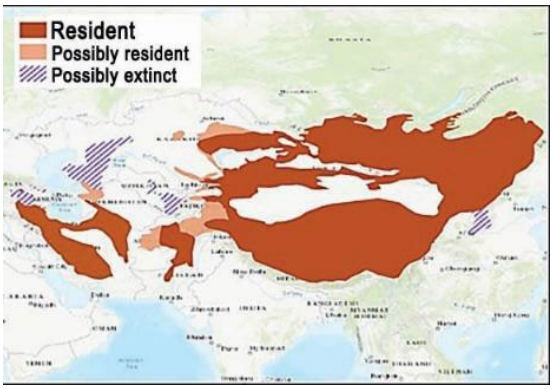
The UN designated 2024 as the International Year of Camelids.

- **Camels, llamas, alpacas, vicuñas and guanacos** are an important source of livelihood for millions of families - most of them pastoralists - in dryland and mountainous rangeland ecosystems around the world.
- The Year is meant to raise the public's and policymakers' awareness of the significant role of camelids in protecting ecosystems, conserving biodiversity, assuring food security, and adapting to climate change.
- The resolution for the International Year of Camelids, proposed by the Government of Bolivia and presented by Ecuador as Country Chair of the Group of Latin American and Caribbean Countries (GRULAC), was approved by the UN General Assembly in 2017 upon recommendation by FAO (Food and Agriculture Organization of the UN).

21. PALLAS' CAT OR MANUL

- Manul cats are a **near-threatened wild cat species** found in the **grasslands and montane steppes of Central Asia.** More commonly known as Pallas' cats, they are distinguished by having round pupils, short legs, and a flat-looking face with wide-set ears.





DISTRIBUTION

- These adorable felines prefer to live in high altitudes of up to 15.000ft and enjoy the cold climate that comes with it.
- Recently DNA testing has confirmed the presence of these elusive cats in and around Mount Everest in Nepal also.

22. INDIAN WOLVES

Wolves in north Gujarat and north Saurashtra will soon be joined by their captive cousins from Junagadh once they finish a basic course in surviving the wild. In a first, the Gujarat Forest Department is set to shift 10 Indian grey wolves from a conservation breeding centre at Junagadh's Sakkarbaug Zoological Park (SZP) to forested areas in north Gujarat and north Saurashtra.



KEY FACTS RELATED TO INDIAN WOLVES

- The Central Zoo Authority (CZA) in collaboration with zoos in India has initiated a conservation breeding program for threatened species in Indian zoos. The Indian wolf is one of the species identified under this initiative.
- The Indian wolf (Canis lupus pallipes) is a subspecies of grey wolf inhabiting semi-arid and arid areas.
- It has a wide distribution range that extends from the Indian subcontinent to Israel.
- The IUCN Redlist considering the wide distribution range of the subspecies has listed it as a species of least concern (Mech and Boitani 2010); however, the Wildlife (Protection) Act of 1972 in view of the threats faced by the animal across its range in India places it in Schedule I Part I.

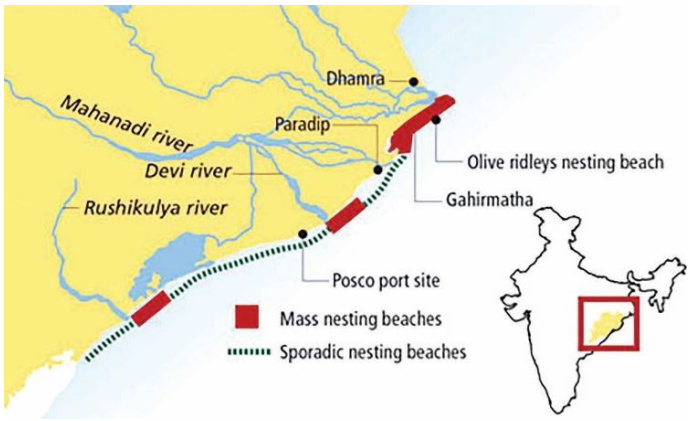
- It is also in appendix 1 of the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES).
- The highest concentration of wolves is currently in Madhya Pradesh (772), followed by Rajasthan (532), Gujarat (494), Maharashtra (396) and Chhattisgarh (320).
- The main prey of wolves are blackbuck and livestock.

23. OLIVE RIDLEY TURTLE

Olive Ridley turtle hatchlings make their way into the Bay of Bengal after being released by the Andhra Pradesh Forest Department as part of the sea turtles community conservation project at the R.K. Beach in Visakhapatnam.

OLIVE RIDLEY

- The Olive ridley turtles are the smallest and most abundant of all sea turtles found in the world, inhabiting warm waters of the Pacific, Atlantic and Indian oceans.
- These turtles are **carnivores** and get their name from their olive-coloured carapace.
- They are best known for their **unique mass nesting** called **Arribada**, where thousands of females come together on the same beach to lay eggs.
- The olive ridley turtles turn up in millions for mass nesting along the Odisha coast every year namely at three river mouths: Dhamara, Devi and Rushikulya.
- **Gahirmatha beach** off Bay of Bengal coast in Kendrapara district is acclaimed as the world's largest nesting ground of these turtles.
- It extends from Dhamra River mouth in the north to Brahmani river mouth in the south.
- The wetland is represented by 3 Protected Areas, the Bhitarkanika National Park, the Bhitarkanika Wildlife Sanctuary and the Gahirmatha Marine Sanctuary.
- The species is listed as **Vulnerable** in the IUCN Red List, **Appendix 1 in CITES**, and **Schedule 1 in Wildlife Protection Act, 1972.**
- They are extensively poached for their meat, shell and leather, and their eggs.
- However, the most severe threat they face is the accidental killing through entanglement in trawl nets and gill nets due to uncontrolled fishing during their mating season around nesting beaches. In recent times, plastic pollution has also emerged as one of the greatest threats to these creatures.



- To reduce accidental killing in India, the Odisha government has made it mandatory for **trawls to use Turtle Excluder Devices (TEDs),** a net specially designed with an exit cover which allows the turtles to escape while retaining the catch.

BRAHMANI – BAITARNI DELTA

The basin consisting of Brahmani and Baitarni extends over states of Odisha, Jharkhand and Chhattisgarh having an area of 51,822 Sq.km. The basin is bounded by the ChhotaNagpur Plateau on the north, by the ridge separating it from Mahanadi basin on the west and the south and by the Bay of Bengal on the east.

The lower reaches of this basin near the deltaic area are subject to floods. Since Mahanadi, Brahmani and Baitarani are interconnected near their delta, the worst flood occurs when there are simultaneous heavy rains in all the three catchments. Prolonged submergence and breaching of embankments are common occurrences during floods. Floods are also caused from cyclonic storms since the coastal areas of the basin are cyclone prone.

Bhitarkanika National Park is situated at the mouth of Brahmani, Baitarni and Dhamra rivers which is famous for saltwater crocodiles. Dense mangrove forests are found here. There also lies Gahirmatha beach and Gahirmatha wildlife sanctuary which is the world's largest ground for Olive Ridley Turtles.

24. INDIAN STAR TORTOISE

ABOUT INDIAN STAR TORTOISE

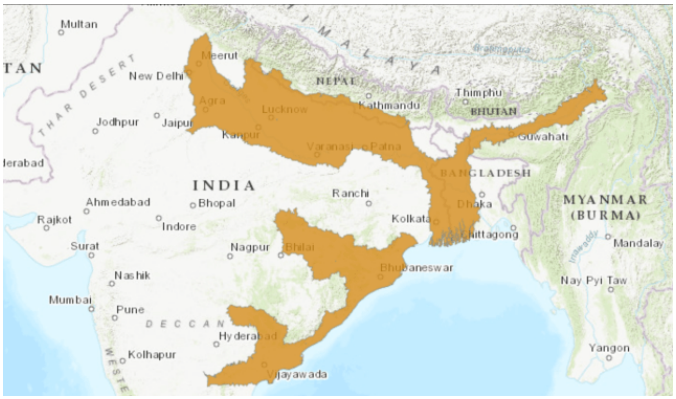


- Indian star tortoises are **endemic** to the Indian subcontinent. They reside in arid pockets of **northwest India** (bordering Pakistan), South India, and Sri Lanka.
- **Features:**
  - **Appearance**: Obsidian shell and striking sun-yellow star patterns adorning it.
  - **Sexual dimorphism:** Adult females are considerably larger than males.
  - Can grow up to 10 inches long.
- **Behaviour:** Mostly Herbivores.
- **Conservation Status**:
  - **CITES:** Appendix I.
  - **IUCN Status:** Vulnerable.
  - **Wildlife (Protection) Act 1972:** Schedule I
    - It is **illegal** to own one in India and unethical since they are vulnerable in the wild.

25. INDIAN TENT TURTLES



Image of Indian Tent Turtle



Locations where Indian Tent Turtles are found

ABOUT INDIAN TENT TURTLES

- Indian Tent Turtle is a species of riverine freshwater turtle native to India, Nepal and Bangladesh.
- Males are smaller than females in size and have longer and thicker tails.
- The size of the carapace ranges from 20 cm to 26.5 cm depending upon the subspecies.
- Due to attractive appearance of the species, they are **illegally traded in the pet market.**
- **Habitat:** Still water pools on river side and slow running water near the river banks.
- These are active swimmers and are **mainly herbivorous.**
- Indian Tent Turtles have three sub-species:
- **Pangshura tentoria:** Occurs in peninsular India and is recorded in Odisha, Maharashtra, Andhra Pradesh, Assam and Madhya Pradesh.

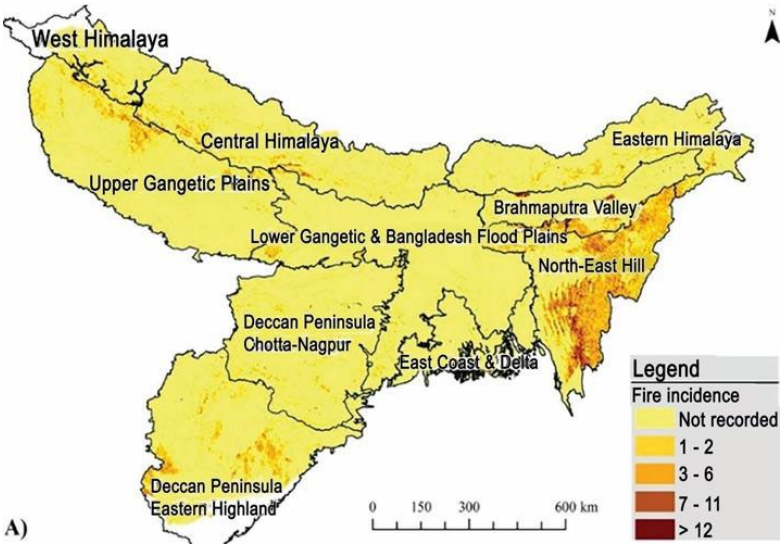
- **Pangshura circumdata:** Occurs in western tributaries of Ganga and the rivers of Gujarat. Thus, found in Rajasthan, MP, UP and Gujarat.
- **Pangshura flaviventer:** Occurs in northern tributaries of Ganga and is recorded from UP, Bihar, West Bengal and Assam.
- **IUCN Status:** Least Concern

26. ELONGATED TORTOISE

- The species, also called the Sal forest tortoise, normally occurs in open deciduous forest patches, including Sal (Shorea robusta) and evergreen forest habitats, dry thorn forests, and savannah grasslands.
- The species is found in the Asian countries of Bangladesh, Bhutan, Cambodia, India, Lao PDR, Malaysia, Myanmar, Nepal, Thailand and Vietnam.



- In and around India, it is found in the Manas-Bhutan transboundary landscape, Corbett-Rajaji-Nepal Terai landscape, and northeast Myanmar and northeast Bangladesh transboundary landscape. There is an isolated population in Odisha also (Chotanagpur region).
- Over 90 percent of the habitat of the critically endangered elongated tortoise falls outside the protected area (PA) network in the Indian subcontinent.
- In India, the Sal forest tortoise is one of the 29 species (24 turtles + 5 tortoises) of freshwater tortoises and turtles.
- It is listed in Appendix II of Convention on International Trade in Endangered Species of Flora and Fauna (CITES).
- This is perhaps the only tortoise species in the Indian subcontinent that has the largest geographical distribution. However, everywhere it is rare.



27. EURASIAN OTTERS

Eurasian Otters have reemerged in the **Gurez Valley** i.e., 123 km north of Srinagar. It represents the first live documentation of species for the last 25 years.



ABOUT EURASIAN OTTERS

- Eurasian otter is a **semi-aquatic mammal** native to Eurasia.
- They have dense and water repellent furs for insulation in cold water.
- **Adaptations:** Webbed feet for swimming, streamlined body and muscular tail for propulsion.

- **Habitat:**
  - Europe, North America, and some parts of Asia.
  - In India, confirmed in Himalayas, Western Ghats, Odisha.
  - Prefers clean freshwater ecosystems.
- **Carnivorous:** Primarily eats fish, amphibians, crustaceans and small mammals.

CONSERVATION STATUS AND THREATS

- **IUCN status:** Near Threatened
- Population decline due to:
  - **Habitat loss** (urbanisation, river pollution)
  - **Poaching** (fur trade, conflicts with fishermen)
  - **Pesticide contamination** affecting fish populations
  - **Climate change** impacting freshwater sources.

28. WESTERN HOOLOCK GIBBON

- Gibbons, the smallest and fastest of all apes, live in tropical and subtropical forests in the south eastern part of Asia.
- The hoolock gibbon, unique to India's northeast, is one of 20 species of gibbons on Earth.
- Over the decades, zoologists thought the northeast housed two species of the ape — the eastern hoolock gibbon (Hoolock leuconedys) found in a specific region of Arunachal Pradesh and the western hoolock gibbon (Hoolock hoolock) distributed elsewhere in the northeast.



- A study led by Hyderabad-based Centre for Cellular and Molecular Biology (CCMB) in 2021 proved through genetic analysis that there is only one species of ape in India. It debunked earlier research that the eastern hoolock gibbon was a separate species based on the colour of its coat.
- The CCMB study concluded that two populations of the western hoolock gibbon and the assumed eastern hoolock gibbon split 1.48 million years ago. It also estimated that the gibbon divergence from a common ancestor occurred 8.38 million years ago.

DESCRIPTION

- It is an ape found in the tropical forest canopy in India, Bangladesh and Myanmar.
- Tailless like other apes, they are nonetheless set apart by their long arms and dense hair.
- The loud and musical calls of Western Hoolock Gibbons can be heard from far away in the forest. Males and females often sing in unison, a series of duet calls that helps the pair to mark their territory.

BEHAVIOUR

- These gibbons are arboreal and only come to the ground in exceptional circumstances.
- They are omnivorous, consuming over 100 species of plants and some invertebrates and birds' eggs, with a diet that can greatly vary based on location.
- Western Hoolock Gibbons form small, monogamous family groups and usually give birth to a single offspring. Babies typically spend their first months of life tightly clung around the mother's waist, followed by gradual weaning over two years. The young gibbon will then stay with his parents until sexual maturity.

THREATS AND CONSERVATION

- They are one of **Asia's most endangered primates(ER)**, with populations expected by the **IUCN** to at least halve over three generations (2001-2015, 2016-2030 and 2031-2045).

- Threats such as hunting for food and medicine and habitat loss have put this species at risk, and habitat protection is critical for their survival.

THE EASTERN HOOLOCK GIBBON



- It is one of two species of Hoolock Gibbon.
- This species is found particularly east of the **Chindwin River in Myanmar** and in southwest Yunnan (China).
- Indian records of the species likely to actually represent the Western Hoolock.

DIFFERENCE BETWEEN EASTERN AND WESTERN HOOLOCK

- Both Hoolock Gibbon species are separated by the **Chindwin River in western Myanmar**, which flows into the Irrawaddy
- The **two species differ** in their fur coloration and DNA sequences, and may have diverged about 1.42 million years ago.
- The Red List maintained by the International Union for Conservation of Nature categorises the eastern hoolock gibbon as **vulnerable.**

29. AMUR FALCON

ABOUT AMUR FALCON



- **Species**: Amur falcon (Falco amurensis), locally known as 'Kahuaipuina'.
- **About Amur Falcons**:
  - **Appearance**: Medium-sized raptors with a distinctive long tail and sharp wings.
  - **Diet**: Primarily insectivorous; known for feeding on locusts and other insects during migration.
  - **Behavior**: Social birds, often seen in large flocks; they roost in trees and are known for their impressive aerial displays.
- **Migration Period**:
  - Arrive from breeding grounds in Northern China, Eastern Mongolia, and Far East Russia in early October.
  - Roost in Tamenglong until the end of November before migrating to wintering grounds in South Africa.
- **Migration Route**:
  - Notable for their long migratory journey, they travel thousands of kilometers across various terrains, demonstrating remarkable navigational skills.
- **Regulation**:
  - Immediate ban on hunting, catching, killing, and selling Amur falcons imposed by the district administration.
  - Owners of air guns must deposit their weapons with village authorities.
- **Legal Framework**:
  - Violations fall under the Wildlife Protection Act, 1972, specifically sections 50 and 51.
- **Awareness and Festivals**:
  - Annual 'Amur Falcon' festival organized to raise awareness.
  - Regular patrolling and conservation programs are implemented.

AMUR FALCON

- IUCN Status – Least concern

- Breeds in South-east Russia and northern China.
- Migrates west through India and across the Arabian Sea to Southern Africa
- Feeds on dragonflies that follow a similar migration path over Arabian Sea.
- 22,000 km journey (longest sea crossing of any raptor)
- Threats: Over harvesting from trapping + Habitat loss from grassland degradation
- They are protected under the Wildlife protection act 1972.
- Nagaland is known as the 'Falcon capital of the world.
- Flyway: East-Asian Australasian and African Eurasian
- ***Doyang lake:*** As per National Register of Large Dams, Nagaland has only one completed dam - Doyang dam which has been constructed on River Doyang, a tributary of River Brahmaputra and located near Wokha city of Nagaland. Doyang dam region attracts many migratory birds during the winter season each year, primary among these is the Amur Falcon.

30. FLYING ARCHITECT: BAYA

ABOUT BAYA WEAVERS

The baya weaver is a fascinating bird species renowned for its intricate nest-building skills and vibrant plumage.



CHARACTERISTICS

- **Size and Appearance**: The baya weaver is a small bird, about 15 cm (5.9 inches) in length. In non-breeding plumage, both males and females resemble female house sparrows. Breeding males are notably colourful with a bright yellow crown, dark brown mask, and a yellow breast with cream buff underparts.

DISTRIBUTION

- **Geographic Range**: Across Indian Subcontinent and Southeast Asia. This includes Pakistan, India (excluding southwest and northeast regions), Sri Lanka, south Nepal, Bhutan, Bangladesh, Myanmar, parts of China, Thailand, Laos, Vietnam, the Malay Peninsula, Borneo, Sumatra, Java, and Bali.
- **Habitat**: They inhabit grasslands, cultivated areas, scrublands, and secondary growth regions. Their nests are typically found in thorny trees or palm fronds, often overhanging water sources to avoid predators.

FEEDING HABITS

- **Diet**: The baya weaver's diet consists mainly of seeds from grasses, including paddy and Guinea grass. They also eat insects like butterflies and occasionally small amphibians, reptiles, and molluscs, particularly when feeding their young.

MATING HABITS

- **Breeding Season**: The breeding season aligns with the monsoon period and is influenced by environmental factors such as day length. Nesting occurs in colonies, often of up to 20–30 pairs.

BEHAVIORS

- **Nest Building**: Males construct elaborate, retort-shaped nests with a central chamber and a long vertical entrance tube. These nests are made from strips of leaves, grasses, and palm fronds, with males making numerous trips to complete them.
- **Social Structure**: Baya weavers are social and gregarious, forming large flocks outside the breeding season. They exhibit complex flight patterns and often roost in reed-beds near water bodies.

31. SEAGULL

Seagulls fly over the Yamuna on a smoggy winter's day in New Delhi.

SEAGULLS



- Seagulls are seabirds, typically medium to large, usually grey, or white, often with black markings on the head or wings.
- Gulls nest in large, densely packed, noisy colonies.
- The gulls have a worldwide cosmopolitan distribution. They breed on every continent, including the margins of Antarctica, and are found in the high Arctic as well.
- Conspicuous and gregarious, gulls are **most abundant as breeders in the Northern Hemisphere,** which has about 30 species in temperate to Arctic regions.
- There is no marked dimorphism among male and female seagulls.
- Gulls are resourceful, inquisitive, and intelligent, the larger species, demonstrating complex methods of communication and a highly developed social structure.
- They not only learn behaviour that proves successful but will educate their young to do so too, especially when it comes to food sources.
- **The black-headed gull**, a dark-headed bird with crimson legs, breeds in Eurasia and Iceland, **winters south in India** and the Philippines, and commonly feeds in fields, where its chief food is insects.
- Seagulls are migratory birds that can be spotted in **the Krishna Wildlife Sanctuary of Andhra Pradesh.** The mangrove wetland also known as Krishna mangroves is famous for Seagulls in India.
- Seagulls can drink both fresh and salt water.

Most animals are unable to do this, but seagulls have a special pair of glands right above their eyes which is specifically designed to flush the salt from their systems through openings in the bill.

32. NAMDAPHA FLYING SQUIRREL

Namdapha flying squirrel resurfaces in Arunachal Pradesh after being missing for 42 years.

NAMDAPHA FLYING SQUIRREL



- The Namdapha flying squirrel is an **arboreal, nocturnal flying squirrel endemic to Arunachal Pradesh in northeast India**, where it is known from a single specimen collected in Namdapha National Park in 1981.
- **No population estimate is available** for B. biswasi, but the known habitat is tall Mesua ferrea jungles, often on hill slopes in the catchment area of Dihing River (particularly on the western slope of Patkai range) in northeastern India.
- **IUCN Status: Critically endangered**
- It is the focus of the **Global Wildlife Conservation Search for Lost Species initiative** along with 25 other 'lost' species.
- It was the sole member in the genus Biswamoyopterus until the description of the **Laotian giant flying squirrel** in 2013.In 2018, Kunming Institute of Zoology at the Chinese Academy of Sciences discovered a new squirrel in the same genus while studying specimens in their collection, called the **Mount Gaoligong flying squirrel**.
- This species looks **very similar to the red giant flying squirrel**. However, it can **be distinguished from the red giant flying squirrel by observing its pale-gray crown, orangish patagium, white underparts, and its characteristic grizzled reddish fur.**

- It is herbivorous and feeds upon fruits, flowers, sap, seeds, nuts, and fungi.

33. PANGOLIN

- There are a **total of eight pangolin species across Africa and Asia.**
  - **Asian species:** Sunda Pangolin, Philippine Pangolin, Chinese Pangolin and 'Indian Pangolin.'
  - **African species:** Long-tailed Pangolin, Tree Pangolin, Giant Pangolin and the Ground Pangolin.
  - Pangolins are the **most trafficked mammal** in the world- with demand **primarily in Asia** and in growing amounts in **Africa-for their meat and scales.**
  - There is also demand in the **US for pangolin products** particularly for their leather to be used in boots, bags and belts.



- **Characteristics:**
  - They have large, protective keratin scales covering their skin, and they are the only known mammals with this feature.
  - They roll into a ball when threatened which can make them easy pickings for poachers.
  - They are primarily **nocturnal**.
  - Their diet consists mainly of ants and termites which they capture using their long tongues.
  - Pangolins have no teeth; they chew with gravel and keratinous spines inside the stomach.
- **Conservation:**
  - **IUCN:** The Indian Pangolin has been classified as **'Endangered'** and the Chinese along with Sunda and Philippine Pangolin as **'Critically Endangered'** by the International Union for Conservation of Nature's Red List of Threatened Species.
  - **Wildlife Protection Act**: Both species are included under India's **Schedule I of the Wildlife Protection Act, 1972.**
  - **CITES:** They are also in **Appendix I of the Convention on International Trade in Endangered Species of Wild Flora and Fauna (CITES), meaning they are most endangered.**
- **India:**
  - **India is home to two species:**
    - the **Indian Pangolin**, found across the subcontinent; and
    - the **Chinese Pangolin**, found across a larger area in south Asia. Bihar, West Bengal, and Assam see the presence of both.

34. GECKO

- Geckos are small, mostly carnivorous lizards that have a wide distribution, found on every continent except Antarctica.
- Belonging to the infraorder Gekkota, geckos are found in warm climates throughout the world. They range from 1.6 to 60 centimeters.
- Geckos are unique among lizards for their vocalizations, which differ from species to species.
- All geckos, except species in the family Eublepharidae lack eyelids; instead, the outer surface of the eyeball has a transparent membrane, the brille.
- Unlike most lizards, geckos are usually nocturnal and have excellent night vision; their color vision in low light is 350 times more sensitive than human eyes.



35. BAOBABS

- Baobabs are long-lived deciduous, small to large trees from 5 to 30 m (20 to 100 ft) tall with broad trunks and compact crowns.
- The baobab is also known as the "upside down tree", a name that originates from several myths.
- They are among the most long-lived vascular plants and have large flowers that are reproductive for a maximum of 15 hours.
- The flowers open around dusk, opening so quickly that movement can be detected by the naked eye, and are faded by the next morning.
- The African baobab boasts the oldest known angiosperm tree: carbon-14 dating places the age of a specimen in Namibia at about 1,275 years.



- They are known as the “Tree of Life,” the species is found throughout the drier regions of Africa and features a water-storing trunk that may reach a diameter of 9 metres and a height of 18 metres.

36. FEW OTHER FLORA & FAUNA IN NEWS

1. **AGARWOOD:** A tree species native to Assam and parts of Northeast India. Also known as Gharuwood and Aloeswood. Uses: For aromatic, medicinal and religious purposes. It is called ‘*Xasi*’ in Assamese and is most famed for its utility in making expensive perfumes. Often termed as ‘liquid gold’ due to its high demand in Arab countries. Only infected Agarwood trees (either by fungal infection or by the borer insect produces dark resin (as a defense mechanism) that makes the aromatic oil. IUCN status: Critically Endangered.
2. **RED SANDERS:** An Indian endemic tree species, with a restricted geographical range in Eastern Ghats especially in Seshachalam forests of Andhra Pradesh. It has no aroma like regular sandalwood tree, the heartwood of it is heavily impregnated with a natural red dye called ‘santalin,’ for which it is valued and considered among the finest luxury woods globally. Red Sanders are known for their rich hue & therapeutic properties, are high in demand across Asia, particularly in China and Japan, for use in cosmetics and medicinal products as well as for making furniture, woodcraft and musical instruments. The Directorate General of Foreign Trade, under the Ministry of Commerce & Industry, has revised its export policy to permit the export of red sanders if it is obtained from cultivated land. IUCN Status: Endangered (Changed) & Appendix II of CITES. The species is cultivated within Sri Lanka, China, around its wild range states and also in Kerala, Maharashtra, Gujarat, Karnataka, Telangana, Tamil Nadu, Odisha and West Bengal within India. Red Sanders is a very slow-growing tree species that attains maturity in natural forests after 25-40 years. Under the foreign trade policy of India, the import of Red Sanders is prohibited, while export is restricted.
3. **NEELAKURINJIIS:** A shrub found in shola forests of Western Ghats between height of 1000 m to 2000 m. Blossoms once in 12 years. The flower being bluish in colour paints, entire mountains blue. Nilgiri mountains is named after it. Last bloomed in 2018. Paliyan tribal people living in Tamil Nadu used it as a reference to calculate their age.
4. **DRAGON FRUIT:** A fruit of cactus species indigenous to Americas (Mexico and Central America). Also known as Pitaya. Cultivated throughout tropical and subtropical regions of the world. Does not require much water and specific soil condition, but slightly acidic soil (pH 5.5-6.5) is preferable for their growth. Contains several antioxidants and rich in fiber & magnesium and a very low-calorie content. Promotes growth of healthy gut bacteria which in turn results in reduction in Fatty liver, inflammation and reduced insulin resistance.
5. **BAMBOO:** Parliament has enacted the Indian Forest Act (IFA) amendment to exempt bamboo grown in non-forest areas from the definition of trees. The amendment aims to exempt bamboo grown in **non-forest areas** from the definition of tree. Bamboo, though, taxonomically a grass, was defined as a tree under the Indian Forest Act, 1927 which meant that the felling and transit of bamboo grown on forest as well non-forest land

for economic use required permit. However, bamboo grown in the forest areas shall continue to be governed by the provisions of Indian Forest Act, 1927.

6. **SNAKE PLANT:** They help to filter indoor air. It is one of the few plants that can convert CO_2 into oxygen at night. This makes it an ideal plant for bedroom décor as it can regulate healthy airflow.
7. **BAMBOO PLANT:** Bamboo palms are good at absorbing formaldehyde, benzene, chloroform and carbon monoxide from the air. They are one of the best plants to remove CO_2 from the air around it.
8. **GAMBUSIA FISH:** It is a freshwater fish species. It is also known as mosquitofish or gambezi. They feed on mosquito larvae and are used as biological control for containing pests. Particularly, they have been used for controlling malaria and dengue. It is also an invasive alien species.
9. **Halari donkey:** Found in Jamnagar & Dwarka districts constituting the Halar region in Saurashtra, Gujarat. They are white in colour. They have a strong built and are large (close to horses). These donkeys are very docile in temperament and are used as pack animals during pastoralist migration and for transportation as donkey carts. Communities associated with: (1) Bharwad and Rabari pastoralist communities who use them as pack animals to carry luggage during migration. (2) Kumbhar who use them in their pottery work. Other species of donkeys found In India are: Spiti Donkey (Found in Spiti region of Himachal Pradesh, they are adapted to cold desert high altitude regions) and Kutchi Donkey (Found in Kutch region, grey in colour and smaller in size as compared to Halari donkeys).

10. WHITE BELLIED HERON

- A bird species, also known as Imperial heron, has been listed as critically endangered in the IUCN red list.
- The heron is listed as a Schedule I species in India's Wild Life Protection Act, 1972, according to the highest legal protection.
- The remaining population inhabits wetlands and rivers in northeastern India, Bhutan and North Myanmar.
- The bird is extinct in Nepal and possibly extinct in Bangladesh too.
- The global population has dwindled to 250, out of which only 50 are possibly left in India.
- India is mainly threatened by human interference and habitat loss, while Hydro power projects in Bhutan have disrupted the nesting grounds.
- Recent sightings at high altitude (in Arunachal Pradesh) for the first time has sparked hope for potential conservation habitats.

11. MUDHOL DOGS

- Also known as Pissouri Hound or Lahori Hound. These dogs are mainly distributed in Bagalkot and Vijayapur districts of Karnataka.
- The KCI (kennel club of India) registers it as a Caravan Hound while the INKC (Indian national kennel club) uses the name Mudhol Hound.
- Known for their endurance and stamina , have been used for hunting and guarding.
- Several trials are being conducted to assess their utility as police dogs.

13. KAPPAPHYCUS ALVAREZII

- It is a fast-growing alga (seaweed) which is known to absorb high amounts of nutrients from seawater. It is indigenous to Indonesia and Philippines and was introduced in India in 1995 for cultivation purposes.
- Commercial significance: Important role in production of industrially lucrative polymer called Carrageenan.
- Global Invasive Species Database of IUCN has placed Kappaphycus in the red list. IUCN has described the Kappaphycus as a destructive invasive species and poses a danger to the coral reefs.
- Kappaphycus is known to adversely impact coral reefs. Its cultivation in the Palk Bay and Gulf of Munnar adversely impacts coral reefs in the region.

14. RHODODENDRONS

- They are known for their showy clusters of large, brightly coloured flowers, and many species are popular ornamental plants in gardens and parks.
- They are evergreen or deciduous shrubs or small trees, with woody stems and broad, leathery leaves.

- **Characteristics:**
  - They exhibit an enormous diversity of size and shape, from prostrate ground covers growing no more than a few inches high to trees more than 100 feet tall.
  - Darjeeling and Sikkim Himalayas comprise only 0.3% of India's geographical area but the region is home to one-third (34%) of all Rhododendron types.
  - In India, Rhododendron ponticum is the state flower of Himachal Pradesh and Jammu and Kashmir, while Rhododendron arboreum is the state flower of Nagaland. Rhododendron niveum and Rhododendron arboreum are the state trees of Sikkim and Uttarakhand, respectively.
- **Distribution:**
  - They are native to the temperate regions of Asia, North America, and Europe, as well as to the tropical regions of Southeast Asia and northern Australia.
  - They occur in a variety of habitats, including alpine regions, coniferous and broadleaved woodlands, temperate rain forests, and even tropical jungle
  - They also require a slightly acid soil to grow well.

37. INDIAN FLYING FOX

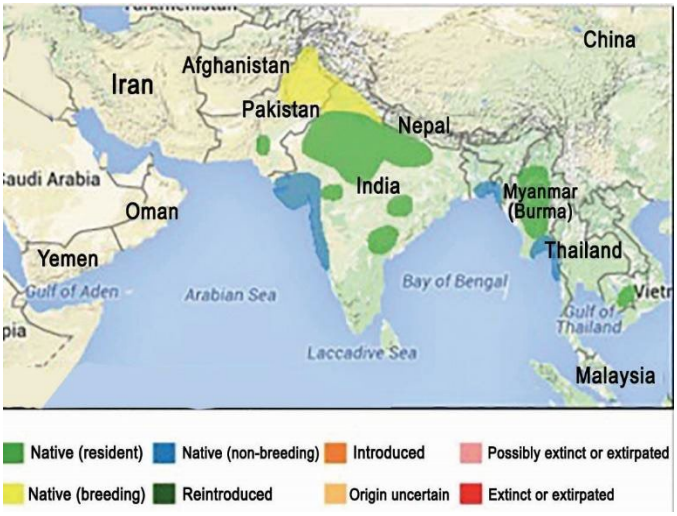
- The **Indian flying fox,** also known as **the greater Indian fruit bat**, is a species of flying fox native to the Indian subcontinent.
- It is **one** of the **world's largest bats**.
- It is nocturnal and feeds mainly on ripe fruits, such as mangoes and bananas, and nectar.
- The Indian flying fox is found across the Indian Subcontinent, including in Bangladesh, Bhutan, India, Tibet, the Maldives, Myanmar, Nepal, Pakistan and Sri Lanka.
- It is threatened by hunting for meat and cutting down habitat trees.
- The IUCN Red List -**Least Concerned category**

38. SPECIES RECOVERY PROGRAM

SPECIES	DETAILS
Asian Wild Buffalo	The wild buffalo was once widely distributed over the tracts of tall grasslands and riverine forests in India and Nepal. The present population of wild buffalo in its entire range is estimated to be lower than 2,000 individuals. Currently, found in North-East India and Chhattisgarh where only a small population survives. **IUCN Status:** Endangered. Schedule I of Wildlife Protection Act, 1972 (Highest Protection). State Animal of Chhattisgarh.
Asiatic Lion	GIR forest, a dry deciduous forest ecosystem in the Saurashtra region of Gujarat, is the abode of the last surviving population of the free ranging Asiatic Lion (Panthera leo persica). The total distribution range of lion in this region is estimated to be around 9000 sq.km in three districts, i.e., Junagadh, Amreli and Bhavnagar, of which GIR National Park, GIR Wildlife Sanctuary, Paniya Wildlife Sanctuary and Mitiyal Wildlife Sanctuary account for about 1,193 sq.km. The conservation initiatives taken so far have resulted in arresting the trend of population decline of lions. **IUCN Status:** Endangered
Brow-Antlered Deer or Sangai	It is a unique animal found only in Manipur, India. The only deer which has adapted itself to swampy habitat. Phumdis of Loktak lake is the residence of this species. **IUCN status:** Critically Endangered

Dugong	Dugong is the only herbivorous mammal that is strictly marine, and the only member of the Order Sirenia found in India. Dugongs are restricted to coastal shallow marine habitats and graze on the sea grass meadows in coastal waters and are called "Sea Cows." In India, it is one of the most seriously endangered species of large mammals. Dugongs are vulnerable to anthropogenic pressures as they are solely dependent on sea grasses in coastal areas, which now have been seriously damaged by mining, trawling etc. Dugongs have also been hunted for their meat, oil, hides, bones and teeth. **IUCN Status:** Vulnerable The Tamil Nadu Government in Palk Bay has established a Dugong conservation reserve. It will be India's first dugong conservation reserve. The site is off the coast of Thanjavur.
Great Indian Bustard	The Bustards are an extremely endangered group of birds dependent on grassland ecosystems. Once upon a time, they used to occur in the arid, semi-arid and moist grasslands across the country. There are four species of Bustards in India: Great Indian Bustard, Lesser Florican, Bengal Florican and Houbara Bustard. They are among the most threatened of the 22 Bustards found in the world. The Great Indian Bustard is now locally extinct from almost 90 per cent of its former range. The present population is estimated to be less than 1000 only. Similarly, perhaps, only less than 2500 Lesser Floricans survive in the whole world. The total global population of Bengal Floricans could be between 400 to 500 individuals. The status of Houbara Bustard is also no more encouraging. These species have depleted, mainly due to the degradation of grasslands. **IUCN Status:** Critically Endangered
Hangul	Kashmir Stag or Hangul is one of the most critically endangered species found in the temperate grasslands of western Himalayas. Dachigam National Park in Kashmir represents one such grassland habitat that supports Hangul, a highly threatened and the only subspecies of the red deer *(Cervus elaphus)* to be found in India, which is now confined only to the Kashmir Valley. **IUCN Status:** Critically Endangered
Jerdon's Courser	It is a nocturnal bird endemic to India. It is found in small geography in the Eastern Ghats of Andhra Pradesh. Its principal habitat is the Sri Lankamalleswara Wildlife Sanctuary, in the Rayalaseema region of Andhra Pradesh. **IUCN Status:** Critically Endangered.
Malabar Civet	The Malabar large spotted civet was once a common species in the coastal districts of Malabar and Travancore in southwest India in the low elevation moist forests of the Western Ghats. By the late 1950s it was reported to be almost 'extinct'. None were seen for a long period of time until 1987, when it was rediscovered about 60 km east of Calicut in Kerala. Extensive deforestation has reduced the Malabar civet's. **IUCN Status:** Critically Endangered
Marine Turtles	Leatherback turtle (Dermochelys coriacea) is one of the most charismatic creatures inhabiting the tropical and temperate waters from Pacific to North Atlantic and throughout the Indian Ocean (Shanker 2003). It is the largest extant marine turtle in the world and follows the longest migratory route known for turtles. The species is currently listed as Vulnerable under the IUCN red list and has been given the highest level of protection under Schedule I (Part II) of the Indian Wildlife protection Act, 1972. In India, Leatherback nesting is specific only to the Andaman and Nicobar archipelago (Namboothri et. al 2010). Pioneering work done by ANET/MCBT, IISc (CES) and Forest department in the past three decades has highlighted Little Andaman and Southernmost Great Nicobar Islands as the potential nesting sites
Nilgiri Tahr	Nilgiri Tahr, a mountain goat, is the highly threatened flagship species occur on the crest lines and ridge forests of the southern Western Ghats. The ideal habitat of this species is the rocky

	outcrops adjacent to the shola-grasslands and other ridge forests. Only less than 2000 individuals of this species are remaining in the wild in the whole world with the major population confined to Eravikulam National Park in Kerala and Grizzled Giant Squirrel Wildlife Sanctuary in Tamil Nadu. **IUCN Status:** Endangered
Swamp Deer	Swamp deer or Barasingha were once abundant throughout tall wet grasslands of North Indian Terai region, Brahamaputra flood plains & Central Indian grasslands bordering sal forests. Currently, swamp deer populations are confined to Uttarakhand, Uttar Pradesh, Assam & Madhya Pradesh in India. It differs from all other Indian deer species as its antlers carry more than three teeth. It has three sub-species: 1. Western Swamp Deer: It is adapted to flooded tall grassland habitat in the Indo-Gangetic plains and terai in India (UP and Uttarakhand) and Nepal. 2. Southern Swamp Deer: It is adapted to hard ground in open sal forests with a grass understorey. It survives only in Kanha National Park. It has also been reintroduced in Satpura National Park. 3. Eastern Swamp Deer: Found only in Assam mostly in Kaziranga & Manas National Park. State Animal of Uttar Pradesh and Madhya Pradesh. **IUCN Status:** Vulnerable
Vultures	Vultures are scavenging birds of prey. They have been divided into New World vultures, which include the Californian and Andean condors, and the Old-World vultures, which include the White-rumped and Red-headed vultures. Old World vultures are found in Europe, Africa, and Asia. There are no vultures in Australia and Antarctica. Distinguishing characteristics of most vultures includes a bald head, devoid of normal feathers and feathery neck. The bare head is supposed to maintain hygiene while feeding on carcasses and also for thermoregulation. Nine species of vultures exist in India, of which five belong to the genus Gyps. Three Gyps vultures, namely White-rumped Vulture, Long-billed Vulture and Slender-billed Vulture are residents, and remaining two, Eurasian Griffon Vulture and Himalayan Griffon Vulture are largely wintering species. Vultures are nature's most efficient scavengers. The Gyps vultures are specialized to feed on the soft tissue of the large ungulate carcasses. They play a vital role in the ecosystem by cleaning up the rotten carcasses left in the open. The population of Gyps vultures in the Indian subcontinent has crashed since the 1990s onwards. The populations of White-rumped Vulture, Long-billed Vulture and Slender-billed Vulture had declined by around 97% during the last two decades. Veterinary use of the non-steroidal anti-inflammatory drug 'diclofenac' is the main cause attributed for this drastic population decline. Government of India has banned the use of diclofenac in veterinary medicine, has initiated Vulture Breeding Program for ex situ conservation and enhanced in situ protection of the remaining populations. Vulture Breeding Centre has been established at Pinjore, Haryana. **Species of Vultures in India and their IUCN Status:** 1. White-rumped Vulture (Critically Endangered) 2. Red-headed Vulture (Critically Endangered) 3. Slender Billed Vulture (Critically Endangered) 4. Indian Vulture (Critically Endangered) 5. Egyptian Vulture (Endangered)

	6. Cinerous Vulture (Near Threatened) 7. Lammergeyer Vulture (Near Threatened) 8. Himalayan Griffon (Near Threatened)
Northern River Terrapin - Species of riverine turtle	Rivers that flow in Eastern India. Hunted for meat and carapace. **IUCN Status:** Critically Endangered
Clouded Leopard	Himalayan foothills from India, South-East Asia and South China. State animal of Meghalaya. It uses its tail for balancing when moving in trees. It can climb down vertical tree trunks headfirst. It rests in trees during the day and hunts by night on the forest floor. **Habitat loss:** poached for its skin and is also a live pet trade. **IUCN Status:** Vulnerable
Red Panda	Closely associated with montane forests with dense bamboo-thicket, it is found in Sikkim, West Bengal, and Arunachal Pradesh. Poached for its meat, and for use in medicines, and as a pet. **IUCN Status:** Endangered
Indian Skimmer	• **IUCN status** – Endangered • **Distribution:** More widespread in winter, the Indian skimmer is found in the coastal estuaries of western and eastern India. It occurs primarily on larger, sandy, lowland rivers, around lakes and adjacent marshes and, in the non-breeding season, in estuaries and coasts.  **Characteristics, Habitat and Behaviour:** • The Indian skimmer grows to a length of 40-43 cm. • It breeds colonially on large, exposed sand-bars and islands. • It feeds on surface-dwelling fish, small crustaceans and insect larvae.
Indian bison	• It belongs to the family of wild oxen and is the tallest living and the second heaviest among oxen. • The Gaur is a social animal. They generally live in groups of about 30 to 40. **Location:** • It is native to the South Asian Region and Southeast Asia. • In India, - Western Ghats, the forests of central India and forest patches in the Northeast.

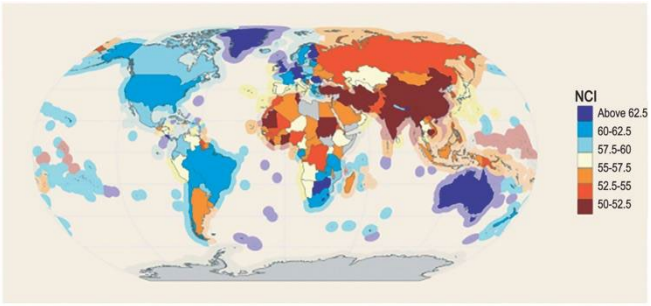
	• 85% of the global population is present in India. • They prefer evergreen forests and moist deciduous forests. However, they can survive in dry deciduous forests also. They are attracted to grounds which are impregnated with salts and minerals. • The Indian Bison is deemed as vulnerable according to the IUCN list. • Hence, the Indian Government has already included the protection of wild bison in the Schedule I of the Wildlife Protection Act, 1972. It calls for the proper regulation in the indiscriminate grazing of cattle around the areas where the gaurs stay.

39. GLOBAL NATURE CONSERVATION INDEX 2024

*India with a score of 45.5 (out of 100) has been ranked **176th** (out of 180 countries) in the Global Nature Conservation Index, 2024.*

ABOUT GLOBAL NATURE CONSERVATION INDEX 2024

- It is the first-ever Nature Conservation Index (NCI).
- **Developed by:** Goldman Sonnenfeldt School of Sustainability and Climate Change at **Ben-Gurion University of the Negev, Israel** and **BioDB.com** (a non-profit website dedicated to maintaining biodiversity data).
- It **evaluates conservation efforts** using four markers — land management, threats to biodiversity, capacity and governance, and future trends.
- The data-driven analysis assesses each country's progress in balancing conservation and development, and is **aimed at** helping governments, researchers, and organisations identify concerns and enhance conservation policies for long-term biodiversity protection.



The Nature Conservation Index (NCI) is a crucial instrumental for evaluating the efficacy of countries' conservation initiatives.

KEY DATA

- **India**'s poor rank is mainly attributed to **inefficient land management** and **rising threats to its biodiversity.**
- **Bhutan** (a global biodiversity hotspot) is the only country in Asia that has performed remarkably, ranking 15th globally with a score of 63.1.
- **Luxembourg** (a small European country situated between Belgium, France, and Germany), with its rich biodiversity ranked 1st with a score of 70.8 out of 100.
- **Europe** dominates the Index with seven of its countries ranking among the top performers.

Click here for INDEX

7 CHAPTER MISCELLANEOUS CONCEPTS

1. REGENERATIVE AGRICULTURE

"Regenerative Agriculture" describes farming and grazing practices that reverse climate change by rebuilding soil organic matter and restoring degraded soil biodiversity – resulting in both carbon drawdown and improving the water cycle.

Regenerative agriculture leads to healthy soil, capable of producing high quality, nutrient-dense food while simultaneously improving, rather than degrading land, and ultimately leading to productive farms and healthy communities and economies.

It is dynamic and holistic, incorporating permaculture (sustainable and self-sufficient agricultural ecosystems) and organic farming practices, including conservation tillage, cover crops, crop rotation, composting, mobile animal shelters and pasture cropping, to increase food production, farmers' income and especially, topsoil.

It has been promoted to counter loss of the world's fertile soil and biodiversity, along with the loss of indigenous seeds and knowledge.

2. PAIRA CROPPING SYSTEM

- This conservation agricultural practice in Odisha, Bihar, Eastern UP.
- It is a relay method of sowing with short-duration pulses/oilseeds, which are broadcast in standing crops of rice ~2 weeks before its harvest.
- IT Does not allow interventions such as tillage, weeding, irrigation, and fertilizer application.
- IT benefits: maximize land use efficiency, boost farmers income, etc.

3. GREEN MANURING

- Growing of plants belonging to a leguminous family and incorporating them into soil after sufficient growth.
- Generally grown as a green manuring crop in India, Dhaincha is used for feeding livestock and soil improvement.
  - Improves soil structure, increases water holding capacity and decreases soil loss by erosion.
  - Reduces weed proliferation and helps reclamation of alkaline soils.
  - It adds organic matter to the soil and stimulates activity of soil microorganisms.

4. INTEGRATED NUTRIENT MANAGEMENT (INM)

- It refers to maintenance of soil fertility and plant nutrient supply at an optimum level for sustaining desired productivity through optimization of benefits from all possible sources of organic, inorganic, and biological components in an integrated manner.

5. ARBORICULTURE

- Arboriculture is a branch of horticulture involved in cultivating and managing and maintaining trees, shrubs and other woody plants in urban and suburban settings.
- Arboriculture differs from urban forestry in the sense that urban forestry deals with the management of forests as systems (or groups of trees) in an urban setting.
- Arboriculture deals with the planting of trees, management through mulching, thinning and tending, 'training', fertilizer application, irrigation, pruning, and disease and pest management.

Amenity Trees: The main focus of Arboriculture is on so-called 'Amenity Trees'. Amenity trees are maintained primarily for landscape purposes for the benefit of human beings. Amenity Trees are found those found in private gardens, public parks and open spaces, schools, churchyards, playgrounds, urban woodlands and nature reserves and alongside roads, railway lines and routes for utilities like electricity pylons.

6. ECOCIDE

- It is human impact on the environment causing mass destruction to the environment.
- It is defined as "as unlawful or wanton acts committed with knowledge that there is a substantial likelihood of severe and either widespread or long-term damage to the environment being caused by those acts".
- It is also referred to as killing one's home or environment. Acts like port expansion projects, deforestation, illegal sand mining, polluting rivers and releasing untreated sewage, etc., destroy fragile natural ecosystems and local livelihoods.
- Commonly cited examples of ecocide include; deforestation during the Vietnam War, the destruction of the environment during the Russian invasion of Ukraine, deforestation in Indonesia and the Amazon rainforest, oil pollution in the Niger Delta and the Chernobyl disaster.
- There is no international law against ecocide that applies in peacetime, but the Rome Statute makes it a crime.
- The Rome Statute of the International Criminal Court (ICC) deals with four atrocities: genocide, crimes against humanity, war crimes, and the crime of aggression. The provision on war crimes is the only statute that can hold a perpetrator responsible for environmental damage, but only if it is intentional and in wartime.

7. RED SNOW

It is a phenomenon caused by Chlamydomonas nivalis, a species of green algae containing a secondary red carotenoid pigment (astaxanthin) in addition to chlorophyll.

- Unlike most species of freshwater algae, it is cryophilic (cold-loving) and thrives in freezing water.
- This alga species, Chlamydomonas Nivalis, exists in snow in the polar and glacial regions and carries a red pigment to keep itself warm.

WHAT WATERMELON SNOW SIGNALS?

The more the algae packed together, the redder the snow. And the darker the tinge, the more the heat absorbed by the snow. Subsequently, the ice melts faster.

While the melt is good for the microbes that need the liquid water to survive and thrive, it's bad for glaciers that are already melting from a myriad of other causes.

These algae change the snow's albedo — which refers to the amount of light or radiation the snow surface can reflect. Changes in albedo lead to more melting. In the melting of snow in the Arctic, key drivers have been snow and ice-albedo.

8. TARBALLS

Tarballs are dark-coloured, sticky balls of oil that form when crude oil floats on the ocean surface. Tarballs are formed by weathering of crude oil in marine environments. They are transported from the open sea to the shores by sea currents and waves.

Tarballs are usually coin-sized and are found strewn on beaches. However, over the years, they have become as big as basketballs and can weigh as high as 6-7 kg.

DO TARBALLS INDICATE AN OIL SPILL?

Most of the time, the presence of several tarballs indicates an oil spill. However, its annual occurrence on the west coast during the monsoon has led marine biologists and experts to demand an investigation into the matter.

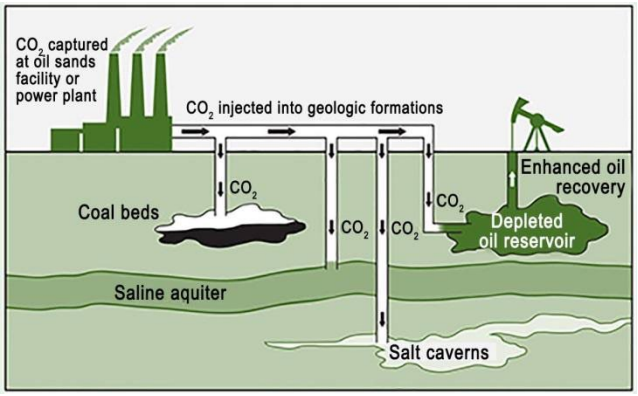
Experts have urged authorities to take stricter vigil and check if ships are dumping burnt oil waste off the western coast of India.

ARE TARBALLS HARMFUL?

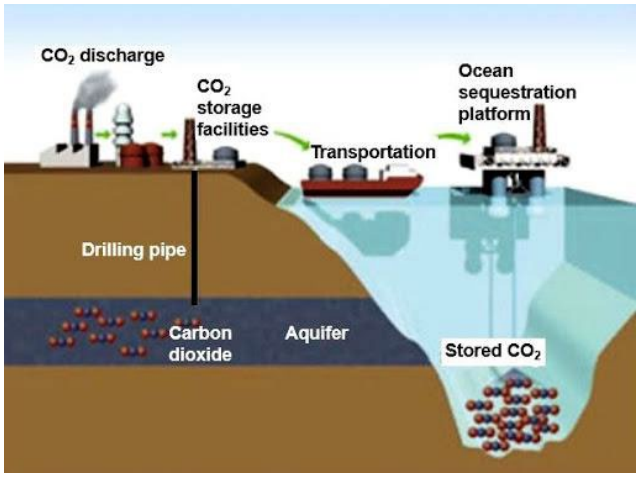
- Tarballs that travel towards the coast can get stuck to the fishing nets installed in the sea, making it difficult for fishermen to clean.
- In addition, it could affect marine life, especially filter feeders like clams and oysters.
- Tarball pollution is a major concern to the global marine ecosystem. Microbes such as bacteria and fungi are known to be associated with tarballs. They presumably play an important role in tarball degradation and some are potential human and animal pathogens.

9. CARBON CAPTURE & STORAGE

Carbon capture and storage (CCS), or carbon capture and sequestration and carbon control and sequestration, is the process of capturing waste carbon dioxide (CO_2), transporting it to a storage site, and depositing it where it will not enter the atmosphere. Usually, the CO_2 is captured from large point sources, such as a cement factory or biomass power plant, and normally it is stored in an underground geological formation. The aim is to prevent the release of large quantities of CO_2 into the atmosphere from heavy industry, and so help to limit climate change.



Although CO_2 has been injected into geological formations for several decades for various purposes, including enhanced oil recovery, the long-term storage of CO_2 is a relatively new concept.



Carbon dioxide can be captured directly from an industrial source, such as a cement kiln, by using a variety of technologies, including absorption, adsorption, chemical looping, membrane gas separation or gas hydrate technologies.

Storage of the CO_2 is envisaged either in deep geological formations, or in the form of mineral carbonates. Pyrogenic carbon capture and storage (PyCCS) is also being researched.

Deep ocean storage is not used, because it could acidify the ocean. A general problem is that long term predictions about submarine or underground storage security are very difficult and uncertain, and there is still the risk that some CO_2 might leak into the atmosphere.

10. GREEN TAX

MORTH has decided to impose additional taxes on old vehicles that are no more fit on the road. This additional tax is called the "Green Tax".

WHAT IS THIS TAX?

- The Ministry of Road Transport and Highways will levy a green tax on transport vehicles older than eight years at the time of renewal of fitness certificate at the rate of 10% to 25% of road tax.
- The revenue collected through the green tax will be kept in a different account and will only be utilized for tackling pollution.

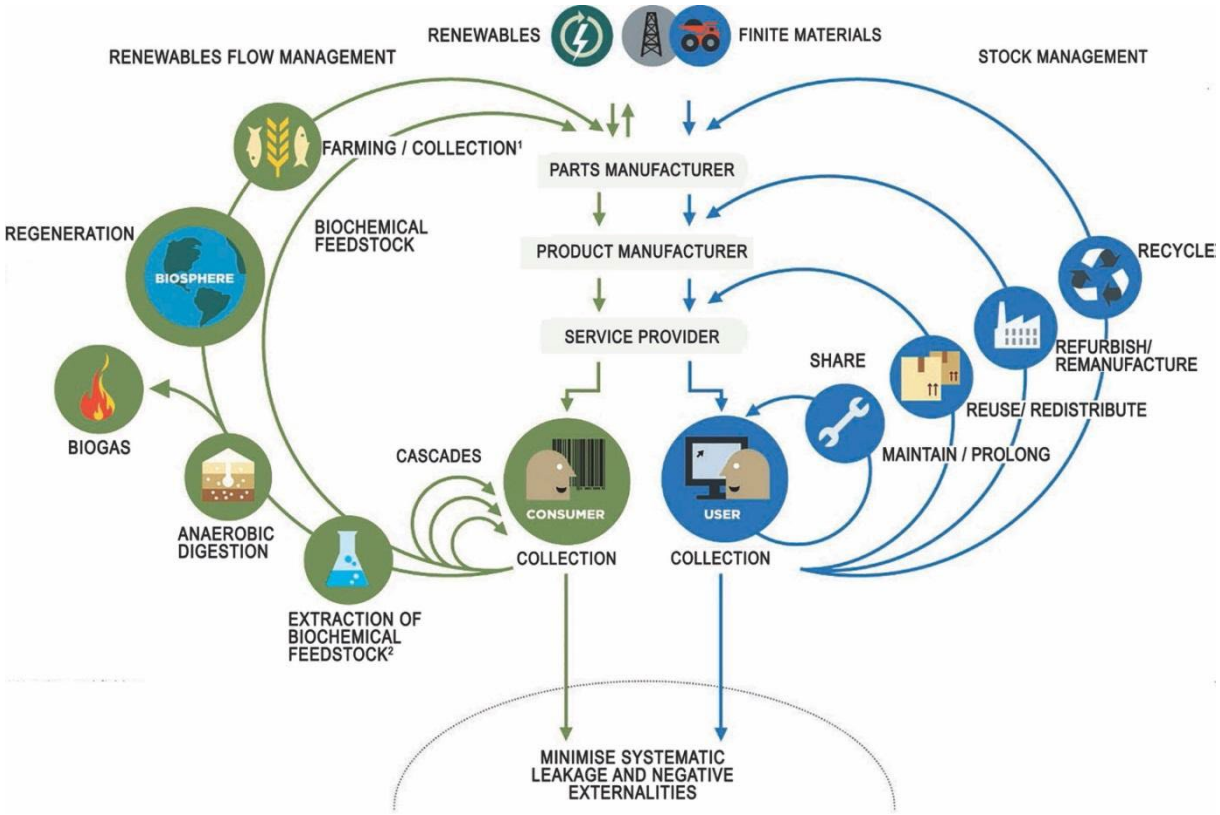
EXEMPTIONS

Vehicles like strong hybrids, electric vehicles and those running on alternate fuels like CNG, ethanol and LPG and vehicles used in farming, such as tractor, harvesters and tillers will be exempted.

BENEFITS OF GREEN TAX

- **Combat Health Hazards by Vehicular Pollution**
- **Follows "Polluter Pays Principle"**
- **Initiates Carbon Pricing in India**: India along with the U.S, China and Japan are some of the countries that are hardest hit by climate impacts. In order to mitigate the condition and reduce the GHG emissions, one smart approach is pricing Carbon such as the introduction of green tax.

11. CIRCULAR ECONOMY



WHAT IS IT?

In the linear economy, raw natural resources are taken, transformed into products and get disposed of. On the opposite, a circular economy model aims to close the gap between the production and the natural ecosystems' cycles – on which humans ultimately depend upon.

This means, on one hand, eliminating waste – composting biodegradable waste or, if it's a transformed and non-biodegradable waste, reusing, remanufacturing and finally recycling it. On the other hand, it also means cutting off the use of chemical substances (a way to help regenerate natural systems) and betting on renewable energy.

12. OCEAN DEOXYGENATION

Ocean deoxygenation is the expansion of oxygen minimum zones in the world's oceans as a consequence of anthropogenic emissions of carbon dioxide.

REASONS FOR OCEAN DEOXYGENATION

- **Ocean warming-driven deoxygenation:** Warmer Ocean water holds less oxygen and is more buoyant than cooler water. This leads to reduced mixing of oxygenated water near the surface with deeper waters, which

naturally contain less oxygen. Warmer water also raises oxygen demand from living organisms. As a result, less oxygen is available for marine life.

- **Excessive growth of algae:** Fertilizer run-off, sewage, animal waste, aquaculture and deposition of nitrogen from the burning of fossil fuels are promoting excessive growth of plant life – a process known as eutrophication, which mostly affects coastal areas.

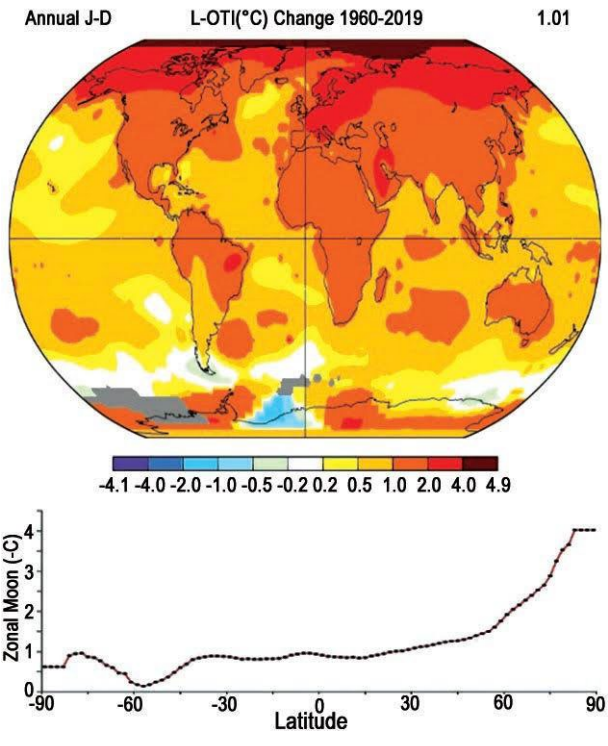
13. OCEAN ALKALINITY ENHANCEMENT (OAE)

- Ocean alkalinity enhancement (OAE), also termed enhanced weathering (EW**), aims to alter seawater chemistry, usually by spreading finely ground alkaline minerals like silicates and carbonates in seawater or on coastal lands.**
- The pulverized minerals dissolve slowly, on the order of years and decades, **adding alkalinity to the ocean so that it can absorb additional CO2 from the atmosphere (Carbon sequestration).** This approach dramatically accelerates natural mineral weathering processes, which normally can take thousands of years.
- When alkalinity increases in seawater, dissolved CO_2 is chemically transformed to bicarbonate and carbonate ions. **This transformation can help de-acidify seawater**, turning the chemical clock of the ocean back to pre-industrial times.
- **Source materials for OAE** — such as lime — To be mined on land or obtained from industrial processes, ground, and then spread on beaches or added to seawater via pipelines or ships. Particles would have to be very small, and locations for mineral addition carefully selected.

14. ARCTIC AMPLIFICATION

Over the past 30 years, the Arctic has warmed at roughly twice the rate as the entire globe, a phenomenon known as Arctic amplification. Most scientists agree that this rapid warming is a signal of human-caused climate change.

This map shows trends in mean surface air temperature over the period 1960 to 2019. Notice that the Arctic is red, indicating that the trend over this 60-year period is for an increase in air temperature of nearly 4° C (7.2° F) across much of the Arctic, which is larger than for other parts of the globe.



REASONS BEHIND ARCTIC AMPLIFICATION

Change in Albedo:

- When bright and reflective ice (with more albedo) melts, it gives way to a darker ocean (lowering albedo); this amplifies the warming trend because the ocean surface absorbs more heat from the Sun than the surface of snow and ice.

Changing Ocean currents:

- Currents are changing because more melting ice is injecting the Arctic Ocean with freshwater. The missing ice also exposes the surface waters to more wind. This mixes up colder freshwater at the surface and warmer saltwater below, raising surface temperatures and further melting ice.

Changing Weather:

- Ocean currents drive the powerful polar jet stream, which moves hot and cold air masses around the Northern Hemisphere. This is a product of the temperature differences between the Arctic and the tropics.
- But as the Arctic warms, the jet stream now undulates wildly north and south. This has been injecting the Arctic with warm air.

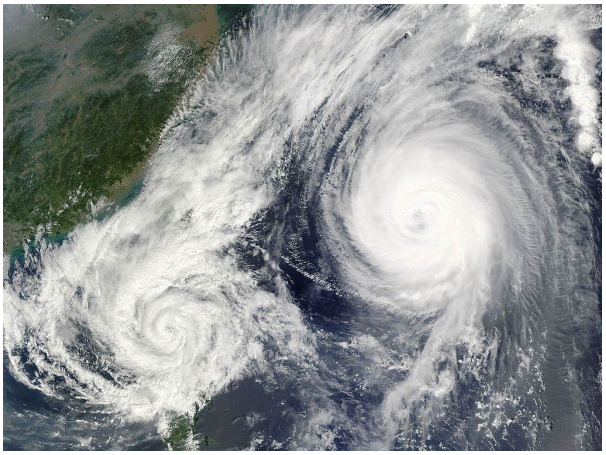
- The abundance of thunderstorms in tropics creates a near-constant flow of heat away from the tropics towards the Arctic.

Change in nature of Ice Cover:

- The quality of ice cover in the Arctic region is changing. New ice is replacing old ice.
- Old ice is more resilient than new ice, which melts faster. New ice contains more salty water and is more subject to melting, whereas old ice has a higher share of water coming from snow precipitation and is a source of freshwater.

15. FUJIWHARA EFFECT

- The Fujiwhara effect or binary interaction, is a phenomenon that occurs when two nearby cyclonic vortices move around each other and close the distance between the circulations of their corresponding low-pressure areas.
- The effect is named after Sakuhei Fujiwhara, the Japanese meteorologist who initially described the effect.
- Binary interaction of smaller circulations can cause the development of a larger cyclone or cause two cyclones to merge into one.
- Extratropical cyclones typically engage in binary interaction when within 2,000 kilometres (1,200 mi) of one another, while tropical cyclones typically interact within 1,400 kilometres (870 mi) of each other.



- When cyclones are in proximity of one another, their centres will circle each other cyclonically (counterclockwise in the Northern Hemisphere and clockwise in the Southern Hemisphere) about a point between the two systems due to their cyclonic wind circulations.
- The two vortices will be attracted to each other, and eventually spiral into the center point and merge.
- It has not been agreed upon whether this is due to the divergent portion of the wind or vorticity advection.
- When the two vortices are of unequal size, the larger vortex will tend to dominate the interaction, and the smaller vortex will circle around it.

16. TROPICALIZATION

Topicalization, also spelled tropicalization, refers to the process by which temperate ecosystems are transformed due to the movement of tropical organisms towards higher latitudes. This transformation occurs as a response to rising ocean temperatures, a consequence of climate change.

Causes:

Rising ocean temperatures: As the Earth's climate warms, the oceans absorb a significant amount of heat. This warming disrupts the delicate balance of marine ecosystems, making previously unsuitable habitats hospitable for tropical species.

Effects:

- Movement of tropical species: As their original habitats become warmer, tropical species are able to expand their ranges towards the poles, colonizing new territories. This can lead to:
- Competition: The influx of tropical species can disrupt the established food webs and compete with native temperate species for resources, potentially leading to population decline or even extinction of the latter.
- Predation: The arrival of new predators from the tropics can further threaten the survival of native species, especially those with no prior defence mechanisms against these new threats.
- Changes in ecosystem function: The introduction of new species can alter the overall functioning of the ecosystem, impacting nutrient cycling, energy flow, and overall biodiversity.

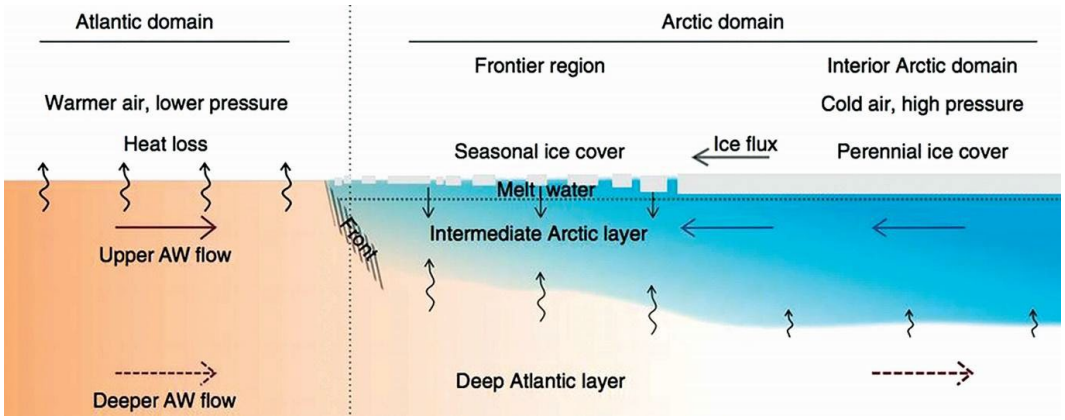
17. ATLANTIFICATION

New research by an international team of scientists explains what's behind a stalled trend in Arctic Ocean sea ice loss since 2007. The findings indicate that stronger declines in sea ice will occur when an atmospheric feature known as the Arctic dipole reverses itself in its recurring cycle.

WHAT IS ATLANTIFICATION?

It refers to the intrusion of Atlantic waters into the Arctic ocean. Scientists have uncovered "hotspots" where some parts of the Barents Sea are starting to more closely resemble the Atlantic. They call this phenomenon "Atlantification". Scientists have observed a rapid decline in ice cover above the Barents Sea in recent decades. According to NASA data, the total area covered by sea ice in this region has fallen by almost half since satellite records began in the early 1980s.

Though the overall scale and impact of Atlantification is still unclear, it could be pushing parts of the Arctic towards a climate "tipping point".

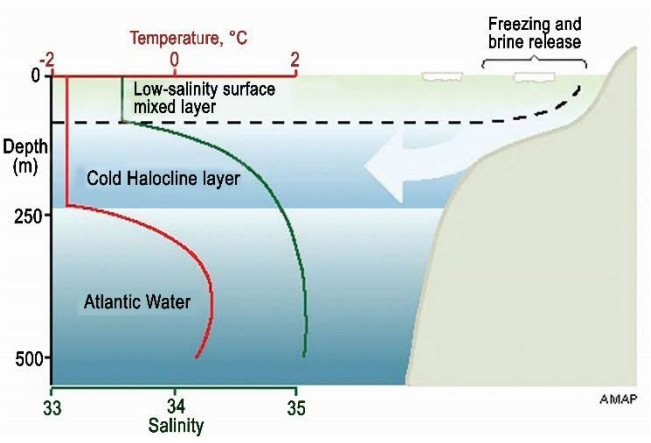


CHARACTERISTICS OF ARCTIC

Unlike the Atlantic and Pacific, the upper waters of the Eurasian Arctic Ocean get warmer as they get deeper. The top of the ocean is typically covered by sea ice. Below this is a layer of cool freshwater, followed by a deeper layer of warmer, saltier water delivered to the Arctic from the Atlantic by ocean currents.

The layers are held in place as a result of differences in water salinity. The cool freshwater is less salty than the warmer Atlantic water and so is more buoyant. Therefore, warmer, saltier water sinks below the fresher top layer. In between, a steep salinity gradient forms. This is known as a "halocline".

The diagram below illustrates the Arctic's unique ocean structure. On the diagram, the red line illustrates increasing water temperature with depth, while the green line illustrates increasing salinity.



The halocline layer protects sea ice from being melted by incoming warm water from the Atlantic.

18. HUMBOLDT ENIGMA

Explorers and naturalists have been asking this question for centuries. Many have also been curious why some areas are more biodiverse than others.

ABOUT HUMBOLDT ENIGMA

- The name derives from **Alexander von Humboldt (1769-1859)** – a polymath who recorded observations on various natural phenomena across the fields known today as geography, geology, meteorology, and biology.

- He suggested **that there was a relationship between temperature, altitude, and humidity on the one hand and biodiversity on the other hand.**
- The world's tropical areas receive more energy from the Sun because of the earth's angle of inclination. So, tropics have greater primary productivity, which then facilitates greater diversity, more ecological niches, creating more complex ecosystems and greater biological diversity.
- The proponents of Humboldt's enigma have held that the Earth's tropical areas by themselves don't contain all the biodiverse regions, that many areas outside the tropics are highly biodiverse. These places are mountains.
- The diversity tends to decrease from the tropics however mountains are exceptions.
- The mountain hosts two processes that generate biodiversity.
  - Geological Process-Like uplift results in new habitats where new species arise, so the habitats are '**cradles.**
  - Species on some climatologically stable mountains persist there for a long time, so these spots are '**museums'** that accumulate many such species over time.
- Coastal tropical sky islands (mountains surrounded by lowlands), like the Shola Sky Islands in the Western Ghats, are a good example.
  - Here, old lineages have persisted on the mountain tops as climates and habitats fluctuated around them in the lower elevations.
  - This is the reason some of the oldest bird species in the Western Ghats, such as the Sholicola and the Montecincla, are housed on the Shola Sky Islands.
- The same mountain can be both a cradle for some species and museum for others, depending on the species' ecologies.
- Another critical force in biodiversity formation is geology.
  - Plants are influenced by the type of soil, which depends on the type of rocks in that area. So high geological heterogeneity often produces unique habitat patches on mountains within similar climate regimes and promotes diversification.

19. POLAR VORTEX?

- Polar vortex is a **circulation of strong, upper-level winds** that normally surround the Northern Pole, moving in a counter-clockwise direction, i.e., a polar low-pressure system.
- These winds **tend to keep the bitter cold air locked in Arctic regions** of the Northern Hemisphere. On occasion, this **vortex can become distorted** and **dip much farther South**, allowing the cold air to spill Southwards.

TYPES OF POLAR VORTEX

- **Tropospheric polar vortex** occurs at the lowest layer of the atmosphere — it extends from the surface up to about 10 km to 15 km — where most weather phenomena occur.
- **Stratospheric polar vortex** occurs at around 15 km to 50 km high. Unlike the tropospheric polar vortex, the stratospheric polar vortex disappears during the summer and is the strongest during the autumn.

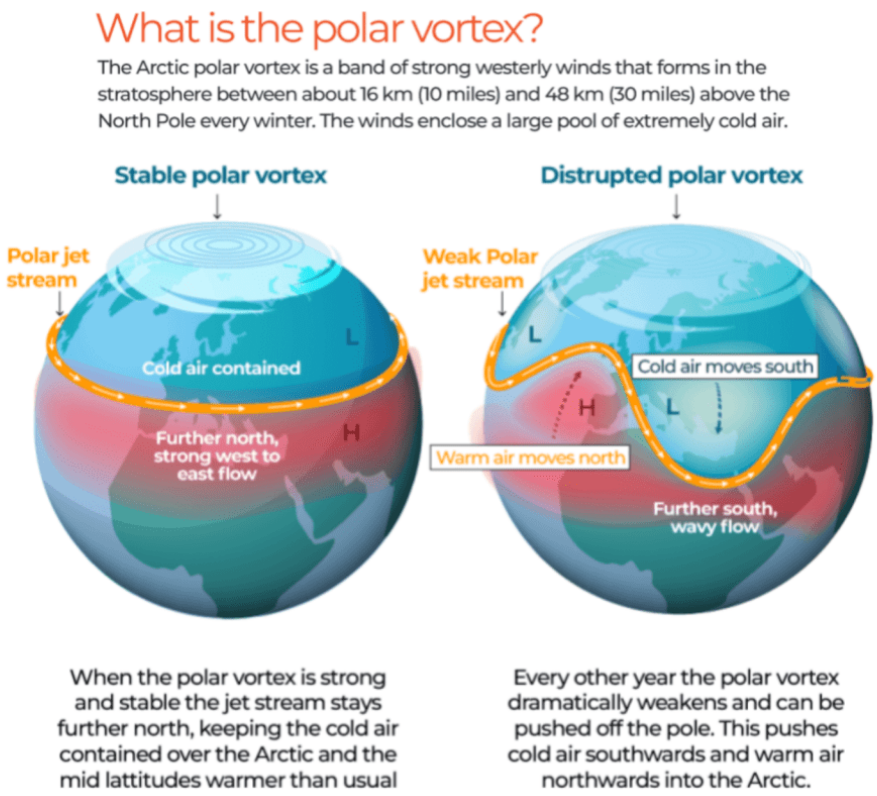
FORMATION OF POLAR VORTEX

- Polar Vortex **forms every winter** because of the temperature difference between Equator and Poles.
- In the polar stratosphere, sunlight basically gets cut off during the late fall and early winter—and that makes it cold, while the equator remains quite warm.
- A jet forms to balance this temperature difference. This jet is what we call the **Polar Vortex** or Polar Night Jet.

DISTORTION OF POLAR VORTEX

- When the pressure difference gets lower, the jet streams get weakened and follow a much wavier path. This change in intensity allows the **dense Arctic cold wind to spill down to the lower latitudes** and lead to **major cold air outbreak** in the US, parts of Europe, and Asia.

- This oscillation is known as the Arctic Oscillation, and it can **switch from a positive phase to negative phase** a few times per year. This oscillation -- namely the Negative phase, where the polar winds are weaker, tends to lead to major cold air outbreaks in one or more regions of the planet.



CLIMATE CHANGE AND POLAR VORTEX

- Scientists are still researching the precise impact of climate change on the polar vortex. Some researchers believe that as the poles are getting warmer at a faster rate than the rest of the Earth, the **polar vortex and jet stream are becoming weaker**. Warmer temperatures make it easier for the polar vortex and jet stream to get disrupted.
- This can be understood through following:
  - Jet streams are propelled forward by temperature differences and the Earth's rotation. Warm air is less dense than cold air and rushes to fill in low-pressure regions. Wider temperature differences create faster moving winds.
  - Earth is warming more quickly at the poles than at the mid-latitude regions, meaning the temperature contrast that drives jet streams has decreased.
  - Complex sequence of events involving sea ice, which is rapidly diminishing in the Arctic. As ice retreats, summertime heat is absorbed by the dark ocean that lies underneath. This heat is released into the atmosphere during winter, spurring winds that can disrupt the polar vortex.

20. SUDDEN STRATOSPHERIC WARMING

STRATOSPHERIC POLAR VORTEX (SPV)

The stratosphere is the layer of the atmosphere from about 10-50 kilometres up. In the winter hemisphere, the pole is tilted away from the sun and is dark 24 hours. At the equator, the stratosphere receives incoming sunlight.

There is, therefore, a large difference in temperature between the high latitude stratosphere and the stratosphere at lower latitudes (a strong temperature gradient). This sets up strong winds blowing in a westerly direction around the cold air over the pole.

This arrangement is known as the Stratospheric Polar Vortex (SPV). This forms every winter. On occasions, this vortex can become disturbed. The temperature can rise by up to 50°C in a few days (although it is still cold) and the winds can weaken, or even reverse.

IF THE WINDS REVERSE, THEN A SUDDEN STRATOSPHERIC WARMING (SSW) IS SAID TO HAVE HAPPENED.

The SSWs happen around six times a decade in the northern hemisphere, but only one has ever been observed in the southern hemisphere.

This is because of the different arrangement of land and sea in the two hemispheres. In the northern hemisphere, there are more regions of land-sea temperature contrast, which can set up atmospheric waves that can disturb the vortex. In the southern hemisphere, there is much more sea, which is continuous around Antarctica.

SUDDEN STRATOSPHERIC WARMING (SSW)

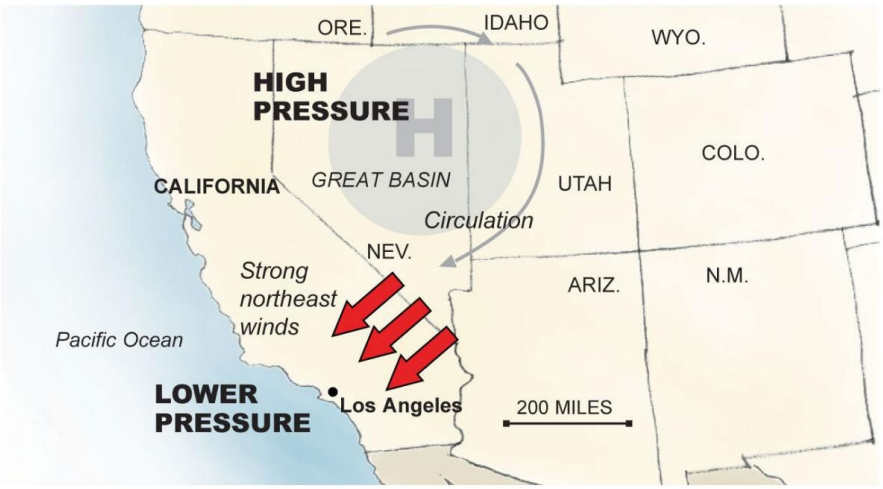
It is an event in which the polar stratospheric temperature rises by several tens of kelvins (up to increases of about 50°C (90°F)) over the course of a few days. The warming is preceded by a slowing then reversal of the westerly winds in the stratospheric polar vortex. In the northern hemisphere SSWs occur about 6 times per decade, and only two SSWs have been observed in the southern hemisphere

There is no consensus about this. SSWs are natural fluctuations in the atmosphere and are not caused by climate change.

Some models predict an increase with climate change, while others predict a decrease, so there is no clear direction. Models are getting better at representing the stratosphere, although there are still deficiencies.

21. SANTA ANA WINDS DRIVE WILDFIRE

- Santa Anas are dry, warm (often hot) local winds that blow westward through **Southern California,** toward the coast.
- The winds are usually seasonal, and typically occur between October and March and peak in December.
- Santa Ana winds are driven by differences in the atmospheric pressure.
  - Santa Ana winds blow when **high pressure builds** over the **Great Basin** (area between the Rocky Mountains and Sierra Nevada) and the **pressure is low over California's coast**.
  - The difference in pressure triggers the movement of powerful winds from the Basin's inland deserts, east and north of Southern California, over the mountains toward the Pacific Ocean.
  - As they flow downwards and cross desserts, they become very dry, warm, and gain speed.
- **Impact:** The **lack of humidity** in Santa Ana winds **dries out vegetation**, making it highly inflammable. The wind also fan the flames and help spread them.

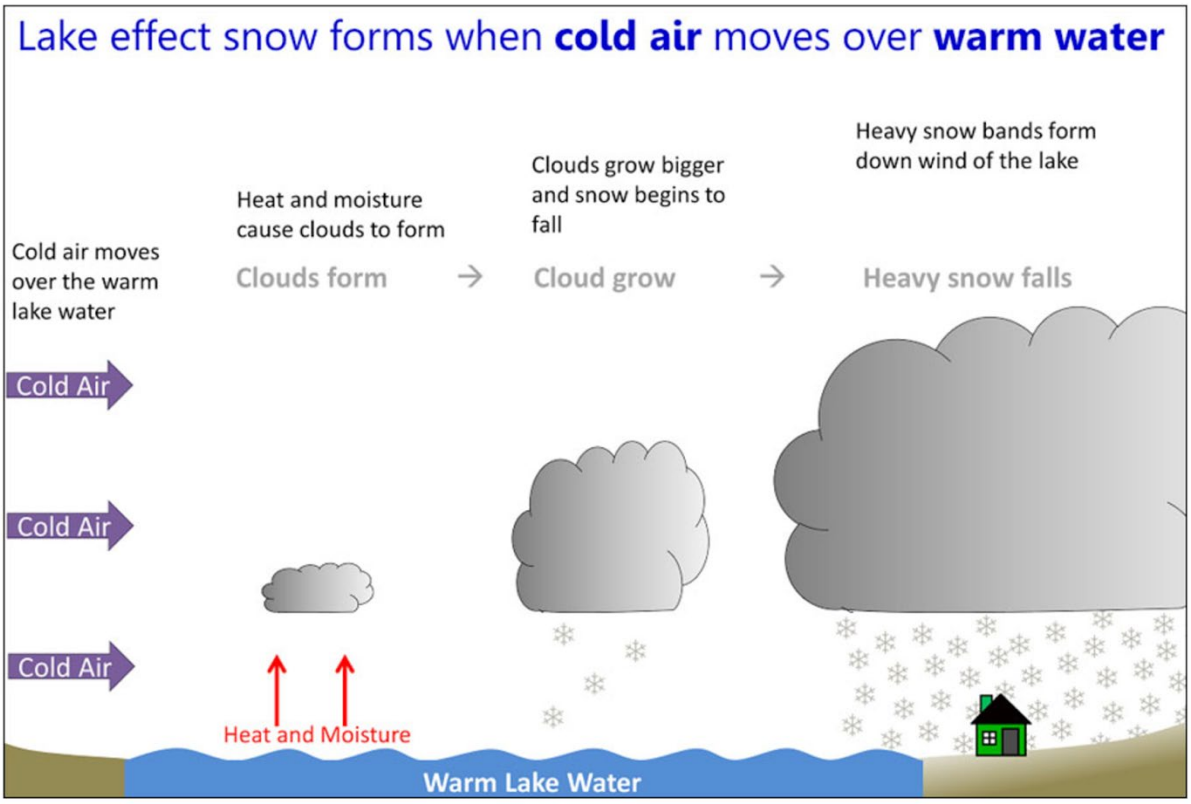


ROLE OF CLIMATE CHANGE

- Wildfire season in California has **lengthened in recent years** in the past two decades and the yearly peak has shifted from August to July.
- The wildfires have also **become more intense** in the past few years. Five of the 10 largest California wildfires (over 20 years) occurred in 2020 alone.
- Rise in global temperatures over the years has led to warmer springs and summers, and early spring snow melts. Such conditions cumulatively cause longer and more intense dry seasons, putting more moisture stress on vegetation. As a result, forests have become more vulnerable to fires.

22. LAKE SNOW EFFECT

- It is a **localised weather phenomenon** where **heavy snowfall occurs near large bodies of water.**
- It is mostly witnessed near the Great Lakes in **North America**, during colder months.
- **Mechanism:**
  - The cold air (often originating from Canada) moves over the warmer surface of the lakes.
  - The lower layer of the air is warmed by the lake's surface and picks up water vapour from the lakes, and rises through colder air.
  - The warm, moist air rises into the colder atmosphere above. Rapid cooling causes the moisture to condense, forming clouds.
  - The clouds grow into a narrow band that results in heavy snowfall, often producing 2–3 inches or more per hour.



23. BLUE TIDES

Observed in several stretches of Maharashtra coast – such as in Juhu Beach. The phenomenon is due to bioluminescence, which is the emission of light by microscopic marine plants called phytoplankton. E.g.: dinoflagellates. The blue light stems from chemical reactions of proteins inside the organism.

WHAT ARE DINOFLAGELLATES?

It is the marine plankton. The population of Dinoflagellates thrives in water based on the sea temperature. Dinoflagellates are one of the largest groups of marine eukaryotes in species. These species are photosynthetic.

WHAT IS BIO-LUMINESCENCE?

The sparkling light appearing in the sea from the microorganisms is called bioluminescence. The Bioluminescence is usually higher in deep living organisms than the shallow species. Bioluminescence is controlled by a circadian clock and only occurs at night. Luminescent and non-luminescent strains can occur in the same species. During night-time the number of species is high. The smaller blooms are not harmful.

Basically, it is an anti-predatory response. Also, bioluminescence helps the microorganisms to gather together easily and form colonies.

HOW IS THE BLUE LIGHT RELATED TO CLIMATE CHANGE?

According to marine experts, the blue light is the sign of climatic changes. It also reveals the low oxygen and high nitrogen content in the sea water. Also, heavy rain causes Bioluminescence.

IS BLUE TIDE HARMFUL?

The smaller blooms of microorganisms are not harmful. On the other hand, the slow-moving larger blooms have an impact on deep sea fishing. Their colonies become larger only when the nitrogen presence is higher and dissolved oxygen content is lower. This environment is highly dangerous for fish as they mainly survive on oxygen. This happens due to fertilizer run off and discharge of untreated sewage into the oceans. Therefore, larger blue tide indicates deteriorating ocean ecosystems.

IS BIO-LUMINESCENCE COMMON IN INDIA?

No, Bioluminescence is not common in India. In India we can find Bioluminescence in Lakshadweep, Goa, Mumbai, Kannada (Udupi).

WHY DO MARINE ORGANISMS GLOW?

Species in the sea glow because of the chemicals in their body or bacteria in the skin. The oxygen reacts to produce a substrate called luciferin that creates blue light in the waves.

24. ZOMBIE FIRES

Zombie Fires in the Arctic region are becoming frequent. This says that the fire regimes in the Arctic are changing rapidly. These regions were once-frozen Tundra.

ABOUT ZOMBIE FIRES

A Zombie fire is a fire from a previous growing season. It can smolder (burn slowly with smoke and no flame) under the ground that is made of carbon rich peat. The fire reignites when the weather warms.

WHAT IS THE ISSUE?

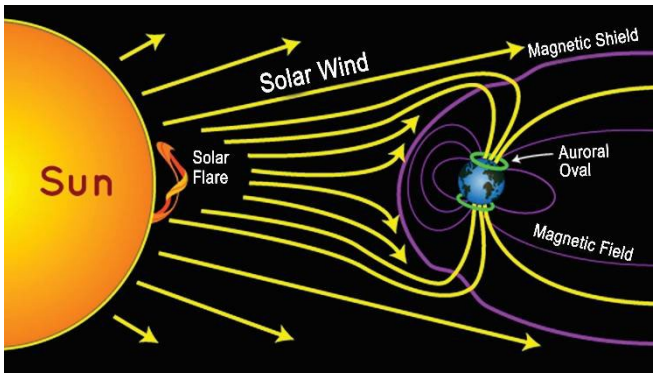
The fires in the Arctics are spreading to areas which were formerly fire-resistant. The north of the Arctic Circle (the Tundra) is drying up. Vegetation in the region such as grass, moss, dwarf shrubs are catching fire. The wildfires in the permafrost region of Siberia in the Arctic are not uncommon. However, in 2019 and 2020, the burning occurred well above the Arctic Circle.

These fires have the potential of turning carbon sinks into carbon sources which in turn increases Global Warming.

25. OBSERVATORIES SPOT RED AURORA OVER LADAKH

The Hanle and Merak observatories in Ladakh, which capture photographs of the night sky, spotted an intense red aurora which is uncommon in India. According to researchers, the auroras appearing in Ladakh were not conventional auroras but a lesser-known auroral activity known as ***SAR arcs.***

ABOUT AURORAS



- Auroras are patterns of bright lights in the sky. They are caused by particles from the sun that enter the Earth's atmosphere and collide with gas particles, such as oxygen and nitrogen.
- When these particles collide, they release energy in the form of light. The different colours of the aurora are produced by different gases. For example, green auroras are caused by oxygen molecules, while red auroras are caused by high-altitude oxygen atoms.
- The phenomenon is usually visible closer to the poles.

ABOUT STABLE AURORAL RED ARCS (SAR-ARCS)

- They are aurora-like phenomena which result in display of vivid shades of red instead of more common green and blue shades in Polar Auroras.
- This event is linked to Coronal Mass Ejection (CME) that occurred on the Sun two days prior. CMEs are massive ejections of plasma and magnetic fields from the Sun's lower atmosphere.
- SAR-Arcs are diffuse, persistent and monochromatic (red in colour) peculiar to mid-latitude regions of Earth. This phenomenon is due to the excitation of atomic oxygen by hot electrons.
- SAR-Arcs are linked to Earth's ring current which consists of millions of Amperes of electrical current, encircling Earth in space near and within geosynchronous orbit. Earth is the only planet in the inner solar system (Mercury, Venus, Earth & Mars) to have such a geomagnetic field and has played a critical role in ensuring habitability of the planet.

ABOUT HANLEY OBSERVATORY

- The Indian Astronomical Observatory (IAO) is a high-altitude astronomy station located in Hanle, India and operated by the Indian Institute of Astrophysics.
- Situated in the Western Himalayas at an elevation of 4,500 meters (14,764 ft), the IAO is one of the world's highest located sites for optical, infrared, and gamma-ray telescopes.
- It is India's first dark sky preserve.
  - A dark sky preserve (DSP) is an area, usually surrounding a park or observatory, that restricts artificial light pollution.

ABOUT MERAK OBSERVATORY

- Merak is situated 176 kilometers east of Leh, on the southern banks of Pangong Tso
- In 2018, a telescope was installed to study the solar chromosphere.

26. GEOMAGNETIC STORMS AND MAGNETOSPHERE

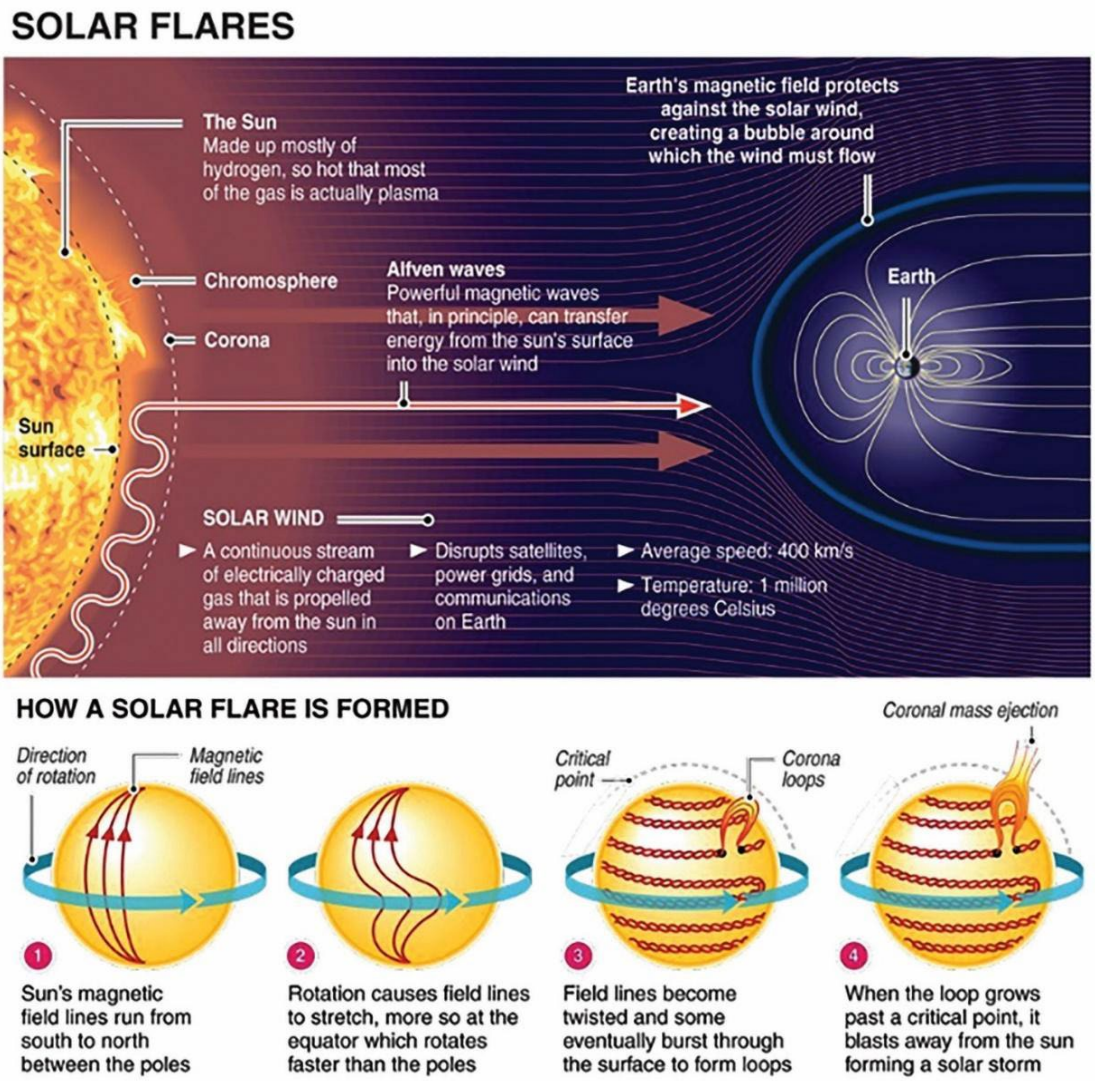
Researchers have traced a very significant increase in special continuous oscillations with pearl-type structures called Geomagnetic Pc1 pearl oscillations on the surface of the Earth in the recovery phase of geomagnetic storms. This study is significant for investigating precipitation particles during geomagnetic storms and can help us understand the radiation hazard to satellites and astronauts.

WHAT ARE GEOMAGNETIC STORMS?

- A geomagnetic storm is a major disturbance of Earth's magnetosphere that occurs when there is a very efficient exchange of energy from the solar wind into the space environment surrounding Earth.
- These storms result from variations in the solar wind that produce major changes in the currents, plasmas, and fields in Earth's magnetosphere.

- The solar wind conditions that are effective for creating geomagnetic storms:
  - Sustained (for several to many hours) periods of high-speed solar wind.
  - Southward directed solar wind magnetic field (opposite the direction of Earth's field) at the dayside of the magnetosphere. This condition is effective for transferring energy from the solar wind into Earth's magnetosphere.
  - High-speed solar wind stream (HSS).
- The largest storms that result from these conditions are associated with solar coronal mass ejections (CMEs) where a billion tons or so of plasma from the sun arrives at Earth.

Storms also result in intense currents in the magnetosphere, changes in the radiation belts, and changes in the ionosphere, including heating the ionosphere and upper atmosphere region called the thermosphere.

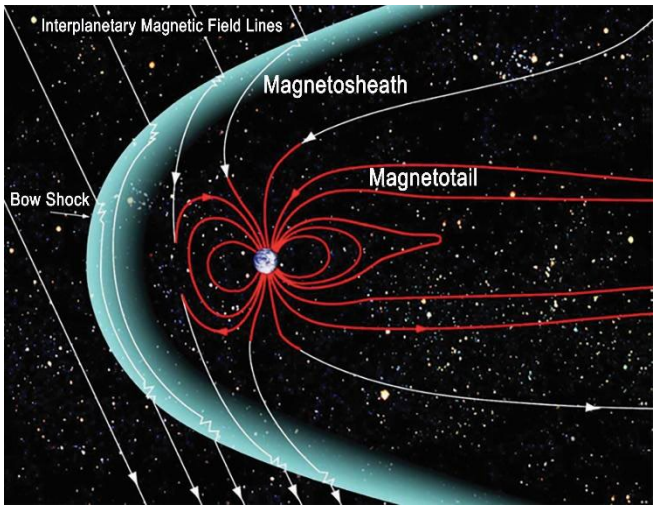


IMPACTS OF SUCH STORMS

- Increase in the density in the upper atmosphere, causing extra drag on satellites in low-earth orbit.
- Horizontal variations in the ionospheric density that can modify the path of radio signals and create errors in the positioning information provided by GPS.
- Storms create beautiful aurora.
- They also can disrupt navigation systems such as the Global Navigation Satellite System (GNSS).
- They can create harmful geomagnetic induced currents (GICs) in the power grid and pipelines.

WHAT IS MAGNETOSPHERE?

- It is the region around a planet dominated by the planet's magnetic field. Other planets in our solar system have magnetospheres, but Earth has the strongest one of all the rocky planets.
- Earth's magnetosphere is a vast, comet-shaped bubble, which shields our home planet from solar and cosmic particle radiation, as well as erosion of the atmosphere by the solar wind - the constant flow of charged particles streaming off the sun.
- It is generated by the convective motion of charged, molten iron, far below the surface in Earth's outer core.
- Constant bombardment by the solar wind compresses the sun-facing side of our magnetic field. The sun-facing side, or dayside, extends about six to 10 times the radius of the Earth. The side of the magnetosphere facing away from the sun - the nightside - stretches out into an immense magnetotail, which fluctuates in length and can measure hundreds of Earth radii, far past the moon's orbit at 60 Earth radii.

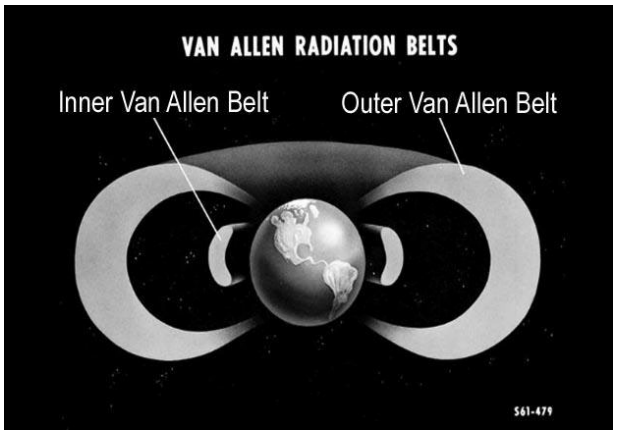


WHAT IS THE DIFFERENCE BETWEEN IONOSPHERE AND MAGNETOSPHERE?

- The ionosphere is a region within the upper mesosphere and thermosphere where solar radiation and particles in the "solar wind" strip electrons from atoms in the atmosphere and create an electrically charged zone of ions. It is used to communicate long distances by radio. Radio waves bounce off the ionosphere in much the same way that light is reflected by a mirror. The ionosphere changes its position from day to night, rising in the night and lowering in the day. This is why radio signals from far off places can be detected at night, while only close-by transmissions are detectable during the day.
- The magnetosphere is contained within the Earth's thermosphere and is the region where the Earth's magnetic field interacts with the charged particles coming from the Sun in the solar wind. These particles become trapped in the magnetic field of the Earth and circulate around the Earth following the Earth's magnetic field. Large bombardments of these charged particles cause the auroras in the ionosphere as they follow the Earth's magnetic field through the upper atmosphere towards the magnetic poles.

WHAT ARE VAN ALLEN RADIATION BELTS?

- The Van Allen radiation belt is a zone of energetic charged particles, most of which originate from the solar wind. The particles are captured by and held around a planet by that planet's magnetic field. It surrounds Earth, containing a nearly impenetrable barrier that prevents the fastest, most energetic electrons from reaching Earth.
- The outer belt is made up of billions of high-energy particles that originate from the Sun and become trapped in Earth's magnetic field, an area known as the magnetosphere. The inner belt results from interactions of cosmic rays with Earth's atmosphere.



27. NEW SEDIMENTARY ROCKS MADE FROM SLAG

In a recent study, researchers documented a new kind of sedimentary rock made from coastal slag deposits in the U.K.

WHAT IS SLAG?

- It is a composite material containing **metal oxides and silicon dioxide** and is a by-product of the **steelmaking process** in the iron and steel industries.
- Slag is a major component of artificial ground.
  - The lithification process hardens industrial waste, including slag, into sedimentary rocks, creating artificial ground.
  - When these sedimentary rocks weather over time, they **release sediments** into the environment.
- Because the rocks are infused with industrial waste, their sediments often **contain toxic metals** that contaminate soil, water, and air.

28. RED MUD

- Produced during Bayer Process for alumina production. It is an insoluble product after bauxite digestion with sodium hydroxide at elevated temperature and pressure.
- It is a mixture of compounds originally present in the parent mineral, bauxite, and of compounds formed or introduced during the Bayer cycle.
- It is disposed as a slurry having a solid concentration in the range of 10-30%, pH in the range of 13 and high ionic strength.

USES

- Metallurgical uses (Iron and Steel production etc.)
- Production of building materials (constructional brick, light weight aggregates, bricks roofing and flooring tiles, cements etc.).
- Ceramics (pottery, sanitary ware, special tiles and glasses, glazes, ferrites)
- pH neutralization - for use in acidic soils as a substitute of limestone, as a treatment for iron deficient soils, in sandy soils to increase phosphorous retention.

29. TEAL CARBON

- India's first 'teal carbon' study undertaken at Keoladeo National Park (KNP). About Teal Carbon
- It refers to carbon stored in non-tidal freshwater wetlands, encompassing carbon sequestered in vegetation, microbial biomass, and dissolved and particulate organic matter.
- It being a color-based terminology, reflects the classification of the organic carbon based on its functions and location rather than its physical properties.
- Other types of carbon
  - **Purple:** Carbon captured through the air or industrial emissions.
  - **Green:** Carbon stored in terrestrial plants.
  - **Black:** Carbon released through the burning of fossil fuels.
  - **Red:** Carbon released through biological particles on snow and ice that reduce albedo.
  - **Grey:** Carbon released through industrial emissions.

30. MANUFACTURED SAND

- Manufactured Sand (M-sand) which is fine aggregate produced by crushing hard Rock to a required size of 150 microns by using crushing, shaping, screening and classifying methods.
- Rocks or quarry stones are blasted and subjected to a series of crushing cycles to reduce the particles to the size of naturally occurring sand.
- The produced sand is then sieved and washed to remove fine particles and impurities and tested for various quality aspects before it is deemed fit as a construction aggregate.

- Such Manufactured Sand obtained must confer to IS Code- 383:2016, and should be
- suitable for construction activity.

COMPARISON BETWEEN NATURAL SAND & M-SAND

NATURAL SAND	M-SAND
Formed due to natural processes of attrition.	Formed due to artificial crushing.
Possess smoother texture and better shape.	Relatively rough texture and uneven shape
Carries entrapped moisture between the particles.	Relatively dry
Relatively rarer	Denser particle packing as it is free from silt and clay particles.
Lower strength	Higher strength and durability
Higher permeability	Lower permeability
Loss to natural ecosystems	Eco friendly as it does not allow destruction of riverbeds.

Sand is classified as a "minor mineral", under The Mines and Minerals (Development and Regulations) Act, 1957 (MMDR Act) and administrative control over minor minerals vests with the State Governments, and accordingly, regulated through State specific rules. Section 23C of the MMDR Act empowers the State Governments to make rules for preventing illegal mining, transportation and storage of minerals and for the purposes connected therewith. Hence, control of illegal mining comes under the legislative and administrative purview of the State Governments. Sand Mining Framework (2018) prepared by the Ministry of Mines envisages alternative sources of sand in the form of Manufactured Sand (M-Sand) from crushed rock fines (crusher dust), sand from Overburden (OB) of coal mines.